



DAL MEL Group

717 Minimum Equipment List

Document Reference: DAL/MEL/1001

Revision: 21.0.0 Date: 18-Feb-2026

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FM.2 Approval Certificate



Approval Certificate

Approval Reference:	Revision 21.0.0 complies with FAA B717 MMEL revision 11, dated 10/09/2017.
Approval Date:	18-Feb-2026
Approval Authority:	FAA
Approved by:	Donald Gaines
Approver Title:	Principal Operations Inspector
Document Title:	717 Minimum Equipment List
Document Reference:	DAL/MEL/1001
Revision:	21.0.0
Revision Date:	18-Feb-2026

FM.3 List of Revisions

Revision Date	Description of Revision	Revision	Publication Date
12-May-2022	Initial Aerodocs Publish.	14.0.0	06-Dec-2022
12-May-2022	Updated 40-40-05 Appendix A links.	14.0.1	22-Feb-2023
24-Apr-2023	Misc. internal requested changes.	15.0.0	24-Apr-2023
24-Apr-2023	Changed target of AMM ref to cloud location.	15.0.1	31-May-2023
24-Apr-2023	Added DAR per Flight Control and Delta POI to support Canadian ADS-B regulatory compliance.	15.0.2	09-Aug-2023
29-Nov-2023	Internal requests.	16.0.0	29-Nov-2023
23-May-2024	Internal requests.	17.0.0	23-May-2024
23-May-2024	Item suspension.	17.0.1	23-May-2024
19-Feb-2025	Internal edit requests.	18.0.0	19-Feb-2025
21-Mar-2025	Internal requests.	19.0.0	21-Mar-2025
07-May-2025	Revising TCAS Only.	20.0.0	07-May-2025
18-Feb-2026	Internal requests.	21.0.0	18-Feb-2026

FM.4 Revision Highlights

Section Number	Description of Change
0.2.1 RNAV / ADS-B Minimum Requirements	24-21-01A removed 1 row in Nav Charts as per Flight Ops.
0.2.2 Low Weather Minimum Requirements	22-22-01 Added Item to Nav Table as per Flight Ops.
22-10-01 Autopilot and Mode Controls	22-10-01L removed DAR as per Dispatch.
22-10-01 Autopilot and Mode Controls	22-10-01M removed DAR as per Dispatch.
22-10-01 Autopilot and Mode Controls	22-10-01A Added (M) procedure link as per Tech Ops Engineering.
(M) PROCEDURES	22-10-01A changed "Maintenance Note" to a Maintenance Procedure (M) as per Tech Ops Engineering.
22-22-01 Automatic Pitch Trim	
22-22-01 Automatic Pitch Trim	22-22-01 added ref link to Nav Charts and DAR as per Flight Ops and Dispatch.
23-12-01 Very High Frequency (VHF) Communication System	23-12-01A Removed DAR as per Dispatch.
23-22-01 Selective Call System (SELCAL)	23-22-01A Removed DAR as per Dispatch.
24-21-01 AC Electrical Power Generation and Control	24-21-01A added link the Theater Restrictions and removed link to Nav Charts as per Flight Ops.
24-21-02 PCDU Generator Control Units (GCU)	24-21-02B Removed DAR as per Dispatch.
(O) PROCEDURES	24-22-01 added new step for FL370 as per Dispatch and Flt Ops.
(O) PROCEDURES	24-22-02 added new step for FL370 as per Dispatch and Flt Ops.
25-99-01 PED Mount	25-99-01B Removed DAR as per Dispatch.
25-99-01 PED Mount	25-99-01C Removed DAR as per Dispatch.
26-21-03 Firex Bottle Discharge Cartridges	26-21-03B added link to 49-10-01B MEL as per Dispatch.
(M) PROCEDURES	27-12-01 re-ordered (M) procedure steps and edited same. Changed format of AMM references and added NOTES for each reference as per Tech Ops Engineering and Dispatch.
(M) PROCEDURES	27-12-01 added graphic as per Tech Ops Engineering.
(M) PROCEDURES	27-24-01 added Prior to Each Dep to match Remarks box.
27-43-01 Alternate Longitudinal Trim Motor Control Relays (Up Trim/Down Trim)	27-43-01 Removed DAR as per Dispatch.
(O) PROCEDURES	27-54-01 Deleted "Operations Note", added step to (O) procedure as per Flight Ops and Performance Engineering.
(M) PROCEDURES	27-54-01 added NOTE to (M) procedure as per Flight Ops and Performance Engineering.
(O) PROCEDURES	27-62-01 Added new step 3. to (O) proc as per Flight Ops.
(O) PROCEDURES	27-67-01 added steps to the (O) procedures as per Flight Ops and Performance Engineering.
28-21-01 Single Point Pressure Refueling System	28-21-01A Removed DAR as per Dispatch.

Section Number	Description of Change
28-21-01 Single Point Pressure Refueling System	28-21-01B Removed DAR as per Dispatch.
28-21-01 Single Point Pressure Refueling System	28-21-01C Removed DAR as per Dispatch.
28-21-01 Single Point Pressure Refueling System	28-21-01D Removed DAR as per Dispatch.
(O) PROCEDURES	28-41-03 changed reference for fueling procedures to the FOM as per Flight Ops.
(O) PROCEDURES	29-22-01 Added (O) Note as per Flight Ops.
(O) PROCEDURES	29-22-01 added (O) procedure step regarding starting Left Eng first as per Flight Ops.
30-72-01 Galley Drain Mast Heaters	30-72-01A Removed DAR as per Dispatch.
30-72-01 Galley Drain Mast Heaters	30-72-01B Removed DAR as per Dispatch.
32-61-02 Proximity Sensing System (PSS) Functions	32-61-02C Removed DAR as per Dispatch.
(O) PROCEDURES	32-61-02A changed the ALT ON call out to XPNDR as per Flight Ops.
(M) PROCEDURES	33-41-01A/B/C/D/E/F updated (M) procedure as per Tech Ops Engineering and Flight Ops.
34-29-01 Non-Stabilized Magnetic Compass (Standby)	34-29-01 Removed DAR as per Dispatch.
(O) PROCEDURES	34-32-01B added Prior to Each Dep to match Remarks box.
34-57-01 ADS-B Out Extended Squitter Transmissions	34-57-01A Removed DAR as per Dispatch.
(O) PROCEDURES	34-63-01A added to (O) procedures as per MCC and Flight Ops.
38-30-01 Potable Water Systems	38-30-01A Removed DAR as per Dispatch.
38-30-02 Lavatory Systems	38-30-02A Removed DAR as per Dispatch.
45-45-01 Central Fault Display Interface Unit (CFDIU)	45-45-01 changed FCMP from a Y to N due to AMM (M) procedure needing to be performed.
46-11-01 Electronic Flight Bag (EFB) System	46-11-01A/B/C Removed DAR as per Dispatch.
49-17-01 APU Air Inlet Door Actuator System	49-17-01B Removed DAR as per Dispatch.
49-44-01 APU Starter Relay	49-44-01 Removed DAR as per Dispatch.
49-51-01 APU Pneumatic Check Valve	49-51-01A Removed DAR as per Dispatch.
49-51-01 APU Pneumatic Check Valve	49-51-01B Added (O) proc link as per Flt Ops.
(O) PROCEDURES	49-51-01B Added (O) proc as per Flt Ops.
49-72-01 APU EGT Probes	49-72-01A Removed DAR as per Dispatch.
(O) PROCEDURES	52-11-01 corrected typo.
(O) PROCEDURES	73-33-01 added Prior to Each Dep to match Remarks box.

FM.5 List of Effective Sections

0 – Introduction / Navigation Equipment / EIS Messages		
Section	Status N=New R=Revised	Revision
0 – Introduction / Navigation Equipment / EIS Messages	R	18-Feb-2026
0.1 – TOPP 40-40-05, Appendix A - MEL/CDL Instructions		07-May-2025
0.2 – Navigation Equipment	R	18-Feb-2026
0.2.1 – RNAV / ADS-B Minimum Requirements	R	18-Feb-2026
0.2.2 – Low Weather Minimum Requirements	R	18-Feb-2026
0.3 – EIS Message List		07-May-2025

11 – Placards		
Section	Status N=New R=Revised	Revision
11 – Placards		07-May-2025
11-00-01 – Aircraft Placards/Markings Repair Installed		07-May-2025

21 – Air Conditioning		
Section	Status N=New R=Revised	Revision
21 – Air Conditioning		07-May-2025
21-21-01 – Radio Rack Cooling Fans		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
21-21-02 – Radio Rack & Standby Radio Rack Fan Check Valves		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
21-22-03 – Ram / Ground Conditioned Air Check Valves		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
21-22-04 – Gasper Fan		07-May-2025
21-22-05 – Conditioned Air Check Valves		07-May-2025
21-22-06 – Cold Air Check Valves		07-May-2025
21-26-01 – Cockpit Instrument Cooling Fan		07-May-2025
(O) – PROCEDURES		07-May-2025
21-29-01 – Ram Air Valve		07-May-2025
21-30-01 – Cabin Pressure Control Systems (Auto And Manual)		07-May-2025
(O) – PROCEDURES		07-May-2025

21 – Air Conditioning		
Section	Status N=New R=Revised	Revision
(M) – PROCEDURES		07-May-2025
21-34-02 – Cabin Pressure Safety Valves		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
21-42-02 – Forward Cargo Compartment Heating System		07-May-2025
(M) – PROCEDURES		07-May-2025
21-51-01 – Air Conditioning Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
21-51-02 – Air Conditioning Flow Control Valve		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
21-52-01 – Pack Flow Indication Systems		07-May-2025
21-53-02 – Compressor Discharge and Turbine Inlet Thermal Switches		07-May-2025
(M) – PROCEDURES		07-May-2025
21-53-03 – Air Conditioning System Shutdown Thermal Switches (190 Degrees F)		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
21-54-01 – Air Conditioning Water Separators		07-May-2025
(O) – PROCEDURES		07-May-2025
21-54-02 – Water Separator 35 Degrees F Thermostats		07-May-2025
(O) – PROCEDURES		07-May-2025
21-54-03 – Water Separator 35 Degrees F Control Valves		07-May-2025
(O) – PROCEDURES		07-May-2025
21-56-01 – Air Conditioning Automatic Shut-Off System		07-May-2025
(O) – PROCEDURES		07-May-2025
21-62-01 – Cockpit / Cabin Compartment Auto Temperature Control Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
21-62-02 – Cockpit / Cabin Compartment Temperature Control Valves		07-May-2025
(O) – PROCEDURES		07-May-2025
21-63-01 – Temperature Control Valve Position Indicating Systems		07-May-2025
21-64-01 – Cockpit/Cabin Temperature Indicating Systems		07-May-2025
(O) – PROCEDURES		07-May-2025

22 – Autoflight		
Section	Status N=New R=Revised	Revision
22 – Autoflight	R	18-Feb-2026
22-00-00 – LLM Autoland Status (Lower Landing Minimums)		07-May-2025
(M) – PROCEDURES		07-May-2025
22-10-01 – Autopilot and Mode Controls	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
(M) – PROCEDURES	N	18-Feb-2026
22-11-01 – Elevator Drive Servo		07-May-2025
22-12-01 – Aileron Drive Servo		07-May-2025
22-13-01 – Rudder Drive Servo		07-May-2025
22-14-01 – Autopilot Control Wheel Disengage Switches		07-May-2025
22-15-01 – Flight Control Panel Speed Mode Controls	R	18-Feb-2026
22-15-02 – Flight Control Panel Heading Mode Control		07-May-2025
22-15-03 – Flight Control Panel Altitude Mode Controls	R	18-Feb-2026
22-15-04 – Flight Control Panel Pitch Mode Controls		07-May-2025
22-22-01 – Automatic Pitch Trim	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
22-22-02 – Mach Trim Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
22-23-01 – Yaw Damper Systems		07-May-2025
(M) – PROCEDURES		07-May-2025
22-30-01 – Autothrottle Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
22-31-01 – Autothrottle Disengage Switches		07-May-2025
22-31-02 – Go-Around Palm Switches		07-May-2025

23 – Communications		
Section	Status N=New R=Revised	Revision
23 – Communications	R	18-Feb-2026
23-12-01 – Very High Frequency (VHF) Communication System	R	18-Feb-2026
23-22-01 – Selective Call System (SELCAL)	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
23-24-01 – ARINC / ACARS Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
23-31-01 – Passenger Address System (PA)		07-May-2025
(O) – PROCEDURES		07-May-2025

23 – Communications		
Section	Status N=New R=Revised	Revision
23-32-01 – Pre-Recorded Passenger Announcement System		07-May-2025
(O) – PROCEDURES		07-May-2025
23-41-01 – Service Interphone System		07-May-2025
(O) – PROCEDURES		07-May-2025
23-41-03 – Maintenance Interphone		07-May-2025
23-51-03 – Audio Control Panel Selector Switchlights		07-May-2025
(O) – PROCEDURES		07-May-2025
23-51-04 – Cockpit Communications Loudspeakers		07-May-2025
23-51-05 – Cockpit Microphones / Headsets		07-May-2025
23-51-06 – Oxygen (Smoke) Mask Microphones		07-May-2025
23-51-09 – Captain / First Officer Push-To-Talk Switches		07-May-2025
(M) – PROCEDURES		07-May-2025
23-71-01 – Cockpit Voice Recorder (CVR) System		07-May-2025

24 – Electrical Power		
Section	Status N=New R=Revised	Revision
24 – Electrical Power	R	18-Feb-2026
24-21-01 – AC Electrical Power Generation and Control	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
(M) – PROCEDURES		07-May-2025
24-21-02 – PCDU Generator Control Units (GCU)	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
24-22-01 – Right Generator Relay (RGR)	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
24-22-02 – Left Generator Relay (LGR)	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
24-22-03 – Auxiliary Power Relay (APR)		07-May-2025
24-24-01 – EMER PWR ON Light (Overhead Panel)		07-May-2025
(O) – PROCEDURES		07-May-2025
24-26-01 – AC Voltage Indicating System		07-May-2025
24-26-02 – AC Frequency Indicating System		07-May-2025
24-26-03 – AC Load Sensing Circuit (Including Loadmeter Indication)		07-May-2025
(O) – PROCEDURES		07-May-2025
24-31-01 – PCDU Transformer/ Rectifiers (TR) Units		07-May-2025

24 – Electrical Power		
Section	Status N=New R=Revised	Revision
(O) – PROCEDURES		07-May-2025
24-41-01 – External Power System		07-May-2025
(M) – PROCEDURES		07-May-2025
24-41-02 – Main External Power Relay (MEPR)		07-May-2025
(M) – PROCEDURES		07-May-2025
24-42-01 – EXT PWR NOT IN USE Light (White)		07-May-2025
24-42-02 – EXT PWR AVAIL Lights		07-May-2025
24-51-02 – EXT Light (Blue)		07-May-2025
24-51-03 – APU Power ON Light		07-May-2025
(O) – PROCEDURES		07-May-2025
24-51-04 – APU Lights (Blue)		07-May-2025
24-51-05 – L/R GEN Lights		07-May-2025
(O) – PROCEDURES		07-May-2025
24-54-01 – Galley Power and Control Systems		07-May-2025
(M) – PROCEDURES		07-May-2025

25 – Equipment / Furnishings		
Section	Status N=New R=Revised	Revision
25 – Equipment / Furnishings	R	18-Feb-2026
25-15-01 – Observer Seat (Including Associated Equipment)		07-May-2025
25-15-02 – Flight Crew Seats		07-May-2025
(M) – PROCEDURES		07-May-2025
25-22-01 – "Fasten Seat Belt While Seated" Signs or Placards (Unlighted)		07-May-2025
25-23-01 – Flight Attendant Seats		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
25-23-03 – Passenger Seats		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
25-25-01 – Storage Bin(s) / Cabin, Galley, and Lavatory Storage Compartments / Closets		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
25-29-01 – Nonessential Equipment and Furnishings (NEF)		07-May-2025
25-36-01 – Galley/Cabin/Lavatory Waste Receptacles Access Doors/Covers		07-May-2025

25 – Equipment / Furnishings		
Section	Status N=New R=Revised	Revision
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
25-42-01 – Exterior Lavatory Door Ashtrays		07-May-2025
25-50-01 – Lower Cargo Compartment Liners		07-May-2025
25-51-02 – Forward Cargo Compartment Thermal Barrier		07-May-2025
25-61-01 – Ventral Exit Door Strap		07-May-2025
25-61-02 – Cabin Emergency Flashlights / Holders		07-May-2025
(O) – PROCEDURES		07-May-2025
25-62-01 – Emergency Medical Equipment		07-May-2025
(O) – PROCEDURES		07-May-2025
25-63-01 – Megaphones		07-May-2025
(O) – PROCEDURES		07-May-2025
25-64-01 – Flotation Devices (Crew and Passengers)	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
25-67-02 – Emergency Locator Transmitter (ELT) Survival Type		07-May-2025
25-99-01 – PED Mount	R	18-Feb-2026

26 – Fire Protection		
Section	Status N=New R=Revised	Revision
26 – Fire Protection	R	18-Feb-2026
26-11-01 – Engine Fire Detection Systems (Detection Loops)		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
26-12-01 – APU Fire Detection System		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
26-12-02 – APU Fire Warning Horn (External Warning)		07-May-2025
(O) – PROCEDURES		07-May-2025
26-12-03 – APU Fire Detection Control Unit		07-May-2025
26-13-01 – Lower Cargo Compartment Smoke Detection System		07-May-2025
(O) – PROCEDURES		07-May-2025
26-14-01 – Lavatory Smoke Detection and Warning Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
26-21-01 – Engine and APU Fire Extinguisher Agent Low Lights		07-May-2025

26 – Fire Protection		
Section	Status N=New R=Revised	Revision
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
26-21-02 – APU Fire Shutoff Controls		07-May-2025
(O) – PROCEDURES		07-May-2025
26-21-03 – Firex Bottle Discharge Cartridges	R	18-Feb-2026
(M) – PROCEDURES		07-May-2025
26-22-01 – Lower Cargo Compartment Fire Suppression System		07-May-2025
(O) – PROCEDURES		07-May-2025
26-25-01 – Lavatory Fire Extinguisher Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
26-26-01 – Portable Fire Extinguishers		07-May-2025

27 – Flight Controls		
Section	Status N=New R=Revised	Revision
27 – Flight Controls	R	18-Feb-2026
27-12-01 – Aileron Trim System	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES	R	18-Feb-2026
27-14-01 – Aileron Position Indication System		07-May-2025
27-15-01 – Aileron Dampers		07-May-2025
27-22-01 – Rudder Trim Position Indicating System		07-May-2025
(M) – PROCEDURES		07-May-2025
27-22-02 – Rudder Center Switch		07-May-2025
27-24-01 – Rudder Position Indication System	R	18-Feb-2026
(M) – PROCEDURES	R	18-Feb-2026
27-25-01 – Rudder Dampers		07-May-2025
27-26-01 – Rudder Limiter Systems (Primary and Secondary)		07-May-2025
27-30-01 – Control Column Disconnect Annunciation System		07-May-2025
(O) – PROCEDURES		07-May-2025
27-34-01 – Elevator Position Indicating System		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
27-38-01 – Stall Protection Systems Stick Shaker Motors		07-May-2025
(M) – PROCEDURES		07-May-2025

27 – Flight Controls		
Section	Status N=New R=Revised	Revision
27-43-01 – Alternate Longitudinal Trim Motor Control Relays (Up Trim/Down Trim)	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
27-43-02 – Alternate Longitudinal Trim Motor Up/Down Limit Switches		07-May-2025
(M) – PROCEDURES		07-May-2025
27-51-01 – Dial-A-Flap System		07-May-2025
27-54-01 – Flap Position Indicating System	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
(M) – PROCEDURES	R	18-Feb-2026
27-61-01 – Spoiler Electronic Control Unit (SECU)		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
27-62-01 – Auto Ground Spoiler System	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
(M) – PROCEDURES		07-May-2025
27-67-01 – Flight Spoilers Actuators	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
(M) – PROCEDURES		07-May-2025

28 – Fuel		
Section	Status N=New R=Revised	Revision
28 – Fuel	R	18-Feb-2026
28-21-01 – Single Point Pressure Refueling System	R	18-Feb-2026
(M) – PROCEDURES		07-May-2025
28-21-03 – Ground Refueling Relay		07-May-2025
28-22-01 – Main Tank Fuel AC Boost Pumps		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
28-22-02 – Center Tank Fuel AC Boost Pumps		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
28-22-03 – DC Start Pump		07-May-2025
(O) – PROCEDURES		07-May-2025
28-22-04 – Crossfeed Valve Position Synoptic Indication		07-May-2025

28 – Fuel		
Section	Status N=New R=Revised	Revision
(O) – PROCEDURES		07-May-2025
28-26-01 – Fuel Return-To-Tank System (RTT)		07-May-2025
(O) – PROCEDURES		07-May-2025
28-41-01 – Cockpit Fuel Quantity Indicating System		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
28-41-02 – Gross Weight / Total Fuel Indication System		07-May-2025
(O) – PROCEDURES		07-May-2025
28-41-03 – Fueling Panel Fuel Quantity Indicators	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
28-42-01 – Dripless Fuel Measuring Sticks (Dipsticks)	R	18-Feb-2026
(M) – PROCEDURES	R	18-Feb-2026
28-43-01 – Main Tank Fuel Level Low Float Switches		07-May-2025
28-44-01 – Pump Pressure Low Annunciation Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
28-45-01 – Fuel Tank Temperature Indicating System		07-May-2025
(O) – PROCEDURES		07-May-2025

29 – Hydraulic Power		
Section	Status N=New R=Revised	Revision
29 – Hydraulic Power	R	18-Feb-2026
29-11-01 – Right Engine Driven Hydraulic Pump		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
29-21-01 – Electric Auxiliary Hydraulic Pump		07-May-2025
(O) – PROCEDURES		07-May-2025
29-22-01 – Hydraulic Power Transfer Unit (PTU)	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
29-31-01 – Right Hydraulic System Pressure Low Annunciation System		07-May-2025
(O) – PROCEDURES		07-May-2025
29-31-02 – Hydraulic Pump Pressure Synoptic Indication Amber Pinwheel (Prior to Engine Start)		07-May-2025
(O) – PROCEDURES		07-May-2025
29-32-01 – Hydraulic Quantity Indicating Systems		07-May-2025
(M) – PROCEDURES		07-May-2025

30 – Ice and Rain Protection		
Section	Status N=New R=Revised	Revision
30 – Ice and Rain Protection	R	18-Feb-2026
30-11-01 – Wing Anti-Ice Pressure Regulator / Shutoff Valve		07-May-2025
30-12-01 – Tail Anti-Ice Pressure Regulator / Shutoff Valve		07-May-2025
30-22-01 – Inlet Cowl Anti-Ice Regulator / Shutoff Valves		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
30-31-01 – Pitot Heaters		07-May-2025
30-32-01 – Static Port Heaters		07-May-2025
30-33-01 – Ram Air Temp Probe Heater		07-May-2025
30-34-01 – Stall Warning Vane Heaters		07-May-2025
30-41-01 – Windshield Heat System		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
30-43-01 – Windshield Wiper System		07-May-2025
30-72-01 – Galley Drain Mast Heaters	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
30-74-01 – Water Service Panel Heater		07-May-2025
30-81-01 – Overwing Ice Detection System		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025

31 – Indicating / Recording System		
Section	Status N=New R=Revised	Revision
31 – Indicating / Recording System		07-May-2025
31-31-01 – Flight Data Recorder (FDR) System		07-May-2025
31-31-02 – Aircraft Condition Monitoring System (ACMS)		07-May-2025
(O) – PROCEDURES		07-May-2025
31-51-01 – Central Aural Warning System (CAWS)		07-May-2025
(O) – PROCEDURES		07-May-2025
31-61-01 – Electronic Instrument System (EIS)		07-May-2025
(O) – PROCEDURES		07-May-2025
31-61-02 – Versatile Integrated Avionics (VIA) Units		07-May-2025

32 – Landing Gear		
Section	Status N=New R=Revised	Revision
32 – Landing Gear	R	18-Feb-2026
32-43-01 – PARKING BRAKES ON Annunciation System		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
32-43-03 – Brake Supply Pressure Indication System		07-May-2025
32-45-01 – Anti-Skid System		07-May-2025
(O) – PROCEDURES		07-May-2025
32-45-02 – Anti-Skid Brake Return Line Shutoff Valves		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
32-47-01 – Brake Temperature Monitoring System		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
32-61-01 – Landing Gear Position Indicating Systems		07-May-2025
32-61-02 – Proximity Sensing System (PSS) Functions	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
(M) – PROCEDURES		07-May-2025

33 – Lights		
Section	Status N=New R=Revised	Revision
33 – Lights	R	18-Feb-2026
33-12-01 – Master Warning Lights		07-May-2025
33-12-02 – Master Caution Lights		07-May-2025
33-13-01 – Cockpit Flight Deck / Flight Compartment and Instrument Lighting Systems		07-May-2025
33-20-01 – Cabin/Handrail Illumination		07-May-2025
33-28-01 – Passenger Notice System (No Smoking/ Fasten Seat Belt/ Return To Cabin)	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
33-31-01 – Cargo Compartment Lighting Systems		07-May-2025
33-32-02 – Maintenance Lights		07-May-2025
33-41-01 – Landing Lights	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES	R	18-Feb-2026
33-42-01 – Anti-Collision Beacon Lights (Fuselage-Red)		07-May-2025

33 – Lights		
Section	Status N=New R=Revised	Revision
33-42-02 – Fixed Forward Position Light Bulbs (Wing Tips)		07-May-2025
33-42-03 – Fixed Aft Position Light Bulbs, (Wing Tips, White)		07-May-2025
(M) – PROCEDURES		07-May-2025
33-42-04 – White Strobe / Anti-Collision Lights, (Forward and Aft Wing Tips)		07-May-2025
(M) – PROCEDURES		07-May-2025
33-43-01 – Wing and Nacelle Illumination Light Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
33-43-02 – Ground Floodlights		07-May-2025
33-44-01 – Tail Illumination (Logo) Lights		07-May-2025
33-51-01 – Emergency Escape Path Marking System Lights		07-May-2025
(O) – PROCEDURES		07-May-2025
33-51-02 – Exterior Emergency Illumination System		07-May-2025
(O) – PROCEDURES		07-May-2025
33-52-01 – Cabin Standby Lighting System		07-May-2025

34 – Navigation		
Section	Status N=New R=Revised	Revision
34 – Navigation	R	18-Feb-2026
34-00-00 – LLM Category II/III Status (Lower Landing Minimums)		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
34-24-01 – Flight Director Systems		07-May-2025
34-25-01 – Navigation		07-May-2025
(O) – PROCEDURES		07-May-2025
34-28-01 – Integrated Standby Instrument System (ISIS)		07-May-2025
34-29-01 – Non-Stabilized Magnetic Compass (Standby)	R	18-Feb-2026
34-31-01 – Marker Beacon System		07-May-2025
34-32-01 – Multi-Mode Receiver (MMR)	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
34-41-01 – Weather Radar Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
34-42-01 – Radio Altimeter Systems		07-May-2025
(M) – PROCEDURES		07-May-2025
34-43-01 – IRS NAV OFF Lights		07-May-2025
(O) – PROCEDURES		07-May-2025

34 – Navigation		
Section	Status N=New R=Revised	Revision
34-45-01 – Ground Proximity Warning System (GPWS)		07-May-2025
(O) – PROCEDURES		07-May-2025
34-46-01 – Traffic Alert / Collision Avoidance System (TCAS)		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
34-47-01 – Windshear Detection and Flight Guidance Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
34-51-01 – VOR Navigation Systems		07-May-2025
34-52-01 – Distance Measuring Equipment (DME) Systems		07-May-2025
34-54-01 – ATC Transponder and Automatic Altitude Reporting Systems	R	18-Feb-2026
34-57-01 – ADS-B Out Extended Squitter Transmissions	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
34-63-01 – VIA Flight Management Systems	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
(M) – PROCEDURES		07-May-2025

35 – Oxygen		
Section	Status N=New R=Revised	Revision
35 – Oxygen	R	18-Feb-2026
35-11-01 – Crew Oxygen System		07-May-2025
(O) – PROCEDURES		07-May-2025
35-21-01 – Passenger Oxygen System	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
35-22-01 – No Mask Annunciation System		07-May-2025
35-31-01 – Portable Oxygen Dispensing Units (Bottle and Mask)		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
35-33-01 – Protective Breathing Equipment (PBE) Seals (Smoke Hood Red Rope Seals)		07-May-2025
(M) – PROCEDURES		07-May-2025

36 – Pneumatic		
Section	Status N=New R=Revised	Revision
36 – Pneumatic		07-May-2025
36-11-01 – Fan Air Valve		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
36-11-03 – High Stage Valve		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
36-11-04 – High Stage Pilot Valve		07-May-2025
(M) – PROCEDURES		07-May-2025
36-13-01 – Pressure Regulator and Shutoff Valve (PRSOV)		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
36-13-02 – Pressure Regulator and Shutoff Pilot Valve		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
36-13-03 – Pneumatic System Controller (PSC) Channels		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
36-13-04 – Intermediate Bleed Pressure Sensor (PI)		07-May-2025
(M) – PROCEDURES		07-May-2025
36-13-05 – Manifold Pressure Sensor (PM)		07-May-2025
(O) – PROCEDURES		07-May-2025
36-13-06 – Intermediate Bleed Temperature Sensor (TI)		07-May-2025
36-13-07 – Manifold Temperature Sensor (TM)		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
36-13-08 – Supply Temperature Sensor (TS)		07-May-2025
(M) – PROCEDURES		07-May-2025
36-14-01 – Ground Connection and Pneumatic Check Valve		07-May-2025
(O) – PROCEDURES		07-May-2025
36-15-01 – Manifold Isolation Valve		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
36-15-02 – Manifold Check Valve		07-May-2025
36-23-01 – Pneumatic Overheat Detection System (PODS)		07-May-2025

38 – Water / Waste		
Section	Status N=New R=Revised	Revision
38 – Water / Waste	R	18-Feb-2026
38-30-01 – Potable Water Systems	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
38-30-02 – Lavatory Systems	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
38-33-01 – Vacuum Waste Logic Control Module		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025

45 – Onboard Maintenance Systems (OMS)		
Section	Status N=New R=Revised	Revision
45 – Onboard Maintenance Systems (OMS)	R	18-Feb-2026
45-45-01 – Central Fault Display Interface Unit (CFDIU)	R	18-Feb-2026
(M) – PROCEDURES		07-May-2025

46 – Information Systems		
Section	Status N=New R=Revised	Revision
46 – Information Systems	R	18-Feb-2026
46-11-01 – Electronic Flight Bag (EFB) System	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025

49 – Airborne Auxiliary Power (APU)		
Section	Status N=New R=Revised	Revision
49 – Airborne Auxiliary Power (APU)	R	18-Feb-2026
49-10-01 – Auxiliary Power Unit (APU)		07-May-2025
(M) – PROCEDURES		07-May-2025
49-11-01 – APU Enclosure and Access Door Acoustic Blankets		07-May-2025
(M) – PROCEDURES		07-May-2025
49-17-01 – APU Air Inlet Door Actuator System	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025

49 – Airborne Auxiliary Power (APU)		
Section	Status N=New R=Revised	Revision
(M) – PROCEDURES		07-May-2025
49-44-01 – APU Starter Relay	R	18-Feb-2026
(M) – PROCEDURES		07-May-2025
49-51-01 – APU Pneumatic Check Valve	R	18-Feb-2026
(O) – PROCEDURES	N	18-Feb-2026
(M) – PROCEDURES		07-May-2025
49-51-02 – APU Bleed Air Modulating Valve	R	18-Feb-2026
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
49-72-01 – APU EGT Probes	R	18-Feb-2026
(M) – PROCEDURES		07-May-2025
49-75-01 – APU Data Memory Module (DMM)		07-May-2025

52 – Doors		
Section	Status N=New R=Revised	Revision
52 – Doors	R	18-Feb-2026
52-11-01 – Door Stop Screws	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
52-54-04 – Enhanced Security Flight Deck Door Automatic Locking System		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
52-54-05 – Boeing / TIMCO Enhanced Security Flight Deck Door Dead Bolt		07-May-2025
52-70-01 – Door Warning Annunciation Systems		07-May-2025
(O) – PROCEDURES		07-May-2025

53 – Fuselage		
Section	Status N=New R=Revised	Revision
53 – Fuselage		07-May-2025
53-00-00 – RVSM Status		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
53-52-01 – Tailcone Unsafe Annunciation System		07-May-2025
(M) – PROCEDURES		07-May-2025

73 – Engine Fuel and Control		
Section	Status N=New R=Revised	Revision
73 – Engine Fuel and Control	R	18-Feb-2026
73-21-01 – Electronic Engine Control Systems (EEC) Short Term Dispatch (STD) Fault		07-May-2025
(M) – PROCEDURES		07-May-2025
73-21-02 – EEC Normal (EPR) Mode		07-May-2025
(O) – PROCEDURES		07-May-2025
73-21-03 – FADEC MODE Switchlights		07-May-2025
73-21-16 – Electronic Engine Control (EEC) Cooling Fan		07-May-2025
73-31-01 – Fuel Flow / Fuel Used Indication Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
73-33-01 – ENG L/R FUEL PRES Annunciation System	R	18-Feb-2026
(O) – PROCEDURES	R	18-Feb-2026
73-34-01 – L/R FUEL FIL PRES DROP Annunciation Systems		07-May-2025
(M) – PROCEDURES		07-May-2025

74 – Engine Ignition		
Section	Status N=New R=Revised	Revision
74 – Engine Ignition		07-May-2025
74-20-01 – Ignition Systems		07-May-2025
(O) – PROCEDURES		07-May-2025

77 – Engine Indicating		
Section	Status N=New R=Revised	Revision
77 – Engine Indicating		07-May-2025
77-32-01 – Engine Vibration Monitoring Systems		07-May-2025

78 – Engine Exhaust		
Section	Status N=New R=Revised	Revision
78 – Engine Exhaust		07-May-2025
78-32-01 – Thrust Reverser Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
78-33-01 – Thrust Reverser System Accumulators		07-May-2025

78 – Engine Exhaust		
Section	Status N=New R=Revised	Revision
(M) – PROCEDURES		07-May-2025
78-33-02 – Thrust Reverser Linear Variable Transformers (LVT)		07-May-2025
78-33-03 – Thrust Reverser Proximity Sensors		07-May-2025
78-33-04 – Maintenance Test Enable Switch (MTES)		07-May-2025

79 – Engine Oil		
Section	Status N=New R=Revised	Revision
79 – Engine Oil		07-May-2025
79-31-01 – Engine Oil Quantity Indicating Systems		07-May-2025
(O) – PROCEDURES		07-May-2025
(M) – PROCEDURES		07-May-2025
79-36-01 – ENG L/R OIL FILTER Annunciation Systems		07-May-2025
(M) – PROCEDURES		07-May-2025

80 – Engine Starting		
Section	Status N=New R=Revised	Revision
80 – Engine Starting		07-May-2025
80-11-01 – START VALVE L/R OPEN Annunciation Systems		07-May-2025
(M) – PROCEDURES		07-May-2025
80-11-02 – Starter Shutoff System		07-May-2025
(M) – PROCEDURES		07-May-2025
80-11-03 – Auto Starter Control System		07-May-2025
(O) – PROCEDURES		07-May-2025
80-11-04 – L/R Engine Start Switch Internal Switchlight		07-May-2025

0 Introduction / Navigation Equipment / EIS Messages

- 0.1** **TOPP 40-40-05, Appendix A - MEL/CDL Instructions**

- 0.2** **Navigation Equipment**
 - 0.2.1** **RNAV / ADS-B Minimum Requirements**
 - 0.2.2** **Low Weather Minimum Requirements**

- 0.3** **EIS Message List**

0 Introduction / Navigation Equipment / EIS Messages

0.1 TOPP 40-40-05, Appendix A - MEL/CDL Instructions

Access the MEL Instructions (TOPP 40-40-05, Appendix A) via the following links:

- If you are viewing this manual in the *Comply365 application*, click [here](#).
- If you are viewing this manual on *Deltanet*, click [here](#)

0.2 Navigation Equipment

0.2.1 RNAV / ADS-B Minimum Requirements

The chart below should be used when making MEL compliance or aircraft equipage decisions prior to the dispatch event. Equipment failures that occur after that event will be referenced in the appropriate AOM. The values referenced below are not intended to reduce the minimum values published on any approach chart or reduce any crew qualification restriction.

To use the chart, locate the MEL number for the system on deferral. Read the columns to the right of the appropriate MEL number to determine the number of systems required for the type of operation under consideration.

MEL Item on Deferral	Number Required				
	ADS-B ⁵	RNAV ENROUTE	RNAV SID/STAR	RNAV (GPS), GPS or VOR using Prof/Nav	RNAV (RNP)
22-10-01: Autopilot ³	0	1	1	1	
AND/OR	OR	OR	OR	AND	N/A
34-24-01: Flight Director Systems	0	1	1	1	
22-10-01: PROF Mode	0	0	1	1	N/A
22-10-01: NAV Mode	0	1	1	1	N/A
22-15-02: FCP Nav Switch	0	0	1	1	N/A
22-15-04: FCP Prof Switch	0	0	1	1	N/A
22-30-01: Autothrottles	0	0	0	0	N/A
22-31-02: Go Around Switches	0	0	0	1	N/A
31-61-01: Nav Display	0	1	1	1	N/A
31-61-01: PFD	0	1	1	1	N/A
34-32-01: GPS	1	0	0	1	N/A
34-42-01: Radio Altimeter	0	1	1	1	N/A
34-45-01: EGPWS	0	0	0	0	N/A
34-52-01: DME	0	1 ⁴	1 ⁴	1 ⁴	N/A
34-54-01A ATC Transponder	1	0	0	0	N/A
34-57-01B ADS-B Out	1	0	0	0	N/A
34-63-01: FMS – MCDU	0	1	1	1	N/A
34-63-01: FMS - NAV Database	0	1 ²	1 ²	1 ²	N/A

1. APU may be used for 1 source.
2. Authorized with 1 or 0 if MEL 34-63-01C Navigation Database “O” procedure is successfully accomplished.
3. Applies to items 22-10-01A, -01B, -01C, -01D, -01E, -01F.
4. All B717 aircraft are equipped with two multi-channel scanning DME units.
5. If ATC authorization for flight without ADS-B Out Squitter Transmissions is obtained, it is not necessary to comply with the requirements of this chart.

0.2.2 Low Weather Minimum Requirements

APPROACH CATEGORIES

The following definitions should be used when making MEL compliance or aircraft equipage decisions prior to the dispatch event. Equipment failures that occur after that event will be referenced in the appropriate QRH (ILS Airborne Equipment Requirements). The values referenced below are not intended to reduce the minimum values published on any approach chart or reduce any crew qualification restriction.

Basic Turbojet Minimums:

A non-precision or precision approach to no less than RVR 4000 (1200m) or $\frac{3}{4}$ SM (1200m) visibility, and a DH of no less than 200 ft or MDH of no less than 250 ft.

Category I Non-ILS:

A non-precision approach to no less than RVR 2400 (550m) or $\frac{1}{2}$ SM (800m) visibility, and a MDH of no less than 250 ft.

Category I ILS:

A precision approach to no less than RVR 1800 (550m) or $\frac{1}{2}$ SM (800m) visibility, and a DH of no less than 200 ft.

Category II:

A precision approach to no less than RVR 1200 (350m) and a DH of no less than 100 ft.

A precision approach to no less than RVR 1000 (300m) and a DH of no less than 100 ft.

Note: "Refer to [Airways Manual](#), [Ops Specs](#), [IAP](#), [Ground Based Equipment](#)"

Category III:

A precision approach to no less than RVR 300 (75m) and an AH of no less than 50 ft.

Note: "Refer to [Airways Manual](#), [Ops Specs](#), [IAP](#), [Ground Based Equipment](#)"

LOW WEATHER MINIMUM APPROACHES: EQUIPMENT REQUIREMENTS

To use the chart, locate the MEL number for the system on MCO. Read the columns to the right of the appropriate MEL number to determine the number of systems required for the type of approach under consideration.

		Number Required for Approach						
MEL Item		Basic Turbojet Min.	CAT I Non-ILS	CAT I ILS	CAT II	CAT III	RVSM	PRM
22-10-01	Autopilot	0	1	1	2	2	1 ¹	0
	OR			OR	AND	AND		
34-24-01:	Flight Director Systems	0	0	1	2	2	0	0
22-11-01	Elevator Drive Servo	0	0	0	1	1	1	0
22-12-01	Aileron Drive Servo	0	0	0	1	1	1	0
22-13-01:	Rudder Drive Servo	0	0	0	1	1	1	0
22-14-01:	Autopilot Control Wheel Disengage Switch. ²	1	1	1	2	2	1	1
22-22-01	Automatic Pitch Trim	0	0	0	1	1	0	0
22-30-01:	Autothrottle Systems	0	0	0	1	1	0	0
22-31-01:	Autothrottle Disengage Switches	0	0	0	1	1	0	0
22-31-02:	Go-Around Palm Switches ³	0	0	0	1	1	0	0
27-14-01:	Aileron Position Indication System	0	0	0	1	1	0	0

CONTINUED ON NEXT PAGE

MEL Item	Number Required for Approach					RVSM	PRM
	Basic Turbojet Min.	CAT I Non-ILS	CAT I ILS	CAT II	CAT III		
27-24-01: Rudder Position Indication System	0	0	0	1	1	0	0
27-26-01: Rudder Limiter System (Primary and Secondary)	0	0	1	2	2	0	0
27-34-01: Elevator Postion Indication System	0	0	0	2	2	0	0
27-54-01: Flap Position Indication System	0	0	1	2	2	0	0
27-61-01: Spoiler Electronic Control Unit	0	0	0	1	1	0	0
27-67-01: Flight Spoiler Actuators	2	2	2	4	4	0	0
30-22-01B: Inlet Cowl Anti-Ice Regulator/Shutoff Valves	0	0	1	2	2	0	0
34-32-01: MMR ILS Function	0	4	1	2	2	0	1
34-42-01: Radio Altimeter Systems	0	0	1	2	2	0	0
34-63-01B: VIA Multi-Function Control Display Units (MCDU)	1	1	1	2	2	0	0
73-21-02: EEC Normal (EPR) Mode.	0	0	2	2	2	0	0

1. RVSM not authorized for MELs 22-10-01B, 22-10-01D and 22-10-01E.
2. At least one Flight Director is operative.
3. Flight Director Go-Around and Windshear guidance are not available with both switches inoperative.
4. Refer to the approach plate for systems that may be required.

0.3 EIS Message List

This section provides a quick reference between the electronic messages which are displayed on the Engine / Alert Display (EAD) or Systems Display (SD), and the 717 Minimum Equipment List. This cross-reference is for guidance only, and may not consider all possible combinations of system faults which might cause a message to be presented.

It should not be presumed that dispatch relief is automatically available for every message with a MEL Item cross-referenced. The provisions of each MEL item must always be complied with whenever that item is placarded inoperative, and this section alone should never be used as any type of authority to dispatch the aircraft.

EIS Alerting Function

The Electronic Instrument System (EIS) includes an alerting function which serves as the primary means of displaying airplane system indications and messages to the flight crew. EIS Messages are categorized into four levels, describes as follows:

<u>Level 3 – Emergency</u>	-- The highest priority Alert message, displayed in red at the top of the message list, preceded by a solid red triangle and boxed in red. Also accompanied by the red Master Warning Light and Tone, and possibly an aural CAWS alert (bell, siren, or voice). Immediate crew action is required.
<u>Level 2 – Abnormal</u>	-- Displayed in amber below any Level 3 messages, boxed in amber and accompanied by amber Master Caution Light and Tone. Immediate crew awareness and future crew action required.
<u>Level 1 – Abnormal/Advisory/Maintenance</u>	-- The lowest priority Alert message, displayed in amber below any Level 3 or Level 2 messages and indented one space to the right. Master Caution Light and Tone may or may not accompany a Level 1 message. Crew awareness and possible future crew action required.
<u>Level 0 – Advisory/Status</u>	-- Reminders of current state of certain normal conditions for crew awareness only. Displayed in cyan below all Level 1, 2 and 3 alerts.

Alerts are normally presented on the EAD, but may be intentionally inhibited by the VIA under certain conditions. All alerts except those classified as Maintenance Alerts can also be shown on the related Systems Display. There are seven different Systems Display presentations, and each is selectable with its respective cue light/switch. These are:

<u>System</u>	<u>Cue</u>
Air Systems	AIR
Configuration	CONFIG
Electrical	ELEC
Secondary Engine Display	ENG
Hydraulic	HYD
Fuel	FUEL
Miscellaneous	MISC

The Cue Light / Switches are located on the Systems Control Panel on the pedestal. When a Level 1, 2, or 3 alert appears on the EAD, pressing its respective Cue Light / Switch calls up the associated systems page and resets the Master Warning or Master Caution (if present). Each Systems Display presentation except MISC shows a schematic of that system, with space below for presentation of the alert and any applicable Consequence messages.

Cross-Referencing EIS Messages to MEL Items

The MEL identifies airplane equipment that may be inoperative. The purpose of the Cross-Reference List is to relate an EIS message indicating an equipment fault or failure to the appropriate MEL item(s). This assists operators in quickly identifying the proper MEL items to consult. All EIS messages are listed in alphabetic order. The list includes the message level, associated Cue, and a description of the condition(s) that result in message display. The last column provides the MEL item(s) to be consulted. Any Alert message that does not have a specific MEL item identified is listed as "None". Messages indicating a system fault or failure with no associated MMEL relief require corrective maintenance action prior to dispatch. Advisory messages do not typically contain a MEL cross-reference, since they are indicative of normal system functionality.

The procedural steps used in determining airplane dispatchability following display of a status (prior to engine start) or alert message are summarized below.

EIS Alert or Status message is displayed

1. Review EIS Cross-Reference List:

- A. If multiple MEL Items are referenced and/or if dispatch capability is otherwise unclear, consult the 717 Fault Isolation Manual for further fault troubleshooting / identification.
- B. If the cause of the fault has been isolated and possible dispatch relief is available, comply with appropriate MEL restrictions and Procedures for that particular item:
 - i. Operations (O) - Modify normal procedures as required.
 - ii. Maintenance (M) - Configure airplane for dispatch by performing required maintenance action.
- C. If dispatch relief is not available:
 - i. Maintenance is required prior to dispatch.

Some MEL Items provide dispatch relief for alerting systems themselves, and it is incumbent upon the operator to discern whether a fault message is attributed to an inoperative indicating system or an actual fault condition. The 717 Fault Isolation Manual provides ample guidance in this respect, and should be consulted whenever a questions exists.

Message	Level	System	Possible Meaning Condition	MMEL Item
A-ICE ALL ON	0	AIR	All air foil and engine anti-ice systems are selected ON.	None
A/P NOT AVAIL	1	MISC	Autopilot is inoperative in both FCC's.	22-10-01 22-11-01 22-12-01
ACARS MESSAGE	0	MISC	An incoming ACARS message is being received.	None
ACARS NO COMM	1	MISC	ACARS communication has been lost.	23-24-01
ACCESS COMPT DOOR	1	MISC	Accessory compartment door is not closed and locked.	52-70-01
ADIRU 1/2 FAULT	1	STATUS	Respective inertial navigation portion of the ADIRU has lost the navigation function, or there is a failure of the baro set input bus from either VIA.	None
ADIRU AUX FAULT	1	STATUS	Inertial navigation portion of the AUX ADIRU (if installed) has lost the navigation function.	34-10-01
AFT BULKHEAD DOOR	1	MISC	Aft bulkhead door is not closed and locked.	52-70-01

<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
AGS FAIL	1	STATUS	The Auto Ground Spoiler (AGS) system has detected a failure which may cause the AGS to not deploy.	27-62-01
AIL TRIM FAIL	1	STATUS	Aileron trim system has failed.	27-12-01
AIR DATA 1/2 FAIL	2	MISC	The respective DEU has detected that the air data bus is not valid.	None
AIR DATA HEAT OFF	1	MISC	The air data probe heater switch is OFF, or a pitot or stall probe heater has failed.	30-31-01 30-34-01
AIR ISOL DISAG	1	AIR	Pneumatic system isolation valve not in the commanded position.	36-15-01
AIR SYS FAULT	1	AIR	A Fan Air Valve has failed open prior to flight	36-11-01
AIR ISOL OPEN	0	AIR	Pneumatic isolation valve is open.	None
AIR SYS L/R NOT OFF	1	AIR	Respective bleed air system has been commanded off, ISOL valve is closed, but manifold pressure is still above 10 psig.	36-13-01
AIR SYS L/R PRES LO	2	AIR	Respective bleed air system is commanded on, ISOL valve is closed, and manifold pressure is less than 10 psi.	36-13-01 36-13-02 36-11-03 36-11-04 36-13-04 36-13-05
AIR SYS L/R OFF	1	AIR	Respective bleed air system has been commanded off, the ISOL valve is closed, and manifold pressure is less than 10 psig.	None
AIRFOIL A-ICE ON	0	AIR	Both wing and tail anti-ice are ON.	None
ANTISKID FAULT	1	STATUS	Antiskid system has detected a fault that could result in a reduction or loss of system redundancy. Anti-skid is still fully functional.	32-45-02
ANTISKID L/R FAIL	2	CONFIG	Respective Anti-skid system has failed or is operating in a degraded mode (at least one wheel has failed).	32-45-01
ANTISKID OFF	1	CONFIG	ANTISKID switch is selected OFF.	None
AOA HEAT L/R FAIL	1	MISC	Respective angle-of-attack probe heater has failed or switch is off in flight.	30-34-01
APU AIR ON	0	AIR	APU is supplying pneumatic power.	None

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<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
APU AIR/ELEC ON	0	ENG	APU is supplying pneumatic and electrical power.	None
APU AUTO SHUTDOWN	1	ENG	APU has detected a fault and has executed automatic shutdown.	49-10-01
APU DOOR OPEN	1	ENG	APU door is open with the APU commanded OFF.	49-17-01
APU EGT HI	2	ENG	APU EGT exceeds 100%.	49-10-01
APU ELEC ON	0	ELEC	APU is supplying electrical power to the tie bus.	None
APU FAULT	2	ENG	The APU has detected a fault which would result in an auto shutdown on the ground, but will inhibit an auto shutdown in flight.	49-10-01
>APU FIRE	3	ENG	A fire has been detected in the APU compartment.	26-12-01 49-10-01
APU ON	0	ENG	APU is operating and available, but not providing pneumatic or electrical power.	None
APU VALVE DISAG	1	AIR	The APU air switch position disagrees with the APU air valve and/or APU load bleed valve positions.	49-51-02
ATC XPDR 1/2 FAIL	1	MISC	Respective ATC transponder has failed.	34-54-01
AUTO TRIM FAIL	1	CONFIG	The autopilot is engaged and the FCC is unable to trim the stabilizer.	22-22-01 27-43-01
AUTOBRAKE FAIL	2	CONFIG	The autobrake system has failed.	32-46-01
AUTOBRAKE HI	0	CONFIG	Indicates auto brakes are selected to HIGH for landing.	None
AUTOBRAKE MAX	0	CONFIG	Indicates auto brakes are selected to MAX for landing.	None
AUTOBRAKE MED	0	CONFIG	Indicates auto brakes are selected to MED for landing.	None
AUTOBRAKE MIN	0	CONFIG	Indicates auto brakes are selected on MIN for landing.	None
AUTOBRAKE OFF	1	CONFIG	Autobrake system is selected OFF with the landing gear down.	None
AUTOBRAKE T/O	0	CONFIG	Indicates auto brakes are selected to T/O mode.	None
AUTOPILOT SINGLE	1	MISC	Indicates autopilot is inoperative in one FCC only.	22-10-01

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<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
AVNCS AIR FLO OFF	2	AIR	Both avionics cooling fans have failed inflight, or the primary fan has failed on the ground.	21-21-01
AVNCS COOL OVRD	1	AIR	Both avionics fans are OFF and the AVIONICS COOLING switch is in OVRD position.	None
BALST FUEL DISAG	2	FUEL	Fuel quantity remaining in the center or auxiliary fuel tank is less than the designated ballast quantity in that tank by more than 200 lbs.	28-41-01
BATT CHARGER FAIL	1	ELEC	Indicates battery charger is inoperative.	None
BATT CHARGING	1	ELEC	Indicates battery charging at 10 amp or higher rate for more than 10 seconds. Alert is normal after extended use of battery such as during APU start. Alert displayed for an extended period may indicate a malfunction.	None
BATT SWITCH OFF	1	ELEC	Main BATT switch is OFF.	None
BATT DISCHARGIN	1	ELEC	Battery is being discharged without emergency power on (battery load > 30 amps for one minute or longer).	None
BLD AIR L/R TEMP HI	2	AIR	Respective Bleed supply temperature sensor (Ts) or manifold temperature sensor (Tm) detects bleed temperature in excess of 535°F (280°C).	36-11-01 36-13-07 36-13-08
BLD AIR L/R TEMP LO	2	AIR	Respective Bleed supply temperature sensor (Ts) or manifold temperature sensor (Tm) detects bleed temperature is less than 390°F with ice protection commanded on.	36-11-01 36-13-07 36-13-08
BLEED AIR L/R FAIL	2	AIR	Respective pressure regulator valve failed, PSC channel failed, or PSC input from Pm or Tm sensor is lost.	36-13-01 36-13-02 36-13-03 36-13-05 36-13-07
BLEED AIR L/R OFF	1	AIR	Respective bleed air valve is closed with air system not commanded off.	36-13-01 36-13-02
BRAKE OVERHEAT	2	CONFIG	One or more brake temperatures are in excess of 260°C.	32-47-01

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<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
BRAKE PRESS LO	1	CONFIG	One or both brake accumulator pressures are less than 1200 psi with the aircraft on the ground or 2000 psi with the aircraft in flight.	None
BUS AC EMER OFF	2	ELEC	AC emergency bus is not powered.	None
BUS AC GS OFF	1	ELEC	AC ground service bus (AC tie bus) is OFF.	None
BUS AC L/R OFF	1	ELEC	Respective main generator AC bus is not powered.	None
BUS DC EMER OFF	2	ELEC	DC emergency bus is not powered.	None
BUS DC XFER OFF	2	ELEC	DC transfer bus has failed.	None
BUS DC L/R OFF	1	ELEC	Respective main DC bus is not powered.	None
BUS TIE L/R LOCKOUT	1	ELEC	Respective AC bus tie is locked open due to a failure.	None
BUS TIE L/R OPEN	1	ELEC	AC bus tie switch is in OPEN position.	None
CAB PRES MAN FAIL	1	AIR	CPC panel or manual control of Outflow Valve (OFV) actuator has failed. The auto mode will continue to operate, however, manual mode is not available.	21-30-01
CAB PRES SYS MAN	1	AIR	Cabin pressurization control system has been selected to MANUAL in the Cabin Press Select Switch.	21-30-01
>CABIN ALTITUDE	3	AIR	Cabin altitude exceeds 10,000 ft.	None
CABIN DOOR	1	MISC	Forward cabin door is not closed and locked.	52-70-01
CABIN DUCT OVHT	2	AIR	Cabin duct temperature exceeds 175°F.	21-53-03
CABIN INFLO LO	2	AIR	Pack flow into cabin is not enough to maintain pressurization.	21-51-01 21-52-01
CABIN PRES HI	2	AIR	Cabin Differential Pressure (DP) exceeds 8.41 psi.	21-34-02
CABIN PRESSURIZED	2	AIR	Cabin Differential Pressure (DP) exceeds 0.5 psid with the aircraft on the ground.	None
CABIN RATE	1	AIR	Cabin rate of climb or descent exceeds comfortable limits.	21-30-01
CABIN TEMP MANUAL	1	AIR	CABIN TEMP control is selected to manual.	21-62-01

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<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
CARGO AGENT 1/2 LO	1	AIR	One or more of the cargo compartment fire agent bottles has been discharged.	26-22-01
CARGO DOOR FWD/ AFT	1	MISC	Respective cargo compartment door is not closed and locked.	52-70-01
>CARGO SMOKE FWD/AFT	3	AIR	Smoke has been detected in the respective cargo compartment.	26-13-01
CARGO SMOKE FWD/ AFT FAIL	1	AIR	Both smoke detectors in a single pan, or two detectors in adjacent pans within the respective compartment are inoperative.	26-13-01
CAWS FAIL	1	MISC	The Central Aural Warning System has failed in both VIA computers.	None
CDU 1/2 MENU REQ	0	MISC	Respective MCDU is selected to a function other than flight management.	None
CKPT DUCT OVHT	2	AIR	Cockpit duct temperature exceeds 175°F.	21-53-03
CKPT TEMP MANUAL	1	AIR	CKPT TEMP control is selected to manual.	21-62-01
CONTROL DISC	1	CONFIG	Control Column positions disagree by greater than approximately 2 1/2°	27-30-01
CPC 1/2 FAIL	1	STATUS	Respective Cabin Pressure Controller (CPC) has failed.	21-30-01
CRG SMK TEST PASS	0	AIR	Cargo smoke test passed.	None
CRG SMK FWD/AFT SGL	1	AIR	Respective cargo smoke detection system is set in the single mode of operation.	26-13-01
CTR FWD/AFT PUMP LO	2	FUEL	Center tank is not empty and the respective pump is commanded on and indicates low pressure with greater than 500 lbs. remaining in the tank.	28-22-02 28-44-01
CTR FWD/AFT PUMP OFF	1	FUEL	Respective tank pump is selected OFF with more than 500 lbs. of usable fuel in center tank.	None
DC TIE SW OPEN	1	ELEC	The DC BUS TIE switch is in the OPEN position.	None
DCU CHANNNEL FAIL	1	STATUS	One Data Concentrator Unit (DCU) channel has failed.	None
DCU FAIL	1	MISC	The Data Concentrator Unit (DCU) has lost data.	None

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<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
DCU FAULT	1	MISC	The Data Concentrator Unit (DCU) has lost data redundancy.	None
DISARM SPOILERS	1	CONFIG	The spoiler handle is armed and the SECU has detected an (AGS) failure which may cause the AGS to not deploy.	27-62-01
DISCH CARGO AGENT	1	AIR	One cargo firex bottle has been discharged, 6 minutes have elapsed, and smoke is still detected in the compartment.	None
DME 1/2 FAIL	1	STATUS	Respective DME receiver has failed.	34-52-01
DOOR OPEN	1	MISC	One or more airplane doors or access hatches are not closed and locked.	52-70-01
DRAIN MAST HEAT	1	MISC	Drain mast heater has failed.	30-72-01
EIS SINGLE SOURCE	1	MISC	A single VIA Channel is driving all six displays. Cross comparison is not available.	None
ELEC COMPT DOOR	1	MISC	The electrical/electronics bay door is closed and not locked.	52-70-01
ELEC FAULT	1	ELEC	The EPCU has detected an internal fault.	None
ELEVATOR SPLIT	2	CONFIG	A split has been detected between the left and right elevator panels, possibly indicating a jammed elevator condition.	27-34-01
EMER LTS DISARM	1	MISC	EMER LT switch is not in the ARM position.	None
EMER POWER ON	1	ELEC	Aircraft battery is powering DC XFER bus and emergency AC and DC busses.	None
EMER POWER TEST	0	ELEC	Emergency power test is in progress.	None
EMER PWR SW OFF	1	ELEC	EMER PWR switch is OFF.	None
EMER PWR TST FAIL	1	ELEC	Emergency power test has failed.	None
ENG A-ICE ON	0	AIR	Both engine anti-ice switches are ON.	None
ENG IGN OVRD ON	0	ENG	Engine ignition has been selected to the ON position.	None
ENG START PUMP ON	0	FUEL	The DC Start Pump is commanded ON.	None
ENG L/R A-ICE DISAG	1	AIR	Respective engine anti-ice valve is open when commanded off, or closed when commanded on.	30-22-01

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<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
ENG L/R A-ICE ON	0	AIR	Respective engine anti-ice switch is ON.	None
ENG L/R COMP STALL	2	ENG	Respective engine compressor stall	None
ENG L/R BRG OVHT	2	ENG	Respective engine aft bearing compartment overheat detected (possible fire).	None
ENG L/R FADEC ALTN	1	ENG	Respective EEC has been selected to the ALTN (N1) Mode.	73-21-02
ENG L/R FUEL FILTER	1	ENG	Respective engine fuel filter is partially clogged and bypass is imminent.	73-34-01
ENG L/R FUEL PRES	1	FUEL	Respective engine inlet fuel pressure is low.	73-33-01
ENG L/R OIL FILTER	1	ENG	Respective engine oil filter is partially clogged and bypass is imminent.	79-36-01
ENG L/R OIL PRES LO	2	ENG	Engine oil pressure is low.	None
ENG L/R OIL TEMP HI	2	ENG	Respective engine oil temperature is over the limit.	None
ENG L/R RPM HI	2	ENG	Respective engine has exceeded the N1 or N2 RPM limit.	None
ENG L/R RPM LO	2	ENG	Respective engine N2 RPM has dropped below 55% with the fuel switch on.	None
ENG L/R START ABORT	1	ENG	Respective EEC has detected an abnormal condition (hung start, hot start, no start, locked rotor, engine surge) and automatically aborted start.	80-11-02 80-11-03
ENG L/R SURGE	1	ENG	Respective engine is experiencing a surge.	None
ENG L/R SYS FAIL	1	ENG	Respective engine EEC has detected a non-dispatchable engine fault condition.	None
ENG L/R SYS FAULT	1	STATUS	Respective EEC has detected a short-term dispatchable fault.	73-21-01
ENG L/R TGT HI	2	ENG	Respective engine Turbine Gas Temperature (TGT) is above redline limit or above amber limit for a minimum of five minutes.	None
ENGINE COOL	0	ENG	Engine has cooled sufficiently after landing to be shut down.	None
>ENGINE L/R FIRE	3	ENG	Engine fire has been detected.	26-11-01

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Message	Level	System	Possible Meaning Condition	MMEL Item
ENGINE VIB HI	1	ENG	Engine vibration parameters exceed 4.0 units.	77-32-01
EXT POWER AVAIL	0	ELEC	External power is plugged in and available.	None
EXT POWER ON	0	ELEC	External power is on and powering the tie bus.	None
F/D G/A ONLY	1	MISC	Autopilot go-around mode is not available.	34-24-01
FCC 1/2 FAIL	1	CONFIG	Both channels of the respective Flight Control Computer (FCC) have failed.	None
FCC 1/2 FAULT	1	CONFIG	One channel of the respective Flight Control Computer has failed.	None
FIRE AGENT 1/2 LO	1	ENG	Respective Engine firex bottle agent has been discharged.	26-21-01
FIRE APU A/B FAULT	1	MAINT	Respective fire detection loop has detected a fault.	26-12-01
FIRE DET APU FAIL	1	ENG	Both APU fire detection loops are inoperative, or a fire/overheat is detected in only one of the two active loops.	26-12-01 49-10-01
FIRE DET L/R FAIL	1	ENG	Both fire detection loops for the respective engine are inoperative, or a fire/overheat is detected in only one of the two active loops of an engine.	26-12-03
FIRE L-A/B FAULT	1	STATUS	Respective left engine fire detection loop is failed.	26-11-01
FIRE R-A/B FAULT	1	STATUS	Respective right engine fire detection loop is failed.	26-11-01
FLAP DISAG	2	CONFIG	Flap position disagrees left to right or with the commanded flap position.	27-54-01
FLT REC FAIL	1	STATUS	Digital Flight Data Recorder or Flight Data Acquisition Unit has failed.	31-31-01
FMS 1/2 FAIL	1	MISC	The respective flight management function of the respective VIA has failed.	34-63-01
FMS DUAL FAIL	1	MISC	Flight management function in both VIA's is failed.	None
FUEL LEVEL LO	1	FUEL	Either main tank has less than 2000 lb/907 kg. remaining, and the quantity of usable fuel remaining in the center tank and aux tanks (if installed) is less than 1000 lbs/405 kg.	28-43-01

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<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
FUEL OFF SCHEDULE	2	FUEL	Either Main tank has less than 2000 lbs. fuel remaining while the center tank has usable fuel in excess of 1000 lbs.	28-22-02
FUEL QTY FAULT	2	FUEL	Loss of fuel quantity information for any tank.	28-41-01
FUEL QTY SYS FAIL	2	FUEL	Selected Fuel Quantity Gaging System (FQGS) channel is inoperative, or stored VIA configuration disagrees with SEM configuration.	28-41-01
FUEL RTT L/R INOP	1	STATUS	Fuel Return-to-Tank System is inoperative	28-26-01 28-45-01
FUEL XFEED ON	0	FUEL	Fuel crossfeed lever is selected open.	None
GALLEY DOOR	1	MISC	The forward galley service door is not closed and locked.	52-70-01
GEAR DOOR OPEN	2	CONFIG	One or more main landing gears are open with the gear down.	None
GEN ALL OFF	2	ELEC	All generators are off and external power is not available.	None
GEN APU OFF	1	ELEC	APU generator switch is in the OFF position.	None
GEN APU OFF	2	ELEC	APU has tripped off-line due to a fault.	24-21-01
GEN L/R OFF	1	ELEC	Respective generator has been selected off by the pilot.	None
GEN L/R OFF	2	ELEC	Respective generator has tripped off line due to a fault.	24-21-01
GPS 1/2 FAIL	1	STATUS	GPS function of the respective MMR has failed.	34-32-01
GPWS FAIL	1	MISC	Indicates the Ground Proximity Warning System has failed.	34-45-01
GPWS FAULT	1	MISC	The Ground Proximity Warning System (GPWS) has detected a fault.	34-45-01
GPWS FLAP OVRD	0	CONFIG	GPWS switch is in the FLAP OVRD position.	None
GPWS TERR OFF	1	SW	The Enhanced Ground Proximity Warning System terrain mode is selected OFF.	34-45-01
HYD AUX PUMP FAIL	1	HYD	Pressure output from Aux hydraulic pump is low with Aux pump commanded ON.	29-21-01

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<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
HYD AUX PUMP OFF	1	HYD	Aux hydraulic pump is not ON with the slats extended and at least one engine operating.	None
HYD L&R FAIL	2	HYD	Both hydraulic systems are detected to have low quantity (<2 qts) or low pressure (<1900 psi.) with the engines above 60% N2 and/or the aircraft in flight.	None
HYD L/R PRES LO	2	HYD	Respective hydraulic system pressure is less than 1900 psi. Message extinguishes when pressure goes above 2400 psi.	29-31-01
HYD PUMP L/R FAIL	1	HYD	Low pressure output from respective engine hydraulic pump with pump commanded ON.	29-11-01
HYD PUMP L/R OFF	1	HYD	Respective hydraulic pump is commanded OFF.	None
HYD L/R QTY LO	2	HYD	Respective hydraulic reservoir fluid quantity is less than 5.5 qts on the ground, or 4 qts in flight.	29-32-01
HYD L/R TEMP HI	2	HYD	Respective hydraulic system fluid temperature is greater than 200°F.	None
HYD TRANS DISAG	1	HYD	Hydraulic power transfer pump is not in commanded position.	29-22-01
HYD TRANS OFF	1	HYD	Hydraulic transfer pump is not ON with the slats extended and at least one engine operating	29-22-01
HYD L/R OFF	1	HYD	Respective engine driven pump, power transfer unit, and aux hydraulic pump (right system only) are commanded OFF.	None
ILS 1/2/3 FAIL	1	STATUS	Indicates respective ILS function of the MMR has failed.	34-32-01
IRS 2 ON BATT	1	MISC	IRS 2 powered by IRS battery. IRS 2 operation beyond 20 minutes cannot be relied on.	None
IRS 1/2 FAIL	2	MISC	Respective inertial navigation portion of the ADIRU has failed.	None
IRS AUX FAIL	2	MISC	Respective inertial navigation portion of the AUX ADIRU has failed.	34-10-01
IRS BATT LO	1	MISC	IRS 2 or IRS Aux battery voltage is below 21 volts.	None
IRS IN ALIGN	0	MISC	One or more of the inertial reference units are in align mode.	None

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<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
IRS OFF	1	MISC	One or more IRU's have been selected OFF in flight.	None
IRS 1/2 NO ALIGN	1	MISC	IRS has failed to complete an alignment.	None
IRS AUX NO ALIGN	1	MISC	IRS has failed to complete an alignment.	34-10-01
LAT FUEL UNBAL	1	FUEL	Left and right main tank fuel quantity differs by more than 1400 lb/630 kg.	28-22-01
LAVATORY SMOKE	1	AIR	Smoke has been detected in a lavatory.	26-14-01
LDG ALTITUDE MAN	1	AIR	Landing field elevation has been set manually.	None
MANUAL G/A ONLY	1	MISC	Autopilot and flight director guidance for go-around is not available.	22-10-01 34-24-01
NO AUTOLAND	1	MISC	Autoland function is not available due to an internal problem in the Flight Control Computer (FCC).	22-10-01
>NO MASKS	3	AIR	Cabin altitude exceeds 14,750 feet and passenger oxygen masks have not automatically deployed.	35-22-01
NO SMOKING	0	MISC	The NO SMOKING lights switch is ON or in Auto with slats extended.	33-28-01
OVERWING DOOR	1	MISC	One or more overwing emergency exit doors is not closed and locked.	52-70-01
PACK OVRD	1	AIR	PACK SHUTDOWN switch is in the OVRD position.	21-56-01
PACK L/R OFF	1	AIR	Respective pack has been commanded OFF and is producing no flow.	21-51-01 21-51-02
PACK L/R OVERHEAT	2	AIR	Respective air conditioning pack has shutoff due to an overheat condition.	21-51-01 21-53-02 21-53-03
PACK L/R FLO DISAG	1	AIR	Respective pack has been selected ON, but no flow condition detected after 20 seconds, or selected to OFF, but there is still flow and pressure after 10 seconds.	21-51-01 21-51-02 21-52-01
PACKS ALL OFF	0	AIR	Packs off takeoff has been selected and the packs have automatically switched off; or packs have been switched off due to engine failure on takeoff.	21-56-01

CONTINUED ON NEXT PAGE

<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
PACKS HIGH FLOW	0	AIR	Pack FLOW switch has been selected to the HIGH flow position.	21-51-02
PARK BRAKE ON	0	CONFIG	On ground, indicates the parking brake is set.	32-43-01 32-45-02
PITOT CAPT FAIL	1	MISC	CAPT pitot probe heater has failed.	30-31-01
PITOT F/O FAIL	1	MISC	F/O pitot probe heater has failed.	30-31-01
PITOT AUX FAIL	1	MISC	AUX pitot probe heater has failed.	30-31-01
PODS A-ICE FAULT	1	STATUS	PODS system has a fault in the wing or tail ice protection system and cannot detect a manifold failure.	36-23-01
PODS FAIL	1	AIR	PODS system has failed, and is unable to detect a manifold failure.	None
PODS FAULT	1	STATUS	One tail compartment sensor loop or controller channel in the PODS system has failed. Manifold failure detection is still operative.	36-23-01
PODS TEST FAIL	1	AIR	The PODS system test has failed.	None
PODS TEST PASS	0	AIR	PODS preflight test is successful (accomplished with FIRE TEST switch).	None
PRED WSHEAR FAIL	1	MISC	Predictive windshear function of the weather radar is inoperative.	34-41-01
PRED WSHEAR FAULT	1	MISC	The interface between the Enhanced Ground Proximity Warning System (EGPWS) and the Predictive Windshear System has been lost.	34-41-01 34-45-01
PRED WSHEAR OFF	0	MISC	Indicates the predictive windshear function of weather radar is not functioning.	34-41-01
PSEU FAIL	1	CONFIG	Proximity sensor system has failed resulting in the loss of data on gear, door and slat position.	None
PSEU FAULT	1	STATUS	PSEU has logged an internal fault or detected a failure of a redundant sensor.	32-61-02
RAT PROBE FAIL	1	MISC	Ram air temperature probe heater has failed or Air Data Heat Switch is off in flight.	30-33-01
REV L/R DEPLOYED	1	ENG	Respective thrust reverser has deployed in flight.	78-32-01 78-33-02 78-33-03

CONTINUED ON NEXT PAGE

<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
REV L/R FAULT	1	ENG	EEC has detected a thrust reverser abnormal condition that may inhibit its operation.	78-32-01
RUD PITOT FAIL	1	MISC	Rudder limiter pitot probe heater has failed or Air Data Heat Switch is off in flight. Possible over/or under restricted rudder.	30-31-01
RUDDER LIM FAIL	2	CONFIG	Both the primary and secondary rudder limiters have failed to the unrestricted position.	None
RUDDER LIM FAULT	1	CONFIG	The redundant rudder limiter system has failed in the unrestricted position.	27-26-01
RUDDER PWR OFF	1	CONFIG	Rudder is in the manual mode either due HYD CONT RUDDER switch selected OFF, or less than 900 psi pressure available in the right hydraulic system.	None
RUDDER RESTRICTED	1	CONFIG	Rudder is restricted more than it should be for the flight condition.	None
SEAT BELTS	0	MISC	The FASTEN SEAT BELTS lights switch is ON, or in AUTO with the aircraft below 10,000 feet.	33-28-01
SECU FAULT	1	STATUS	Spoiler system has detected a fault.	27-61-01
SEL CAB PRES MAN	1	AIR	Both auto cabin pressure controllers have failed and the system must be selected to MANUAL.	21-30-01
SEL CTR PUMPS OFF	1	FUEL	Center fuel tank has less than 70 lbs usable fuel remaining and CTR Tank pump switches are on.	None
SEL CTR PUMPS ON	1	FUEL	Center fuel tank has more than 500 lbs usable fuel remaining and CTR TANK pump switches are off.	None
SELECT FADEC ALTN	2	ENG	An engine has reverted to the N1 mode.	73-21-02
SET LDG ALTITUDE	1	AIR	Cabin pressurization system has lost landing field elevation data from the FMS and must be set manually.	None
SINGLE LAND	1	MISC	Dual land is not available.	22-10-01
SLAT DISAG	2	CONFIG	The slat position disagrees left to right or with the commanded slat position.	None
SPEEDBRAKE	0	CONFIG	Speed brakes are extended.	None

CONTINUED ON NEXT PAGE

<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
SPEEDBRAKE DISAG	1	CONFIG	Speedbrakes are extended and throttles are advanced to a high power setting.	None
SPEEDBRAKE/FLAP	1	CONFIG	Speedbrakes are extended with flaps in the landing range.	None
SPOILER FAULT	1	CONFIG	A partial or total spoiler system failure has occurred, or the spoiler system test is being conducted with the airplane in flight.	None
SPOILER INBD/OTBD FAIL	1	CONFIG	A pair of spoiler panels has shut down due to a detected SECU or actuator failure.	None
STAB OUT OF TRIM	1	CONFIG	Stabilizer is not responding to autopilot trim commands.	22-22-01 27-43-01
STAB TRIM OFF	1	CONFIG	STABILIZER TRIM Shutoff switch is selected OFF.	None
STAIRWAY DOOR FWD	1	MISC	Indicates FWD stairway door is not closed and locked.	None
STALL WARN FAIL	1	STATUS	Indicates stall warning (stick shaker, SSRS and stick pusher) will not function.	None
STALL WARN FAULT	1	STATUS	The stall warning system has lost two channels of stall warnings, and/or one channel of stick pusher.	None
START AIR PRES LO	1	AIR	Pressure in the manifold is less than 25 psi during an attempted engine start.	None
START VLV L/R OPEN	2	ENG	Respective engine start valve is open and the engine is above idle rpm.	80-11-01 80-11-02
STATIC L/R HEAT	1	STATUS	Respective static plate heater has failed or Air Data Heat Switch is off in flight.	30-32-01
STICK PUSHER FAIL	1	STATUS	Stick pusher has failed in a passive mode.	None
TAIL A-ICE DISAG	2	AIR	Tail anti-ice valve is open when commanded off, or closed when commanded on.	30-12-01
TAIL A-ICE ON	0	AIR	Tail anti-ice is selected ON and wing and engine anti-ice is OFF; or wing and tail are selected ON and system is in alternating mode.	None

CONTINUED ON NEXT PAGE

<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
TAIL A-ICE OFF	2	AIR	Tail anti-ice has been commanded off by the PODS system due to an overheat or manifold failure.	36-23-01
TAILCONE UNLOCK	2	MISC	Tailcone is not locked.	53-52-01
>TAIL MANF FAIL	3	AIR	PODS system has detected an overheat condition in the tail anti-ice manifold and has failed to close the tail ice-protection shutoff valve.	None
>TAIL TEMP L/R HI	3	AIR	PODS has detected an overheat condition or manifold failure in the tail compartment.	None
TANK L/R PUMPS OFF	1	FUEL	Respective fwd and aft main tank pumps are commanded OFF with respective engine FUEL switch ON.	None
TERRAIN FAIL	1	MISC	The Terrain mode of the Enhanced Ground Proximity Warning System (EGWS) has failed.	34-45-01
TERRAIN OVRD	0	MISC	Terrain mode of the Enhanced Ground Proximity Warning System (EGWS) is selected to OVRD.	34-45-01
TERRAIN NOT AVAIL	1	MISC	The Terrain mode of the Enhanced Ground Proximity Warning System (EGWS) has failed.	34-45-01
TNK L/R FWD/AFT PMP OFF	1	FUEL	Respective main tank pump switch is OFF with respective engine FUEL switch ON.	None
TNK L/R FWD/AFT PUMP LO	1	FUEL	Respective main tank pump is commanded ON, but output pressure is low.	28-22-01 28-44-01
TANK L/R PUMPS LO	1	FUEL	Low fuel pressure output is detected from affected main tank with all respective main tank pumps commanded ON.	28-22-01 28-44-01
TR L/R/C FAIL	1	ELEC	Respective transformer-rectifier unit has failed (voltage low).	24-31-01
UNABLE RNP	1	MISC	FMS required navigational performance not within limits for particular phase of flight.	None
VHF-3 VOICE	0	MISC	VHF-3 receiver is in the voice rather than the data mode.	23-12-01
VIA CONFIG DISAG	1	MISC	Configuration of VIA -1 and VIA-2 do not agree.	None
VIA FAIL	1	MISC	A VIA has failed in a non- dispatchable state.	None

CONTINUED ON NEXT PAGE

<u>Message</u>	<u>Level</u>	<u>System</u>	<u>Possible Meaning Condition</u>	<u>MMEL Item</u>
VIA FAULT	1	STATUS	A VIA has a failure of a redundant Graphics Generator or Display Channel, or the Engine Comparison monitor has failed.	31-61-02
VOR 1/2 FAIL	1	STATUS	Respective VOR receiver has failed.	34-51-01
WING A-ICE DISAG	2	AIR	Wing anti-ice valve is open when commanded off, or closed when commanded on.	30-11-01
WING A-ICE ON	0	AIR	Wing anti-ice is selected ON and tail and engine anti-ice is OFF; or wing and tail are selected ON and system is in alternating.	None
WING A-ICE OFF	2	AIR	Wing anti-ice has been commanded off by the PODS system due to an overheat or manifold failure.	36-23-01
WING ICE DET PASS	0	AIR	Upper surface ice detection system test has passed successfully.	None
WING ICE DETECTED	1	AIR	Ice buildup has been detected on left or right wing surface.	None
WING ICE L/R FAIL	1	AIR	Respective overwing ice detection system has failed.	30-81-01
>WING MANF FAIL	3	AIR	PODS system has detected an overheat condition in the wing anti-ice manifold and has failed to close the wing ice-protection shutoff valve.	None
WSHEAR DET FAIL	1	MISC	The reactive windshear detection and guidance system has failed in both FCC's.	34-47-01
WSHLD A-ICE OFF		MISC	Windshield Anti-ice switch has been selected OFF.	None
WSHLD ANTI-FOG ON	0	MISC	Windshield ANTI-FOG switch has been selected ON.	None
WSHLD HEAT FAIL	2	MISC	Windshield heater has overtemped.	30-41-01
YAW DAMP FAIL	1	CONFIG	Yaw damper function has been shut off by the FCC.	22-23-01

11 Placards

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11-00-01	Aircraft Placards/Markings Repair Installed
11-00-01	Aircraft Placards/Markings Repair Installed

11-00-01 Aircraft Placards/Markings Repair Installed

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
11-00-01	Aircraft Placards/Markings Repair Installed "Contact MCC" Administrative Control Item	N	-	-	Item may be used to track repair provided: - MCC determines that placard is FAA required, - Install repair per TOPP 40-40-05, - Enter in logbook repair is applied per TOPP 40-40-05, and - Enter in logbook that repair is deferred per MEL 11-00-01. Note: If MCC determines that placard is not FAA required, use TOPP 40-40-08 (NEF).

MEL ITEM	TAB LOCATION	PLACARD TEXT
11-00-01	Aircraft Logbook	_____ REPAIRED PER TOPP 40-40-05

21 Air Conditioning

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21-21-01	Radio Rack Cooling Fans
	RADIO RACK COOLING FANS
21-21-01A	Radio Rack Cooling Fans
21-21-01B	Primary Fan (Left)
21-21-01C	Standby Fan (Right)
21-21-02	Radio Rack & Standby Radio Rack Fan Check Valves
21-21-02A	Radio Rack & Standby Radio Rack Fan Check Valves
21-21-02B	Radio Rack & Standby Radio Rack Fan Check Valves
21-22-03	Ram / Ground Conditioned Air Check Valves
21-22-03A	Ram / Ground Conditioned Air Check Valves
21-22-03B	Ram / Ground Conditioned Air Check Valves
21-22-04	Gasper Fan
21-22-04	Gasper Fan
21-22-05	Conditioned Air Check Valves
21-22-05	Conditioned Air Check Valves
21-22-06	Cold Air Check Valves
21-22-06	Cold Air Check Valves
21-26-01	Cockpit Instrument Cooling Fan
21-26-01	Cockpit Instrument Cooling Fan
21-29-01	Ram Air Valve
21-29-01A	Ram Air Valve
21-29-01B	Ram Air Valve
21-30-01	Cabin Pressure Control Systems (Auto And Manual)
21-30-01A	Cabin Pressure Control Systems (Auto and Manual)
21-30-01B	Cabin Pressure Control Systems (Automatic)
21-30-01C	Cabin Pressure Control Systems (Automatic)
21-30-01D	Cabin Pressure Control Systems (Manual)
21-34-02	Cabin Pressure Safety Valves
21-34-02	Cabin Pressure Safety Valves
21-42-02	Forward Cargo Compartment Heating System
21-42-02	Forward Cargo Compartment Heating System
21-51-01	Air Conditioning Systems
21-51-01	Air Conditioning Systems
21-51-02	Air Conditioning Flow Control Valve
21-51-02A	Air Conditioning Flow Control Valve
21-51-02B	Air Conditioning Flow Control Valve (Normal Mode)
21-52-01	Pack Flow Indication Systems
21-52-01	Pack Flow Indication Systems
21-53-02	Compressor Discharge and Turbine Inlet Thermal Switches
21-53-02	Compressor Discharge and Turbine Inlet Thermal Switches

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21-53-03 Air Conditioning System Shutdown Thermal Switches (190 Degrees F)

- 21-53-03A Air Conditioning System Shutdown Thermal Switches (190 Degrees F)
- 21-53-03B Air Conditioning System Shutdown Thermal Switches (190 Degrees F)

21-54-01 Air Conditioning Water Separators

- 21-54-01 Air Conditioning Water Separators

21-54-02 Water Separator 35 Degrees F Thermostats

- 21-54-02A Water Separator 35 Degrees F Thermostats
- 21-54-02B Water Separator 35 Degrees F Thermostats

21-54-03 Water Separator 35 Degrees F Control Valves

- 21-54-03 Water Separator 35 Degrees F Control Valves

21-56-01 Air Conditioning Automatic Shut-Off System

- 21-56-01 Air Conditioning Automatic Shut-Off System

21-62-01 Cockpit / Cabin Compartment Auto Temperature Control Systems

- 21-62-01A Cockpit / Cabin Compartment Auto Temperature Control Systems
- 21-62-01B Cockpit / Cabin Compartment Auto Temperature Control Systems

21-62-02 Cockpit / Cabin Compartment Temperature Control Valves

- 21-62-02 Cockpit / Cabin Compartment Temperature Control Valves

21-63-01 Temperature Control Valve Position Indicating Systems

- 21-63-01 Temperature Control Valve Position Indicating Systems

21-64-01 Cockpit/Cabin Temperature Indicating Systems

- 21-64-01 Cockpit/Cabin Temperature Indicating Systems

21-21-01 Radio Rack Cooling Fans

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
RADIO RACK COOLING FANS					
21-21-01A	Radio Rack Cooling Fans "DISP.APPR.REQ'D"	Y	2	0	C
(O) (M) Both may be inoperative provided: - Both Air Conditioning Systems are available for pressurized flight, - Avionics cooling selector switch remains in OVRD, - Outside air temperature does not exceed 90 degrees F, and - Ground operation of electronic equipment is limited to a maximum of 45 minutes. - E&E compartment door is closed PRIOR TO EACH DEPARTURE. Note: No live animals or temperature sensitive cargo may be loaded in the forward cargo compartment. Repetitive Check Required (Yellow Placard)					
21-21-01B	Primary Fan (Left) "DISP.APPR.REQ'D"	N	1	0	C
(O) (M) May be inoperative provided Standby Fan is operative. Note: No live animals or temperature sensitive cargo may be loaded in the forward cargo compartment.					
21-21-01C	Standby Fan (Right)	N	1	0	C
May be inoperative provided Primary Fan is operative.					

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-21-01A	Avionics Cooling OVRD / Fan Switch	Primary / Standby Radio Rack Cooling Fans INOP
21-21-01B	Avionics Cooling OVRD / Fan Switch	Primary Radio Rack Cooling Fan (Left) INOP
21-21-01C	Avionics Cooling OVRD / Fan Switch	Standby Radio Rack Cooling Fan (Right) INOP

(O) PROCEDURES

21-21-01A

For both Primary and Standby Fans inoperative:

- Limit ground operation of electronic equipment to 45 minutes.
- Ensure that AVNCS COOLING Selector Switch is in OVRD.

Note: "AVNCS AIR FLO OFF" may be displayed after the (M) Procedures are completed.

21-21-01B

If " AVIONICS AIR FLO OFF" message is illuminated in flight, select AVIONICS COOLING selector switch to OVRD.

(M) PROCEDURES

21-21-01A

For both Primary and Standby Fans inoperative:

1. Verify both Air Conditioning Systems are available for pressurized flight.
2. Select AVIONICS COOLING selector switch to OVRD.
3. Pull and secure RADIO RACK FAN and STANDBY RADIO RACK FAN circuit breakers:

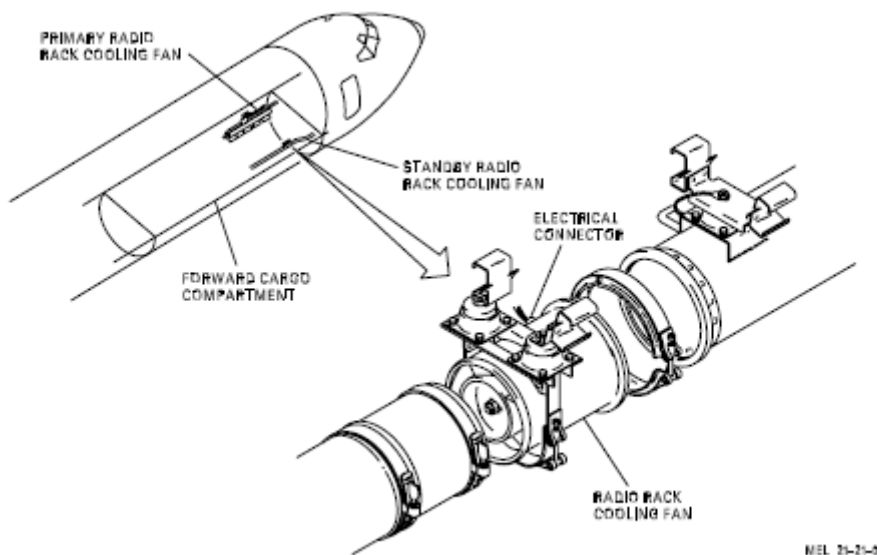
<u>Circuit Breaker (CB#)</u>	<u>Panel</u>	<u>Location</u>
RADIO RACK FAN ØA (B1-78)	EPC CBP	H-4
RADIO RACK FAN ØB (B1-79)	EPC CBP	H-5
RADIO RACK FAN ØC (B1-80)	EPC CBP	H-6
RADIO RACK FAN STANDBY ØA (B1-859)	GS Panel	A-02
RADIO RACK FAN STANDBY ØB (B1-860)	GS Panel	B-02
RADIO RACK FAN STANDBY ØC (B1-861)	GS Panel	C-02

4. Limit ground operation of electronic equipment to 45 minutes. The 45 minute time limit can be reset by using either of the following procedures:
 - A. Apply ground conditioned air directly into the E&E compartment through the E&E compartment door for at least 15 minutes. To reduce the possibility of exceeding the 45 minute time limit requirement, it is recommended that ground conditioned air be applied directly into the E&E compartment until push back.
 - B. Removing all aircraft electrical power or operating using only Ground Service electrical power with avionics turned off for at least 15 minutes.
5. **PRIOR TO EACH DEPARTURE**, confirm E&E compartment door is closed prior to push back.

21-21-01B

1. Check that RADIO RACK FAN CAUTION (B1-82 CBP Location U-29) circuit breaker is in.
2. In the avionics compartment, check that the RIGHT RADIO FAN OFF light on the GEN CONT RACK is not on (verifies Standby Fan operation).
3. Pull and secure the RADIO RACK FAN circuit breakers (all three phases) (H-4,H-5,H-6 EPC Panel).
4. Verify that the AVNCS AIR FLO OFF Alert light in the cockpit is illuminated.

Figure 21-1: Radio Rack Cooling Fans



21-21-02 Radio Rack & Standby Radio Rack Fan Check Valves

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
21-21-02A	Radio Rack & Standby Radio Rack Fan Check Valves	N	2	1	C (O) (M) One may be inoperative CLOSED provided associated fan circuit breaker remains open.
21-21-02B	Radio Rack & Standby Radio Rack Fan Check Valves "DISP.APPR.REQ'D"	N	2	0	C (O) (M) Both may be inoperative provided: - Both Air Conditioning Systems are available for pressurized flight, - Avionics Cooling Selector Switch remains in OVRD, - Outside Air Temperature does not exceed 90 degrees F, and - Ground operation of electronic equipment is limited to a maximum of 45 minutes. Note: No live animals or temperature sensitive cargo may be loaded in the forward cargo compartment.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-21-02A	Radio Rack OVRD / Fan Switch	___ Fan Check Valve INOP
21-21-02B	Radio Rack OVRD / Fan Switch	Primary / Standby Radio Rack Fan Check Valve(s) INOP

(O) PROCEDURES

21-21-02A

If AVNCS AIR FLO OFF message illuminates in flight, select AVNCS COOLING switch to OVRD.

21-21-02B

- Limit ground operation of electronic equipment to 45 minutes. The 45 minute time limit can be reset by using either of the following procedures:
 - Apply ground conditioned air directly into the E&E compartment through the E&E compartment door for at least 15 minutes. To reduce the possibility of exceeding the 45 minute time limit requirement, it is recommended that ground conditioned air be applied directly into the E&E compartment until push back.
 - Removing all aircraft electrical power or operating using only Ground Service electrical power with avionics turned off for at least 15 minutes.
- Ensure that AVNCS COOLING Selector Switch is in OVRD.
- If the E&E compartment door was opened, confirm E&E compartment door is closed prior to push back.

(M) PROCEDURES

21-21-02A

- For the Primary Radio Rack Fan Check Valve inoperative in the closed position:

- Check that RADIO RACK FAN CAUTION circuit breaker (B1-82 EPC CBP Location U-30) is in.
- Pull and secure the RADIO RACK FAN circuit breakers (all three phases).

<u>Circuit Breaker (CB#)</u>	<u>Panel</u>	<u>Location</u>
RADIO RACK FAN ØA (B1-78)	EPC CBP	H-4
RADIO RACK FAN ØB (B1-79)	EPC CBP	H-5
RADIO RACK FAN ØC (B1-80)	EPC CBP	H-6

- Verify that the AVNCS AIR FLO OFF Alert appears on the EAD.
- In the avionics compartment, check that the RIGHT RADIO FAN OFF light on the GEN CONT RACK is not on. This verifies operation of the Standby Fan.

- For the Standby Radio Rack Fan Check Valve inoperative in the closed position:

- Pull and secure the STANDBY RADIO RACK FAN circuit breakers (EPC CBP J-4, J-5, J-6) (all three phases).

<u>Circuit Breaker (CB#)</u>	<u>Panel</u>	<u>Location</u>
RADIO RACK FAN STANDBY ØA (B1-859)	GS CBP	A-02
RADIO RACK FAN STANDBY ØB (B1-860)	GS CBP	B-02
RADIO RACK FAN STANDBY ØC (B1-861)	GS CBP	C-02

- Verify that the AVNCS AIR FLO OFF Alert does not appear on the EAD. This verifies operation of the Primary Radio Rack Fan.

21-21-02B

For both Radio Rack Fan Check Valve and Standby Radio Rack Fan Check Valve inoperative in the closed position, or one of both inoperative open:

- Verify both Air Conditioning Systems are available for pressurized flight.
- Select AVNCS Cooling switch to OVRD.
- With both check valves inoperative closed, pull and secure the RADIO RACK FAN) and RADIO RACK FAN STANDBY circuit breakers.

<u>Circuit Breaker (CB#)</u>	<u>Panel</u>	<u>Location</u>
RADIO RACK FAN ØA (B1-78)	EPC CBP	H-4
RADIO RACK FAN ØB (B1-79)	EPC CBP	H-5
RADIO RACK FAN ØC (B1-80)	EPC CBP	H-6
RADIO RACK FAN STANDBY ØA (B1-859)	GS CBP	A-02
RADIO RACK FAN STANDBY ØB (B1-860)	GS CBP	B-02
RADIO RACK FAN STANDBY ØC (B1-861)	GS CBP	C-02

- Limit ground operation of electronic equipment to 45 minutes. The 45 minute time limit can be reset by using either of the following procedures:
 - Apply ground conditioned air directly into the E&E compartment through the E&E compartment door for at least 15 minutes. To reduce the possibility of exceeding the 45 minute time limit requirement, it is recommended that ground conditioned air be applied directly into the E&E compartment until push back.
 - Removing all aircraft electrical power or operating using only Ground Service electrical power with avionics turned off for at least 15 minutes.

21-22-03 Ram / Ground Conditioned Air Check Valves

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
21-22-03A	Ram / Ground Conditioned Air Check Valves "DISP.APPR.REQ'D"	N	2	1	C (O) (M) May be inoperative OPEN provided: 1. The Ground Conditioned Air Connector Door is verified closed and latched PRIOR TO EACH DEPARTURE, 2. Cockpit Instrument Cooling Fan is operative, 3. Associated Air Conditioning System is considered inoperative and deferred per MEL 21-51-01, and 4. The airplane remains at or below FL 250. Repetitive Check Required (Yellow Placard)
21-22-03B	Ram / Ground Conditioned Air Check Valves	Y	2	0	C May be inoperative CLOSED.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-22-03A	Cabin Pressure Control Panel	___ Ram / Ground A/C Check Valve INOP OPEN
21-22-03B	Cabin Pressure Control Panel	___ Ram / Ground A/C Check Valve(s) INOP CLOSED

(O) PROCEDURES

21-22-03A

Refer to (O) PROCEDURES for single Air Conditioning System operation listed in [MEL 21-51-01](#).

(M) PROCEDURES

21-22-03A

PRIOR TO EACH DEPARTURE, verify Ground Conditioned Air Connector Door is closed and latched.

21-22-04 Gasper Fan

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	Y	1	0	D
21-22-04	Gasper Fan	Y	1	0	D

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-22-04	Gasper Booster Fan Switch	Gasper Fan INOP

MAINTENANCE NOTE:

The fan may be secured, if necessary, by pulling and securing the associated Gasper Fan circuit breakers, as follows:

<u>Circuit Breaker (CB#)</u>	<u>Panel</u>	<u>Location</u>
GASPER BOOSTER FAN (B1-1170)	EPC CBP	J-27
GASPER BOOSTER FAN CONTROL (B1-1171)	EPC CBP	C-3

21-22-05 Conditioned Air Check Valves

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
21-22-05	Conditioned Air Check Valves	Y	2	0	C	May be inoperative OPEN provided both Ram / Ground Conditioned Air Check Valves are operative if the Air Conditioning Systems are to be used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-22-05	Cabin Pressure Control Panel	____ Conditioned Air Check Valve(s) INOP OPEN

21-22-06 Cold Air Check Valves

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
21-22-06	Cold Air Check Valves	Y	2	0	C	May be inoperative OPEN provided both Ram / Ground Conditioned Air Check Valves are operative if the Air Conditioning Systems are to be used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-22-06	Cabin Pressure Control Panel	____ Cold Air Check Valve(s) INOP OPEN

21-26-01 Cockpit Instrument Cooling Fan

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			1	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	Y			C	REMARKS OR EXCEPTIONS
21-26-01	Cockpit Instrument Cooling Fan	Y			C	(O) May be inoperative provided both Air Conditioning Systems are operative and are used whenever the EIS Display Units are operating.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-26-01	FLOW Indicator (Upper Ctr Inst Pnl)	Cockpit Instrument Cooling Fan INOP

(O) PROCEDURES

1. Operate both Air Conditioning Systems whenever electrical power is being supplied to the 115 VAC left and / or Right Busses.
2. If practical, heat output of Display Units may be reduced by rotating DU Brightness Control Knobs for any unnecessary Displays fully counter-clockwise to the OFF position.

21-29-01 Ram Air Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION	N	1	0	C	REMARKS OR EXCEPTIONS
21-29-01A	Ram Air Valve	N	1	0	C	May be inoperative provided: 1. Ram Air Valve is OPEN, and 2. All Ground Air Conditioning Check Valves are operative.
21-29-01B	Ram Air Valve	N	1	0	C	May be inoperative provided: 1. Ram Air Valve is CLOSED, and 2. Both Air Conditioning Systems are operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-29-01A	RAM AIR Switch	Ram Air Valve INOP OPEN
21-29-01B	RAM AIR Switch	Ram Air Valve INOP CLOSED

21-30-01 Cabin Pressure Control Systems (Auto And Manual)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
21-30-01A	Cabin Pressure Control Systems (Auto and Manual) "DISP.APPR.REQ'D"	Y	3	0	C (O) (M) May be inoperative provided: 1. Flight is conducted in an unpressurized configuration, 2. Cabin Air Outflow Valve remains OPEN, and 3. Extended overwater flight is prohibited.
21-30-01B	Cabin Pressure Control Systems (Automatic)	Y	2	1	C May be inoperative.
21-30-01C	Cabin Pressure Control Systems (Automatic)	Y	2	0	C (O) (M) May be inoperative provided, Manual pressurization control is verified operative PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)
21-30-01D	Cabin Pressure Control Systems (Manual) "DISP.APPR.REQ'D"	Y	1	0	C (O) (M) May be inoperative for unpressurized flight provided: 1. Flight is conducted in an unpressurized configuration, 2. Cabin Air Outflow Valve remains OPEN, and 3. Extended overwater flight is prohibited.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-30-01A	Cabin Pressure Control Panel	____ Both Cabin Pressure Controller Systems INOP
21-30-01B	Cabin Pressure Control Panel	Auto Cabin Pressure Controller Systems INOP
21-30-01C	Cabin Pressure Control Panel	Auto Cabin Pressure Controller Systems INOP
21-30-01D	Cabin Pressure Control Panel	Manual Cabin Pressure Controller Systems INOP

(O) PROCEDURES

21-30-01A, 21-30-01D

1. Refer to [QRH](#), Non-Normal Checklist, Section 2, Air Systems, Unpressurized Flight.
2. Limit flight altitude to a maximum of 10,000 ft. MSL.

21-30-01C

Refer to [QRH](#), Non-Normal Checklist, Section 2, Air Systems, Manual Pressurization Control.

(M) PROCEDURES

21-30-01A, 21-30-01D

1. Select MANUAL pushbutton on Cabin Pressure Control Panel in the cockpit.
2. Toggle CLIMB / DESC switch in both directions, and observe corresponding Valve movement on the adjacent valve position indicator.
3. At the Outflow Valve, verify Valve is fully open by observing position to be approximately 90 degrees from flush.

21-30-01C

PRIOR TO EACH DEPARTURE:

1. Select MANUAL pushbutton on Cabin Pressure Control Panel in the cockpit.
2. Toggle CLIMB / DESC switch in both directions, and observe corresponding Valve movement on the adjacent valve position indicator.

21-34-02 Cabin Pressure Safety Valves

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
21-34-02	Cabin Pressure Safety Valves "DISP.APPR.REQ'D"	Y	2	0	C (O) (M) May be inoperative provided: 1. Flight is conducted in an unpressurized configuration, 2. The Cabin Air Outflow Valve remains OPEN, and 3. Extended overwater flight is prohibited.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-34-02	Cabin Pressure Control Panel	____ Cabin Pressure Safety Valve(s) INOP

(O) PROCEDURES

1. Refer to [QRH](#), Non-Normal Checklist, Section 2, Air Systems, Unpressurized Flight.
2. Limit flight altitude to a maximum of 10,000 ft. MSL.

(M) PROCEDURES

1. Select MANUAL pushbutton on Cabin Pressure Control Panel in the cockpit.
2. Move CLIMB / DESC switch to CLIMB position and observe corresponding valve movement on the adjacent valve position indicator.
3. At the Outflow Valve, verify Valve is fully open by observing position to be approximately 90° from flush.

21-42-02 Forward Cargo Compartment Heating System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	Y	1	0	C
21-42-02	Forward Cargo Compartment Heating System "DISP.APPR.REQ'D"	Y	1	0	C
					(M) May be inoperative provided system is deactivated. Note: No live animals or temperature sensitive cargo may be loaded in the forward cargo compartment.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-42-02	AVIONICS COOLING Switch	FWD Cargo Compartment Heater System INOP

(M) PROCEDURES

Pull and secure the three (3) CARGO COMPARTMENT HEATER circuit breakers:

<u>Circuit Breaker (CB#)</u>	<u>Panel</u>	<u>Location</u>
CARGO COMPARTMENT HEATER 0A (B1-655)	EPC	J-12
CARGO COMPARTMENT HEATER 0A (B1-656)	EPC	J-13
CARGO COMPARTMENT HEATER 0A (B1-657)	EPC	J-14

21-51-01 Air Conditioning Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY				
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS	
21-51-01	Air Conditioning Systems "DISP.APPR.REQ'D"	N	2	1	C	(O) (M) May be inoperative provided: 1. Upon initial MEL application, MCC shall review the 7 day history, of the operative pack and pneumatic bleed system associated with the operative pack to ensure no known conditions exist that could affect proper operation of the pressurization system, 2. Cockpit Instrument Cooling Fan is operative, 3. Affected Flow Control Valve is manually CLOSED, and 4. The aircraft remains at or below FL 250.	

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-51-01	Associated Pack Switch	___ A/C System INOP

(O) PROCEDURES

Note: Higher engine idle speeds will be noted with one pack switch in the off position (Pack Inoperative), therefore careful planning must be done during the descent phase in order to meet altitude crossing restrictions.

For takeoff with operative pack ON:

1. With APU or Engine Bleed Air, verify the inoperative pack is producing no flow.
 - A. Select the Gasper Fan Switch to the OFF position.
 - B. Select the inoperative Pack Switch to AUTO.
 - C. Select the operative Pack Switch to OFF.
 - D. Verify by feel at the open Air Conditioning outlets that there is no air flow.

Note: Some air flow is to be expected from the air vents due to the avionics cooling fan.

2. Prior to takeoff, select both Pack Switches to AUTO.

Note: The PACK L/R FLOW DISAG alert associated with the inoperative Pack will be displayed on the EAD Air Page.

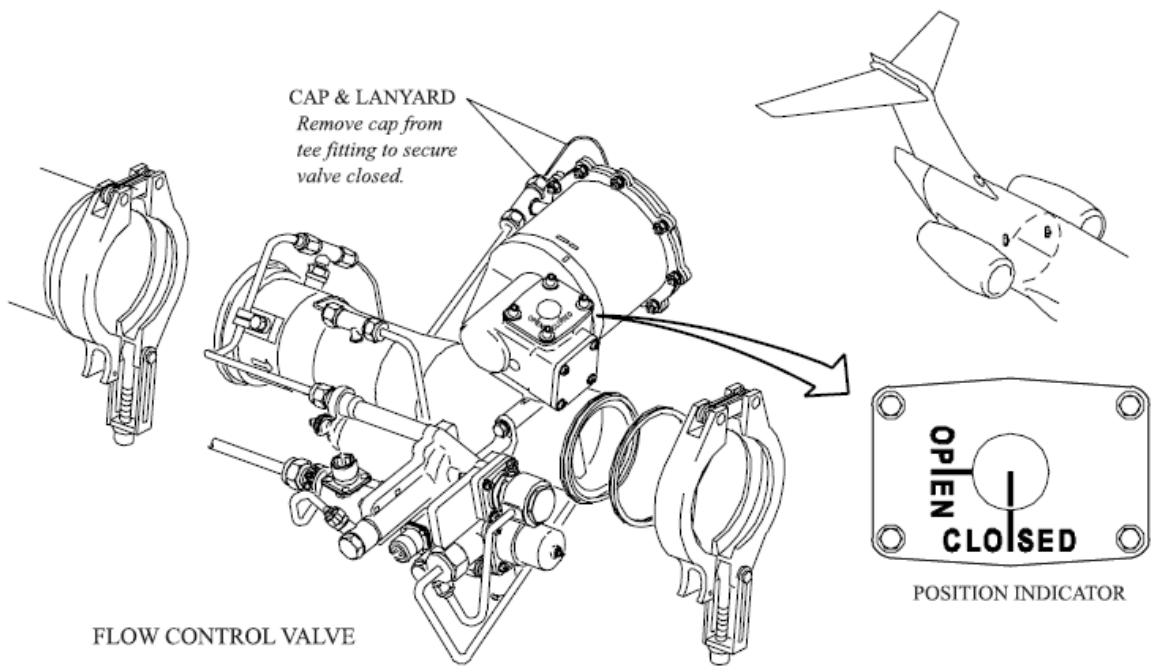
- A. After first power reduction following takeoff, select Pack Switch for the inoperative Pack to OFF. Verify associated PACK L/R FLOW DISAG Alert extinguishes.
- B. Limit flight altitude to a maximum of FL250.

(M) PROCEDURES

 [Link to Learning Video](#)

1. Select the affected L or R PACK Switch on the Cockpit Overhead Panel to OFF.
2. Secure the Flow Control Valve for the affected Air Conditioning System in the Closed position as follows:
 - A. Gain access to the valve, located in the aft accessory compartment upstream of the Primary Heat Exchanger.
 - B. Remove the lanyard cap from the Flow Control Valve Actuator tee fitting.
 - C. Verify position indicator on the Valve body indicates the Valve is CLOSED.
3. With APU or Engine Bleed Air, verify the inoperative Pack is producing no flow.
 - A. Select the Gasper Fan Switch to the OFF position.
 - B. Select the inoperative Pack Switch to AUTO.
 - C. Select the operative Pack Switch to OFF.
 - D. Verify by feel at the open Air Conditioning outlets that there is no air flow.

CONTINUED ON NEXT PAGE

Figure 21-2: Flow Control Valve


21-51-02 Air Conditioning Flow Control Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
21-51-02A	Air Conditioning Flow Control Valve "DISP.APPR.REQ'D"	N	2	1	C (O) (M) May be inoperative provided: 1. Cockpit Instrument Cooling Fan is operative, 2. Affected Flow Control Valve is CLOSED, 3. Associated Air Conditioning System is not used, and 4. The airplane remains at or below FL 250.
21-51-02B	Air Conditioning Flow Control Valve (Normal Mode) "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided High Flow mode of affected Flow Control Valve is operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-51-02A	L or R PACK Switch	1. ____ A/C Flow Control Valve INOP 2. Do NOT Use ____ A/C System
21-51-02B	Associated A/C SUPPLY Switches	____ A/C Flow Control Valve in HIGH

(O) PROCEDURES

21-51-02A

Refer to (O) PROCEDURES for one Air Conditioning System inoperative given in [MEL 21-51-01](#).

21-51-02B

- Place FLOW Selector Switch on the overhead panel to HIGH.
- Both Air Conditioning Packs must be OFF for takeoff.
- After takeoff, restore Air Conditioning Systems after a minimum of 400 feet AGL is attained.
- In the event of an engine failure during takeoff, do not restore Air Conditioning Packs until obstacle clearance is assured, and not before flap retraction.

21-51-02B	AIR CONDITIONING FLOW CONTROL VALVE (NORMAL MODE)
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	Increase BLOCK FUEL by 0.5% per affected valve.
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF None LANDING None

(M) PROCEDURES

21-51-02A

Refer to (M) PROCEDURES for one Air Conditioning System inoperative given in [MEL 21-51-01](#).

21-52-01 Pack Flow Indication Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
21-52-01	Pack Flow Indication Systems	Y	2	0	C	May be inoperative provided associated Air Conditioning System is operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-52-01	Below Captains N.D. Panel	____ A/C Flow Indication INOP

21-53-02 Compressor Discharge and Turbine Inlet Thermal Switches

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
21-53-02	Compressor Discharge and Turbine Inlet Thermal Switches	N	4	0	C	(M) May be inoperative provided the Air Conditioning System Shutdown Thermal Switch is operative on the associated Air Conditioning System.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-53-02	L or R PACK Switch	____ Compressor Discharge / Turbine Inlet Thermal Switch(es) INOP

OPERATIONS NOTE:

Do not use affected pack for takeoff and landing.

(M) PROCEDURES

1. Access the affected Thermal Switch in the aft accessory compartment.
2. Remove the connector from the Switch. Coil and stow the connector.
3. If compressor Discharge and Turbine Inlet Thermal Switches are both inoperative and in the same Air Conditioning System, proper operation of the Air Conditioning System Shutdown Thermal Switch in the affected Air Conditioning System may be verified as follows:
 - A. Access the associated Air conditioning Thermal Switch above the ceiling panel in the passenger compartment.
 - B. Remove the Thermal Switch from the Conditioned Air Duct.
 - C. With the associated Air Conditioning System operating, use a heat gun to supply heat to the sensor in excess of 198 degrees F.
 - D. Verify the associated Air Conditioning System shuts down within 90 seconds.

21-53-03 Air Conditioning System Shutdown Thermal Switches (190 Degrees F)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
21-53-03A	Air Conditioning System Shutdown Thermal Switches (190 Degrees F) "DISP.APPR.REQ'D"	N	2	1	C (O) May be inoperative provided: 1. Cockpit Instrument Cooling Fan is operative, 2. Associated Air Conditioning System is considered inoperative and deferred per MEL 21-51-01, and 3. The airplane remains at or below FL250.
21-53-03B	Air Conditioning System Shutdown Thermal Switches (190 Degrees F)	N	2	1	C (M) May be inoperative provided Cabin and Cockpit Duct Overheat Alerts are operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-53-03A	L or Right PACK Switch	____ A/C INOP
21-53-03B	L or Right PACK Switch	____ A/C System Auto Shutdown INOP

(O) PROCEDURES

21-53-03A

Refer to (O) PROCEDURES for one Air Conditioning System inoperative given in MEL 21-51-01.

(M) PROCEDURES

21-53-03B

1. Access the affected Thermal Switch above the ceiling panel in the passenger compartment.
2. Remove the connector from the Switch. Coil and stow the connector.

21-54-01 Air Conditioning Water Separators

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
21-54-01	Air Conditioning Water Separators "DISP.APPR.REQ'D"	N	2	1	C	(O) May be inoperative provided: 1. Cockpit Instrument Cooling Fan is operative, 2. Associated Air Conditioning System is considered inoperative and deferred by MEL 21-51-01, and 3. The airplane remains at or below FL 250.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-54-01	L or R PACK Switch	1. ____ A/C Water Separator INOP 2. Do NOT Use ____ A/C System

(O) PROCEDURES

Refer to (O) PROCEDURES for one Air Conditioning System inoperative given in MEL 21-51-01.

21-54-02 Water Separator 35 Degrees F Thermostats

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY			
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS	
21-54-02A	Water Separator 35 Degrees F Thermostats	Y	2	1	C	May be inoperative provided the associated Air Conditioning Pack otherwise operates normally.	
21-54-02B	Water Separator 35 Degrees F Thermostats "DISP.APPR.REQ'D"	N	2	1	C	(O) May be inoperative provided: 1. Cockpit Instrument Cooling Fan is operative, 2. Associated Air Conditioning System is considered inoperative and deferred per MEL 21-51-01, and 3. The airplane remains at or below FL 250.	

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-54-02A	L or R PACK Switch	____ Water Separator 35° F Thermostats INOP
21-54-02B	L or R PACK Switch	1. ____ Water Separator 35° F Thermostats INOP 2. Do NOT Use ____ A/C System

(O) PROCEDURES

21-54-02B

Refer to (O) PROCEDURES for one Air Conditioning System inoperative given in [MEL 21-51-01](#).

21-54-03 Water Separator 35 Degrees F Control Valves

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
21-54-03	Water Separator 35 Degrees F Control Valves "DISP.APPR.REQ'D"	N	2	1	C	(O) May be inoperative provided: 1. Cockpit Instrument Cooling Fan is operative, 2. Associated Air Conditioning System is considered inoperative and deferred per MEL 21-51-01, and 3. The airplane remains at or below FL 250.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-54-03	L or R PACK Switch	1. ____ Water Separator 35° F Valve INOP 2. Do NOT Use ____ A/C System

(O) PROCEDURES

Refer to (O) PROCEDURES for one Air Conditioning System inoperative given in MEL 21-51-01

21-56-01 Air Conditioning Automatic Shut-Off System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			1	0	C	REPAIR CATEGORY
ITEM NO.	NAME/DESCRIPTION	Y				REMARKS OR EXCEPTIONS
21-56-01	Air Conditioning Automatic Shut-Off System	Y				(O) May be inoperative provided: 1. Both Air Conditioning Systems are selected OFF prior to takeoff, and 2. Air Conditioning Systems are not selected ON until a minimum altitude of 400 feet AGL is attained.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-56-01	L or R PACK Switch	A/C Auto Shut-off System INOP

(O) PROCEDURES

- Prior to takeoff select the L and R PACK Switches to OFF.
- For normal takeoff, after a minimum of 400 feet AGL is attained, select L and R PACK Switches to ON one at a time.
- In the event of an engine failure on takeoff, select L or R PACK Switch on the remaining side to AUTO only after obstacle clearance is assured, and not before flap retraction.

21-62-01 Cockpit / Cabin Compartment Auto Temperature Control Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			1	C	REPAIR CATEGORY
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
21-62-01A	Cockpit / Cabin Compartment Auto Temperature Control Systems "DISP.APPR.REQ'D"	N	2	1	(O) May be inoperative provided: 1. Cockpit Instrument Cooling Fan is operative, 2. Associated Air Conditioning System is considered inoperative and deferred per MEL 21-51-01, and 3. The airplane remains at or below FL 250.
21-62-01B	Cockpit / Cabin Compartment Auto Temperature Control Systems	Y	2	0	(M) May be inoperative provided associated Manual Temperature Control System is operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-62-01A	1. Associated CKPT / CABIN TEMP Selector 2. L or R PACK Switch	1. ____ Compartment Auto Temperature Control Systems INOP 2. Do NOT Use ____ A/C System
21-62-01B	Associated CKPT / CABIN TEMP Selector	____ Compartment Auto Temperature Control Systems INOP

(O) PROCEDURES

21-62-01A

1. Refer to (O) PROCEDURES for one Air Conditioning System inoperative given in MEL 21-51-01.
2. Adjust temperature control manually in flight as required to maintain cockpit/cabin comfort.

(M) PROCEDURES

21-62-01B

1. Select associated CKPT TEMP or CABIN TEMP Selector Knob to MANUAL STOP.
2. Toggle the selector knob toward HOT and COLD. Verify valve position movement on the AIR Synoptic Page corresponds with knob selection.

21-62-02 Cockpit / Cabin Compartment Temperature Control Valves

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
21-62-02	Cockpit / Cabin Compartment Temperature Control Valves "DISP.APPR.REQ'D"	N	2	1	C (O) May be inoperative provided: 1. Cockpit Instrument Cooling Fan is operative, 2. Associated Air Conditioning System is considered inoperative and deferred per MEL 21-51-01, and 3. The airplane remains at or below FL 250.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-62-02	L or R PACK Switch	1. ____ Temperature Control Valve INOP 2. Do NOT Use ____ A/C System

(O) PROCEDURES

Refer to (O) PROCEDURES for one Air Conditioning System inoperative given in [MEL 21-51-01](#).

21-63-01 Temperature Control Valve Position Indicating Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
21-63-01	Temperature Control Valve Position Indicating Systems	Y	2	0	C	May be inoperative provided the associated Automatic Temperature Control System is operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-63-01	Below Captains N.D. Panel	____Temperature Control Valve position INOP

21-64-01 Cockpit/Cabin Temperature Indicating Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
21-64-01	Cockpit/Cabin Temperature Indicating Systems	Y	2	0	C (O) May be inoperative provided: 1. Both Air Conditioning System Shutdown Thermal Switches are operative, and 2. Cockpit/Cabin Compartment Auto Temperature Control Systems are operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
21-64-01	Cabin TEMP SEL Switch	CABIN TEMP Indicator INOP

(O) PROCEDURES

- Toggle the CKPT TEMP and/or CABIN TEMP Selector knob toward Hot and Cold. Verify valve position movement on the AIR Synoptic Page corresponds with knob selection.
- Operate the Air Conditioning Systems with the CKPT TEMP and CABIN TEMP Selector Knobs in AUTO.

Note: *If packs operated properly in AUTO mode during prior flight, this test may be omitted*

22 Autoflight

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22-00-00L LLM Autoland Status (Lower Landing Minimums)

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22-14-01 Autopilot Control Wheel Disengage Switches

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FLIGHT CONTROL PANEL PITCH MODE CONTROLS
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22-31-02 Go-Around Palm Switches

22-00-00 LLM Autoland Status (Lower Landing Minimums)

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION			REMARKS OR EXCEPTIONS
22-00-00L	LLM Autoland Status (Lower Landing Minimums) "DISP.APPR.REQ'D" Administrative Control Item	N	-	- (M) Aircraft restricted from autopilot coupled Autoland due to chronic problem, or no Autoland in 30 days. 1. This Item Number is only used for deferring the Autoland status of the aircraft and MCC authorization is required prior to its usage or removal. 2. Practice Autolands are permitted if authorized by MCC.

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-00-00L	Adjacent to each PFD.	Autoland Not Authorized

(M) PROCEDURES

1. Create a new log page in the aircraft logbook using the appropriate MEL Item Number (contact MCC for instructions).
2. The deferral code LLM must also be entered on the log page and the placard when deferring this item.
3. MCC authorization is required prior to its usage or removal.

22-10-01 Autopilot and Mode Controls

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
22-10-01A	Autopilot "DISP.APPR.REQ'D"	Y	2	1	C (M) May be inoperative provided RNAV operations do not require their use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements . CAT II/III Not Authorized
22-10-01B	Autopilot "DISP.APPR.REQ'D"	Y	2	0	B May be inoperative provided: 1. Enroute operations and/or approach minimums do not require its use, 2. At least one Autothrottle is operative, and 3. RNAV operations do not require their use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements . CAT II/III Not Authorized AUTOLAND Not Authorized RVSM Operations Not Authorized
22-10-01C	Altitude Hold Mode "DISP.APPR.REQ'D"	Y	2	1	C May be inoperative provided RNAV operations do not require their use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements .
22-10-01D	Altitude Hold Mode "DISP.APPR.REQ'D"	Y	2	0	B May be inoperative provided RNAV operations do not require their use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements . RVSM Operations Not Authorized
22-10-01E	Altitude Preselect Mode "DISP.APPR.REQ'D"	Y	2	1	C May be inoperative provided RNAV operations do not require their use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements . RVSM Operations Not Authorized
22-10-01F	Vertical Speed Mode "DISP.APPR.REQ'D"	Y	2	0	C May be inoperative provided RNAV operations do not require their use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements .
22-10-01G	IAS/Mach Hold Mode	Y	2	0	C
22-10-01H	PROF Mode "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided RNAV operations do not require their use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements .

CONTINUED ON NEXT PAGE

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD						REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION						
22-10-01I	NAV Mode (FMS) “DISP.APPR.REQ’D”	Y	2	0	C	May be inoperative provided RNAV operations do not require their use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements.	
22-10-01J	ILS Capture “DISP.APPR.REQ’D”	Y	2	0	C	May be inoperative provided approach minimums do not require its use, refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized	
22-10-01K	Autoland Mode “DISP.APPR.REQ’D”	Y	2	0	C	May be inoperative provided approach minimums do not require its use, refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized	
22-10-01L	Heading Select Mode	Y	2	1	C		
22-10-01M	Heading Select Mode	Y	2	0	B		
22-10-01N	Heading Hold Mode	Y	2	0	C		

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-10-01A	Near AUTOFLIGHT Push Button	___ Autopilot INOP
22-10-01B	Near AUTOFLIGHT Push Button	___ Autopilot(s) INOP
22-10-01C	Near Altitude Select Knob	___ ALT Hold INOP
22-10-01D	Near Altitude Select Knob	___ ALT Hold INOP
22-10-01E	Near Altitude Select Knob	___ Altitude Preselect INOP
22-10-01F	Near Vertical Speed/FPA Select Wheel	___ Vert Speed INOP
22-10-01G	Near Speed Select Knob	___ IAS / MACH Hold INOP
22-10-01H	Near PROF Push Button	___ PROF INOP
22-10-01I	Near NAV Push Button	___ NAV INOP
22-10-01J	Near APPR/LAND Push Button	___ ILS Capture INOP
22-10-01K	Near APPR/LAND Push Button	___ Autoland INOP
22-10-01L	Near HDG/TRK Select Knob	___ HDG Select INOP
22-10-01M	Near HDG/TRK Select Knob	___ HDG Select INOP
22-10-01N	Near HDG/TRK Select Knob	___ HDG Hold INOP

CONTINUED ON NEXT PAGE

(O) PROCEDURES

22-10-01H

Refer to the [FCTM](#), Initial Climb - All Engines, Pitch Modes, NO PROF, for Takeoff without PROF engaged.

(M) PROCEDURES

22-10-01A

Pull and secure the appropriate Autopilot circuit breakers to prevent nuisance disconnects of the remaining system.

<u>Circuit Breaker (CB#)</u>	<u>Panel</u>	<u>Location</u>
AUTOPILOT-1 (B1-353)	EPC	E-19
AUTOPILOT-2 (B1-354)	EPC	E-7

22-11-01 Elevator Drive Servo

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
22-11-01	Elevator Drive Servo "DISP.APPR.REQ'D"	Y	1	0 B	May be inoperative provided: 1. Autopilot is not used, 2. Enroute operations and approach minimums do not require use of Autopilot, and 3. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>RVSM Operations</u> Not Authorized <u>AUTOLAND</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-11-01	Flight Control Panel	1. Elevator Drive Servo INOP 2. Do No Use Autopilot

22-12-01 Aileron Drive Servo

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
22-12-01	Aileron Drive Servo "DISP.APPR.REQ'D"	Y	1	0 B	May be inoperative provided: 1. Autopilot is not used, 2. Enroute operations and approach minimums do not require use of Autopilot, and 3. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized <u>RVSM Operations</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-12-01	Flight Control Panel	Aileron Drive Servo INOP

22-13-01 Rudder Drive Servo

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
22-13-01	Rudder Drive Servo "DISP.APPR.REQ'D"	Y	1	0	C
					May be inoperative provided: 1. Autopilot is not used for Takeoff, Approach, Landing, and Go-Around, 2. Enroute operations and approach minimums do not require use of Autopilot, and 3. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>RVSM Operations</u> Not Authorized <u>AUTOLAND</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-13-01	Flight Control Panel	Rudder Drive Servo INOP

22-14-01 Autopilot Control Wheel Disengage Switches

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
22-14-01	Autopilot Control Wheel Disengage Switches "DISP.APPR.REQ'D"	Y	2	1	C May be inoperative provided: 1. Pilot Controlling the airplane with the autopilot has the operative switch, 2. Autopilot is not used below 1500feet AGL, 3. Approach minimums do not require the use of the Autopilot, and 4. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized Note: CAT I (ILS) is authorized if at least one Flight Director System is operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-14-01	Flight Control Panel	____ Autopilot Control Wheel Disengage Switch INOP

22-15-01 Flight Control Panel Speed Mode Controls

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
FLIGHT CONTROL PANEL SPEED MODE CONTROLS						
22-15-01A	IAS/MACH Selector Pushbutton	Y	1	0	C	May be inoperative provided the airspeed bug is in the IAS Mode on each PFD.
22-15-01B	IAS/MACH Display (Glareshield)	Y	1	0	C	May be inoperative provided the airspeed bug is operative on each PFD.
22-15-01C	Speed Select Knob Push Function	Y	1	0	C	
22-15-01D	FMS SPD Mode Select Pushbutton	Y	1	0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-15-01A	Flight Control Panel	IAS/MACH Push Button INOP
22-15-01B	Flight Control Panel	IAS/MACH Display INOP
22-15-01C	Flight Control Panel	Speed Select Knob Push Button INOP
22-15-01D	Flight Control Panel	FMS SPD Mode Select Push Button INOP

22-15-02 Flight Control Panel Heading Mode Control

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
FLIGHT CONTROL PANEL HEADING MODE CONTROL						
22-15-02A	HDG/TRK Mode Selector Pushbutton	Y	1	0	C	May be inoperative provided Heading Mode is displayed on each ND.
22-15-02B	HDG/TRK Display	Y	1	0	C	May be inoperative provided Heading and Track Mode information is displayed on each ND.
22-15-02C	HDG/TRK Selector Knob Push Function	Y	1	0	C	
22-15-02D	Bank Angle Limit Knob Select Function	Y	1	0	C	May be inoperative provided the Automatic Bank Angle Limit Function is operative.
22-15-02E	FMS NAV Mode Select Pushbutton “DISP.APPR.REQ'D”	Y	1	0	C	May be inoperative provided RNAV operations do not require its use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements.

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-15-02A	Flight Control Panel	HDG/TRK Push Button INOP
22-15-02B	Flight Control Panel	HDG/TRK Display INOP
22-15-02C	Flight Control Panel	HDG/TRK Select Knob Push Button INOP
22-15-02D	Flight Control Panel	Bank Angle Limit Select Knob INOP
22-15-02E	Flight Control Panel	FMS NAV Mode Push Button INOP

22-15-03 Flight Control Panel Altitude Mode Controls

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
FLIGHT CONTROL PANEL ALTITUDE MODE CONTROLS					
22-15-03A	FEET/METER Selector Pushbutton	Y	1	0	C May be inoperative provided: 1. Altitude Display on each PFD is in FEET, and 2. Altitude Select Display on each PFD is operative.
22-15-03B	Altitude Display	Y	1	0	C May be inoperative provided: 1. PROF Mode is not used, and 2. Altitude Select Display on each PFD is operative.
22-15-03C	Altitude Select Knob Push Function “DISP.APPR.REQ’D”	Y	1	0	C May be inoperative provided enroute operations do not require its use. <u>RVSM Operations</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-15-03A	Flight Control Panel	FEET/METER Push Button INOP
22-15-03B	Flight Control Panel	Altitude Display INOP
22-15-03C	Flight Control Panel	Altitude Display Knob Push Function INOP

OPERATIONS NOTE:

If PROF Mode is not used , careful attention will be needed for altitude crossing restraints. Refer to [VOL 1](#) for normal takeoff without PROF and engine fire/failure takeoff without PROF.

22-15-04 Flight Control Panel Pitch Mode Controls

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
FLIGHT CONTROL PANEL PITCH MODE CONTROLS						
22-15-04A	V/S-FPA Selector Pushbutton	Y	1	0	C	May be inoperative provided Vertical Speed Scale is displayed on each PFD.
22-15-04B	Vertical Speed / Flight Path Angle Display Window	Y	1	0	C	May be inoperative provided Vertical Speed Scale is displayed on each PFD.
22-15-04C	Pitch Selector Wheel “DISP.APPR.REQ’D”	Y	1	0	C	May be inoperative provided approach minimums do not require its use.
22-15-04D	FMS PROF Mode Select Pushbutton “DISP.APPR.REQ’D”	Y	1	0	C	May be inoperative provided RNAV operations do not require its use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements.

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-15-04A	Flight Control Panel	V/S-FPA Push Button INOP
22-15-04B	Flight Control Panel	Vertical Speed / Flight Path Angle Display INOP
22-15-04C	Flight Control Panel	Pitch Selector Wheel INOP
22-15-04D	Flight Control Panel	FMS PROF Mode Select Pushbutton INOP

22-22-01 Automatic Pitch Trim

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD			Y	2	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION						REMARKS OR EXCEPTIONS
22-22-01	Automatic Pitch Trim "DISP.APPR.REQ'D"				C	(O) May be inoperative provided: 1. It is verified that Manual Alternate Trim is operative, 2. Alternate procedures are used if Autopilot is to be used for approach and landing, 3. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements, and 4. Mach Trim Systems is considered inoperative and deferred per 22-22-02	

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-22-01	Flight Control Panel	____ Automatic Pitch Trim INOP

(O) PROCEDURES

1. Verify Manual Alternate Trim functions properly by operating ALT LONG TRIM Switches on the center pedestal and observing corresponding movement of the Stabilizer on the CONFIG Synoptic page.
2. In flight, trim aircraft manually prior to engaging Autopilot.
3. If the AUTO TRIM FAIL Alert appears, re-trim the stabilizer by either of the following methods:
 - A. Operate the ALT LONG TRIM Switches on the center pedestal (which does not require autopilot disconnect), or
 - B. Disconnect the Autopilot, manually trim the aircraft using the Control Wheel Trim Switches, and then re-engage the Autopilot.
4. If the Autopilot is to be used for landing, manually trim the aircraft and track the glideslope at final approach speed prior to Autopilot engagement.

22-22-02 Mach Trim Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
22-22-02	Mach Trim Systems "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided: 1. Flight speed is limited to Mach 0.78 or less, and 2. Aircraft is not operated within the boundaries of a Critical Terrain Box.

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-22-02	Flight Control Panel	Mach Trim INOP

(O) PROCEDURES

1. Maximum operating speed with inoperative Mach Trim is 0.78 Mach.
2. Program KIAS/Mach speed restriction in FMS.

22-23-01 Yaw Damper Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY
ITEM NO.	NAME/DESCRIPTION				
22-23-01	Yaw Damper Systems	Y	2	0	C
(M) May be inoperative. Note: Fuel burn may be affected unless Yaw Damper Actuator is parked in neutral position.					

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-23-01	Flight Control Panel	____ Yaw Damper System INOP

(M) PROCEDURES

1. Pull and secure the circuit breakers for the Yaw Damper System as follows:

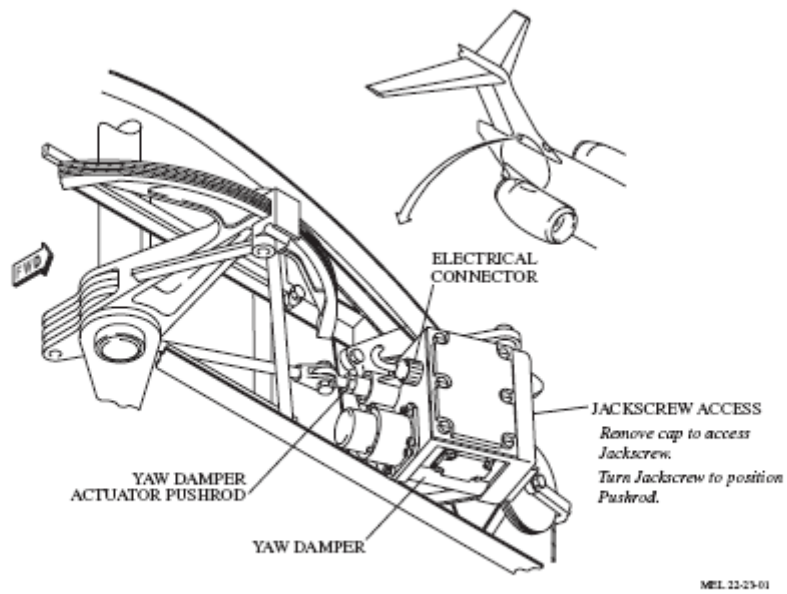
<u>Circuit Breaker (CB#)</u>	<u>Panel</u>	<u>Location</u>
YAW DAMPER-1 (B1-345)	EPC CBP	G-21
YAW DAMPER-2 (B1-346)	EPC CBP	G-8
YAW DAMPER (B1-1603)	EPC CBP	B-13

2. Aircraft trim drag will be minimized if the Yaw Damper Actuator is in its neutral position. If desired, the following procedure may be used to verify Yaw Damper Actuator position, and to reposition the Actuator.

- A. Turn the Aux Hydraulic Pump ON to pressurize the right hydraulic system.
- B. Align the rudder pedals and move rudder trim to neutral, as verified on the CONFIG Synoptic Page.

Note: *Nose wheel will move when rudder pedals are pushed. If necessary, disable nose wheel steering to prevent nose wheel movement during this procedure.*

- C. Visually check that rudder is aligned with the upper trailing edge of the stabilizer.
- D. If rudder is not aligned, the Yaw Damper Actuator may be neutralized as follows (refer to figure on following page): *If this procedure is necessary, maintenance will be required and Flight Crew MAY NOT PLACARD.*
 - i. Access the Yaw Damper Actuator in the aft canted bulkhead of the aft accessory compartment.
 - ii. Remove cap directly behind Actuator Pushrod to access actuator Jackscrew.
 - iii. Turn Jackscrew to reposition Actuator Pushrod.

Figure 22-1: Yaw Damper Systems


22-30-01 Autothrottle Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
22-30-01	Autothrottle Systems "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided approach minimums do not require their use, refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements . Note: One, two, or no throttles may move when one or both autothrottle systems are deferred. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-30-01	AUTOFLIGHT Push Button	____ Autothrottle INOP

(O) PROCEDURES

Brief the possibility of one or both throttle(s) remaining stationary (with autothrottles on) during all phases of flight, to include during initial takeoff power application, windshear events, and go-arounds. The threat of one throttle advancing with one remaining stationary requires highlighting in all briefings with a thorough mitigation strategy planned and understood by both the Pilot Flying and the Pilot Monitoring.

22-31-01 Autothrottle Disengage Switches

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
22-31-01	Autothrottle Disengage Switches "DISP.APPR.REQ'D"	Y	2	0	C May be inoperative provided Autothrottles are not used, refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized

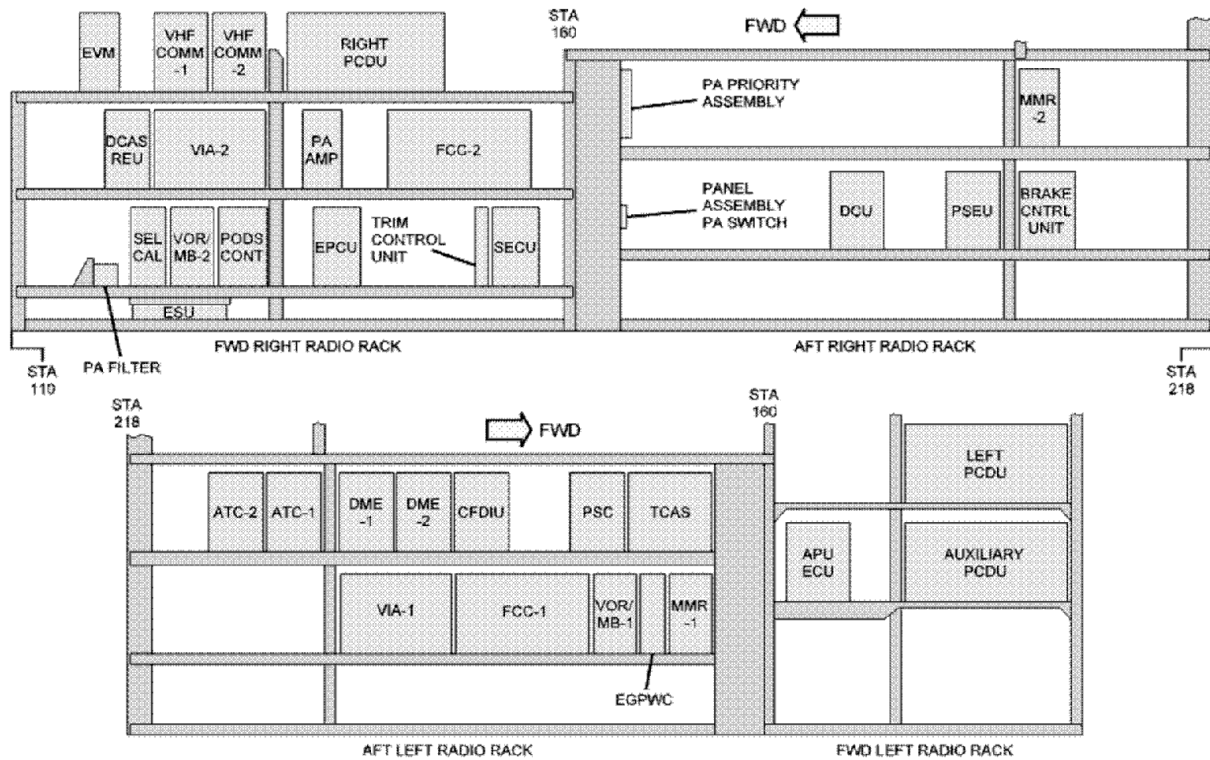
MEL ITEM	TAB LOCATION	PLACARD TEXT
22-31-01	Throttle Control Module	____ Autothrottle Switch(s) INOP

22-31-02 Go-Around Palm Switches

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
22-31-02	Go-Around Palm Switches "DISP.APPR.REQ'D"	Y	2	0 C	May be inoperative provided: 1. Approach procedures do not require their use, and 2. RNAV operations do not require their use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized Note: <i>Flight Director Go-Around and Windshear guidance are not available with both Go-Around Switches inoperative.</i>

MEL ITEM	TAB LOCATION	PLACARD TEXT
22-31-02	Throttle Control Module	___Go Around Switch(es) INOP

23 Communications



ELECTRICAL/ELECTRONICS COMPARTMENT RACKS

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	ARINC / ACARS Systems
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23-51-06 Oxygen (Smoke) Mask Microphones

23-51-06 Oxygen (Smoke) Mask Microphones

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Captain / First Officer Push-To-Talk Switches

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23-71-01 Cockpit Voice Recorder (CVR) System

23-71-01 Cockpit Voice Recorder (CVR) System

23-12-01 Very High Frequency (VHF) Communication System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
23-12-01A	Very High Frequency (VHF) Communication System (No. 2)	Y	1	0	C	May be inoperative provided #1 VHF Comm and #3 VHF Comm systems are operative.
23-12-01B	Very High Frequency (VHF) Communication System (No. 3) “DISP.APPR.REQ’D”	Y	1	0	C	May be inoperative provided: 1. #1 VHF Comm and #2 VHF Comm Systems are operative, and 2. ACARS is considered inoperative and deferred per MEL 23-24-01A .

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-12-01A	#2 VHF Comm Control Panel	#2 VHF Comm INOP
23-12-01B	#3 VHF Comm Control Panel	#3 VHF Comm INOP

23-22-01 Selective Call System (SELCAL)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
23-22-01A	Selective Call System (SELCAL)	Y	1	0	D	(O) May be inoperative provided procedures do not require its use.
23-22-01B	Selective Call System (SELCAL)	Y	1	0	C	(O) May be inoperative provided cockpit crew continuously monitors appropriate radio frequencies as required.
23-22-01C	Channels	Y	1	0	D	(O) May be inoperative provided procedures do not require its use.
23-22-01D	Channels	Y	1	0	C	(O) May be inoperative provided cockpit crew continuously monitors appropriate radio frequencies as required.

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-22-01A	Audio Control Panel	SELCAL INOP
23-22-01B	Audio Control Panel	SELCAL INOP
23-22-01C	Audio Control Panel	SELCAL Channels INOP
23-22-01D	Audio Control Panel	SELCAL Channels INOP

(O) PROCEDURES

23-22-01A, 23-22-01C

If ACARS is inoperative, a listening watch of company frequency is required.

23-22-01B, 23-22-01D

Cockpit crew must maintain a continuous listening watch on appropriate radio frequencies as required.

23-24-01 ARINC / ACARS Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
ARINC / ACARS Systems					
23-24-01A	ARINC Communications Addressing and Reporting Systems (ACARS) "DISP.APPR.REQ'D"	Y	1	0	C (O) May be inoperative provided cockpit crew continuously tunes and monitors appropriate radio frequencies as required. Note: Any portion of the system that is operative may be used.
23-24-01B	ACARS Communications Management Unit "DISP.APPR.REQ'D"	Y	1	0	C (O) May be inoperative provided procedures do not require its use.
23-24-01C	ACARS Printer "DISP.APPR.REQ'D"	Y	1	0	C May be inoperative provided required data is obtained from an alternate source prior to pushback. Note: Any portion of the system that is operative may be used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-24-01A	MCDU	ACARS INOP
23-24-01B	MCDU	ACARS INOP
23-24-01C	ACARS Printer	ACARS Printer INOP

(O) PROCEDURES

23-24-01A, 23-24-01B

1. A hard copy closeout AWABS should be available at the departure gate.
2. Continuously monitor appropriate company voice radio frequencies. Refer to [Airway Manual, Communications](#), for communications requirements.
3. An Engine Performance Report should be completed (if possible, submit via ACARS, workload permitting); Refer to [VOL. 1 Supplemental Procedures, Engine Performance Report](#).
4. Crew must complete required reports and operational information by voice communication.
5. Flight Duty Period (FDP) Limit:
 - A. Utilize LATT Worksheet OR Contact Flight Control to determine LATT/FDP Limit and if FDP extension is required to complete the flight.
Pilots must monitor FDP Limit prior to OFF event.
 - B. At or within 15 minutes of FDP Limit and if an FDP extension is required to complete the flight prior to IN, contact ATL radio via VHF or telephone and advise Dispatcher of pilot(s) employee number and FDP Accept/Reject.
Dispatch will send reply in Flight History for electronic documentation.

23-31-01 Passenger Address System (PA)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
23-31-01A	Passenger Address System (PA)	Y	1	0	B (O) May be inoperative provided: 1. Alternate, Normal, and Emergency procedures and/or operating restrictions are used, and 2. Flight Attendant alerting system (audio and visual) operates normally. 3. Passenger notice system (NO SMOKING/FASTEN SEAT BELT/RETURN TO CABIN) is operative. <i>Note: Any station function(s) that operate normally may be used.</i>
23-31-01B	Lavatory Speakers	Y	3	0	C (O) May be inoperative provided alternate procedures are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-31-01A	Audio Control Panel	PA System INOP
23-31-01B	Aircraft Logbook	____ Lav Speaker INOP

(O) PROCEDURES

23-31-01A

1. Reference [QRH](#), Section 5, PA Inoperative Non-Normal Checklist.
2. If PA is not capable of being used to perform passenger briefing, utilize alternate procedure as follows:
 - A. Before Takeoff the cabin attendant shall make a direct voice communication with all passengers in each cabin. The powered megaphone may be used for making these announcements.
 - B. The flight crew shall be notified when pre-flight announcements have been completed.
3. In an emergency the flight crew shall notify the cabin crew who shall establish direct voice contact with each passenger. The powered megaphone may be used for making announcements.

23-31-01B

1. Prior to departure, the pilot in command will notify flight attendants of inoperative lavatory speaker(s).
2. Flight attendants must give instructions to lavatory occupants verbally when lavatory speakers are inoperative.

23-32-01 Pre-Recorded Passenger Announcement System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
23-32-01	Pre-Recorded Passenger Announcement System	Y	1	0	D	(O) May be inoperative provided alternate procedures are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-32-01	FWD Flight Attendant Panel	Pre-Recorded Passenger Announcement System INOP

(O) PROCEDURES

Before takeoff the cabin attendant shall make a direct voice communication with all passengers in each cabin. The PA system or powered megaphone may be used for making these announcements. The flight crew shall be notified when pre-flight announcements have been completed.

23-41-01 Service Interphone System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
SERVICE INTERPHONE SYSTEM					
23-41-01A	Flight Deck-to-Cabin Cabin-to-Flight Deck Functions	Y	2	1	B
					(O) May be inoperative provided: - Operative station has an operative flight attendant seat (in addition to flight deck, FWD position must be operative), and - Alternate communications procedures between the affected Flight Attendant stations(s) are used. Note: Any station function(s) that operate normally may be used.
23-41-01B	Cabin-to-Cabin Functions(s)	Y	2	0	B
					(O) May be inoperative provided alternate communications procedures between the affected Flight Attendant stations(s) are used. Note: Any station function(s) that operate normally may be used.
23-41-01C	Flight Deck-to-Ground (Flight Interphone)	Y	1	0	C
					(O) Flight interphone flight deck to ground / ground to flight deck function may be inoperative provided: - Alternate procedures are used, and - Nose gear / forward fuselage service interphone jack operates normally.
23-41-01D	Flight Deck-to-Ground (Flight Interphone)	Y	1	0	B
					(O) May be inoperative provided alternate procedures are used.
23-41-01E	Flight Deck-to-Ground (Service Interphone)	Y	1	0	C
					(O) Service interphone flight deck to ground / ground to flight deck function may be inoperative provided: - Alternate procedures are used, and - Nose gear / forward fuselage flight interphone jack operates normally.
23-41-01F	Flight Deck-to-Ground (Service Interphone)	Y	1	0	B
					(O) May be inoperative provided alternate procedures are used.
23-41-01G	Flight Deck Call Visual Alerting System (Call Light)	Y	1	0	B
					(O) May be inoperative provided: - Flight deck audio alerting system operates normally, and - Refer to FOM Chapter 17, Emergency Cabin Signal Note: The flight deck audio alerting must always be operative.

CONTINUED ON NEXT PAGE

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY			
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS	
SERVICE INTERPHONE SYSTEM							
23-41-01H	Flight Attendant Visual Alerting System (Call Light)	Y	1	0	B	(O) May be inoperative provided: 1. PA system operates normally, 2. If affected visual alerting system is used for lavatory smoke detector alerting, an alternate lavatory smoke detector alert (audio or visual) is installed and operates normally, and 3. Alternate procedures for contacting flight attendants are used. Note: Passenger-to-Attendant Call System is considered a Nonessential Equipment and Furnishings (NEF) item. Note: Any visual alerting system function(s) that operates normally may be used.	
23-41-01I	Flight Attendant Audio Alerting System (Chime)	Y	1	0	B	(O) May be inoperative provided: 1. PA system operates normally, 2. If affected audio alerting system is used for lavatory smoke detector alerting, an alternate lavatory smoke detector alert (audio or visual) is installed and operates normally, and 3. Alternate procedures for contacting flight attendants are used. Note: Passenger-to-Attendant Call System is considered a Nonessential Equipment and Furnishings (NEF) item. Note: Any Alerting system function(s) that operates normally may be used.	
23-41-01J	Pilot to Mechanic Call System (Mechanic Call)	Y	1	0	C	(O) May be inoperative provided Alternate, Normal and Emergency operations procedures are used.	
23-41-01K	Mechanic to Pilot Call System (Pilot Call)	Y	1	0	C	(O) May be inoperative provided Alternate, Normal and Emergency operations procedures are used.	
23-41-01L	Flight Deck Handset	Y	1	0	C	(O) May be inoperative provided: 1. Flight Deck to cabin communication operates normally, and 2. Alternate Procedures are used.	
23-41-01M	Cabin Attendant Handset	Y	2	1	B	(O) Aft Cabin Attendant Handset may be inoperative provided: 1. Forward Cabin Attendant Handset operates normally and is located at operative forward Flight Attendant Seat, and 2. Alternate communications procedures between the affected Flight Attendant station(s) are used. Note: Any handset(s) function(s) that operate normally may be used.	

CONTINUED ON NEXT PAGE

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-41-01A	1. Captain's Audio Selector Panel 2. Affected Flight Attendant Handset (If Applicable)	____ Flight Deck-to-Cabin, Cabin-to-Flight Deck Function(s) INOP
23-41-01B	Affected Flight Attendant Handset(s)	Cabin-to-Cabin Function(s) INOP
23-41-01C	Flight Interphone Jack External Power Panel	Flight Interphone Flight Deck to Ground Function INOP
23-41-01D	Flight Interphone Jack External Power Panel	Flight Interphone Flight Deck to Ground Function INOP
23-41-01E	Service Interphone Jack External Power Panel	Service Interphone Flight Deck to Ground Function INOP
23-41-01F	Service Interphone Jack External Power Panel	Service Interphone Flight Deck to Ground Function INOP
23-41-01G	Flight Deck Call Light	Flight Deck Call Light INOP
23-41-01H	FWD Flight Attendant Panel	Flight Attendant Call Light INOP
23-41-01I	FWD Flight Attendant Panel	Flight Attendant Chime INOP
23-41-01J	Call Switch	Mechanic Call INOP
23-41-01K	1. Captain's Audio Control Panel Interphone Jack (Ext. Pwr Pnl)	Pilot Call INOP
23-41-01L	1. Flight Deck Handset	Flight Deck Handset INOP
23-41-01M	Affected Flight Attendant Handset	Cabin Attendant Handset INOP

(O) PROCEDURES

23-41-01A, 23-41-01B

Refer to [QRH](#), Non-Normal Checklist, Section 5, Communications, Cabin Interphone System Inoperative.

23-41-01C, 23-41-01D, 23-41-01E, 23-41-01F

1. Use operative flight or service interphone flight deck-to-ground function for communication with ground personnel.
2. Refer to [FOM Chapter 30, Ramp Operations](#).

23-41-01G

Refer to [FOM Chapter 17, Emergency Cabin Signal](#).

23-41-01H, 23-41-01I

Use Cabin Interphone System and/or PA System to communicate with cabin crew.

23-41-01J, 23-41-01K

Refer to [FOM Chapter 30, Ramp Operations](#).

23-41-01L

Use the Cabin Interphone / PA function of the Captain or F/O Audio Select Panel.

23-41-01M

Captain to brief Flight Attendants to use PA and/or face-to-face communication with attendant at location with inoperative handset.

23-41-03 Maintenance Interphone

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
23-41-03A	Maintenance Interphone	Y		1	0	C
23-41-03B	Maintenance Interphone Jacks	Y	11	2	D	

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-41-03A	MAINTENANCE INTERPHONE Switch	Maintenance Interphone System INOP
23-41-03B	MAINTENANCE INTERPHONE Switch	____ Maintenance Interphone Jack(s) INOP

23-51-03 Audio Control Panel Selector Switchlights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
23-51-03A	Audio Control Panel Selector Switchlights	Y	21	0	D
					(O) May be inoperative provided associated function is verified to be selectable PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)
23-51-03B	Audio Control Panel Selector Switchlights	Y	21	0	D
					May be inoperative provided associated function is not used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-51-03A	Affected Audio Control Panel	____ Audio Control Panel Selector Switchlights INOP
23-51-03B	Affected Audio Control Panel	____ Audio Control Panel Selector Switchlights INOP

(O) PROCEDURES

23-51-03A

PRIOR TO EACH DEPARTURE:

If the function associated with an inoperative switchlight is to be used for the subsequent departure, verify it is selectable by switching in and out of it.

23-51-04 Cockpit Communications Loudspeakers

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
23-51-04	Cockpit Communications Loudspeakers	Y	2	1	C	May be inoperative provided an operative headset is provided for each person on cockpit duty.

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-51-04	Associated Audio Selector Panel	____ Cockpit Loudspeakers INOP

23-51-05 Cockpit Microphones / Headsets

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
COCKPIT MICROPHONES / HEADSETS						
23-51-05A	Observer’s Handheld Microphone	Y	1	0	D	
23-51-05B	Cockpit Handheld Microphones	Y	3	0	C	May be inoperative or missing provided associated handset / boom microphone operates normally.
23-51-05C	Observer’s Headset Boom Microphone “Contact MCC”	Y	1	0	D	
23-51-05D	Headset Boom Microphones “Contact MCC”	Y	3	0	A	May be inoperative provided: 1. Associated handheld microphone is installed, operates normally, and 2. Repairs are made within three (3) flight days.
23-51-05E	Observer’s Cockpit Headset	Y	1	0	D	
23-51-05F	Captain’s and First Officer’s Cockpit Headsets	Y	2	1	C	May be inoperative provided associated flight deck speaker operates normally.

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-51-05A	Observer's Audio Control Panel	Observer's Handheld Microphone(s) INOP
23-51-05B	Associated Audio Control Panel	____ Handheld Microphone(s) INOP
23-51-05C	Observer's Audio Control Panel	Observer's Boom Microphone(s) INOP
23-51-05D	Associated Audio Control Panel	____ Boom Microphone(s) INOP
23-51-05E	Observer's Audio Selector Panel	Observer's Headset INOP
23-51-05F	Associated Audio Control Panel	____ Headset(s) INOP

23-51-06 Oxygen (Smoke) Mask Microphones

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
23-51-06	Oxygen (Smoke) Mask Microphones	Y	3	2	D	Observer's mask microphone may be inoperative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-51-06	Associated Audio Selector Panel	Observer's Oxygen Mask Microphone INOP

23-51-09 Captain / First Officer Push-To-Talk Switches

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
	Captain / First Officer Push-To-Talk Switches				
23-51-09A	Control Wheel PTT Switches	N	2	0	C (M) May be inoperative provided: 1. Associated Audio Selector Panel PTT switch operates normally, and 2. The affected switch is deactivated.
23-51-09B	Audio Selector Panel PTT Switches	Y	3	0	C (M) May be inoperative provided: 1. If the switch is on the Captain's or First Officer's Audio Control Panel, the associated Control Wheel PTT Switch is operative, and 2. It is verified that the affected switch is failed in a non-transmitting state.

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-51-09A	Associated Audio Control Panel	____ Control Wheel PTT Switch(es) INOP
23-51-09B	Affected Audio Control Panel	____ Audio Selector Panel PTT Switch(es) INOP

(M) PROCEDURES

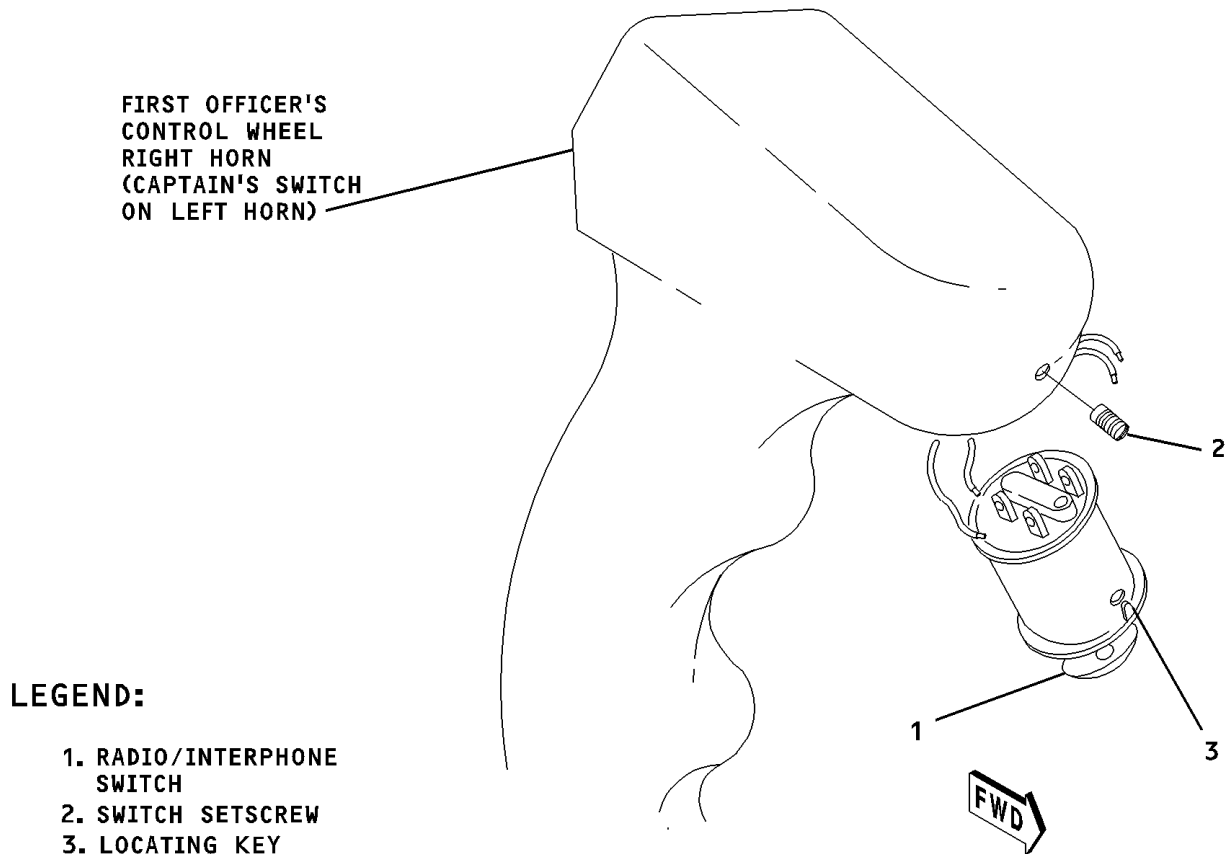
23-51-09A

1. Pull and secure the following circuit breakers:

Circuit Breaker (CB#)	Panel	Location
CAPT DCAS (B10-487)	Overhead CBP	A-8
F/O DCAS (B10-488)	Upper ECP CBP	T-34
OBSI1 DCAS (B10-489)	GCP B Main Power	G-7
DCAS SERV/MAINT INTERPHONE (B1-1712)	Overhead CBP	A-7

2. On the control wheel, remove the set screw.
3. Pull switch downward until electrical connections are out of control wheel horn and are accessible for removal.
4. Tag and disconnect wires from switch terminal, cap / cover exposed wires and stow within control wheel.
5. Reinsert deactivated switch into control wheel, and secure by tightening setscrew.
6. Reset the circuit breakers opened in step 1 above.

Figure 23-1: Control Wheel PTT Switches



SWITCH INSTALLATION - TYPICAL

23-51-09B

1. With electrical power on the airplane, select the PA Transmit switch on the associated Audio Control Panel.
2. Plug a headset into the associated Jack Panel.
3. Without keying the Control Wheel PTT Switch, speak into the headset boom microphone.
4. Verify voice transmission is not heard over the PA system. If voice transmission is NOT audible, affected switch is electrically failed in the OPEN position and dispatch is permitted.

Note: Similar procedures can also be performed using the Flight Interphone System in lieu of the PA System.

23-71-01 Cockpit Voice Recorder (CVR) System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
23-71-01	Cockpit Voice Recorder (CVR) System "Contact MCC"	Y	1	0	A	May be inoperative provided: 1. Flight Data Recorder (FDR) operates normally, and 2. Repair is made within three (3) flight days.

MEL ITEM	TAB LOCATION	PLACARD TEXT
23-71-01	COCKPIT VOICE RECORDER Panel	CVR INOP

24 Electrical Power

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	PCDU Generator control units (GCU)
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24-51-02	EXT Light (Blue)
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24-51-03A	APU Power ON Light
24-51-03B	APU Power ON Light

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24-51-04 APU Lights (Blue)

24-51-04A APU Lights (Blue)
24-51-04B APU Lights (Blue)

24-51-05 L/R GEN Lights

24-51-05A L/R GEN Lights
24-51-05B L/R GEN Lights

24-54-01 Galley Power and Control Systems

24-54-01 Galley Power and Control Systems

24-21-01 AC Electrical Power Generation and Control

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
AC Electrical Power Generation and Control					
24-21-01A	Engine Integrated Drive Generator (IDG) Systems "DISP.APPR.REQ'D"	N	2	1	B
					(O) (M) May be inoperative provided: 1. APU Generator System is operative and furnishing power to the associated bus, 2. Bus Tie Switch on the affected side is in AUTO, 3. APU and remaining (operative) IDG log book history for past 30 days is reviewed and approved by MCC prior to deferral, 4. APU oil quantity and affected IDG are serviced to full prior to deferral and ONCE EACH FLIGHT DAY (Log Book entry required), and 5. Performance corrections are applied. Theater Restrictions May Apply Repetitive Check Required (Yellow Placard)
24-21-01B	APU Generator System "DISP.APPR.REQ'D"	Y	1	0	C
					(O) May be inoperative provided both Engine Generator Systems are operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-21-01A	Associated Eng Generator Switch	____ Engine Generator INOP
24-21-01B	APU Generator Switch	APU Generator INOP

(O) PROCEDURES

24-21-01A

1. Dispatch with Engine IDG inoperative:
 - A. Verify associated inoperative generator GEN switch is OFF.
 - B. Verify both L and R BUS TIE Switches are in AUTO.
 - C. Verify APU is operating and is furnishing power to the affected bus.
2. Maximum Operating Altitude is FL 370.

24-21-01A	ENGINE INTEGRATED DRIVE GENERATOR (IDG) SYSTEMS
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	Increase BLOCK FUEL by 220 lbs. per hour for APU usage.
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF None LANDING None

24-21-01B

Start engine per starting procedure in [Vol I, Supplemental Procedures, Starting with APU Air or Ground Air Source](#) (External AC electrical power available).

(M) PROCEDURES

24-21-01A

1. Make sure that the APU oil level is at the FULL mark on the sight glass. Service to FULL if necessary.
2. ONCE EACH FLIGHT DAY:
 - A. Check BOTH IDGs for oil level within the operational band.
 - B. Make sure that the APU oil level is at the FULL mark on the sight glass. Service to FULL if necessary.
 - C. Accomplish required log book entry.

24-21-02 PCDU Generator Control Units (GCU)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
PCDU Generator control units (GCU)					
24-21-02A	L/R PCDU GCU "DISP.APPR.REQ'D"	N	2	1	B
					(O)(M) May be inoperative provided: - AC Bus Tie and DC Tie Switches are in AUTO, - IDG or APU Generator associated with the remaining GCUs are operative, - Remaining GCUs are fully operative and are furnishing power to the affected bus, and - APU oil quantity and affected IDG are serviced to full prior to deferral and ONCE EACH FLIGHT DAY (Log Book entry required), Repetitive Check Required (Yellow Placard)
24-21-02B	APU PCDU GCU	Y	1	0	C
					(O) May be inoperative provided: - AC Bus Tie and DC Tie Switches are in AUTO, - IDG associated with the remaining GCUs are operative, and - Remaining GCUs are fully operative and are furnishing power to the affected bus.

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-21-02A	Associated Eng Generator Switch and APU Master Switch	____ Generator GCU(s) INOP
24-21-02B	APU Generator Switch	1. ____ APU Generator GCU INOP 2. Do Not Use APU Generator

(O) PROCEDURES

24-21-02A

- Verify both L and R BUS TIE Switches are in AUTO.
- If L or R PCDU GCU is inoperative, verify APU is operating and is furnishing power to the affected bus.
- If L or R PCDU GCU is inoperative, verify the Bus Tie Relay associated with the inoperative GCU opens and closes by cycling the affected L or R BUS TIE Switch from AUTO to OPEN and back to AUTO. EAD should display appropriate BUS AC L/R OFF message with the Bus Tie Relay open, and message should extinguish when the switch is placed back to AUTO.
- Maximum Operating Altitude is FL 370.

24-21-02A	L/R PCDU GCU
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	Increase BLOCK FUEL by 220 lbs. per hour for APU usage.
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF None LANDING None

24-21-02B

Verify both L and R BUS TIE Switches are in AUTO.

(M) PROCEDURES

24-21-02A

1. Make sure that the APU oil level is at the FULL mark on the sight glass. Service to FULL if necessary.
2. ONCE EACH FLIGHT DAY:
 - A. Check BOTH IDGs for oil level within the operational band.
 - B. Make sure that the APU oil level is at the FULL mark on the sight glass. Service to FULL if necessary.
 - C. Accomplish required log book entry.

24-22-01 Right Generator Relay (RGR)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY			
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS	
24-22-01	Right Generator Relay (RGR) “DISP.APPR.REQ’D”	Y	1	0	B	(O) May be inoperative provided: 1. RGR failure isolates the Right Generating System output from the rest of the electrical distribution system, 2. APU Generator System is operative and furnishing power to the Right AC Bus, and 3. LGR operates normally.	

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-22-01	Right Eng Generator Switch	Right Generator Relay INOP

(O) PROCEDURES

1. Verify both L and R BUS TIE Switches are in AUTO.
2. Verify APU is operating and is furnishing power to the affected bus.
3. Maximum Operating Altitude is FL 370.
4. Apply the following Performance Corrections:

24-22-01	RIGHT GENERATOR RELAY (RGR)
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	Increase BLOCK FUEL by 220 lbs. per hour for APU usage.
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF None LANDING None

24-22-02 Left Generator Relay (LGR)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
24-22-02	Left Generator Relay (LGR) “DISP.APPR.REQ’D”	Y	1	0	B (O) May be inoperative provided: 1. LGR failure isolates the Left Generating System output from the rest of the electrical distribution system, 2. APU Generator System is operative and furnishing power to the Left AC Bus, and 3. RGR operates normally.

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-22-02	Left Eng Generator Switch	Left Generator Relay INOP

(O) PROCEDURES

- Verify both L and R BUS TIE Switches are in AUTO.
- Verify APU is operating and is furnishing power to the affected bus.
- Maximum Operating Altitude is FL 370.
- Apply the following Performance Corrections:

24-22-02	LEFT GENERATOR RELAY (LGR)
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	Increase BLOCK FUEL by 220 lbs. per hour for APU usage.
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF None LANDING None

24-22-03 Auxiliary Power Relay (APR)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			1	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	Y			C	REMARKS OR EXCEPTIONS
24-22-03	Auxiliary Power Relay (APR) “DISP.APPR.REQ’D”	Y			C	May be inoperative provided: 1. APU Generator System is considered inoperative, and 2. APU GEN Switch remains off.

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-22-03	APU Generator Switch	1. Auxiliary Power Relay INOP 2. Do Not Use APU Generator

24-24-01 EMER PWR ON Light (Overhead Panel)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
24-24-01	EMER PWR ON Light (Overhead Panel)	Y	1	0	C	(O) May be inoperative provided EMER PWR ON message on EAD is verified operative PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-24-01	EMER PWR Switch	EMER PWR ON Light INOP

(O) PROCEDURES

PRIOR TO EACH DEPARTURE:

1. Verify EMER PWR ON message on EAD is not displayed when EMER PWR switch is in the OFF and ARM positions.
2. Rotate EMER PWR Switch to ON and observe EMER PWR ON and EMER PWR TEST Alerts both appear on EAD.

24-26-01 AC Voltage Indicating System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
24-26-01	AC Voltage Indicating System	Y	4	0	C

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-26-01	Electrical Control Panel	AC Voltage Indication on SD Electrical Page INOP

24-26-02 AC Frequency Indicating System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
24-26-02	AC Frequency Indicating System	Y	4	0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-26-02	Electrical Control Panel	AC Frequency Indication on SD Electrical Page INOP

24-26-03 AC Load Sensing Circuit (Including Loadmeter Indication)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
24-26-03	AC Load Sensing Circuit (Including Loadmeter Indication) "DISP.APPR.REQ'D"	Y	3	2 B	(O) May be inoperative provided: - The associated generator is not used, and - If APU is used, performance penalties are applied.

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-26-03	Associated Eng Generator Switch and APU Start Switch	____ AC Load Sensing Circuit(s) INOP

(O) PROCEDURES

- Verify both L and R BUS TIE Switches are in AUTO.
- If L or R GEN AC Load Indication is inoperative, verify APU is operating and is furnishing power to the affected bus.
- Apply the following Performance Corrections:

24-26-03	AC LOAD SENSING CIRCUIT (INCLUDING LOADMETER INDICATION)
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	Increase BLOCK FUEL by 220 lbs. per hour for APU usage.
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF None LANDING None

24-31-01 PCDU Transformer/ Rectifiers (TR) Units

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
24-31-01	PCDU Transformer/ Rectifiers (TR) Units	Y	3	2	B (O) May be inoperative provided: 1. AC Bus Tie and DC Tie Switches are in AUTO , 2. IDG or APU Generator associated with the remaining TR Units are operative, and 3. Remaining TR Units are fully operative and are furnishing power to the affected DC bus.

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-31-01	Electrical Control Panel	____ Transformer / Rectifier INOP

(O) PROCEDURES

Verify the DC BUS TIE Switch is in the AUTO Position.

24-41-01 External Power System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
24-41-01	External Power System "DISP.APPR.REQ'D"	Y	1	C	(M) May be inoperative provided: 1. Procedures are used to isolate affected components from the rest of the electrical distribution system, and 2. External Power Receptacle is placarded "DO NOT CONNECT ELECTRICAL POWER".

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-41-01	1. External Power Switch on the Electrical Control Panel 2. External Power Receptacle	1. External Power INOP 2. DO NOT CONNECT ELECTRICAL POWER.

(M) PROCEDURES

1. Pull and secure the External Power Control circuit breakers on the External Power Panel.

Circuit Breaker (CB#)

EXTERNAL POWER 0A (B1-1149)

EXTERNAL POWER 0B (B1-1150)

EXTERNAL POWER 0C (B1-1151)

Location

External Power Panel

External Power Panel

External Power Panel

2. Apply a placard near the external power receptacle which states, "DO NOT CONNECT ELECTRICAL POWER".

24-41-02 Main External Power Relay (MEPR)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
24-41-02	Main External Power Relay (MEPR) "DISP.APPR.REQ'D"	Y	1	C	(M) May be inoperative provided: 1. Procedures are used to isolate the MEPR from the rest of the electrical distribution system, and 2. External Power Receptacle is placarded "DO NOT CONNECT ELECTRICAL POWER".

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-41-02	1. Near EXT PWR AVAIL Light on Overhead Panel 2. External Power Receptacle	1. Main External Power Relay INOP 2. DO NOT CONNECT ELECTRICAL POWER.

(M) PROCEDURES

1. Pull and secure the External Power Control circuit breakers on the External Power Panel.

Circuit Breaker (CB#)

EXTERNAL POWER 0A (B1-1149)

EXTERNAL POWER 0B (B1-1150)

EXTERNAL POWER 0C (B1-1151)

Location

External Power Panel

External Power Panel

External Power Panel

2. Apply a placard near the external power receptacle which states, "DO NOT CONNECT ELECTRICAL POWER".

24-42-01 EXT PWR NOT IN USE Light (White)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
24-42-01	EXT PWR NOT IN USE Light (White)	Y	1	0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-42-01	Ground Power Panel	EXT PWR NOT IN USE Light INOP

24-42-02 EXT PWR AVAIL Lights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
24-42-02	EXT PWR AVAIL Lights	Y	2	0 C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-42-02	Ground Power Panel, and/or Electrical Control Panel	____ EXT PWR AVAIL Light(s) INOP

24-51-02 EXT Light (Blue)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
24-51-02	EXT Light (Blue)	Y	2	0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-51-02	Electrical Control Panel	____ EXT PWR In Use Light(s) INOP

24-51-03 APU Power ON Light

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
24-51-03A	APU Power ON Light	Y	1	0	C	(O) May be inoperative provided APU GEN OFF message is verified to be operative PRIOR TO EACH DEPARTURE if APU Generator is to be used. Repetitive Check Required (Yellow Placard)
24-51-03B	APU Power ON Light	Y	1	0	C	May be inoperative provided equivalent indication is available on Electrical Systems Synoptic Page.

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-51-03A	APU POWER ON Light	APU PWR ON Light INOP
24-51-03B	APU POWER ON Light	APU PWR ON Light INOP

(O) PROCEDURES

24-51-03A

If APU is to be used for electrical power, **PRIOR TO EACH DEPARTURE** verify GEN APU OFF message appears on the EAD when the APU GEN Switch is moved to OFF after starting the APU.

24-51-04 APU Lights (Blue)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
24-51-04A	APU Lights (Blue)	Y	2	0	C	May be inoperative provided the L/R AC BUS OFF messages are operative if APU electrical power is to be used.
24-51-04B	APU Lights (Blue)	Y	2	0	C	May be inoperative provided equivalent information is available on Electrical Systems Synoptic Page.

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-51-04A	Electrical Control Panel In Use Light(s)	____ APU PWR In Use Light(s) INOP
24-51-04B	Electrical Control Panel In Use Light(s)	____ APU PWR In Use Light(s) INOP

24-51-05 L/R GEN Lights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
24-51-05A	L/R GEN Lights	Y	2	0	C (O) May be inoperative provided the BUS AC L/R OFF messages are verified to be operative.
24-51-05B	L/R GEN Lights	Y	2	0	C (O) May be inoperative provided equivalent indication is available on Electrical Systems Synoptic Page.

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-51-05A	Electrical Control Panel	____ L/R GEN Light(s) INOP
24-51-05B	Electrical Control Panel	____ L/R GEN Light(s) INOP

(O) PROCEDURES

24-51-05A

Start the affected engine and with APU Generator OFF, select affected L or R GEN Switch to OFF and observe associated BUS AC L or R OFF message on EAD.

24-51-05B

Confirm the affected bus is powered by observing Bus indication on ELEC Synoptic page is green with associated GEN Switch ON.

24-54-01 Galley Power and Control Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
24-54-01	Galley Power and Control Systems	Y	3	0	C (M) May be inoperative provided affected galley power is deactivated.

MEL ITEM	TAB LOCATION	PLACARD TEXT
24-54-01	Electrical Control Panel	___ Galley Power INOP

(M) PROCEDURES

Pull and secure the circuit breaker(s) corresponding to affected galley/system located in the Right and Left Generator Bus panels of the Left EPC, lower inboard section.

Galley 1 EPC Circuit Breaker Panel

B1-877

B1-878

B1-879

Galley 2 EPC Circuit Breaker Panel

B1-880

B1-881

B1-882

Galley 7 EPC Circuit Breaker Panel

B1-886

B1-887

B1-888

25 Equipment / Furnishings

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25-61-02A	Cabin Emergency Flashlights / Holders
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25-62-01 Emergency Medical Equipment

EMERGENCY MEDICAL EQUIPMENT

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25-63-01 Megaphones

25-63-01	Megaphones
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25-64-01 Flotation Devices (Crew and Passengers)

FLOTATION DEVICES (CREW AND PASSENGERS)

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25-67-02 Emergency Locator Transmitter (ELT) Survival Type

25-67-02A	Emergency Locator Transmitter (ELT) Survival Type
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25-99-01 PED Mount

25-99-01A	PED Mount
25-99-01B	PED Mount
25-99-01C	PED Mount

25-15-01 Observer Seat (Including Associated Equipment)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
25-15-01A	Observer Seat (Including Associated Equipment) "DISP.APPR.REQ'D"	Y	1	0	A May be inoperative provided: 1. A passenger seat in the passenger cabin is available to an FAA inspector for the performance of official duties, and 2. Repairs are made within two (2) flight days.
25-15-01B	Observer Seat (Including Associated Equipment) "Contact MCC"	Y	1	0	A May be inoperative provided: 1. Required minimum safety equipment (safety belt and oxygen) is available, 2. Seat is acceptable to the FAA inspector for the performance of official duties, and 3. Repairs are made within two (2) flight days. Note: <i>These provisos are intended to provide for occupancy of the above seats by an FAA inspector when the minimum safety equipment (oxygen and safety belt) is functional and the inspector determines the conditions to be acceptable.</i> Note: <i>The pilot-in-command will determine if the minimum safety equipment is functional for other persons authorized to occupy any observer seat(s).</i>

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-15-01A	Near Inoperative Equipment or Aircraft Logbook	____ Observer's Seat INOP
25-15-01B	Near Inoperative Equipment or Aircraft Logbook	____ Observer's Seat INOP

25-15-02 Flight Crew Seats

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
FLIGHT CREW SEATS						
25-15-02A	Lumbar Supports	Y	2	0	C	May be inoperative in the retracted position provided the seat is acceptable to the affected crewmember.
25-15-02B	Thigh Supports	Y	4	0	C	May be inoperative in the retracted position provided the seat is acceptable to the affected crewmember.
25-15-02C	Armrests	Y	4	2	B	One armrest on each seat may be inoperative or missing provided: 1. Egress is not impaired, and 2. Seat is acceptable to affected crewmember.
25-15-02D	Pneumatic Height Adjustment	Y	2	0	C	May be inoperative provided the seat height can be manually adjusted to the proper position.
25-15-02E	Recline	N	2	0	A	(M) May be inoperative provided: 1. Seat(s) is secured in an upright position acceptable to affected crewmember(s), and 2. Repairs are made within two flight days.
25-15-02F	Headrest	Y	2	0	C	May be inoperative provided seat is acceptable to the affected crewmember.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-15-02A	Aircraft Logbook	____ Seat Lumbar Support(s) INOP
25-15-02B	Aircraft Logbook	____ Seat Thigh Support(s) INOP
25-15-02C	Aircraft Logbook	____ Seat Armrest(s) INOP
25-15-02D	Aircraft Logbook	____ Pneumatic Height Adjustment INOP or Missing
25-15-02E	Aircraft Logbook	____ Seat Recline INOP
25-15-02F	Aircraft Logbook	____ Headrest INOP

(M) PROCEDURES

25-15-02E

1. Deactivate and secure recline mechanism in an upright position that is acceptable to the affected pilot.
2. Verify that the seat back cannot be moved.

25-22-01 "Fasten Seat Belt While Seated" Signs or Placards (Unlighted)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
25-22-01	"Fasten Seat Belt While Seated" Signs or Placards (Unlighted)	Y	-	-	C	One or more signs or placards may be illegible or missing provided a legible sign or placard is visible from each occupied passenger seat.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-22-01	Aircraft Logbook	____ Fasten Seat Belts While Seated Signs or Placards Missing / Illegible

25-23-01 Flight Attendant Seats

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
FLIGHT ATTENDANT SEATS					
25-23-01A	Flight Attendant Seats (Single or Dual Position) "Contact MCC" "DISP.APPR.REQ'D" "Revenue Passengers Not Authorized"	Y	-	0	C (O) May be inoperative or missing provided: 1. MCC must provide authorization prior to use, 2. No revenue passengers or cargo are carried, 3. A maximum of 19 persons are carried as authorized by 14 CFR for non-passenger carrying operations, and 4. Alternate procedures are used.
25-23-01B	Required Flight Attendant Seats (Single or Dual position) (Refer to In-flight Service Onboard Manual for Required Seat Locations) "DISP.APPR.REQ'D"	Y	-	-	B (O) (M) One seat position or assembly (dual position) may be inoperative provided: 1. Affected seat position or seat assembly is not occupied, 2. Flight Attendant(s) displaced by inoperative seat(s) occupies either an adjacent flight attendant seat or the passenger seat which is most accessible to the inoperative seat(s), so as to most effectively perform assigned duties, 3. Alternate procedures are used for displaced Flight Attendant(s) (Ref. In-flight Service Onboard Manual), 4. Folding type seat stows automatically or is secured in the retracted position, 5. ONCE EACH FLIGHT DAY (Log Book Entry Required) insure seat has remained in fully stowed position. If seat has been secured in the fully stowed position with tape, insure tape is in good condition, and 6. Passenger seat assigned to flight attendant(s) is placarded "FOR FLIGHT ATTENDANT ONLY". Note: An automatic folding seat that will not stow automatically is considered inoperative. Note: A seat position with an inoperative or missing restraint system is considered inoperative. Note: If one side of a dual seat assembly is inoperative and a flight attendant is displaced to the adjacent seat, the adjacent seat must operate normally. Repetitive Check Required (Yellow Placard)

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NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
	FLIGHT ATTENDANT SEATS				
25-23-01C	Non-Required Flight Attendant Seats (Single or Dual Position) (Refer to In-flight Service Onboard Manual for Required Seat Locations) "DISP.APPR.REQ'D"	Y	1	0	C
					(M) May be inoperative provided: - Affected seat position or seat assembly is not occupied, - Folding type seat stows automatically or is secured in the retracted position, and - ONCE EACH FLIGHT DAY (Log Book Entry Required) insure seat has remained in fully stowed position. If seat has been secured in the fully stowed position with tape, insure tape is in good condition. Note: An automatic folding seat that will not stow automatically is considered inoperative. Note: A seat position with an inoperative or missing restraint system is considered inoperative. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-23-01A	Aircraft Logbook	____ Flight Attendant(s) Seat INOP
25-23-01B	Aircraft Logbook	____ Flight Attendant(s) Seat INOP
25-23-01C	Aircraft Logbook	____ Flight Attendant(s) Seat INOP

(O) PROCEDURES

25-23-01A

- A maximum of 19 persons authorized by 14 CFR 121.583 may be carried. Persons being carried must be approved by the Flight Control Dispatcher.
- Captain to brief persons authorized to be carried on flight on safety/evacuation procedures.

25-23-01B

The flight attendant assigned to the affected seat position must occupy either a passenger seat or an excess flight attendant seat, whichever is most accessible to the inoperative seat. The flight attendant must be able to reach the assigned emergency exit in essentially the same time as from the normally assigned seat, i.e., a 2 or 3 second time difference is considered "essentially the same time".

(M) PROCEDURES

25-23-01B, 25-23-01C

Insure the Flight Attendant Seat automatically retracts without assistance to the fully stowed position.

If the seat will not automatically retract without assistance to the fully stowed position, secure the inoperative seat in the fully stowed position using the following procedure: (Maintenance is Required and Flight Crew May NOT Placard):

- Secure seat in the retracted (fully stowed) position using high visibility tape.
- Placard the seat using high visibility tape or a section of 2 inch wide masking tape with the words "UNSERVICABLE – DO NOT USE" printed in 1 inch black letters.

ONCE EACH FLIGHT DAY:

Ensure seat has remained in fully stowed position. If seat has been secured in the fully stowed position with tape, ensure tape is in good condition.

25-23-03 Passenger Seats

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
PASSENGER SEATS					
25-23-03A	Passenger Seats "DISP.APPR.REQ'D"	Y	-	0	C
May be inoperative provided: 1. Seat does not block an Emergency Exit, 2. Seat does not restrict any passenger from access to the main aircraft aisle, and 3. The affected seat(s) are blocked and placarded "DO NOT OCCUPY." Note: A seat with an inoperative seat belt is considered to be inoperative. Note: Inoperative seats do not affect the required number of Flight Attendants. Note: Affected seats may include the seat behind and / or adjacent outboard seats. Note: MCC must contact Flight Control for coordination with ACS and the ROC to remove seat from inventory until repaired.					
25-23-03B	Recline Mechanism	N	-	0	D
(M) May be inoperative and seat occupied provided seat is secured in the full up-right position.					
25-23-03C	Recline Mechanism	Y	-	0	D
May be inoperative and seat occupied provided seat is immovable in the full up-right position.					
25-23-03D	Under Seat Baggage Restraining Bars	Y	-	0	C
(O) May be inoperative provided: 1. Baggage is not stowed under seat with inoperative restraining bar, 2. Associated seat is placarded "DO NOT STOW BAGGAGE UNDER THIS SEAT", and 3. Procedures are used to alert Cabin Crew of inoperative restraining bar.					
25-23-03E	Arm Rest With Recline Mechanism	N	-	0	D
(M) May be inoperative or missing and seat occupied provided: 1. Arm Rest does not block an Emergency Exit, 2. Arm Rest does not restrict any passenger from access to main aircraft aisle, and 3. If Arm Rest is missing, seat is secured in the full upright position.					
25-23-03F	Arm Rest Without Recline Mechanism	Y	-	0	D
May be inoperative or missing and seat occupied provided: 1. Arm Rest does not block an Emergency Exit, and 2. Arm Rest does not restrict any passenger from access to main aircraft aisle.					

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MEL ITEM	TAB LOCATION	PLACARD TEXT
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25-23-03B	Aircraft Logbook	Recline Mechanism Seat(s) ____ INOP
25-23-03C	Aircraft Logbook	Recline Mechanism Seat(s) ____ INOP
25-23-03D	Aircraft Logbook	Underseat Baggage Restraining Bar Seat(s) ____ INOP
25-23-03E	Aircraft Logbook	Arm Rest Seat(s) ____ INOP
25-23-03F	Aircraft Logbook	Arm Rest Seat(s) ____ INOP

(O) PROCEDURES

25-23-03D

1. Advise cabin crew to ensure baggage is not stowed under seats with inoperative or missing Restraining Bars.
2. Advise cabin crew to inform passengers occupying seat with affected bar and passenger behind the seat with the affected bar not to stow baggage under the seat.

(M) PROCEDURES

25-23-03B

Secure the inoperative seat in the upright position.

25-23-03E

1. Secure the associated seat in the upright position.
2. If necessary, remove the inoperative armrest.

25-25-01 Storage Bin(s) / Cabin, Galley, and Lavatory Storage Compartments / Closets

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
25-25-01A	Storage Bin(s) / Cabin, Galley, and Lavatory Storage Compartments / Closets	Y	-	0	C (O) (M) May be inoperative provided: 1. Affected compartment(s) is secured CLOSED, 2. Associated bin or compartment is prominently placarded "UNSERVICABLE – DO NOT USE", 3. Any emergency equipment located in the affected compartment is considered inoperative and associated MEL relief is applied, and 4. Affected compartment is not used for storage of any items(s) except for those permanently affixed. Note: For overhead storage compartments, if no partitions are installed, the entire overhead storage compartment is considered one compartment.
25-25-01B	Storage Bin(s) / Cabin, Galley, and Lavatory Storage Compartments / Closets	N	-	0	C (O) (M) May be inoperative provided: 1. Affected door(s) is removed or, for retractable doors, secured in the retracted (fully open) position, 2. Affected bin, compartment or closet is not used for storage of any items, except those permanently affixed, 3. Affected bin or compartment is prominently placarded "UNSERVICABLE – DO NOT USE", 4. Procedures are used to alert crew members and passengers of inoperative bins, compartments and closets, and 5. Passengers are briefed that associated bin or compartment is not used. Note: For overhead storage compartments, if no partitions are installed, the entire overhead storage compartment is considered inoperative. Note: Any emergency equipment located in the associated compartment (permanently affixed) is available for use.
25-25-01C	Storage Compartment Key Locks	Y	-	0	D (M) May be inoperative in the unlocked position provided door(s) can be secured by other means.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-25-01A	Aircraft Logbook	____ Overhead Storage Bin(s) / Cabin and Galley Storage Compartment / Closets INOP
25-25-01B	Aircraft Logbook	____ Overhead Storage Bin(s) / Cabin and Galley Storage Compartment / Closets INOP
25-25-01C	Aircraft Logbook	____ Storage Compartment Key Lock(s) INOP

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(O) PROCEDURES

25-25-01A

Captain to brief Lead Flight Attendant/Purser:

1. Which bins or storage compartments are inoperative and cannot be used.
2. Flight Attendant/Purser to brief passengers which bins or storage compartments cannot be used.
3. Any emergency equipment located in the associated bin or storage compartment (permanently affixed) is NOT available for use.

25-25-01B

Captain to brief Lead Flight Attendant/Purser:

1. Which bins or storage compartments are inoperative and cannot be used.
2. Flight Attendant/Purser to brief passengers which bin or compartment cannot be used.
3. Any emergency equipment located in the associated compartment (permanently affixed) is available for use.

(M) PROCEDURES

25-25-01A

1. Empty compartment (except for permanently affixed items).
2. For inoperative bins: Using suitable pressure sensitive tape (i.e., bag bin tape, nylon or fiberglass reinforced tape, not masking tape, etc.), secure compartment CLOSED.
3. Placard compartment using "Inoperative Cabin Appearance" placard or equivalent.
4. The following items are considered "emergency equipment" and may not be relocated:
 - Fire Extinguishers
 - Megaphones
 - Flashlights
 - EEMK
 - AED
 - PBE
 - Passenger Oxygen Bottles
 - First Aid Kits
5. For cabin equipment items that are relocated, comply with the following procedure:
 - A. Location placarding (if installed) of the affected item is removed or obscured.
 - B. Install temporary location placard on the outside of compartment / bin where the item was located denoting the new location.
 - C. Install temporary location placard on the outside of compartment / bin where the item has been relocated.
 - D. Note location of relocated equipment on aircraft logbook placard.

25-25-01B

To remove the affected door(s) accomplish the following:

1. Removed the affected door.
2. Place door hardware in parts bag and attach to removed door.
3. Placard door "MUST REMAIN WITH AIRPLANE" and stow in available closet or cargo compartment as applicable. Door is to remain in closet or cargo compartment until it is removed by Maintenance for repair or re-installation.
4. Place a strip of Red Colored OM2 UNSERVICABLE – DO NOT USE tape (or equivalent*) diagonally across the door opening.

*An acceptable alternate to the OM2 tape is a section of 2 inch wide masking tape with the words "UNSERVICABLE – DO NOT USE" printed legibly in 1 inch black letters.

25-25-01C

To secure the affected compartment(s) closed, accomplish the following:

1. Secure affected compartment(s) CLOSED by using the latch on the door.

2. If the latch on the door is inoperative, secure the affected compartment(s) CLOSED by placing nylon reinforced strapping tape (packing tape) across the compartments(s).

25-29-01 Nonessential Equipment and Furnishings (NEF)

NUMBER INSTALLED		NUMBER REQUIRED FOR DISPATCH		REPAIR CATEGORY	REMARKS OR EXCEPTIONS
ITEM NO.	NAME/DESCRIPTION	FLIGHT CREW MAY PLACARD			
25-29-01	Nonessential Equipment and Furnishings (NEF)	-	-		May be inoperative, damaged or missing provided that item(s) is deferred in accordance with the NEF deferral program. Note: Do not use this MEL item to defer the discrepancy. Defer the discrepancy under the NEF program per TOPP 40-40-08. Note: Exterior Lavatory Door Ash Trays are not considered NEF Items.

25-36-01 Galley/Cabin/Lavatory Waste Receptacles Access Doors/Covers

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
25-36-01A	Galley/Cabin Waste Receptacles Access Doors/Covers	Y	-	-	C
					(O) (M) May be inoperative provided: 1. Associated waste container is empty, 2. Receptacle access is secured to prevent waste introduction into the compartment, and 3. Procedures are used to ensure that sufficient Galley/Cabin waste receptacles are available to accommodate all waste that may be generated on a flight. Note: This MEL item is NOT applicable to galley waste carts.
25-36-01B	Lavatory Waste Receptacles Access Doors/Covers "DISP.APPR.REQ'D"	Y	3	2	C
					(M) May be inoperative provided: 1. Associated waste container is empty, 2. Lavatory is used only by crewmembers, and 3. Associated lavatory entrance door is locked closed and placarded: INOPERATIVE – DO NOT ENTER. Note: These provisions are not intended to prohibit lavatory use or inspection by crewmembers.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-36-01A	Aircraft Logbook	____ Galley/Cabin Waste Receptacles Access Doors / Covers INOP
25-36-01B	Aircraft Logbook	____ Lavatory Waste Receptacles Access Doors / Covers INOP

(O) PROCEDURES

25-36-01A

Refer to the [In-Flight Service On-Board Manual](#), Section 5 Regulations/ Guidelines, Trash.

(M) PROCEDURES

25-36-01A

1. Galley/Cabin Waste Receptacle Access Doors/Covers
 - A. Empty the affected waste container.
 - B. Secure associated access door closed.

25-36-01B

1. Lavatory Waste Receptacle Access Doors/Covers
 - A. Empty the affected waste container.
 - B. Secure associated access door closed.
 - C. Close and lock the lavatory door and placard: INOPERATIVE - DO NOT ENTER.

25-42-01 Exterior Lavatory Door Ashtrays

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
25-42-01	Exterior Lavatory Door Ashtrays "Contact MCC"	Y	3	2	A Up to 50 percent may be missing or inoperative provided they are replaced within three calendar days. Note: Interior lavatory ashtrays are considered Non-essential Equipment and Furnishings (NEF) items. Note: This item is AD related.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-42-01	Aircraft Logbook	____ Exterior Lavatory Door Ashtrays Missing / INOP

25-50-01 Lower Cargo Compartment Liners

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
25-50-01	Lower Cargo Compartment Liners "DISP.APPR.REQ'D"	Y	-	0	C May be inoperative or missing provided only non-combustible materials are carried in affected cargo compartment. Note: Refer to (TOPP 40-40-05- MEL/CDL Instructions)

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-50-01	Aircraft Logbook	Lower ____ Cargo Compartment INOP

25-51-02 Forward Cargo Compartment Thermal Barrier

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
25-51-02	Forward Cargo Compartment Thermal Barrier "DISP.APPR.REQ'D"	Y	1	0	D	May be damaged or missing. Note: No live animals or temperature sensitive cargo should be carried in the forward cargo compartment.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-51-02	Aircraft Logbook	Forward Cargo Compartment Thermal Barrier INOP

25-61-01 Ventral Exit Door Strap

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			1	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
25-61-01	Ventral Exit Door Strap	Y	0	C	May be inoperative or missing provided: 1. Flight Attendant is positioned on the rear door jumpseat, and 2. A passenger announcement is made to stay clear of rear door until the door is OPEN.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-61-01	Aircraft Logbook	Rear Entrance Door Strap INOP

25-61-02 Cabin Emergency Flashlights / Holders

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
25-61-02A	Cabin Emergency Flashlights / Holders	Y	4	0	C	May be inoperative or missing provided cabin crewmember assigned to affected position has an operative flashlight readily available. Note: <i>Cell phone flashlights do not qualify and may not be used as an emergency flashlight.</i>
25-61-02B	Cabin Emergency Flashlights / Holders "Contact MCC" "DISP.APPR.REQ'D" "Revenue Passengers Not Authorized"	Y	4	0	C	(O) May be inoperative or missing provided: 1. MCC must provide authorization prior to use, 2. No revenue passengers or cargo are carried, 3. A maximum of 19 persons are carried as authorized by 14 CFR for non-passenger carrying operations, and 4. Alternate procedures are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-61-02A	Aircraft Logbook	___ Cabin Emergency Flashlight / Holder(s) INOP / Missing
25-61-02B	Aircraft Logbook	___ Cabin Emergency Flashlight / Holder(s) INOP / Missing

(O) PROCEDURES

25-61-02B

1. A maximum of 19 persons authorized by 14 CFR 121.583 may be carried. Persons being carried must be approved by Flight Control Dispatcher.
2. Captain to brief persons authorized to be carried on flight on safety/evacuation procedures.

25-62-01 Emergency Medical Equipment

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
EMERGENCY MEDICAL EQUIPMENT					
25-62-01A	Automatic External Defibrillator (AED) and/or Associated Equipment "Contact MCC"	Y	1	0	A (O) May be incomplete, missing or inoperative provided: 1. AED is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit, and 2. Repair or replacements are made within 1 flight cycle.
25-62-01B	AED Bracket / Lock	Y	1	0	C
25-62-01C	Enhanced Emergency Medical Kit (EEMK) and/or Associated Equipment "Contact MCC"	Y	2	0	A (O) May be incomplete, missing or inoperative provided: 1. EEMK is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit, and 2. Repair or replacements are made within 1 flight cycle.
25-62-01D	Enhanced Emergency Medical Kit (EEMK) and/or Associated Equipment "Contact MCC"	Y	2	1	D (O) May be incomplete, missing or inoperative provided EEMK is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
25-62-01E	EEMK Bracket / Lock	Y	1	0	C
25-62-01F	First Aid Kits (FAK) and/or Associated Equipment "Contact MCC"	Y	2	1	A (O) May be incomplete, missing or inoperative provided: 1. First Aid Kit is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit, and 2. Repair or replacements are made within 1 flight cycles.
25-62-01G	First Aid Kit Seals "Contact MCC"	Y	2	0	D (O) May be missing or broken provided the kit contains the minimum required contents PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)
25-62-01H	Red Pouch (Medical Accessory Pouch)	Y	1	0	C May be inoperative, missing, or contain less than minimum required contents.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-62-01A	Aircraft Logbook	AED Incomplete / Missing / INOP
25-62-01B	Aircraft Logbook	AED Incomplete / Missing / INOP
25-62-01C	Aircraft Logbook	E EMK Incomplete / Missing / INOP
25-62-01D	Aircraft Logbook	EEMK Incomplete / Missing / INOP
25-62-01E	Aircraft Logbook	EEMK BRACKET / LOCK INOP
25-62-01F	Aircraft Logbook	____ First Aid Kit Incomplete / Missing / INOP
25-62-01G	Aircraft Logbook	First Aid Kit Seal(s) Missing / INOP
25-62-01H	Aircraft Logbook	Red Pouch (Medical Accessory Pouch) INOP / Missing

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(O) PROCEDURES

25-62-01A, 25-62-01C, 25-62-01D, 25-62-01F

After use, reseal with green seal and record in the CDL.

25-62-01G

If seal is broken, **PRIOR TO EACH DEPARTURE**, verify the First Aid Kit contains the following:

- | | |
|--|---|
| 2 boxes Antiseptic Swabs(10 per box) | 5 boxes 40 Inch Triangular Bandages (1 per box) |
| 8 boxes 4 Inch Compress Bandages (1 per Box) | 1 box Ammonia Inhalants(10 per Box) |
| 4 boxes 4 inch Gauze Bandages (1 per box) | 2 boxes Wire Splints (1 per box) |
| 1 pair Bandage Scissors | 2 rolls Adhesive Tape |
| 1 box 1 inch Adhesive Bandages (16 per box) | 1 pair Protective Gloves |

25-63-01 Megaphones

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD			Y	2	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION						REMARKS OR EXCEPTIONS
25-63-01	Megaphones "Contact MCC" "DISP.APPR.REQ'D" "Revenue Passengers Not Authorized"				C	(O) May be inoperative or missing provided: 1. MCC must provide authorization prior to use, 2. No revenue passengers or cargo are carried, 3. A maximum of 19 persons are carried as authorized by 14 CFR for non-passenger carrying operations, and 4. Alternate procedures are used.	

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-63-01	Aircraft Logbook	____ Megaphones Missing / INOP

(O) PROCEDURES

1. A maximum of 19 persons authorized by 14 CFR 121.583 may be carried. Persons being carried must be approved by Flight Control Dispatcher.
2. Captain to brief persons authorized to be carried on flight on safety/evacuation procedures.

25-64-01 Flotation Devices (Crew and Passengers)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
FLOTATION DEVICES (CREW AND PASSENGERS)					
25-64-01A	Life Vests (Passenger/Tourist Class/ Observer) "DISP.APPR.REQ'D"	Y	-	-	D May be inoperative or missing provided extended overwater operations are not conducted.
25-64-01B	Life Vests "DISP.APPR.REQ'D"	Y	-	-	D May be inoperative or missing provided associated seat is blocked off, placarded DO NOT OCCUPY and deferred per MEL item Observer MEL 25-15-01A , Flight Attendant MEL 25-23-01A , Passenger MEL 25-23-03A .
25-64-01C	Spare Life Vest	Y	-	-	D May be inoperative or missing.
25-64-01D	Infant Life Vests (If Installed)	Y	-	-	D (O) May be inoperative or missing.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-64-01A	Aircraft Logbook	____ Life Vest(s) INOP / Missing
25-64-01B	Aircraft Logbook	____ Life Vest(s) INOP / Missing
25-64-01C	Aircraft Logbook	____ Spare Life Vest(s) INOP / Missing
25-64-01D	Aircraft Logbook	____ Infant Life Vest(s) INOP / Missing

Note: Unserviceable life vests must be removed from the cabin or stored in a location where they can not be mistaken for serviceable vests.

(O) PROCEDURES

25-64-01D

On aircraft equipped with single cell adult life vests, an infant life vest is normally available for each occupant under 35 pounds for extended overwater operations. In the event that there are not enough infant life vests onboard for the extended overwater operation and all attempts have been made to locate and board additional infant life vests, notify an Airport Customer Service (ACS) Agent so that they can initialize their passenger handling procedures.

25-67-02 Emergency Locator Transmitter (ELT) Survival Type

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
25-67-02A	Emergency Locator Transmitter (ELT) Survival Type	Y	2	1	D May be inoperative or missing.
25-67-02B	Emergency Locator Transmitter (ELT) Survival Type "DISP.APPR.REQ'D"	Y	2	0	D May be inoperative or missing provided: 1. Departures to a country other than the United States or Canada are Not Authorized, 2. Aircraft is not operated on extended overwater flights, and 3. Operations solely within the states of Alaska and Hawaii over uninhabited areas are Not Authorized.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-67-02A	Aircraft Logbook	Emergency Locator Transmitter INOP/MISSING
25-67-02B	Aircraft Logbook	Emergency Locator Transmitter INOP/MISSING

25-99-01 PED Mount

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
25-99-01A	PED Mount Administrative Control Item	Y	-	-	Reconfigure cockpit for two (2) operative PED mounts. See TOPP 40-40-05 for instructions.
25-99-01B	PED Mount Administrative Control Item	Y	-	-	Reconfigure cockpit for one (1) operative PED mount. See TOPP 40-40-05 for instructions.
25-99-01C	PED Mount Administrative Control Item	Y	-	-	Reconfigure cockpit for zero (0) operative PED mounts. See TOPP 40-40-05 for instructions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
25-99-01A	Aircraft Logbook	Reconfigured PED Mounts Per TOPP 40-40-05
25-99-01B	Aircraft Logbook	Reconfigured PED Mounts Per TOPP 40-40-05
25-99-01C	Aircraft Logbook	Reconfigured PED Mounts Per TOPP 40-40-05

26 Fire Protection

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26-11-01 Engine Fire Detection Systems (Detection Loops)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			2	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
26-11-01	Engine Fire Detection Systems (Detection Loops) "Contact MCC"	Y	4	C	(O) (M) ONE COMPLETE LOOP (A OR B) ON EACH ENGINE MAY BE INOPERATIVE.

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-11-01	ASSOCIATED LOOP SELECTOR SWITCH(ES)	____ ENG FIRE DETECTION LOOP(S) ____ INOP

(O) PROCEDURES

1. Verify LOOPS switch is in the position of the operative Loop.
2. Verify proper operation of operative fire detection loops by performing Fire Detection System check as normal during Cockpit Preflight Check.

(M) PROCEDURES

1. Select associated engine LOOPS switch to operative loop.
2. Perform FIRE TEST and FAULT TEST as necessary to verify integrity of remaining loops.

26-12-01 APU Fire Detection System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
APU FIRE DETECTION SYSTEM					
26-12-01A	APU Fire Detection Loops	Y	2	1	C (O) (M) One complete Loop (A or B) may be inoperative.
26-12-01B	APU Fire Detection Loops "DISP.APPR.REQ'D"	Y	2	0	C (O) (M) Both Loops (A and B) may be inoperative provided APU is considered inoperative and deferred per MEL 49-10-01.

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-12-01A	APU Loop Selector Switch	____ APU Fire Detection Loop INOP
26-12-01B	APU Loop Selector Switch	1. Both APU Fire Detection Loops INOP 2. Do NOT Use APU

(O) PROCEDURES

26-12-01A

1. Verify LOOPS switch is in the position of the operative Loop.
2. Verify proper operation of operative fire detection loops by performing Fire Detection System check as normal during Cockpit Preflight Check.

26-12-01B

Utilize ground cart or other means to accomplish engine start, and for ground service power.

(M) PROCEDURES

26-12-01A

1. Select the APU LOOPS switch to the operative loop.
2. Perform FIRE TEST and FAULT TEST as necessary to verify integrity of remaining loop.

26-12-01B

Pull and secure APU CONTROL circuit breaker (B1-29: Lower ECP, Z-39).

26-12-02 APU Fire Warning Horn (External Warning)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
26-12-02	APU Fire Warning Horn (External Warning)	Y	1	0	C	(O) May be inoperative provided APU Fire Warning System is monitored in the cockpit during APU operation.

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-12-02	APU Start Switch	APU Fire Warning Horn INOP

(O) PROCEDURES

When APU is operating on the ground, monitor its operation in the cockpit.

26-12-03 APU Fire Detection Control Unit

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
26-12-03	APU Fire Detection Control Unit "DISP.APPR.REQ'D"	Y	1	0	C	May be inoperative provided APU is not used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-12-03	1. APU External Panel 2. APU Master Start Switch	1. APU Fire Detection INOP 2. Do not use APU

MAINTENANCE NOTE

Should the Left or Right Engine Fire Detection Control Units (FDCU) become inoperative, they may be repositioned to the APU Fire Detection Control Unit position as follows. Maintenance will be required and Flight Crew may NOT placard.

1. Pull and secure the following circuit breakers.

<u>Circuit Breaker (CB#)</u>	<u>Panel</u>	<u>Location</u>
FIRE DETECTORS RIGHT ENGINE LOOP A (B1-59)	Lower EPC CBP	W-37
FIRE DETECTORS RIGHT ENGINE LOOP B (B1-191)	Lower EPC CBP	W-38
FIRE DETECTORS LEFT ENGINE LOOP A (B1-282)	Lower EPC CBP	W-39
FIRE DETECTORS LEFT ENGINE LOOP B (B1-281)	Lower EPC CBP	W-40
FIRE DETECTORS APU LOOP A (B1-323)	Lower EPC CBP	W-35
FIRE DETECTORS APU LOOP B (B1-324)	Lower EPC CBP	W-36

2. Gain access to the three FDCUs located in the left forward rack of the E&E compartment.
 - A. For aircraft 9500-9510, and 9563-9566, the units are mounted on the X-40 panel from top to bottom in the following order; Right Engine, Left Engine, and APU.
 - B. For aircraft 9511-9562, 9567-9586, and 9594, the left and right engine units are mounted at the forward end of the left radio rack while the APU unit is located on the top shelf of the forward left radio rack.
3. Remove the electrical connector from the APU FDCU and from the inoperative FDCU.
4. Remove the APU FDCU and the inoperative FDCU from the rack by removing the four screws, washers, and lockwashers.
5. Swap the APU FDCU with the inoperative unit, and reinstall both units in their new respective positions in the rack. Secure the units in the rack with the four screws, washers, and lockwashers.
6. Attach the electrical connector to the Units.
7. Reset the circuit breakers pulled in step 1 above.

26-13-01 Lower Cargo Compartment Smoke Detection System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD						REPAIR CATEGORY
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
26-13-01A	Lower Cargo Compartment Smoke Detection System “DISP.APPR.REQ’D”	Y	-	0	C	(O) May be inoperative provided procedures are used to ensure the associated cargo compartment remains empty, or is verified to contain only empty cargo handling equipment, ballast and/or Fly Away Kits. Refer to (TOPP 40-40-05 - MEL/CDL Instructions)
26-13-01B	Lower Cargo Compartment Smoke Detector Units	Y	12	6	C	(O) One detector at each location may be inoperative and the remaining portions of the Smoke Detection System may still be used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-13-01A	Cargo Smoke Detection Panel	Cargo Smoke Detection System INOP
26-13-01B	Cargo Smoke Detection Panel	___ Cargo Smoke Detector INOP

(O) PROCEDURES

26-13-01A

PRIOR TO EACH DEPARTURE, Captain is to ensure that no cargo is loaded in the associated cargo compartment. Refer to (TOPP 40-40-05 - MEL/CDL Instructions) for a list of approved ballast and Fly Away Kit items.

26-13-01B

1. If inoperative smoke detector is in the forward cargo compartment, place the FWD switch on the Cargo Smoke Control Panel to SINGLE.
2. If inoperative smoke detector is in the aft cargo compartment, place the AFT switch on the Cargo Smoke Control Panel to SINGLE.

26-14-01 Lavatory Smoke Detection and Warning Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
26-14-01	Lavatory Smoke Detection and Warning Systems "DISP.APPR.REQ'D"	Y	3	0	C (O) (M) For each lavatory, the Lavatory Smoke Detection System may be inoperative provided: 1. Associated Lavatory waste receptacle is empty, 2. Associated Lavatory door is locked CLOSED and placarded, "INOPERATIVE - DO NOT ENTER", and 3. Associated Lavatory is only used by crewmembers. Note: These provisos are not intended to prohibit lavatory use or inspections by crewmembers.

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-14-01	Aircraft Logbook	____ Lavatory Smoke Detection and Warning System(s) INOP

(O) PROCEDURES

1. Inform cabin crewmembers they may enter the lavatory to perform inspections as necessary, including verification that the lavatory waste receptacle is empty.
2. Verify door is locked after any Lavatory inspection.

(M) PROCEDURES

1. Verify that the affected lavatory trash receptacle is empty.
2. Close and lock the associated lavatory door.
3. Placard lavatory door "INOPERATIVE - DO NOT ENTER".

26-21-01 Engine and APU Fire Extinguisher Agent Low Lights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
26-21-01	Engine and APU Fire Extinguisher Agent Low Lights "Contact MCC"	N	4	2	C (O) (M) May be inoperative provided: 1. Associated bottle pressure is verified normal PRIOR TO EACH DEPARTURE (Logbook Entry Required), 2. At least one Agent Low Light indicating system is operative on the flight compartment upper instrument panel, and 3. Procedures are used for discharging the bottle with the operative Agent Low light first. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-21-01	Affected AGENT LOW Light and APU Ground Panel	____ Agent Low Light INOP

(O) PROCEDURES

If extinguisher system is to be used, discharge bottle with operative AGENT LOW light first.

(M) PROCEDURES

PRIOR TO EACH DEPARTURE (Logbook Entry Required):

1. Verify bottle pressure is adequate by conducting the following pressure switch continuity test. Successful completion of the test verifies the bottle is pressurized.

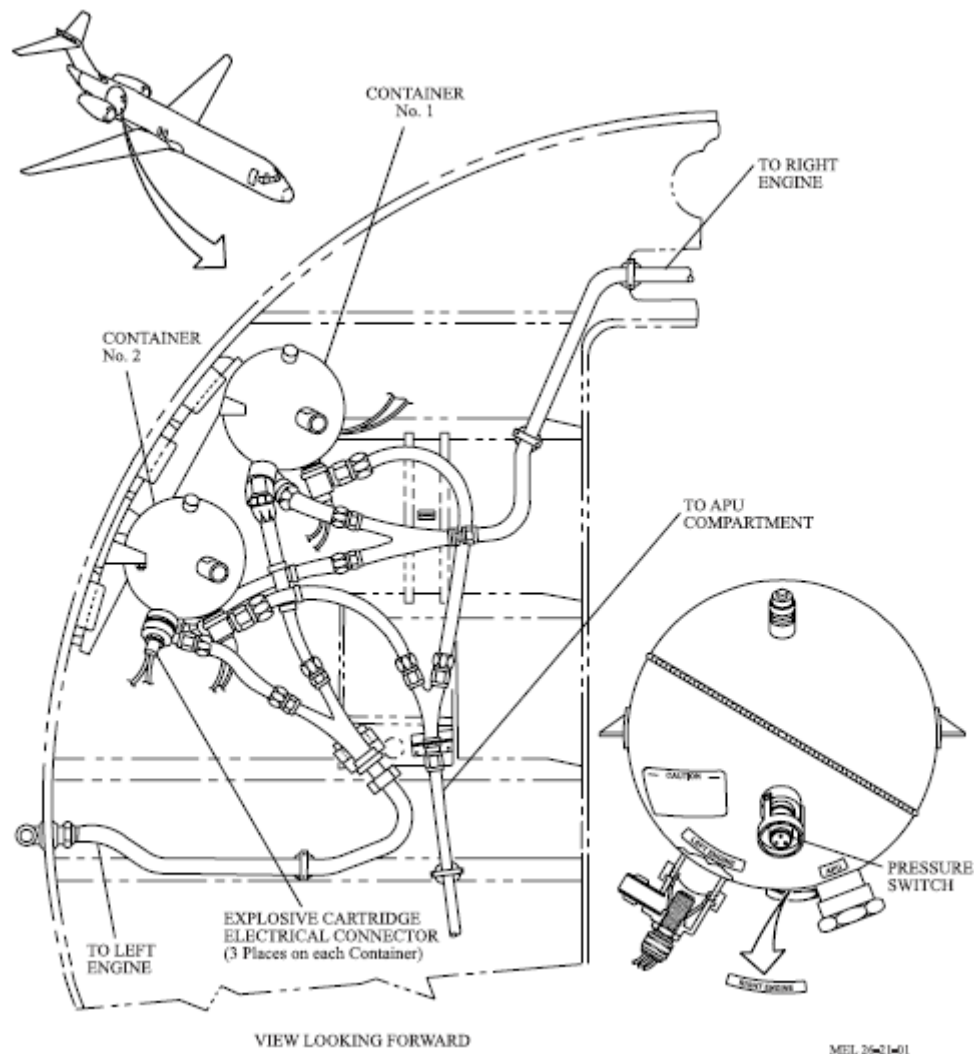
Procedure I

- A. Disconnect the pressure switch electrical connector at the bottle.
- B. Insert 1/16" Allen wrench in test socket of pressure switch. Using light finger pressure, actuate pressure switch by turning Allen wrench.
- C. Verify pressure switch functions as follows:
 - i. Pins B to C - no continuity
 - ii. Pins B to A - continuity
- D. Release Allen wrench and verify switch returns to its normal position. Remove Allen wrench.
- E. Verify pressure switch functions as follows:
 - i. Pins B to C - continuity
 - ii. Pins B to A - no continuity

Procedure II

2. If previous procedure is unsuccessful, or as an alternative, weigh the affected fire agent container. The weight of the container (without discharge head and squibs) should be within 1/2 lb. of the weight stamped on the container.

CONTINUED ON NEXT PAGE

Figure 26-1: Engine and APU Fire Extinguisher Bottles


26-21-02 APU Fire Shutoff Controls

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
	APU FIRE SHUTOFF CONTROLS					
26-21-02A	Flight Deck "DISP.APPR.REQ'D"	Y	1	0	C	May be inoperative provided APU is not used.
26-21-02B	Exterior	Y	1	0	C	(O) May be inoperative provided APU operation is monitored in the cockpit.

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-21-02A	APU Start Switch	1. Flight Deck APU Fire Shutoff Controls INOP 2. Do NOT Use APU
26-21-02B	APU Start Switch	Exterior APU Fire Shutoff Controls INOP

(O) PROCEDURES

26-21-02B

When APU is operating on the ground, monitor its operation in the cockpit.

26-21-03 Firex Bottle Discharge Cartridges

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
26-21-03A	Firex Bottle Discharge Cartridges	N	6	5	C	(M) May be inoperative provided: 1. Affected Cartridge is in the APU position, and 2. APU Firex Control Switches (Cockpit and External Ground Control Panel) are placarded appropriately so as to prevent inadvertent use of the affected switch.
26-21-03B	Firex Bottle Discharge Cartridges "DISP.APPR.REQ'D"	Y	6	4	C	May be inoperative provided: 1. Affected Cartridges are in the APU position, and 2. APU is considered inoperative and deferred per MEL 49-10-01.

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-21-03A	Aircraft Logbook	____ Bottle Discharge Cartridges INOP
26-21-03B	Aircraft Logbook	____ Bottle Discharge Cartridges INOP

(M) PROCEDURES

26-21-03A

- Verify the affected Discharge Cartridge is in the APU position.
- Verify the APU Firex Controls have been placarded appropriately, as noted above in the Remarks or Exceptions column.

Note: If the affected Discharge Cartridge is in other than the APU position, it may be repositioned to the APU position per the instructions of the 717 Aircraft Maintenance Manual. Refer to [AMM Section 26-21-02](#).

26-22-01 Lower Cargo Compartment Fire Suppression System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
26-22-01	Lower Cargo Compartment Fire Suppression System "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided procedures are used to ensure the associated cargo compartment remains empty, or is verified to contain only empty cargo handling equipment, ballast and/or Fly Away Kits. Refer to (TOPP 40-40-05 - MEL/CDL Instructions)

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-22-01	Cargo Fire Control Panel	___ FWD or AFT Cargo Compartment Fire suppression System INOP

(O) PROCEDURES

PRIOR TO EACH DEPARTURE, Captain is to ensure that no cargo is loaded in the associated cargo compartment. Refer to (TOPP 40-40-05 - MEL/CDL Instructions) for a list of approved ballast and Fly Away Kit items

MAINTENANCE NOTE:

1. If either bottle has been discharged, both compartments must remain empty.
2. If one of the Discharge Cartridges becomes inoperative but both Firex bottles are fully charged, the unaffected compartment may still be used. Do not load cargo into the affected compartment, and placard the CARGO SMOKE Panel appropriately so as to prevent discharging agent into the affected compartment.

26-25-01 Lavatory Fire Extinguisher Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
26-25-01A	Lavatory Fire Extinguisher Systems	Y	3	0	C For each lavatory, the Lavatory Fire Extinguisher System may be inoperative provided the associated Lavatory Smoke Detector System operates normally.
26-25-01B	Lavatory Fire Extinguisher Systems "DISP.APPR.REQ'D"	Y	3	0	C (O) (M) For each lavatory, the Lavatory Fire Extinguisher System may be inoperative provided: 1. Associated Lavatory Waste receptacle is empty, 2. Associated Lavatory Door is locked CLOSED and placarded, "INOPERATIVE-DO NOT ENTER", and 3. Associated Lavatory is used only by crewmembers. Note: These provisos are not intended to prohibit lavatory use or inspections by crewmembers.

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-25-01A	Aircraft Logbook	___ Lavatory Fire Extinguisher System(s) INOP
26-25-01B	Aircraft Logbook	___ Lavatory Fire Extinguisher System(s) INOP

(O) PROCEDURES

26-25-01B

1. Inform cabin crewmembers they may enter the lavatory to perform inspections as necessary, including verification that the lavatory waste receptacle is empty.
2. Verify door is locked after any Lavatory inspection.

(M) PROCEDURES

26-25-01B

1. Verify that the affected lavatory trash receptacle is empty.
2. Close and lock the associated lavatory door.
3. Placard lavatory door "INOPERATIVE - DO NOT ENTER".

26-26-01 Portable Fire Extinguishers

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			3	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
26-26-01	Portable Fire Extinguishers	Y	-	D	Any in excess of 3 may be inoperative or missing provided: 1. The inoperative fire extinguisher is tagged inoperative, removed from the installed location, and placed out of sight so it can not be mistaken for a functional unit, and 2. Required distribution is maintained.

MEL ITEM	TAB LOCATION	PLACARD TEXT
26-26-01	Aircraft Logbook	____ Portable Fire Extinguisher INOP

27 Flight Controls

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27-67-01

Flight Spoilers Actuators

27-67-01

Flight Spoilers Actuators

27-12-01 Aileron Trim System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-12-01	Aileron Trim System "Contact MCC"	N	1	0	C (O) (M) May be inoperative provided Aileron Trim Tabs are verified to be in neutral position PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-12-01	Aileron Trim Switches	___ Aileron Trim INOP

(O) PROCEDURES

PRIOR TO EACH DEPARTURE: Verify that the aileron trim tabs are in the neutral position.

(M) PROCEDURES

PRIOR TO EACH DEPARTURE:

Verify that the aileron trim tabs are in the neutral position.

 [Link to Learning Video](#)

1. Verify Aileron Trim Tabs are in neutral position.
 - A. Try this electrically first by activating switches on center pedestal, Ref [AMM 27-11-01-720-801](#), or
 - B. Ref [AMM 27-12-03-820-801C1](#) manual adjustment, or

Note: RII required for Rig/Adjustment of Aileron Actuators. Reference [TOPP 40-30-05](#), Section IX, Appendix K.

 - C. Ref [AMM 27-12-03-720-801](#) Aileron Trim Tab Actuator Adj/Test

Note: Multiple turns are required for manual adjustments.

Deactivate both Aileron Trim Actuators as follows:

1. Gain access to the Aileron Trim Actuators through the lower aileron hinge access doors located just forward of each Aileron Trim Tab (ref Access Doors 530XB and 630XB).
2. Disconnect, cap and stow the electrical connector from the disconnect panel.
3. If either Aileron Trim Tabs are not faired with the Aileron at the trailing edge, open the adjacent access door (530YB, 630YB) and manually adjust the trim tab actuator as shown in AMM Practices and Procedures.

Figure 27-1: Trim Tab Actuator

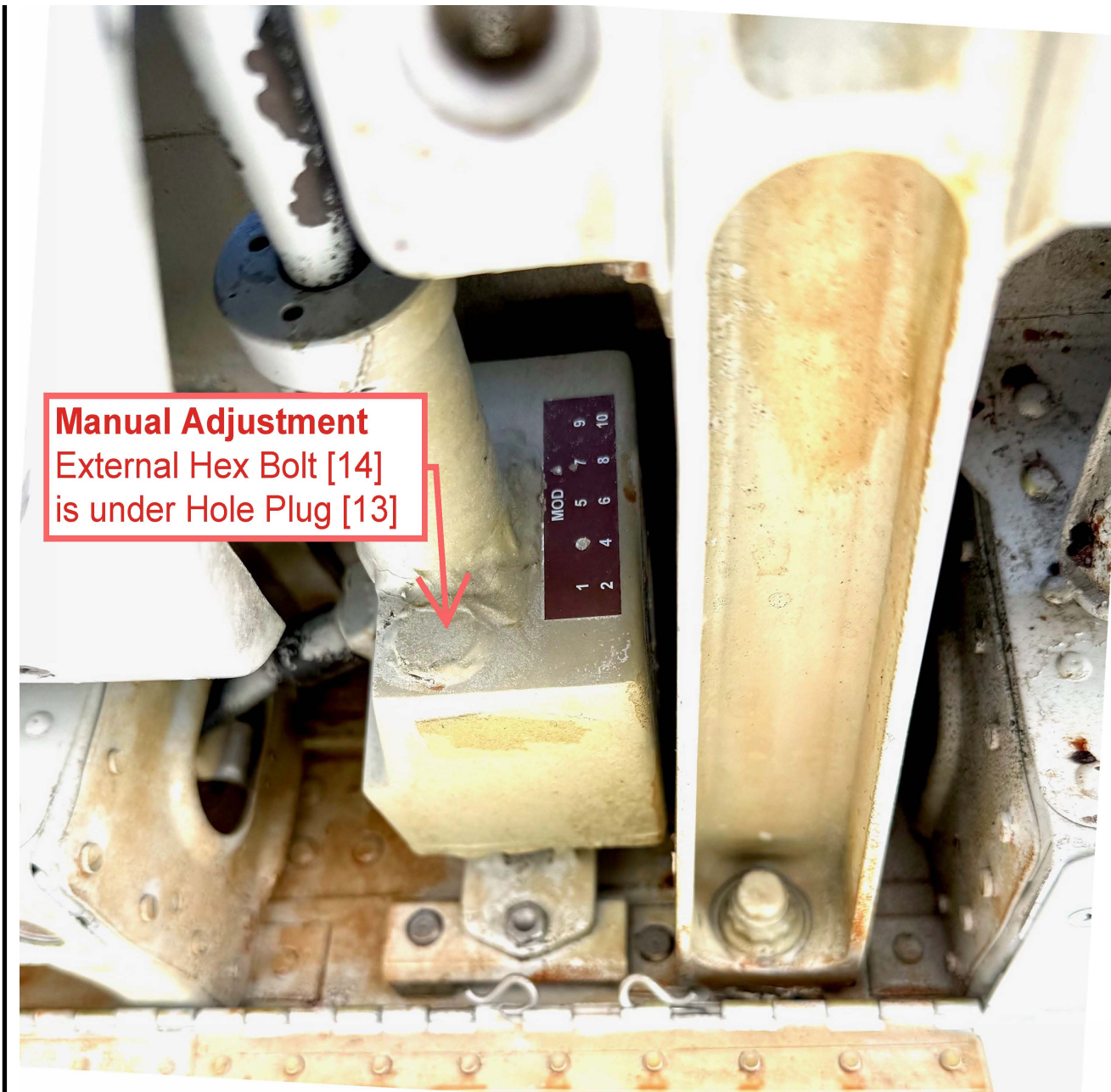
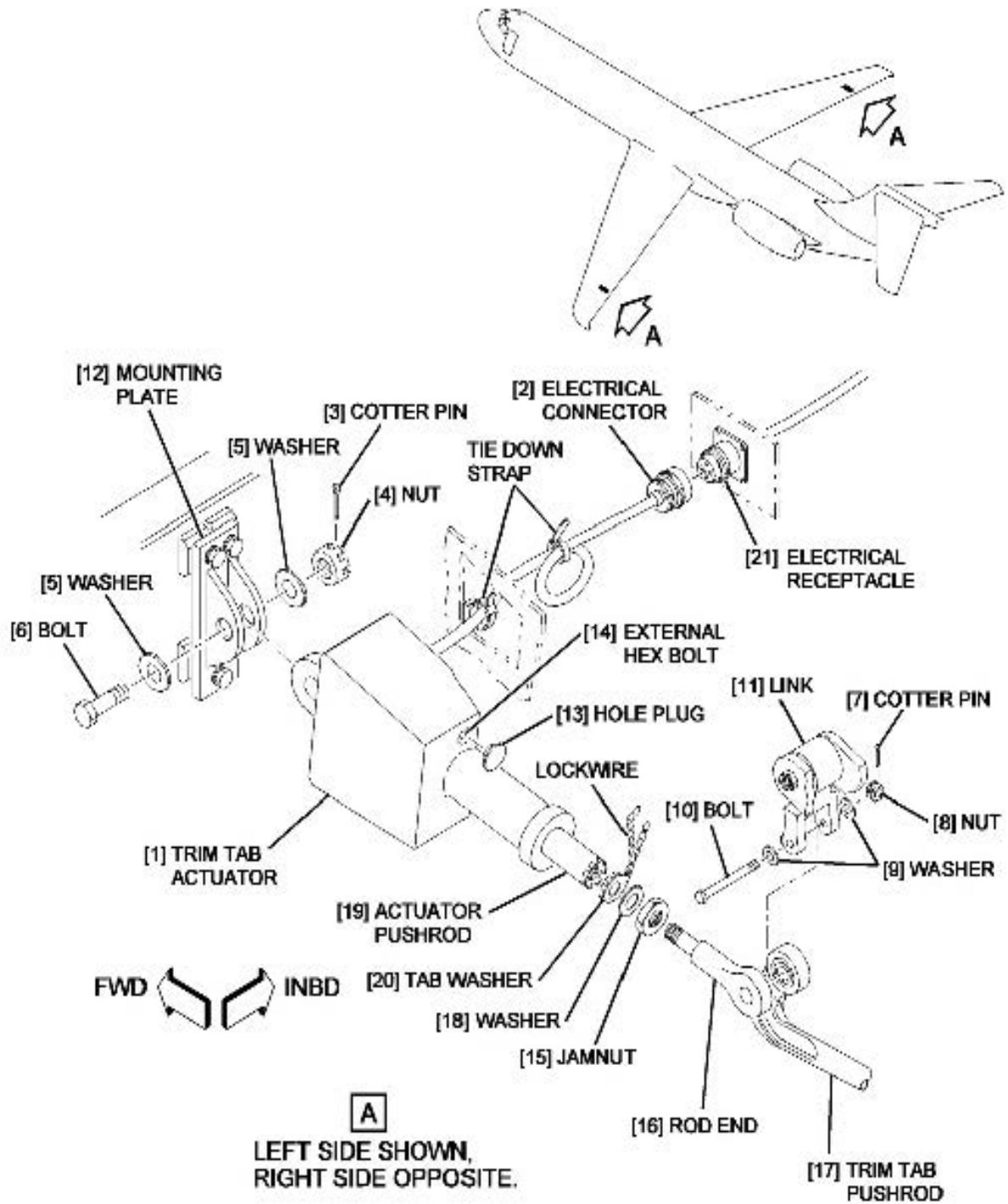


Figure 27-2: Aileron Trim System



27-14-01 Aileron Position Indication System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-14-01	Aileron Position Indication System "DISP.APPR.REQ'D"	Y	2	0	C Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-14-01	Near Aileron Position Switches	____ Aileron Position Indication System INOP

27-15-01 Aileron Dampers

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-15-01	Aileron Dampers	Y	4	2 C	One may be inoperative on each aileron.

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-15-01	Near Aileron Trim Switches	____ Aileron Damper(s) INOP

27-22-01 Rudder Trim Position Indicating System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-22-01	Rudder Trim Position Indicating System	Y	2	0	C
					(M) May be inoperative provided Rudder is verified to be in neutral position PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-22-01	Near Rudder Trim Knob	____ Rudder Trim Position Indicating INOP

(M) PROCEDURES

PRIOR TO EACH DEPARTURE:

1. Set the AUX Hydraulic System Switch on the Hydraulics Overhead Panel to the ON position.
2. Push RDR CTR Pushbutton on the pedestal.
3. Visually confirm Rudder Trim tab is faired with the rudder.
4. Set the AUX Hydraulic System Switch on the Hydraulics Overhead Panel to the OFF position.

27-22-02 Rudder Center Switch

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
27-22-02	Rudder Center Switch	Y	1	0	C	May be inoperative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-22-02	Near Rudder Center Push Button	____ Rudder Center Switch INOP

27-24-01 Rudder Position Indication System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-24-01	Rudder Position Indication System "DISP.APPR.REQ'D"	Y	1	0 C	(M) May be inoperative provided proper Rudder movement is verified PRIOR TO EACH DEPARTURE, refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-24-01	Near Center Display Panel	____ Rudder Position Indication INOP

(M) PROCEDURES

| PRIOR TO EACH DEPARTURE:

1. Coordinate with flight crew prior to each takeoff to observe control surface check.
2. Visually confirm rudder moves in the direction commanded by the flight crew through the rudder pedals.

27-25-01 Rudder Dampers

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
27-25-01	Rudder Dampers	Y	2	1	C	May be inoperative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-25-01	Near Rudder Trim Knob	____ Rudder Damper INOP

27-26-01 Rudder Limiter Systems (Primary and Secondary)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-26-01	Rudder Limiter Systems (Primary and Secondary) "DISP.APPR.REQ'D"	Y	2	1 C	May be dispatched with RUDDER LIM FAULT displayed on the STATUS Page, refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. Note: The rudder will continue to be restricted/unrestricted as normal in this condition. Note: This item may not be used to defer any Rudder system failures or any Alerts other than RUDDER LIM FAULT. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-26-01	Near Rudder Trim Knob	____ Rudder Limiter INOP

27-30-01 Control Column Disconnect Annunciation System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	Y	1	0	C
27-30-01	Control Column Disconnect Annunciation System	Y	1	0	C
					REMARKS OR EXCEPTIONS (O) May be inoperative provided Control Columns are verified to be connected PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-30-01	L or R Instrument Panel	Control Column Disconnect Ann.Sys. INOP

(O) PROCEDURES

PRIOR TO EACH DEPARTURE:

1. From either the Captain's or First Officer's seat, grasp both Control Columns.
2. Apply a steady force of 20 to 30 lbs (approximate) on one control column while restraining movement of the opposite control column. Verify both columns move in tandem, with no relative difference in movement between columns.
3. If only one column moves when force is applied, or if a relative difference in column movement of one inch or greater exists, columns may be disconnected. Maintenance action is required prior to departure. Dispatch is not authorized with disconnected control columns.

27-34-01 Elevator Position Indicating System

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH		REPAIR CATEGORY	REMARKS OR EXCEPTIONS
ITEM NO.	NAME/DESCRIPTION	FLIGHT CREW MAY PLACARD				
27-34-01	Elevator Position Indicating System "DISP.APPR.REQ'D"	N	2	0	C	(O) (M) May be inoperative provided: 1. Elevator movement is verified by performing a Full Deflection check PRIOR TO EACH DEPARTURE, 2. Aircraft remains at or below FL220, 3. Flight speed is restricted to 240 KIAS, 4. Aircraft is not operated within the boundaries of a Critical Terrain Box, 5. Autoland is not used, and 6. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. Theater Restrictions May Apply <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-34-01	Below Captains N.D. Display	____ Elevator Position Indicating INOP

(O) PROCEDURES

1. Plan flight so as not to exceed 240 KIAS, inform ATC as required.
2. Maximum permitted altitude is FL220.
3. Coordinate with ground crew to move the elevator and verify proper movement of the control surfaces.
4. Program KIAS/Mach speed restriction in FMS.

(M) PROCEDURES

PRIOR TO EACH DEPARTURE:

1. Verify elevator travel is available through its full range of corresponding control movement:
 - A. Coordinate with flight crew to perform control surface check, and position an observer to verify elevator tab and elevator surface movement.
 - B. Verify correct elevator control tab movement occurs when the control column is pulled full aft.
 - C. Verify correct elevator surface movement occurs when control column is pushed full forward.

27-38-01 Stall Protection Systems Stick Shaker Motors

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH		REPAIR CATEGORY	
FLIGHT CREW MAY PLACARD					REMARKS OR EXCEPTIONS	
ITEM NO.	NAME/DESCRIPTION					
27-38-01	Stall Protection Systems Stick Shaker Motors "Contact MCC"	N	2	1	C	(M) May be inoperative provided: - Affected Stick Shaker Motor is deactivated, - Stall Protection System aural and visual alerts are not indicated other than a STALL WARN FAULT, and - STALL WARN FAULT is verified to be the result of the inoperative Stick Shaker Motor PRIOR TO EACH DEPARTURE (Logbook Entry Required). Note: This item may not be used to disposition any Stall Protection System failures other than STALL WARN FAULT when displayed as a result of an inoperative Stick Shaker Motor. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-38-01	Near Captain's STICK PUSHER Light	___ Stick Shaker INOP

(M) PROCEDURES

- For an inoperative Stick Shaker Motor, pull and secure the appropriate circuit breaker for the affected,

<u>Circuit Breaker (CB#)</u>	<u>Panel</u>	<u>Location</u>
F/O STICK SHAKER (B1-026)	Lower EPC	R-33
CAPTAIN'S STICK SHAKER (B1-487)	Lower EPC	P-33

- PRIOR TO EACH DEPARTURE**, perform a FCC-1 Stall Warning System Return-to-Service (RTS) and a FCC-2 Stall Warning System Return-to-Service (RTS) per [AMM task 22-00-00-741-815](#).

Note: During each Stall Warning System Return-to-Service (RTS), at the end of Stick Pusher Test a menu will be displayed on the MCDU advising of an optional Stick Pusher Engage Test. Select the CONTINUE prompt to not run this test.

- Verify at the end of each FCC-1 and FCC-2 Stall Protection System RTS only one of the following failures is present, SHAKER MOTOR - CAPT or SHAKER MOTOR - F/O. This confirms STALL WARN FAULT was generated by opening the associated circuit breaker of the inoperative Stick Shaker Motor. Dispatch with any other failure(s) is not permitted.

27-43-01 Alternate Longitudinal Trim Motor Control Relays (Up Trim/Down Trim)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-43-01	Alternate Longitudinal Trim Motor Control Relays (Up Trim/Down Trim)	Y	2	0	C (O) (M) May be inoperative provided: 1. It is verified that the associated relay is not energized, 2. Manual Alternate Trim System operates normally, and 3. Mach Trim System is considered inoperative and deferred per MEL 22-22-02.

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-43-01	Flight Control Panel	____ Alternate Longitudinal Trim Motor Control Relay (Up Trim / Down Trim) INOP

(O) PROCEDURES

1. Verify Manual Alternate Trim functions properly by moving ALT LONG TRIM Switches on the center pedestal and observing corresponding movement of the Stabilizer on the CONFIG Synoptic page.
2. In flight, trim aircraft manually prior to engaging Autopilot.
3. If the AUTO TRIM FAIL Alert appears, return the stabilizer by either of the following methods:
 - A. Operate the ALT LONG TRIM Switches on the center pedestal (which does not require autopilot disconnect), or
 - B. Disconnecting the autopilot, manually trim the aircraft using the Control Wheel Trim Switches, and then re-engage the Autopilot.
4. If the Autopilot is to be used for landing, manually trim the aircraft and track the glideslope at final approach speed prior to Autopilot engagement.

(M) PROCEDURES

1. Operate the Primary Trim Switches on both Control Wheel horns and confirm corresponding movement of the Stabilizer. Stabilizer position and movement can be observed on the CONFIG Synoptic page.
2. Defer the Mach Trim System per [MEL 22-22-02](#).

27-43-02 Alternate Longitudinal Trim Motor Up/Down Limit Switches

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-43-02	Alternate Longitudinal Trim Motor Up/Down Limit Switches	Y	2	0	C (M) May be inoperative provided Alternate Trim System otherwise functions normally.

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-43-02	Near Alternate Trim Switches	___ Alternate Longitudinal Trim Motor Limit Switch(s) INOP

(M) PROCEDURES

1. Operate the Primary Trim Switches on either Control Wheel horn and verify stabilizer travels through its complete range from 12° ANU to 2.6° AND. Stabilizer position and movement can be observed on the CONFIG Synoptic page.
2. Repeat step 1. using the ALT LONG TRIM Switches on the pedestal.

Note: Rate of movement using Primary Trim Switches is approximately 1/3 degree per second, and stabilizer movement through the full range of travel requires approximately 48 seconds. Rate of movement using ALT LONG TRIM Switches is approximately 1/5 degree per second, and stabilizer movement through the full range of travel requires approximately 2 minutes, 40 seconds.

27-51-01 Dial-A-Flap System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			1	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	Y			C	REMARKS OR EXCEPTIONS
27-51-01	Dial-A-Flap System "DISP.APPR.REQ'D"	Y				May be inoperative provided: 1. Movable detent is in the STOWED position, 2. Fixed Flap Detent System is operates normally, and 3. Both Flap Position Indicating Systems are operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-51-01	Near Dial-A-Flap FLAP T.O. SEL Window or Wheel	Dial-A-Flap System INOP

OPERATIONS NOTE:

Data for a Flaps 13 or Flaps 18 takeoff will be required.

27-54-01 Flap Position Indicating System

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION			REMARKS OR EXCEPTIONS
27-54-01	Flap Position Indicating System "DISP.APPR.REQ'D"	N	2	1 C
				(O) (M) One may be inoperative provided: - It is verified that flaps are operative through their normal operating range, - Visual inspection verifies that flaps are in takeoff position and no asymmetry exists PRIOR TO EACH DEPARTURE (Log Book Entry Required), and - Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-54-01	Below Captains N.D. Display	_____ Flap Indicating System INOP

(O) PROCEDURES

- Stay at gate for WDR.
- Prior to takeoff, coordinate with maintenance to verify flaps are in commanded position.
- Do not move flap/slat handle again until after takeoff. If flap/slat handle is moved (e.g., during de-icing operations), reconfirm with maintenance that flaps are again selected to their proper position before takeoff.

(M) PROCEDURES

 [Link to Learning Video](#)

PRIOR TO EACH DEPARTURE:

- Verify flaps can be operated through their entire range of motion.
- After flaps are set for takeoff, visually confirm flap position to be correct by inspecting flap position at the inboard flap hinge:

Note: *Flight Crew will not obtain the command position of the flaps until departure time.*

- Remove panels 144AZ and 145AZ, part number 5924897-85 inside the wheel well.
- Use scribe marks at the hinge, and coordinate with flight crew to ensure flaps are at commanded position, and no asymmetry exists.
- Re-install panel part number 5924897-85.

Figure 27-3: Flap Position Indicating System

27-61-01 Spoiler Electronic Control Unit (SECU)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-61-01A	Spoiler Electronic Control Unit (SECU) "DISP.APPR.REQ'D"	Y	1	0	C (O) May be dispatched with SECU FAULT displayed on the STATUS page provided appropriate performance data is used, refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements . <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized Note: The SECU will continue to operate normally in this condition. Note: This item may not be used to disposition any Spoiler system failures or any Alerts other than SECU FAULT.
27-61-01B	Spoiler Electronic Control Unit (SECU) "DISP.APPR.REQ'D"	Y	1	0	C (M) SECU -04 (and subs) may be dispatched with SECU FAULT displayed on the STATUS page provided: 1. SECU Fault Review is performed PRIOR TO EACH DEPARTURE, 2. SECU Ground Spoiler Logic Fault (Fault Code 276026) is verified NOT the cause of SECU FAULT alert, and 3. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements . <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized Note: The SECU will continue to operate normally in this condition. Note: This item may not be used to disposition any Spoiler system failures or any Alerts other than SECU FAULT. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-61-01A	1. Below Spoiler/Speedbrake Lever 2. Below Spoiler/Speedbrake Lever 3. Below Captains N.D.	1. Auto Ground Spoiler System INOP 2. SECU Fault Display INOP 3. Do Not Use Autopilot Autoland mode
27-61-01B	1. Below Spoiler/Speedbrake Lever 2. Below Spoiler/Speedbrake Lever 3. Below Captains N.D.	1. Auto Ground Spoiler System INOP 2. SECU Fault Display INOP 3. Do Not Use Autopilot Autoland mode

CONTINUED ON NEXT PAGE

(O) PROCEDURES

27-61-01A

27-61-01A	SPOILER ELECTRONIC CONTROL UNIT (SECU)
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	None
NOTES	Minimum takeoff and landing runway length is 6,000 ft.
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF 1. Reduce RATOW by 19,200 lbs. 2. Reduce T/O CLIMB LIMIT by 19,200 lbs. LANDING 1. Reduce RWY ALLOWABLE LANDING WT by 18,100 lbs. 2. Reduce LANDING CLIMB LIMIT WT by 18,100 lbs.

(M) PROCEDURES

27-61-01B

PRIOR TO EACH DEPARTURE:

SECU Fault Review (perform steps in the following order),

1. Select MENU on either MCDU to access the main menu page.
2. Select CFDS.
3. Select LRU MAINTENANCE.
4. Press PAGE Key twice to access Page 3.
5. Select SECU.
6. Select FAULT REVIEW.
 - A. If the following SECU Fault Code and Maintenance Message is present, this is a SECU Ground Spoiler Logic Fault. Notify Flight Operations Office/Dispatch Office of a SECU Ground Logic Fault that AFM Appendix 3K must be observed. The aircraft may then be dispatched by applying the first proviso of this MMEL item and no further maintenance action is required.

Fault Code

276026

Maintenance Message

SECU INTERNAL FAULT

- B. If the SECU Fault Code is other than shown above, repeat this Maintenance Procedure prior to each flight.

27-62-01 Auto Ground Spoiler System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-62-01	Auto Ground Spoiler System "DISP.APPR.REQ'D"	Y	1	0	C (O) (M) May be inoperative provided: - Appropriate performance penalties are applied, and - A SECU Fault Review is performed PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-62-01	Near Speed Brake Lever	Auto Ground Spoiler INOP

(O) PROCEDURES

Note: Procedures will require the manual use of Ground Spoilers (RTO etc.)

- Do not arm Auto Spoilers for takeoff or for landing.
- Green box will not be present for takeoff or landing.
 - For takeoff, a white box around "SPOILERS" will be displayed.
 - For landing, a white box around "SPOILERS" will be displayed.
- Spoilers can be manually extended to full deployment by moving the speedbrake lever up-aft.

27-62-01	AUTO GROUND SPOILER SYSTEM
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	None
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF Reduce RATOW by 1,000 lbs. Reduce T/O CLIMB LIMIT by 1,000 lbs. LANDING Reduce RWY ALLOWABLE LANDING WT by 9,100 lbs. Reduce LANDING CLIMB LIMIT WT by 9,100 lbs.

(M) PROCEDURES

PRIOR TO EACH DEPARTURE, perform a SECU Fault Review as follows:

1. Verify the Spoiler Handle is disarmed (down).
2. Push the MENU on either MCDU to access the main menu page.
3. From the Main MENU Page, select CFDS.
4. From the CFDS INITIAL MENU Page, select LRU MAINTENANCE.
5. From the CFDS LRU MAINTENANCE Page 1, press the PAGE key twice to access Page 3.
6. From the CFDS LRU MAINTENANCE Page 3, select SECU.
7. From the SECU MAINTENANCE Page, select CURRENT FAULTS to review specific fault messages.
8. If either of the following SECU Fault Messages are present, dispatch is not authorized

<u>Fault Code</u>	<u>Maintenance Message</u>
276211 AGS	FAULT ARM SWITCH 1
276212 AGS	FAULT ARM SWITCH 2
9. Arm (raise) the Spoiler Handle, and then press the REFRESH key on the MCDU. Again verify neither of the above SECU Fault Messages are present.
10. Disarm the Spoiler Handle.

27-67-01 Flight Spoilers Actuators

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
27-67-01	Flight Spoilers Actuators "DISP.APPR.REQ'D"	N	4	2	A (O) (M) One symmetrical pair (either both inboard or both outboard) may be inoperative provided: 1. Associated symmetrical pair of spoilers are deactivated, 2. Remaining spoilers are verified to operate normally PRIOR TO EACH DEPARTURE (Logbook Entry Required), 3. Ailerons are verified to move freely PRIOR TO EACH DEPARTURE, 4. Aircraft remains at or below FL 250, 5. Performance penalties are applied, 6. Repairs are made within three flight cycles, and 7. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> NOT AUTHORIZED <u>AUTOLAND</u> NOT AUTHORIZED Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
27-67-01	Near Speed Brake Lever	___ Flight Spoilers INOP

(O) PROCEDURES

1. Obtain FAR Landing Distances from ODM, Abnormal Section, MEL Support.
2. Limit flight altitude to a maximum of FL 250.
3. During exterior pre-flight control check, move the ailerons to verify that ailerons move freely.
4. During normal pre-flight controls check, verify on the configuration display the spoiler white box appears on each wing when the control wheel is turned in the corresponding direction. Spoiler boxes will not display full deflection (Spoiler box will not shade green on CONFIG page).
5. Spoilers may be armed for take-off.
6. Do not arm or deploy spoilers for landing.
7. The pilot flying must assume manual control of aircraft prior to the final approach fix (i.e. the autopilot must be off).
8. Prior to landing, an outlined "white box" with SPOILERS will be displayed in the essential items checklist on the EAD. A green landing checklist complete box (outlined green box) will not be displayed.
9. Do not manually deploy spoilers after touchdown or landing roll-out.

27-67-01	FLIGHT SPOILER ACTUATORS
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	None
NOTES	Minimum takeoff and landing runway length is 6,000 ft. The limiting crosswinds for takeoff and landing is 15 kts.
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF 1. Reduce RATOW by 19,200 lbs. 2. Reduce T/O CLIMB LIMIT by 19,200 lbs. LANDING 1. Reduce RWY ALLOWABLE LANDING WT by 29,350 lbs. 2. Reduce LANDING CLIMB LIMIT WT by 29,350 lbs.

(M) PROCEDURES

1. If the failed spoiler actuator(s) is an inboard actuator, disconnect the actuator servo valve electrical connectors from the disconnect brackets: P1-3315C (LIB) and P1-3316C (RIB), for BOTH inboard actuators. Coil and stow the connector.

Note: *It may be necessary to open the applicable Bent Up Trailing Edge (BUTE) doors (530MB, 530RB, 630MB, 630RB) to properly disconnect and stow the connector. OR*

2. If the failed spoiler actuator(s) is an outboard actuator, for BOTH outboard actuators: disconnect the actuator servo valve electrical connectors from the disconnect bracket: P1-3317C (LOB) and P1-3318C (ROB). Coil and stow the connector.

Note: *It may be necessary to open the applicable Bent Up Trailing Edge (BUTE) doors (530MB, 530RB, 630MB, 630RB) to properly disconnect and stow the connector.*

3. PRIOR TO EACH DEPARTURE:

- A. Verify the following spoiler related alerts are NOT displayed
 - i. If deactivating outboard spoilers verify SPOILER INBD FAIL is not displayed, or If deactivating inboard spoilers verify SPOILER OUTBD FAIL is not displayed.
 - ii. SECU FAULT
 - iii. SPOILER FAULT
- B. Observe the spoiler panels (and/or monitor the SECU Sensor Readout under the MCDU LRU Maintenance Menu) while commanding the spoilers to extend using the speed brake lever or control wheel; identify the spoiler panel that is non-responsive.
- C. Perform a Return-To-Service (RTS) test of the Spoiler System per [AMM task 27-60-00-741-801](#). For dispatch, the only permitted faults reported by the RTS are the following which correspond to the failed spoiler actuator(s): 276151 (Left Inboard), 276161 (Left Outboard), 276171 (Right Inboard) or 276181 (Right Outboard).

CAUTION

FOR DISPATCH, EITHER BOTH INBOARD ACTUATORS OR BOTH OUTBOARD ACTUATORS MUST BE FUNCTIONAL.

- D. Verify remaining spoilers operate normally.
 - i. Turn the hydraulic system on.
 - a. On one MCDU, monitor the spoiler actuator positions via the SECU Sensor Readout to verify correct functioning of the remaining pair of spoiler panels.
 1. MCDU: Menu - LRU Maintenance - Page until "SECU" is present
 2. Select "SECU"
 3. Select "SENSOR READOUT"
 4. Page until page 3 of 5 is present: "SPOILER POSITIONS 3/5"
 - b. Move the speed brake lever to the full Flight Spoiler position.
 1. Verify spoiler position in Sensor Readout is 35 +/- 3 degrees at full flight spoiler for operating pair spoilers only.
 - c. Move the speed brake lever to the full Ground Spoiler position.
 1. Verify spoiler position in Sensor Readout is 60 +/- 4 degrees at full ground spoiler for operating pair spoilers only.
 - ii. Return speed brake lever to the full forward and disarm position.
 - iii. Turn the control wheel clockwise to full right wing down position and verify the remaining operable right wing spoiler achieves a minimum indication of 56 degree on the SECU Sensor Readout. Repeat for the left wing down position and the remaining operable left wing spoiler by turning the control wheel counter clockwise.

Note: *A strong force at the control wheel may be required to attain and hold the minimum value of 56 degrees; there may be a slight delay in displaying the sensor readout.*

- E. Turn off hydraulic power.
- F. Accomplish required logbook entry.

28 Fuel

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28-21-01 Single Point Pressure Refueling System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD						REPAIR CATEGORY
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
28-21-01A	Single Point Pressure Refueling System	Y	1	0	C	(M) May be inoperative provided an accepted alternate refueling procedures is used.
28-21-01B	Single Point Pressure Refueling System Fuel Fill Valves	Y	3	0	C	(M) May be inoperative provided an accepted alternate refueling procedures is used.
28-21-01C	Single Point Pressure Refueling System Auto Fill Shutoff	Y	1	0	C	(M) May be inoperative and Pressure Fueling System may be used provided Manual Fill Shutoff System is operative.
28-21-01D	Single Point Pressure Refueling System Auto Fill Shutoff	Y	1	0	C	May be inoperative provided fueling operations are conducted using Gravity Fill System.

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-21-01A	1. Aircraft Logbook 2. Pressure Fueling Control Panel	Single Point Pressure Fueling System INOP
28-21-01B	1. Aircraft Logbook 2. Pressure Fueling Control Panel	Single Point Pressure Refueling System Fuel Fill Valves INOP
28-21-01C	1. Aircraft Logbook 2. Pressure Fueling Control Panel	Single Point Pressure Fueling System Auto Fill Shutoff INOP
28-21-01D	1. Aircraft Logbook 2. Pressure Fueling Control Panel	Single Point Pressure Refueling System Auto Fill Shutoff INOP

(M) PROCEDURES

28-21-01A, 28-21-01B, 28-21-01C

Load fuel as follows:

1. Add fuel through the single point refueling receptacle.
2. If any of the fuel fill valves are electrically inoperative, manually open and close valves as required to achieve proper distribution.
3. If the auto shutoff function is inoperative, manually terminate fueling operation when tanks are full.

28-21-03 Ground Refueling Relay

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
28-21-03	Ground Refueling Relay	Y	1	0	C

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-21-03	1. Aircraft Logbook 2. Pressure Fueling Control Panel	Ground Refueling Relay INOP

MAINTENANCE NOTE:

The underwing refueling panel may not be operative in this case.

28-22-01 Main Tank Fuel AC Boost Pumps

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
28-22-01	Main Tank Fuel AC Boost Pumps "DISP.APPR.REQ'D"	Y	4	2	C (O) (M) May be inoperative provided: 1. Inoperative Fuel Pumps are not in the same tank, 2. When two Fuel Pumps are inoperative, the remaining Fuel Pumps are powered from separate AC Bus Systems, 3. The combination of the Left FWD and Right AFT boost pumps may not be inoperative, 4. If the flight is to be conducted above FL 170, then the Fuel Return-To-Tank System for the respective engine must be operative, and 5. Increase planned trip fuel for each inoperative fuel pump as follows: 580 lbs. for each inoperative forward main tank pump. 330 lbs. for each inoperative aft main tank pump. Note: Added fuel is to be considered unusable for the purpose of flight planning.

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-22-01	Associated Fuel Boost Pump Switch	___ Main Tank Fuel AC Boost Pump(s) INOP

(O) PROCEDURES

1. Observe additional fuel requirements for flight planning, as noted in the remarks above.
2. If two pumps are inoperative, go-around is not recommended if there is less than 1,000 lbs. remaining in either main tank.

(M) PROCEDURES

1. Set the associated Fuel Boost Pump Switch to OFF.
2. Pull and secure the associated 115 VAC Phase A-C Fuel Boost Pump circuit breaker(s) for the inoperative pump(s).

Note: When 2 pumps are inoperative, use the following chart to comply with proviso b):

Fuel Boost Pump	CB Panel	Location	AC Bus System
Left Forward	Upper EPC	J-23	Right 115 VAC
Left Aft	Upper EPC	H-23	Left 115 VAC
Right Forward	Upper EPC	H-19	Left 115 VAC
Right Aft	Upper EPC	J-19	GS 115 VAC

28-22-02 Center Tank Fuel AC Boost Pumps

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
28-22-02A	Center Tank Fuel AC Boost Pumps "DISP.APPR.REQ'D"	Y	2	1	B (O) (M) May be inoperative with USABLE fuel carried in the Center Tank provided: 1. Cockpit Fuel Quantity Indication System is fully operative, 2. The Main Tank FUEL LEVEL LO and FUEL OFF SCHEDULE annunciation systems are operative, 3. The center tank pump low pressure annunciation system for the operative center tank pump is operative, and 4. If the flight is to be conducted above FL 170, then both Fuel Return-To-Tank Systems must be operative.
28-22-02B	Center Tank Fuel AC Boost Pumps "DISP.APPR.REQ'D"	Y	2	0	C (M) May be inoperative provided: 1. All Cockpit Fuel Quantity Indicators are operative, and 2. Any fuel in Center Tank is considered unusable.

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-22-02A	Associated Fuel Boost Pump Switch	___ Center Tank Fuel AC Boost Pump INOP
28-22-02B	Associated Fuel Boost Pump Switch	___ Center Tank Fuel AC Boost Pump(s) INOP

(O) PROCEDURES

28-22-02A

Modify normal fuel burn procedures as follows:

1. Select Fuel Pump Switches for all operative Boost Pumps to ON for takeoff.
2. After takeoff, when conditions permit and at least by top of climb, select Main Tank PUMP Switches OFF, one tank at a time.
3. Verify ENG L/R FUEL PRESS alerts do not appear on EAD.
4. Select FUEL Synoptic Page to verify fuel quantity in the center tank is decreasing.
5. When Center Tank indicates 500 pounds remaining, turn one boost pump in one Main Tank ON.
6. When Center Tank is empty, turn remaining Main Tank Boost Pumps ON.

(M) PROCEDURES

28-22-02A, 28-22-02B

1. Set the associated Fuel Boost Pump Switch to OFF.
2. Pull and secure the associated 115 VAC Phase A-C Fuel Boost Pump circuit breaker(s) for the inoperative pump(s).

Fuel Boost Pump	CB Panel	Location	AC Bus System
Center Forward	Upper EPC	H-21	Left 115 VAC
Center Aft	Upper EPC	J-21	Right 115 VAC

28-22-03 DC Start Pump

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
28-22-03	DC Start Pump	Y	1	0	C (O) May be inoperative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-22-03	DC Start Pump Switch	DC Start Pump INOP

(O) PROCEDURES

Use APU or external power to supply AC power to the Main Tank Fuel Pumps for engine start.

28-22-04 Crossfeed Valve Position Synoptic Indication

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
28-22-04	Crossfeed Valve Position Synoptic Indication	Y	1	0	C
					(O) May be inoperative provided crossfeed valve operation is verified PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-22-04	Crossfeed Valve Position Synoptic Indication	____ Crossfeed Valve Position Synoptic Indication INOP

(O) PROCEDURES

PRIOR TO EACH DEPARTURE, verify proper operation of the fuel crossfeed valve, as follows:

1. Select FUEL X-FEED Lever to ON.
2. Place Fuel Switch for one engine to ON.
3. Observe applicable ENG L/R FUEL PRES message on EAD appears.
4. Turn on one boost pump in the opposite-side main tank (e.g., if Left Fuel Switch is ON, select Right FWD Pump Switch to ON). Observe ENG L/R FUEL PRES message extinguishes.
5. Close FUEL X-FEED Lever, and verify ENG L/R FUEL PRES message reappears.
6. Turn off Fuel Switch and main tank fuel boost pump.

28-26-01 Fuel Return-To-Tank System (RTT)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
28-26-01A	Fuel Return-To-Tank System (RTT)	Y	2	1	C May be inoperative provided both Main Tank Fuel Boost Pumps in the associated Main Tank are operative.
28-26-01B	Fuel Return-To-Tank System (RTT) "DISP.APPR.REQ'D"	Y	2	0	C May be inoperative provided the airplane remains at or below FL 170.
28-26-01C	Fuel Return-To-Tank System (RTT) "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided: 1. The APU is operative and is running for any portion of the flight above FL 170, 2. All four Main Tank Fuel Boost Pumps are operative, and 3. Both Center Tank Fuel Boost Pumps are operative if usable fuel is carried in the Center Tank.

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-26-01A	Below Captains N.D. Panel	___ Fuel Return-To-Tank System INOP
28-26-01B	Below Captains N.D. Panel	Both Fuel Return-To-Tank Systems INOP
28-26-01C	Below Captains N.D. Panel	Both Fuel Return-To-Tank Systems INOP

OPERATIONS NOTE:

If only one Return-To-Tank system is affected, flight crew might observe a split in engine oil temperature. This is normal, and does not require special attention or crew procedures as long as Oil Temperature remains within limits.

(O) PROCEDURES

28-26-01C

28-26-01C FUEL RETURN-TO-TANK SYSTEM (RTT)	
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	Increase BLOCK FUEL by 220 lbs. per hour for APU usage.
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF None LANDING None

28-41-01 Cockpit Fuel Quantity Indicating System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY			
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS	
COCKPIT FUEL QUANTITY INDICATING SYSTEM							
28-41-01A	Channels (A & B)	Y	2	1	C	(O) May be inoperative provided: 1. All Cockpit Fuel Quantity Displays are operative on the remaining channel, 2. Both Fuel Used Indications are operative, and 3. All Fuel Tank Boost Pumps are operative.	
28-41-01B	Left & Right Main Tank Displays "DISP.APPR.REQ'D"	Y	2	1	C	(O) (M) One may be inoperative on both channels provided: 1. Fuel quantity in the associated tank is verified by an alternate method PRIOR TO EACH DEPARTURE, 2. Remaining Main Tank and Center Tank Fuel Quantity Displays in the cockpit are operative on both channels, 3. Fuel Flow Indication for associated engine is operative, and 4. Both Fuel Used Indications are operative, Note: Gross Weight / Total Fuel Indication will not be available. Note: PROF may not be available if UFOB is not entered after engine start. Note: "LAT FUEL UNBAL" message may be shown. Repetitive Check Required (Yellow Placard)	
28-41-01C	Center Tank Display "DISP.APPR.REQ'D"	Y	1	0	C	(O) (M) May be inoperative on both channels provided: 1. Fuel quantity in the associated tank is verified by an alternate method PRIOR TO EACH DEPARTURE. 2. Any fuel in the Center Tank is considered UNUSABLE and is included in the zero fuel weight and center of gravity calculation as ballast fuel, 3. Both Cockpit Main Tank Fuel Quantity Displays are operative on both channels, and 4. Both Fuel Used Indications are operative. Note: Gross Weight / Total Fuel Indication will not be available. Note: PROF may not be available if UFOB is not entered after engine start. Repetitive Check Required (Yellow Placard)	

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NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
	COCKPIT FUEL QUANTITY INDICATING SYSTEM				
28-41-01D	Center Tank Display "DISP.APPR.REQ'D"	Y	1	0	C (O) (M) May be inoperative with fuel in the Center Tank considered USABLE provided: 1. Both Center Tank Fuel Boost Pumps and their associated center tank fuel boost pump pressure low annunciation systems are operative, 2. Both Cockpit Main Tank Fuel Quantity Displays are operative on both channels, 3. Both Fuel Used Indications are operative, 4. Quantity of fuel loaded into the Center Tank is verified by an alternate method PRIOR TO EACH DEPARTURE, and 5. Procedures are used to verify when fuel in the Center Tank has been depleted. <i>Note: Gross Weight / Total Fuel Indication will not be available.</i> <i>Note: PROF may not be available if UFOB is not entered after engine start.</i> Repetitive Check Required (Yellow Placard)
28-41-01E	Fuel System Test	Y	1	0	C (O) May be inoperative provided: 1. All Cockpit Fuel Quantity Displays are operative, and 2. Both Fuel Used Indications are operative.
28-41-01F	Fuel Quantity Channel A/B Selector Switch	Y	1	0	C (O) May be inoperative provided: 1. All Cockpit Fuel Quantity Displays are operative on the remaining channel, 2. Both Fuel Used Indications are operative, and 3. All Fuel Tank Boost Pumps are operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-41-01A	Below Captains N.D. Panel	Channel ____ of Fuel Quantity Indicating System INOP
28-41-01B	Below Captains N.D. Panel	____ Wing Main Tank Display INOP
28-41-01C	Below Captains N.D. Panel	Center Tank Display INOP
28-41-01D	Below Captains N.D. Panel	Center Tank Display INOP
28-41-01E	Below Captains N.D. Panel	Fuel System Test INOP
28-41-01F	Below Captains N.D. Panel	Fuel Quantity Channel A/B Selector Switch INOP

CONTINUED ON NEXT PAGE

(O) PROCEDURES

28-41-01A

1. Should remaining channel or any other portions of the Fuel Quantity Indication System fail in flight, keep remaining boost pumps ON for the duration of the flight.
2. Monitor fuel used indication.

28-41-01B

PRIOR TO EACH DEPARTURE, verify alternate fueling procedures have been accomplished.

1. Disregard fuel totalizer and gross weight indication.
2. In flight, when center tank is empty, monitor fuel used indication.
3. Enter UFOB on FMS INIT page 2 (LSK 1R) after engine start, otherwise PROF may not be available.

28-41-01C, 28-41-01D

PRIOR TO EACH DEPARTURE, verify alternate fueling procedures have been accomplished.

1. Disregard fuel totalizer and gross weight indication.
2. With unusable fuel in center tank:
 - A. Turn off center tank pumps.
 - B. In flight, monitor fuel used indications and cross-check versus main tank quantity decrease.
 - C. If fuel used indication does not match indicated fuel depletion from main tanks, turn on center tank pumps.
3. With usable fuel in the center tank:
 - A. In flight, monitor fuel used indications and verify no fuel transfer from the main tanks until fuel used equals quantity of fuel in the center tank.
 - B. After center is empty, leave one center tank pump switch ON for the duration of the flight.
4. Enter UFOB on FMS INIT page 2 (LSK 1R) after engine start, otherwise PROF may not be available.

28-41-01E

1. Use proper fuel management procedures, such as awareness of individual pump pressure low alerts, FUEL OFF SCHEDULE alert and FUEL LOW alert.

28-41-01F

1. Should remaining channel or any other portions of the Fuel Quantity Indication System fail in flight, keep remaining boost pumps ON for the duration of the flight.
2. Monitor fuel used indication.

(M) PROCEDURES

28-41-01B, 28-41-01C, 28-41-01D

1. Install second placard in a conspicuous location at the fueling station and place placard tab on or adjacent to inoperative indicator.
2. For alternate fueling procedures, refer to the following sections of the [Fueling Procedures Manual](#):
 - B717.40 B-717-200 Inoperative Gauge
 - B717.90.0 B-717-200 Fuel Stick Charts
 - B717.90.X Fuel Stick Charts

28-41-02 Gross Weight / Total Fuel Indication System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
28-41-02	Gross Weight / Total Fuel Indication System "DISP.APPR.REQ'D"	Y	1	0	C (O) May be inoperative provided alternate procedures for determining fuel quantity and gross weight are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-41-02	Below Captains N.D. Panel	Fuel Tank Totalizing System and Gross Weight Indicator INOP

(O) PROCEDURES

1. Add individual fuel quantities displayed for each tank to obtain total fuel loaded. Cross check total quantity onboard with dispatch personnel.
2. If any of the individual tank displays is also inoperative, confirm with dispatch or fueling personnel that total fuel onboard has been verified using an alternate method, e.g., dipsticks.
3. Add total fuel on board to Zero Fuel Weight from load sheet to obtain gross weight.

28-41-03 Fueling Panel Fuel Quantity Indicators

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
28-41-03	Fueling Panel Fuel Quantity Indicators "DISP.APPR.REQ'D"	Y	3	0	C (O) May be inoperative provided fuel quantity in associated tank is verified by an alternate accepted method PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-41-03	1. Aircraft Logbook 2. Pressure Fueling Control Panel	____ Fueling Panel Fuel Quantity Indicator(s) INOP

(O) PROCEDURES

Install second placard in a conspicuous location at the fueling station and place tab on or adjacent to inoperative indicator.

PRIOR TO EACH DEPARTURE:

1. Verify alternate fueling procedures have been accomplished.
2. For alternate fueling procedures, refer to [FOM 14.4.1.7 – Inoperative Gauge Refueling](#)

28-42-01 Dripless Fuel Measuring Sticks (Dipsticks)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD			Y	9	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION						REMARKS OR EXCEPTIONS
28-42-01	Dripless Fuel Measuring Sticks (Dipsticks) "DISP.APPR.REQ'D"				C	(M)	May be inoperative or missing provided fuel quantity is determined by other acceptable means.

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-42-01	Aircraft Logbook	____ Dipstick(s) INOP

(M) PROCEDURES

During fueling determine quantity of fuel in the affected tank by:

1. Using quantity gages in cockpit or at the wing refueling station, and/or
2. Loading a known quantity of fuel into the affected tank.

Note: Use fuel quantity indicating system to fuel airplane. If quantity indications are inoperative, refer to MEL item [28-41-01](#) or [28-41-03](#) as applicable.

28-43-01 Main Tank Fuel Level Low Float Switches

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
28-43-01	Main Tank Fuel Level Low Float Switches	Y	2	0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-43-01	Below Captains N.D. Panel	Main Tank Fuel Level Low Float Switches INOP

MAINTENANCE NOTE:

1. If the Float Switch is failed Closed, the FUEL OFF SCHEDULE Alert (Level 2) will be present on the EAD. This must be inhibited prior to dispatch, as follows. Maintenance will be required and Flight Crew May NOT Placard.
 - A. Gain access to the wiring for the affected Low Fuel Level Float Switch through the appropriate access panel listed below:

Affected Float Switch	Access Panel	Terminal Strip	Wire Identification	Lug on Terminal Strip
Left (S1-577)	510 BB	S3-1	FRNS1577-20	1
		GND	FRNS1577A-20N	715D
Right (S1-578)	610 BB	S3-91	FRNS1578-20	5
		GND	FRNS1578A-20N	815D

- B. Remove the appropriate Low Fuel Level Float Switch wire from the Terminal Strip, and the associated Ground Wire from the Ground Terminal Strip. Insulate, coil and stow wires as appropriate.
- C. Verify associated FUEL LEVEL LOW and / or FUEL OFF SCHEDULE alerts are not present on the EAD.

28-44-01 Pump Pressure Low Annunciation Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD			Y	6	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	REMARKS OR EXCEPTIONS					
28-44-01	Pump Pressure Low Annunciation Systems	(O) May be inoperative provided affected pump is functionally checked PRIOR TO EACH DEPARTURE. Note: If affected tank is not carrying usable fuel, its respective boost pump need not be checked. Repetitive Check Required (Yellow Placard)					

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-44-01	Below Captains N.D. Panel	____ Pump Pressure Low Annunciations INOP

(O) PROCEDURES

PRIOR TO EACH DEPARTURE:

1. Center tank Pump Pressure Low Annunciation System inoperative:
 - A. After engine start, select both center tank pump switches and main tank pump switches in one main tank to OFF, one at a time.
 - B. Verify appropriate ENG L/R FUEL PRES message appears on EAD.
 - C. Select boost pump switch associated with inoperative annunciation system to ON, and verify ENG L/R FUEL PRES message extinguishes.
 - D. Return all boost pumps to ON, as appropriate.
2. Main tank Pump Pressure Low Annunciation System inoperative:
 - A. After engine start, select center and main tank pump switches in affected tank to OFF, one at a time.
 - B. Verify appropriate ENG L/R FUEL PRES message appears on EAD.
 - C. Select boost pump switch associated with inoperative annunciation system to ON, and verify ENG L/R FUEL PRES message extinguishes.
 - D. Return all boost pumps to ON, as appropriate.

28-45-01 Fuel Tank Temperature Indicating System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
28-45-01A	Fuel Tank Temperature Indicating System	Y	2	1	C May be inoperative provided both Main Tank Fuel Boost Pumps in the associated Main Tank are operative.
28-45-01B	Fuel Tank Temperature Indicating System "DISP.APPR.REQ'D"	Y	2	0	C May be inoperative provided the airplane remains at or below FL 170.
28-45-01C	Fuel Tank Temperature Indicating System "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided: - The APU is operative and is running for any portion of the flight above FL 170. - All four Main Tank Fuel Boost Pumps are operative, and - Both Center Tank Fuel Boost Pumps are operative if usable fuel is carried in the Center Tank.

MEL ITEM	TAB LOCATION	PLACARD TEXT
28-45-01A	Below Captains N.D. Panel	Fuel Temperature Indicator INOP
28-45-01B	Below Captains N.D. Panel	Fuel Temperature Indicator INOP
28-45-01C	Below Captains N.D. Panel	Fuel Temperature Indicator INOP

(O) PROCEDURES

28-45-01C

- Refer to [Operations Manual, Volume 1, Supplementary Procedures, Adverse Weather](#), regarding ice on upper wing surface.
- Utilize Overwing Ice Detection System to comply with inspection requirements, if necessary.
- If the Overwing Ice Detection System is inoperative, flight crew should physically inspect the upper wing surface for ice (in accordance with AFM limitations) when there is frost or ice on the inboard portion of the wing lower surface, or when conditions exist which may be conducive to ice formation, i.e., ambient temperature less than 50 °F (10 °C) and visible moisture (fog, rain, drizzle, etc.) is present.
- Fuel Return-To-Tank System will be inoperative for an inoperative Fuel Tank Temperature Indication. If only one System is affected, flight crew might observe a split in engine oil temperature. This is normal, and does not require special attention or crew procedures as long as Oil Temperature remains within limits.

28-45-01C	FUEL TANK TEMPERATURE INDICATION
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	Increase BLOCK FUEL by 220 lbs. per hour for APU usage.
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF None LANDING None

29 Hydraulic Power

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29-11-01 Right Engine Driven Hydraulic Pump

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD						REPAIR CATEGORY
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
29-11-01A	Right Engine Driven Hydraulic Pump "DISP.APPR.REQ'D"	Y	1	0	C	(O) May be inoperative provided: 1. Left Engine Driven Hydraulic Pump, Electrical Auxiliary Hydraulic Pump and Hydraulic Power Transfer Unit are operative, and 2. Performance penalties are applied.
29-11-01B	Right Engine Driven Hydraulic Pump "DISP.APPR.REQ'D"	N	1	0	C	(O) (M) May be dispatched with Right Engine Driven Hydraulic Pump removed provided: 1. Left Engine Driven Hydraulic Pump, Electrical Auxiliary Hydraulic Pump and Hydraulic Power Transfer Unit are operative, 2. Appropriate Hydraulic Dispatch Kit installed, and 3. Performance penalties are applied.

MEL ITEM	TAB LOCATION	PLACARD TEXT
29-11-01A	Hydraulics Control Panel	____ Right Engine Driven Hydraulic Pump INOP
29-11-01B	Hydraulics Control Panel	____ Right Engine Driven Hydraulic Pump INOP

CONTINUED ON NEXT PAGE

(O) PROCEDURES

29-11-01A, 29-11-01B

1. Verify all remaining hydraulic pumps are selected ON for all takeoffs and landings. The AUX hydraulic pump should remain ON for the duration of each flight, but the TRANS hydraulic pump may be selected OFF during climb, cruise, descent.
2. Apply the following performance corrections:

29-11-01A RIGHT ENGINE DRIVEN HYDRAULIC PUMP	
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	None
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF 1. Reduce RATOW by 16,800 lbs. 2. Reduce T/O CLIMB LIMIT by 16,800 lbs.
	LANDING None

29-11-01B RIGHT ENGINE DRIVEN HYDRAULIC PUMP	
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	None
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF 1. Reduce RATOW by 16,800 lbs. 2. Reduce T/O CLIMB LIMIT by 16,800 lbs.
	LANDING None

(M) PROCEDURES

29-11-01B

1. Depressurize the Right Hydraulic System.
2. Open RT EDP Circuit Breaker as follows:

<u>Circuit Breaker (C/B#)</u>	<u>Panel</u>	<u>Location</u>
RIGHT HYD PUMP CONTROL (B1-52)	Lower EPC	T-26

Note: Ensure RT Hydraulic System Switch on the Hydraulics Overhead Panel is in the ON position before opening this C/B.

3. Deactivate and remove the right Engine Driven Hydraulic Pump (EDP) as follows:
 - A. Disconnect hoses from the hydraulic fittings. Plug and secure hydraulic hoses to engine brackets using the following:
 - i. 1 Plug - P/N MS21913-12 for Suction Hose.
 - ii. 1 Plug - P/N MS21913-10 for Pressure Hose.
 - iii. 1 Plug - P/N MS21913-6 for Case Drain Hose.
 - B. Disconnect cap and secure the disconnected electrical connector per Standard Wiring Practice Manual, SWPM 20-10-04.
 - C. Disconnect fittings from affected pump and place in appropriately labeled plastic bag. (Place bag in PFM box or on flight deck to facilitate pump reactivation.)
 - D. While holding hydraulic pump, loosen the V-band clamp, Abex P/N 24974, and then remove pump (including splined shaft) from mounting plate on Engine Accessory Gear Box.
4. Install cover assembly, P/N L29001-2 on Engine Accessory Gear Box by using the V-band clamp to hold the cover assembly in position.
 - A. Install the clamp in the best position for access to the nut. Ensure the V-band clamp will not touch the adjacent parts or structure as it is tightened.
 - B. Use a mallet with a soft face to lightly tap around the outer surface of the clamp to equally apply the tension and then torque the clamp to 45-55 in-lbs.
5. Set the RT Hydraulic System Switch on the Hydraulics Overhead Panel to the OFF position.
6. Leak check the associated hydraulic lines by use of the Aux Hydraulic Pump.
7. Close RIGHT HYD PUMP CONTROL C/B (B1-52) (ref: step 2).
8. Motor affected engine per 717 AMM and visually check the Engine Accessory Gearbox area and pump cover assembly for leaks.
9. Check the following hydraulic filters in the aft accessory compartment for extension of their "Delta P" indicators (DPI's):
 - A. Right Engine Hydraulic Pressure Filter
 - B. Right Engine Hydraulic Case Drain Filter
10. If DPI's are not extended, the procedure is complete.
11. If DP's are extended perform the following:
 - A. Check and replace the affected filter element.
 - B. Check the affected filter element for contamination.

CAUTION

IF CONTAMINATION IS FOUND IN ANY FILTER ELEMENT, FLUSH THE AFFECTED LINES AND VALVES TO ELIMINATE CONTAMINATION. THIS ACTION MAY BE POSTPONED ONLY UNTIL AIRCRAFT ARRIVES AT THE NEXT MAINTENANCE BASE.

29-21-01 Electric Auxiliary Hydraulic Pump

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			1	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	Y			C	REMARKS OR EXCEPTIONS
29-21-01	Electric Auxiliary Hydraulic Pump "DISP.APPR.REQ'D"	Y	1	0	C	(O) May be inoperative provided: 1. Right Engine Driven Hydraulic Pump, Left Engine Driven Hydraulic Pump and Hydraulic Power Transfer Unit are operative, and 2. Performance penalties are applied.

MEL ITEM	TAB LOCATION	PLACARD TEXT
29-21-01	Aircraft Logbook	____ Electric Auxiliary Hydraulic Pump INOP

(O) PROCEDURES

1. Verify all remaining hydraulic pumps are selected ON for all takeoffs and landings.
2. Verify adequate brake pressure prior to push back. It may be necessary to motor an engine to build required brake pressure.
3. Apply the following performance corrections:

29-21-01	ELECTRIC AUXILIARY HYDRAULIC PUMP
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	None
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF 1. Reduce RATOW by 16,800 lbs. 2. Reduce T/O CLIMB LIMIT by 16,800 lbs. LANDING None

29-22-01 Hydraulic Power Transfer Unit (PTU)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
29-22-01	Hydraulic Power Transfer Unit (PTU) "DISP.APPR.REQ'D"	Y	1	0	C (O) May be inoperative provided performance penalties are applied.

MEL ITEM	TAB LOCATION	PLACARD TEXT
29-22-01	Hydraulics Control Panel	Hydraulic Power Transfer Unit (PTU) INOP

(O) PROCEDURES

1. Verify all remaining hydraulic pumps are selected ON for all takeoffs and landings.
 - A. **Note:** HYD PRES LO message will extinguish when both engines are started.
2. If MEL 49-10-01 (APU) and 29-22-01 (Hyd Power Transfer Unit) are applied concurrently, the **LEFT** engine should be started first.
3. Apply the following Performance Corrections:

29-22-01	HYDRAULIC POWER TRANSFER UNIT (PTU)
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	None
NOTES	AWABS Takeoff data is provided for runways with a slope between +1.0% and -1.0%. The limiting crosswinds for takeoff and landing is 15 kts.
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	Manual WDR not permitted with MEL 29-22-01.

29-31-01 Right Hydraulic System Pressure Low Annunciation System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			1	REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION	Y		0	C	REMARKS OR EXCEPTIONS
29-31-01	Right Hydraulic System Pressure Low Annunciation System	Y	1	0	C	(O) May be inoperative provided the Right Engine-Driven Hydraulic Pump is verified to operate normally.

MEL ITEM	TAB LOCATION	PLACARD TEXT
29-31-01	Below Captains N.D. Panel	____ Hydraulic Pressure Indicating System(s) INOP

(O) PROCEDURES

1. After engine start, verify RUD POWER OFF alert is not present on EAD.
2. If RUD POWER OFF alert appears in flight, refer to appropriate [QRH](#) Abnormal Procedure for this alert, and also to Abnormal Procedure for HYD R PRES LO.

29-31-02 Hydraulic Pump Pressure Synoptic Indication Amber Pinwheel (Prior to Engine Start)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
29-31-02	Hydraulic Pump Pressure Synoptic Indication Amber Pinwheel (prior to engine start)	Y	3	0	C	(O) May be inoperative provided the affected pump is verified operative by digital readout on the Hydraulic Synoptic Page.

MEL ITEM	TAB LOCATION	PLACARD TEXT
29-31-02	Below Captains N.D. Panel	____ Hydraulic Pump Pressure Synoptic Indication Amber Pinwheel INOP

(O) PROCEDURES

1. If the Aux Hydraulic Pump is affected, select only the Aux Hydraulic Pump to ON (with electrical power available) and the remaining Hydraulic Pump Switches to OFF (including Transfer Pump). Observe the following:
 - A. Verify the R HYD PRESS LOW alert is not displayed.
 - B. Select the HYD Synoptic Page, and observe the Right hydraulic system pressure to be within normal range.
2. If the Left or Right Engine Driven Pump is affected, observe the following after engine start:
 - A. Select only the affected Hydraulic Pump Switch to ON and the remaining Hydraulic Pump Switches (including Transfer Pump) to OFF.
 - B. Verify the L or R HYD PRESS LOW alert is not displayed for the operating hydraulic system.
 - C. Select the HYD Synoptic Page, and observe system pressure to be within normal range.
3. Set Hydraulic Pumps as normal.

29-32-01 Hydraulic Quantity Indicating Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
29-32-01	Hydraulic Quantity Indicating Systems	N	2	0	C
					(M) May be inoperative provided associated reservoir is checked PRIOR TO EACH DEPARTURE (Log Book Entry Required). Repetitive Check Required (Yellow Placard)

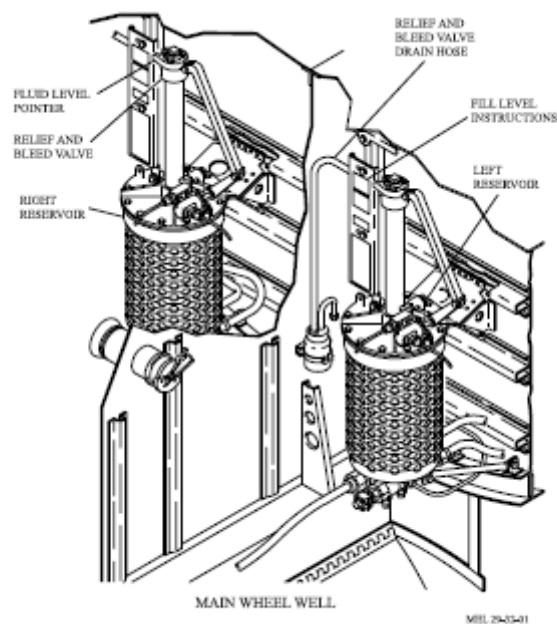
MEL ITEM	TAB LOCATION	PLACARD TEXT
29-32-01	Below Captains N.D. Panel	____ Hydraulic Quantity Indicating System(s) INOP

(M) PROCEDURES

PRIOR TO EACH DEPARTURE:

1. Check each hydraulic system reservoir for proper fluid level before each flight.
2. Applicable hydraulic system circuit breaker (B1-63 Left Hydraulic Oil Quantity/B1-64 Right Hydraulic Oil Quantity) may be pulled and collared to prevent nuisance MASTER CAUTION lights.

Figure 29-1: Hydraulic Quantity Indicating Systems



30 Ice and Rain Protection

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30-11-01 Wing Anti-Ice Pressure Regulator / Shutoff Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			1	REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION	Y		0	C	REMARKS OR EXCEPTIONS
30-11-01	Wing Anti-Ice Pressure Regulator / Shutoff Valve "DISP.APPR.REQ'D"	Y	1	0	C	May be inoperative CLOSED provided: 1. Airfoil Ice Protection System is not used, and 2. Flight is not made in known or forecast icing conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-11-01	Wing Anti-Ice Switch	Wing Anti-Ice Pressure Regulator/Shutoff Valve INOP

30-12-01 Tail Anti-Ice Pressure Regulator / Shutoff Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			1	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	Y			C	REMARKS OR EXCEPTIONS
30-12-01	Tail Anti-Ice Pressure Regulator / Shutoff Valve “DISP.APPR.REQ’D”	Y	1	0	C	May be inoperative CLOSED provided: 1. Airfoil Ice Protection System is not used, and 2. Flight is not made in known or forecast icing conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-12-01	Tail Anti-Ice Switch	Tail Anti-Ice Pressure Regulator/Shutoff Valve INOP

30-22-01 Inlet Cowl Anti-Ice Regulator / Shutoff Valves

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
30-22-01A	Inlet Cowl Anti-Ice Regulator / Shutoff Valves "DISP.APPR.REQ'D"	N	2	0	C (M) May be inoperative CLOSED provided: 1. Valve is secured in the CLOSED position, and 2. Flight is not made in known or forecast icing conditions.
30-22-01B	Inlet Cowl Anti-Ice Regulator / Shutoff Valves "DISP.APPR.REQ'D"	N	2	0	C (O) (M) May be inoperative OPEN provided: 1. Valve is secured in the OPEN position, 2. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements, and 3. Performance penalties are applied. CAT II/III Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-22-01A	Associated Engine Anti-Ice Switch	___ Inlet Cowl Anti-Ice Regulator / Shutoff Valve INOP Closed
30-22-01B	Associated Engine Anti-Ice Switch	___ Inlet Cowl Anti-Ice Regulator / Shutoff Valve INOP OPEN

(O) PROCEDURES

30-22-01B

- Obtain Takeoff Thrust EPR, Go-Around EPR and Maximum Continuous EPR settings from [ODM, Abnormal Section, MEL Support](#).
- Set EPR manually.
- Autothrottles must not be used below 25,000 feet. (Autoland is not authorized)
- For Climb and Cruise Thrust below 25,000 feet, normal ratings may be used provided EPR does not exceed Max Continuous Thrust settings from the [ODM](#).
- If approach minimums are predicated on the use of flight director guidance for missed approach, disconnect autothrottles immediately after TOGA selection upon go-around initiation.
- When affected engine anti-ice switch is OFF, a Level 1 ENG L/R A-ICE DISAG message will be displayed on EIS.

30-22-01B INLET COWL ANTI-ICE REGULATOR / SHUTOFF VALVES	
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	Reduce ENROUTE LIMIT WT by 27,400 lbs. Increase BLOCK FUEL by 3.1%.
NOTES	Make a notation on the Flight Plan when Enroute Climb is limiting. Takeoff performance limit weights decrease significantly per degree of temperature increase when OAT is between 45°F and 50°F (7°C to 10°C). Use GWT function in AWABS to anticipate performance impact of a possible OAT increase.
AWABS	Performance corrections have been applied. Refer to ODM – Abnormal section for EPR settings.
TAKEOFF PERFORMANCE WORKSHEET	Manual WDR not permitted with MEL 30-22-01B.

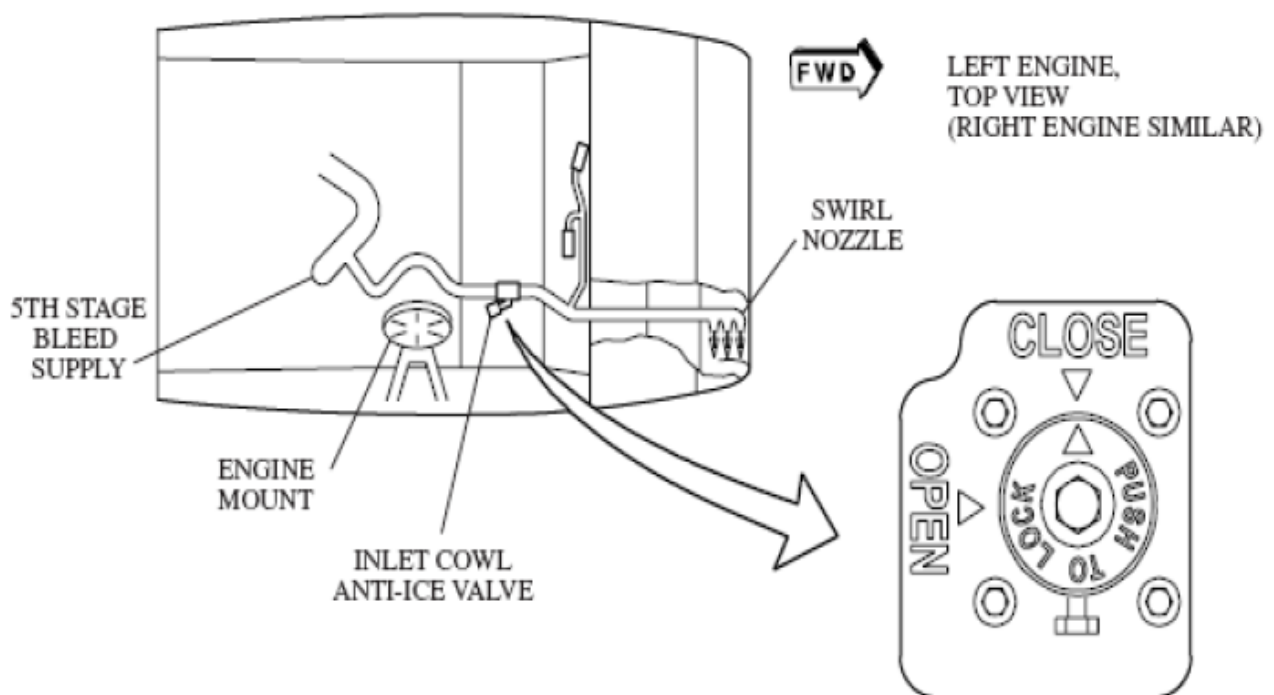
(M) PROCEDURES

 [Link to Learning Video](#)

30-22-01A, 30-22-01B

1. Gain access to the inoperative valve inside the engine cowl.
2. Observe indicated position of valve, and rotate the manual control knob on the valve housing to position the valve in the desired position (OPEN or CLOSED).
3. Lock the Valve into position:
 - A. Loosen the locking collar lock screw.
 - B. Push the locking collar down to engage the four sided cutout.
 - C. Resecure the locking collar lock screw.

Figure 30-1: Inlet Cowl Anti-Ice Regulator / Shutoff Valves



30-31-01 Pitot Heaters

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
	PITOT HEATERS				
30-31-01A	Captains and F/O Pitot Heaters "DISP.APPR.REQ'D"	Y	2	1	B May be inoperative provided: 1. Flight is made in VMC conditions, 2. Flight is not operated at night, and 3. Flight is not made in visible moisture or in known or forecast icing conditions.
30-31-01B	Aux Pitot Heater "DISP.APPR.REQ'D"	Y	1	0	B May be inoperative provided flight is not made in visible moisture or in known or forecast icing conditions.
30-31-01C	Rudder Limiter Pitot Heater "DISP.APPR.REQ'D"	Y	1	0	C May be inoperative provided flight is not made in known or forecast icing conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-31-01A	Ice Protection Panel	____ Pitot Heater INOP
30-31-01B	Ice Protection Panel	Auxiliary Pitot Heater INOP
30-31-01C	Ice Protection Panel	Rudder Limiter Pitot Heater INOP

30-32-01 Static Port Heaters

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
30-32-01	Static Port Heaters "DISP.APPR.REQ'D"	Y	4	0	C May be inoperative except when arrival and departure airport temperatures are +5 degrees C or below and runways are covered with slush or standing water.

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-32-01	Ice Protection Panel	____ Static Port Heater(s) INOP

30-33-01 Ram Air Temp Probe Heater

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
30-33-01	Ram Air Temp Probe Heater "DISP.APPR.REQ'D"	Y	1	0	C	May be inoperative provided flight is not made in known or forecast icing conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-33-01	Ice Protection Panel	Ram Air Temp Probe Heater INOP

30-34-01 Stall Warning Vane Heaters

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
30-34-01	Stall Warning Vane Heaters "DISP.APPR.REQ'D"	Y	2	0	C	May be inoperative provided flight is not made in known or forecast icing conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-34-01	Ice Protection Panel	____ Stall Warning Vane Heater(s) INOP

30-41-01 Windshield Heat System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
WINDSHIELD HEAT SYSTEM					
30-41-01A	Anti-Fog System (Includes 3 windshields, 2 clearviews)	Y	1	0	C (M) May be inoperative provided Windshield Anti-Ice System is operative.
30-41-01B	Anti-Ice System (Includes left, center, and right windshields) "DISP.APPR.REQ'D"	Y	1	0	C (O) (M) May be inoperative provided: 1. Anti-Fog System is operative, 2. Flight is not made in known or forecast icing conditions, and 3. Appropriate performance corrections are applied.

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-41-01A	Windshield ANTI-FOG Switch	Windshield / Window Anti-Fog System INOP
30-41-01B	Windshield ANTI-ICE Switch	Windshield Anti-Ice System INOP

(O) PROCEDURES

30-41-01B

1. If Windshield Anti-Ice System is inoperative, do not exceed 315 knots below 10,000 feet.
2. Program KIAS/Mach speed restriction in FMS.

Note: Above 10,000 feet -- no restrictions
 Below 10,000 feet -- do not exceed 315 KIAS

(M) PROCEDURES

30-41-01A

Dispatch with Windshield Anti-Fog System inoperative on one or more panes:

1. Select Windshield Anti-Ice switch to ON.
2. Conduct a hand feel check to verify Windshield Anti-Ice operation on the affected panes.
3. Select Windshield Anti-Ice switch to OFF after completion of check.

Note: Anti-Fog System will not come on unless cockpit temperature is below 90°F.

30-41-01B

Dispatch with Windshield Anti-Ice System inoperative on one or more panes:

1. Select Windshield Anti-Fog switch to ON.
2. Conduct a hand feel check to verify Windshield Anti-Fog system operation.
3. Select Windshield Anti-Fog switch to OFF after completion of check.

Note: Anti-Fog System will not come on unless cockpit temperature is below 90°F.

30-43-01 Windshield Wiper System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
	WINDSHIELD WIPER SYSTEM					
30-43-01A	Windshield Wiper System “DISP.APPR.REQ’D”	Y	2	0	C	May be inoperative provided airplane is not operated in precipitation within 5 nautical miles of the airport of takeoff or intended landing.
30-43-01B	High Speed	Y	2	0	C	May be inoperative provided associated Low Speed is operative.
30-43-01C	Low Speed	Y	2	0	C	May be inoperative provided associated High Speed is operative.
30-43-01D	Park Control	Y	2	0	C	May be inoperative provided the blade(s) can be positioned in a location that will not obstruct forward vision.

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-43-01A	Windshield WIPER Switch	___ Windshield Wiper System(s) INOP
30-43-01B	Windshield WIPER Switch	___ Windshield Wiper(s) HIGH Speed INOP
30-43-01C	Windshield WIPER Switch	___ Windshield Wiper(s) SLOW Speed INOP
30-43-01D	Windshield WIPER Switch	___ Windshield Wiper(s) PARK Control INOP

30-72-01 Galley Drain Mast Heaters

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
30-72-01A	Galley Drain Mast Heaters	N	2	0	D (O) (M) May be inoperative provided: 1. Drain line to associated drain mast is disconnected, or turned OFF, and 2. Associated galley sink(s) are blocked off to prevent their use.
30-72-01B	Drain Mast Heat Annunciation	N	-	-	C (M) May be inoperative provided drain mast heaters are operating normally on the ground and air modes PRIOR TO EACH DEPARTURE (Log Book Entry Required). Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-72-01A	Aircraft Logbook	____ Galley / Lav Drain Mast Heater(s) INOP
30-72-01B	Aircraft Logbook	Drain Mast Heat Annunciation INOP

(O) PROCEDURES

During walkaround, verify affected Drain Mast is free from ice.

Note: *The forward drain mast is also connected to drain holes around Forward Entrance and Service Doors. If the Forward Drain Mast Heater is inoperative and precipitation exists at departure airport, Forward Drain Mast should be verified free from ice after Forward Entrance and Service Doors are closed.*

(M) PROCEDURES

30-72-01A

1. Disconnect and plug drain line from affected Lavatory Sink.
2. If the Forward Drain Mast Heater is inoperative, disconnect and plug the drain line from the galley sink. Also plug the drain from the Ice Bin Drawer.
3. Shut off water supply to affected galley and lavatory sinks.
4. Advise Flight Crew which lav/galley sinks and coffee makers will be affected.

30-72-01B

PRIOR TO EACH DEPARTURE (Log Book Entry Required).

Verify drain mast heaters are operating normally. Verify temperature increase of at least 100 degrees in the air mode from the ground mode.

30-74-01 Water Service Panel Heater

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			1	REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION	Y		0	C	REMARKS OR EXCEPTIONS
30-74-01	Water Service Panel Heater	Y				

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-74-01	Aircraft Logbook	Water Service Panel Heater INOP

30-81-01 Overwing Ice Detection System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
30-81-01	Overwing Ice Detection System	Y	1	0	C	(O) (M) May be inoperative provided alternate procedures are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
30-81-01	Ice Protection Panel	Overwing Ice Detection System INOP

(O) PROCEDURES

1. Verify that no ice, frost, or other contaminants are present on the upper surface of the wing prior to departure. Refer to [Operations Manual, Vol. 1, Supplemental Procedures, Tactile Wing Inspection](#).
2. De-ice aircraft as necessary to ensure the wings are clear of ice prior to takeoff.

Note: Cold soaked fuel in contact with the upper surface can cause the formation of ice that is not easily visible. This clear ice could shed and be ingested into one or both engines during takeoff.

Note: The wing upper surfaces must be physically checked for ice when the airplane has been exposed to conditions conducive to ice formation or when there is frost or ice on the inboard portion of the wing lower surface.

(M) PROCEDURES

Pull and secure WING ICE DETECTION circuit breaker (B1-995: EPC Panel, location M-29)

31 Indicating / Recording System

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31-31-01A	Flight Data Recorder (FDR) System
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31-31-02	Aircraft Condition Monitoring System (ACMS)
31-51-01	Central Aural Warning System (CAWS)
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31-51-01A	APU Fire Warning Horn
31-51-01B	Cabin Altitude Warning Tone
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	ELECTRONIC INSTRUMENT SYSTEM (EIS)
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31-61-01J	SISP Pushbutton Internal Switchlights
31-61-02	Versatile Integrated Avionics (VIA) Units
31-61-02	Versatile Integrated Avionics (VIA) Units

31-31-01 Flight Data Recorder (FDR) System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
31-31-01A	Flight Data Recorder (FDR) System "DISP.APPR.REQ'D"	Y	1	0	A May be inoperative provided: 1. Cockpit Voice Recorder (CVR) operates normally, 2. Airplane may not depart from ATL unless: A. FDR failure occurs after pushback, but prior to takeoff, or B. FDR repair was attempted but not successful, 3. In those cases where repair is attempted but not successful, the aircraft may be dispatched on a flight or series of flights, but shall not continue beyond ATL where repair must be accomplished prior to dispatch, 4. It is required that the MCC initiate a Corrective Action Required (CAR) for next designated airport (ATL), and 5. Repairs are made within three flight days.
31-31-01B	DFDR Recording Parameters required by 14 CFR "Contact MCC"	N	88	85	A Up to three (3) recording parameters may be inoperative provided: 1. Cockpit Voice Recorder (CVR) operates normally, and 2. Repairs are made within twenty calendar days.
31-31-01C	DFDR Recording Parameters not required by 14 CFR "Contact MCC"	N	54	0	A May be inoperative provided repairs are made prior to the completion of the next heavy maintenance check.

MEL ITEM	TAB LOCATION	PLACARD TEXT
31-31-01A	FLT RCDR Switch	Flight Data Recorder INOP
31-31-01B	FLT RCDR Switch	Flight Data Recorder ____ Parameter INOP
31-31-01C	FLT RCDR Switch	Flight Data Recorder ____ Parameter INOP

31-31-02 Aircraft Condition Monitoring System (ACMS)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
31-31-02	Aircraft Condition Monitoring System (ACMS)	Y	1	0	D (O) May be inoperative provided engine trend data is entered using ACARS or manually entered onto Engine Performance Report Forms.

MEL ITEM	TAB LOCATION	PLACARD TEXT
31-31-02	Near Either Center EIS Display	Auto Engine Trend Data Report not Transmitting - Record Data in Engine Performance Report Form

(O) PROCEDURES

Refer to [Operations Manual, Volume 1, Supplementary Procedures, 5.7.11 – Engine Performance Report](#).

Ensure that a minimum of two (2) Engine Performance Report Report Forms are installed in the aircraft logbook. If not, contact MCC.

Note: *If an Engine Performance Report Form is not available and MCC determines that there is no requirement for ENCON readings during that flight segment, the Engine Performance Report Form may be boarded at the destination. It is desired to have the ENCON data available once per flight day.*

31-51-01 Central Aural Warning System (CAWS)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
	CENTRAL AURAL WARNING SYSTEM (CAWS)				Note: The CAWS must be operative for dispatch except that portions may be inoperative as listed below.
31-51-01A	APU Fire Warning Horn	Y	1	0	C (O) May be inoperative provided system is monitored in the cockpit during APU operations.
31-51-01B	Cabin Altitude Warning Tone "DISP.APPR.REQ'D"	Y	1	0	C May be inoperative provided the airplane remains at or below 10,000 feet MSL.
31-51-01C	SELCAL Chime	Y	1	0	C May be inoperative provided SELCAL light is operative if procedures require use of SELCAL.
31-51-01D	Voice Warnings (Vocal)	Y	1	0	C (O) May be inoperative provided procedures do not require its use.
31-51-01E	Advisory Callouts	Y	1	0	C (O) May be inoperative provided: 1. Alternate procedures are used, and 2. Windshear Detection and Avoidance System (Predictive) operates normally.

MEL ITEM	TAB LOCATION	PLACARD TEXT
31-51-01A	APU Start Switch	CAWS APU Fire Warning Horn INOP
31-51-01B	CABIN PRESS Panel	CAWS Cabin Altitude Warning INOP
31-51-01C	Audio Control Panels	CAWS SELCAL Chime INOP
31-51-01D	Engine and Alert Display	CAWS Voice Warning Function INOP
31-51-01E	Left and Right Navigation Displays	CAWS Advisory Callouts INOP

CONTINUED ON NEXT PAGE

(O) PROCEDURES

31-51-01A

If APU Fire Warning Horn is inoperative, monitor the APU Fire annunciator on the overhead annunciator panel when the APU is in use.

Note: *If an APU Fire is detected, the APU will automatically shut down and the APU FIRE Warning (Level 3 Alert) and Master Warning will annunciate.*

31-51-01D

Consult flight instruments, nav instruments, or EAD for indications.

31-51-01E

1. Review Standard Callout procedures. Refer to [Operations Manual, Volume 1, Normal Procedures, Introduction, Standard Callouts](#).
2. Preflight briefing to include review of terrain and/or obstructions in the proximity of the departure and arrival airports.
3. Pilot monitoring flight path during takeoff, approach and landing, will alert the pilot flying if any of the following conditions exist:
 - A. Excessive descent rate.
 - B. Excessive terrain closure rate.
 - C. Altitude loss after takeoff or go-around.
 - D. Unsafe terrain clearance while not in the landing mode.
 - E. Deviation from glideslope.

31-61-01 Electronic Instrument System (EIS)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
ELECTRONIC INSTRUMENT SYSTEM (EIS)					
31-61-01A	Display Units	Y	6	5	B Any one Display Unit may be inoperative provided appropriate information is displayed on the remaining displays, refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements.
31-61-01B	Remote Light Sensor	Y	1	0	C May be inoperative provided DU brightness is acceptable to the flight crew.
31-61-01C	Source Input Select Panel (SISP) Pushbuttons (Flight Director OFF)	Y	2	1	C (O) May be inoperative provided the same Pushbutton is operative on the opposite side of the cockpit.
31-61-01D	Source Input Select Panel (SISP) Pushbuttons (Flight Director CAPT ON 2 F/O ON 1)	Y	2	1	C (O) May be inoperative provided the same Pushbutton is operative on the opposite side of the cockpit.
31-61-01E	Source Input Select Panel (SISP) Pushbuttons (Air Data CAPT ON 2 F/O ON 1)	Y	2	1	C (O) May be inoperative provided the same Pushbutton is operative on the opposite side of the cockpit.
31-61-01F	Source Input Select Panel (SISP) Pushbuttons (Inertial Reference System CAPT ON AUX F/O ON AUX)	Y	2	1	C (O) May be inoperative provided the same Pushbutton is operative on the opposite side of the cockpit.
31-61-01G	Source Input Select Panel (SISP) Pushbuttons (VOR Nav Radio CAPT ON 2 F/O ON 1)	Y	2	1	C (O) May be inoperative provided the same Pushbutton is operative on the opposite side of the cockpit.
31-61-01H	Source Input Select Panel (SISP) Pushbuttons (Approach/ILS CAPT ON 2 F/O ON 1)	Y	2	1	C (O) May be inoperative provided the same Pushbutton is operative on the opposite side of the cockpit.
31-61-01I	Source Input Select Panel (SISP) Pushbuttons (EIS Source Lights (CAPT ON 2, F/O ON 1))	Y	4	2	C (O) May be inoperative provided the same Pushbutton is operative on the opposite side of the cockpit.
31-61-01J	SISP Pushbutton Internal Switchlights	Y	20	10	C (O) Individual Switchlights may be inoperative provided the same Pushbutton Switchlight is operative on the opposite side of the cockpit.

CONTINUED ON NEXT PAGE

MEL ITEM	TAB LOCATION	PLACARD TEXT
31-61-01A	Near Associated Display Panel	____ Display Unit INOP
31-61-01B	Near Associated Display Panel	Remote Light Sensor INOP
31-61-01C	Near Associated Display Panel	____ Source Input Select Panel Push Buttons Flight Director OFF INOP
31-61-01D	Near Associated Display Panel	____ Source Input Select Panel Push Buttons Flight Director CAPT ON 2 F/O ON 1 INOP
31-61-01E	Near Associated Display Panel	____ Source Input Select Panel Push Buttons Air Data CAPT ON 2 F/O ON 1 INOP
31-61-01F	Near Associated Display Panel	____ Source Input Select Panel Push Buttons Internal Reference System CAPT ON AUX F/O ON AUX INOP
31-61-01G	Near Associated Display Panel	____ Source Input Select Panel Push Buttons VOR Nav Radio CAPT ON 2 F/O ON 1 INOP
31-61-01H	Near Associated Display Panel	____ Source Input Select Panel Push Buttons Approach/ILS CAPT ON 2 F/O ON 1 INOP
31-61-01I	Near Associated Display Panel	____ Source Input Select Panel Push Buttons EIS Source Lights (CAPT ON 2, F/O ON 1) INOP
31-61-01J	Near Associated Display Panel	____ Source Input Select Panel Push Buttons Internal Switchlights INOP

(O) PROCEDURES

31-61-01C thru 31-61-01J

1. With a SISP Pushbutton or Pushbutton Switchlight inoperative, select in and out of the opposite-side button to verify its operation.
2. With a SISP Pushbutton Switchlight inoperative, select the pushbutton and verify the same pushbutton on the opposite side of the cockpit illuminates.

31-61-02 Versatile Integrated Avionics (VIA) Units

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY			
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS	
31-61-02	Versatile Integrated Avionics (VIA) Units	Y	2	0	C	May be dispatched with the VIA FAULT message displayed on the STATUS Page. Note: <i>The VIA will continue to operate normally in this condition.</i> Note: <i>This item may not be used to disposition any VIA system failures or any Alerts other than VIA FAULT.</i>	

MEL ITEM	TAB LOCATION	PLACARD TEXT
31-61-02	VIA Units Display	VIA INOP

32 Landing Gear

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32-45-01	Anti-Skid System
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	PROXIMITY SENSING SYSTEM (PSS) FUNCTIONS
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32-61-02B	Passenger Notice System Slat Monitor
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32-43-01 PARKING BRAKES ON Annunciation System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
32-43-01	PARKING BRAKES ON Annunciation System	N	1	C	(O) (M) May be inoperative provided: 1. Alternate procedures are used for brake release, and 2. Alternate maintenance procedures are used to verify proper operation of the Anti-Skid and Takeoff Warning Systems.

MEL ITEM	TAB LOCATION	PLACARD TEXT
32-43-01	Below Captains N.D. Panel	PARKING BRAKES ON Annunciator System INOP

(O) PROCEDURES

1. Do not turn on ANTISKID during preflight.
2. Prior to taxi,
 - A. verify that PARK BRAKE control is released and brake pedals have normal travel.
 - B. turn on ANTISKID and verify the ANTISKID OFF switch light is extinguished.
3. Verify that no Anti-Skid messages are on EAD.
4. When parking brakes are required, turn anti-skid OFF before setting the PARKING BRAKE.

(M) PROCEDURES

1. PARKING BRAKE ON annunciation OFF with parking brakes set:
 - A. Chock aircraft wheels.
 - B. Release parking brake.
 - C. Place ANTI-SKID switch to ARM.
 - D. Open (pull) PARKING BRAKE CONTROL circuit breaker (B1-216: EPC CBP Location S-42).
 - E. Verify all ANTI-SKID annunciations appear on EAD.
 - F. Close (reset) PARKING BRAKE CONTROL circuit breaker.
 - G. Verify all ANTI-SKID annunciation are off.
2. PARKING BRAKES ON annunciation ON with parking brakes released:
 - A. Chock aircraft wheels.
 - B. Open (pull) PARKING BRAKE CONTROL circuit breaker.
 - C. Gain access to Parking Brake Switch (S1-130) located under floor panels below pilot's rudder pedals. Disconnect connector from parking brake switch, and jumper connector pin A to pin B.
 - D. Close (reset) PARKING BRAKE CONTROL circuit breaker.
 - E. Perform Anti-Skid Return-to-Service test, and verify all ANTI-SKID annunciations illuminate on the EAD.
 - F. When test is complete, verify PARKING BRAKE ON annunciation and ANTI-SKID annunciations are all OFF.
 - G. If test is unsuccessful or as an alternate procedure, access Brake Return Shutoff Valves located in each wheel well (refer to figure shown in Item [32-45-02](#) of this Section). Disconnect electrical connector from valves and move manual lever to OPEN position, and lockwire lever in this position.

32-43-03 Brake Supply Pressure Indication System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
32-43-03	Brake Supply Pressure Indication System	Y	2	0	C	May be inoperative provided Hydraulic Pressure Indication Systems are operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
32-43-03	Below Captains N.D. Panel	Brake Supply Pressure Indication System INOP

32-45-01 Anti-Skid System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
32-45-01	Anti-Skid System "Contact MCC" "DISP.APPR.REQ'D" "Revenue Passengers Not Authorized"	Y	1	0 B	(O) May be inoperative provided: - MCC must provide authorization prior to use, - No revenue passengers or cargo are carried, - A maximum of 19 persons are carried as authorized by 14 CFR for non-passenger carrying operations, - Both Thrust Reversers are operative, - Flight crew is briefed by Flight Ops, and Performance penalties are applied for antiskid inoperative operations.

MEL ITEM	TAB LOCATION	PLACARD TEXT
32-45-01	ANTI-SKID Switch	Anti-Skid System INOP

(O) PROCEDURES

- Refer to limitations and performance data in **ODM**, Ferry Flight Planning and Operation section.
- Operations are restricted to dry runways only.
- Reduced Takeoff Thrust (TO FLX) is not permitted, but Derate may be used.
- Rejected takeoff performance data with the antiskid inoperative is based on the following: engine thrust to idle and immediate application of all available reverse thrust while verifying extension of ground spoilers, manual braking technique (i.e. gradually increasing brake pedal force until a moderate deceleration is felt) will be used.
- Normal landing procedures are to be followed, except that the manual braking procedure is to be used in lieu of normal anti-skid braking. This is, upon initiation of brakes, gradually increase brake pedal force until a moderate deceleration is felt..
- Plan a Flaps 40 Landing.

32-45-01	ANTISKID SYSTEM
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	None
NOTES	1. AWABS Takeoff data is provided for DRY runway conditions only. Takeoffs using SLP-WET, ICY, QCTR or HCTR data are not authorized.
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	Manual WDR not permitted with MEL 32-45-01.

32-45-02 Anti-Skid Brake Return Line Shutoff Valves

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD			N	2	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION						REMARKS OR EXCEPTIONS
32-45-02	Anti-Skid Brake Return Line Shutoff Valves				C	(O) (M)	May be inoperative provided: 1. Valve is secured in the OPEN position, and 2. Anti-Skid System is operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
32-45-02	Logbook	Anti-Skid Brake Return Line Shutoff Valves INOP

(O) PROCEDURES

1. If ANTISKID FAIL message is present on the EAD prior to takeoff and after Parking Brake release cycle the Antiskid Switch to OFF then ON to extinguish the alert.
2. Verify AUX Hydraulic Pump is ON whenever Parking Brake is set.

Note: The ANTISKID FAIL message may appear on the EAD each time the Parking Brake is set.

Note: Selecting CONFIG on the Systems Display Control Panel will clear the message from the EAD. The message should extinguish upon parking brake release. If it remains on, the CFDS should be interrogated for further information and possible maintenance action.

Note: May be dispatched with the ANTISKID FAULT message displayed on the STATUS Page.

3. Coordinate with ground crew to chock wheels immediately after parking, as required.

(M) PROCEDURES

 [Link to Learning Video](#)

1. Perform Brake Control Unit fault review using the CFDS to determine which Valve(s) is affected.

Note: *This failure will be initially indicated by the presence of either the ANTISKID FAULT or ANTISKID FAIL message on the EAD. CFDS interrogation is required to determine the specific cause of the initial Alert.*

2. Access the inoperative Valve(s) in the respective MLG wheel well.
3. Move the manual lever(s) to the OPEN position, and lockwire the lever(s) in this position.
4. For one Valve inoperative, perform one of the following procedures:

A. PROCEDURE 1 - PREFERRED

- i. Pull the PARK BRAKE CONTROL circuit breaker (B1-216: EPC CBP Location S-42).
- ii. Access the inoperative Valve in the respective MLG wheel well.
- iii. Disconnect the electrical connector from the Valve.
- iv. Using 16 gauge wire / pin connectors, connect a jumper from Pin C to Pin D of the electrical connector. Secure the electrical connector as appropriate.
- v. Reset PARK BRAKE CONTROL circuit breaker.

Note: *The Parking Brake may be set as normal in this case. However, wheel chocks should still be used whenever practical.*

B. PROCEDURE 2 – To be used if no jumper wire is available.

Note: *It is normal for the Antiskid Fail message to appear on the EAD when Procedure 2 is accomplished. Selecting CONFIG on the Systems Display Control Panel will clear the message from the EAD.*

- i. Pull and secure the PARK BRAKE CONTROL circuit breaker (B1-216: EPC CBP Location S-42).
- ii. Advise maintenance personnel and flight crew to leave AUX Hydraulic Pump running whenever aircraft is parked using the Parking Brake.

CAUTION

BRAKE ACCUMULATORS WILL ONLY HOLD THE PARKING BRAKE FOR A SHORT DURATION. IT IS IMPERATIVE THAT A HYDRAULIC PUMP REMAIN RUNNING WHILE THE PARKING BRAKE IS SET.

- iii. Put wheel chocks in place as soon as possible after parking.

5. For two Valves inoperative:

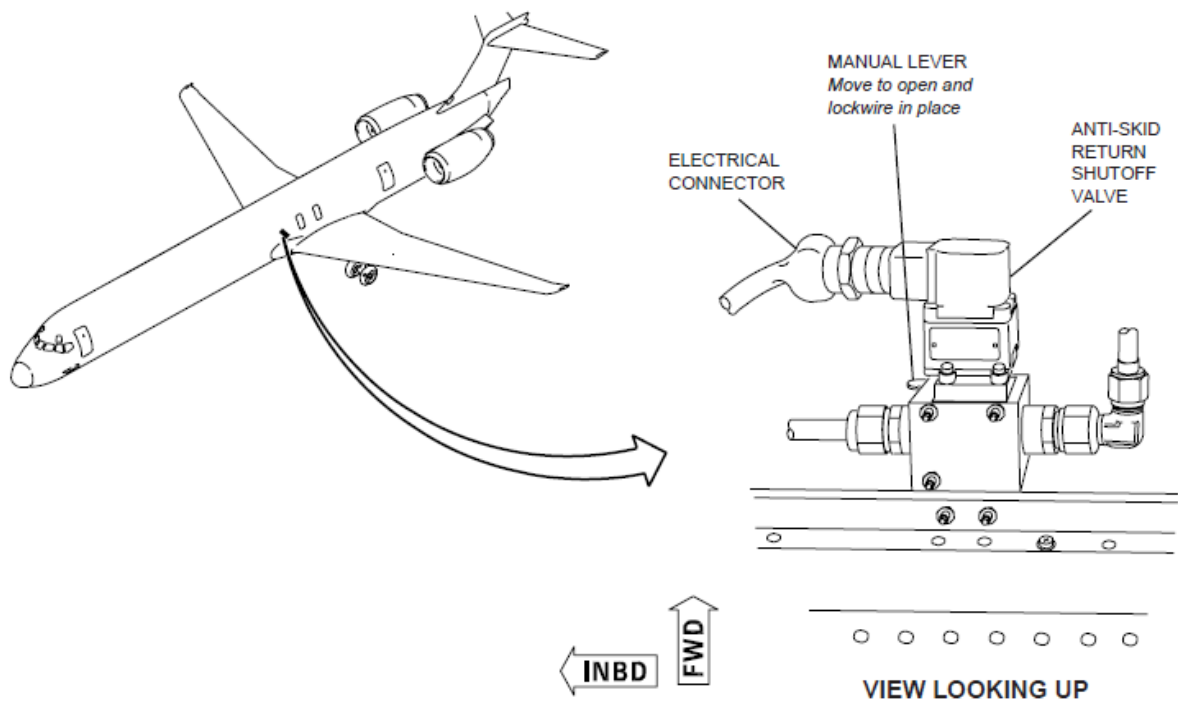
- A. Pull and secure the PARK BRAKE CONTROL circuit breaker (B1-216: EPC CBP Location S-42).
- B. Advise maintenance personnel and flight crew to leave AUX Hydraulic Pump running whenever aircraft is parked using the Parking Brake.

CAUTION

BRAKE ACCUMULATORS WILL ONLY HOLD THE PARKING BRAKE FOR A SHORT DURATION. IT IS IMPERATIVE THAT A HYDRAULIC PUMP REMAIN RUNNING WHILE THE PARKING BRAKE IS SET.

- C. Put wheel chocks in place as soon as possible after parking.

Figure 32-1: Anti-Skid Brake Return Line Shutoff Valves



32-47-01 Brake Temperature Monitoring System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
32-47-01	Brake Temperature Monitoring System "DISP.APPR.REQ'D"	Y	1	0	C (O) (M) May be inoperative provided an alternative method(s) is used to determine brake temperatures if quick turn-around procedures are to be used PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
32-47-01	Below Captains N.D. Panel	Brake Temperature Monitoring System INOP

(O) PROCEDURES

1. Quick turn-around is permitted if landing weight is at or below the weight determined from the 717-200 ODM Quick Turn-Around Landing Weight.
2. Do not set the Parking Brake if landing weight exceeds the weight determined from the ODM Chart listed in step 1. above. Coordinate with ground crew to chock wheels immediately after parking as required.
3. If a high speed rejected take-off is conducted, refer to the ODM Abnormal Section, Brake Cooling.

(M) PROCEDURES

PRIOR TO EACH DEPARTURE:

Visually verify that the associated tire(s) thermal fuse plugs have not melted and that the tire(s) remains inflated.

Note: If actual landing weight (after wind and slope corrections) has exceeded maximum quick turn-around weight, wait at least 50 minutes after block-in to perform this inspection.

32-61-01 Landing Gear Position Indicating Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
32-61-01A	Landing Gear Position Indicating Systems Synoptic Display	Y	3	0	C	May be inoperative provided all Landing Gear Position Indication Lights on Upper Instrument Panel are operative.
32-61-01B	Landing Gear Position Indicating Systems Upper Instrument Panel Lights	Y	3	0	C	May be inoperative provided all Landing Gear Position Indications on Synoptic Display are operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
32-61-01A	Below Captains N.D. Panel	Landing Gear Position Indicating Systems Synoptic Display INOP
32-61-01B	Below Captains N.D. Panel	Landing Gear Position Indicating Systems Upper Instrument Panel Lights INOP

32-61-02 Proximity Sensing System (PSS) Functions

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
	PROXIMITY SENSING SYSTEM (PSS) FUNCTIONS				Note: PSEU FAULT Alert will be present for any of the dispatchable failure conditions listed below. Note: PSS function failures associated with PSEU FAIL Alert are nondispatchable.	
32-61-02A	TCAS / Landing Gear Position Interface	N	1	0	C	(O) (M) May be inoperative provided PSEU is interrogated at least ONCE EACH FLIGHT DAY for verification of all PSS faults (Log Book Entry Required). Repetitive Check Required (Yellow Placard)
32-61-02B	Passenger Notice System Slat Monitor	N	1	0	C	(O) (M) May be inoperative provided: - Automatic Passenger Notice System is placarded inoperative and deferred per MEL 23-32-01, and - PSEU is interrogated at least once each flight day for verification of all PSS faults (Log Book Entry Required). Repetitive Check Required (Yellow Placard)
32-61-02C	ACARS Door Position Monitor	N	1	0	C	(O) (M) May be inoperative provided: - Alternate procedures are used for recording aircraft block times, and - PSEU is interrogated at least once each flight day for verification of all PSS faults (Log Book Entry Required). Repetitive Check Required (Yellow Placard)
32-61-02D	PSEU Program Pin Monitor	N	1	0	C	(M) May be inoperative provided PSEU is interrogated at least once each flight day for verification of all PSS faults (Log Book Entry Required). Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
32-61-02A	Below Captains N.D. Panel	TCAS / Landing Gear Position Interface INOP
32-61-02B	Below Captains N.D. Panel	Passenger Notice System Slat Monitor INOP
32-61-02C	Lower Aft Pedestal	ACARS Door Position Monitor INOP
32-61-02D	Lower Aft Pedestal	PSEU Program Pin Monitor INOP

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(O) PROCEDURES

32-61-02A

1. Prior to departure, verify Function Selector Knob on the TCAS / Transponder Control Panel is positioned to XPNDR. The TCAS STANDBY message should be present on both Primary Flight Displays, unless the TCAS System is also inoperative.

Note: *If TCAS is inoperative and has been deactivated per the instructions of Item 34-46-01 of this section, TCAS STANDBY message will not appear on the Primary Flight Displays.*

2. After takeoff, rotate Function Selector Knob to TA/RA position.
3. On approach, prior to gear extension, rotate Function Selector Knob to XPNDR position.

Note: *In this dispatch condition, false Traffic Alerts and/or Resolution Advisories may occur if TCAS is operating when the landing gear is extended. In any case, the normal TCAS inhibits will still be in effect for takeoff and landing.*

32-61-02B

1. Advise Cabin Attendants that the Automatic Passenger Notice System will be inoperative.
2. Operate the Passenger Notice System manually.

32-61-02C

ACARS will not report aircraft block times automatically. Alternate procedures for recording / reporting block times should be reported to the station by radio communication.

(M) PROCEDURES

32-61-02A, 32-61-02B, 32-61-02C, 32-61-02D

Some PSS function failures associated with the PSEU FAULT alert are non-dispatchable. Interrogate the PSEU using an MCDU to access the CFDS to determine dispatchability of a PSS failure, as follows:

1. Select the CFDS on the MCDU main menu page.
2. Select the LRU prompt.
3. Use the PAGE key to scroll the LRU Menu until the PSEU prompt is displayed, then select the PSEU prompt.
4. Select the CURRENT FAULTS prompt to identify PSS failures currently logged in the CFDS maintenance buffer.
5. ONCE EACH FLIGHT DAY, verify dispatchability of the PSEU FAULT alert using the following table:

Fault Code	Description	Dispatchability
326065	PSEU PIN PROGRAMMING INVALID	32-61-02D
326081	R GEAR DOWN LOCK SENSOR A FAULT	NONE
326082	N GEAR DOWN SENSOR A FAULT	NONE
326083	N GEAR LOCKED SENSOR A FAULT	NONE
326085	L GEAR DOWN LOCK SENSOR A FAULT	NONE
326099	R GEAR DOWN LOCK SENSOR B FAULT	NONE
326100	N GEAR DOWN SENSOR B FAULT	NONE
326101	N GEAR LOCKED SENSOR B FAULT	NONE
326102	L GEAR DOWN LOCK SENSOR B FAULT	NONE
326146	PROGRAM PIN 1 INPUT VOLTAGE FAULT	32-61-02D
326147	PROGRAM PIN 2 INPUT VOLTAGE FAULT	32-61-02D
326148	PROGRAM PIN 3 INPUT VOLTAGE FAULT	32-61-02D
326149	PROGRAM PIN 4 INPUT VOLTAGE FAULT	32-61-02D
326150	PROGRAM PIN 5 INPUT VOLTAGE FAULT	32-61-02D
326151	PROGRAM PIN 6 INPUT VOLTAGE FAULT	32-61-02D
326166	TCAS INTERFACE OVERCURRENT	32-61-02A
326167	FASTEN SEAT BELT RELAY INTERFACE OVERCURRENT	32-61-02B
326168	FCC / SECU LEFT WOW OVERCURRENT	NONE
326182	FCC / SECU RIGHT WOW OVERCURRENT	NONE
326185	ACARS FWD CABIN DOOR INTERFACE OVERCURRENT	32-61-02C
326186	ACARS FWD GALLEY DOOR INTERFACE OVERCURRENT	32-61-02C
326187	FWD STAIR DOOR RELAY INTERFACE OVERCURRENT	NONE
326214	TCAS INTERFACE OPEN / GROUND	32-61-02A
326215	FASTEN SEAT BELT RELAY INTERFACE OPEN / GROUND	32-61-02B
326216	FCC / SECU LEFT WOW OPEN / GROUND	NONE
326224	FCC / SECU RIGHT WOW OPEN / GROUND	NONE
326227	ACARS FWD CABIN DOOR INTERFACE OPEN / GROUND	32-61-02C
326228	ACARS FWD CABIN DOOR INTERFACE OPEN / GROUND	32-61-02C
326229	FWD STAIR DOOR RELAY OPEN / GROUND	NONE

33 Lights

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33-42-03	Fixed Aft Position Light Bulbs, (Wing Tips, White)
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33-42-03B	Fixed Aft Position Light Bulbs, Non LED,
33-42-03C	Fixed Aft Position LED Lights,
33-42-04	White Strobe / Anti-Collision Lights, (Forward and Aft Wing Tips)
33-42-04A	White Strobe / Anti-Collision Lights, Non LED,
33-42-04B	White Strobe / Anti-Collision Lights, LED,

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WING AND NACELLE ILLUMINATION LIGHT SYSTEMS

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- 33-43-01B Nacelle Illumination Lights

33-43-02 Ground Floodlights

- 33-43-02 Ground Floodlights

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- 33-44-01 Tail Illumination (Logo) Lights

33-51-01 Emergency Escape Path Marking System Lights

EMERGENCY ESCAPE PATH MARKING SYSTEM LIGHTS

- 33-51-01A Photoluminescent Marking System and Lights
- 33-51-01B Photoluminescent Marking System and Lights

33-51-02 Exterior Emergency Illumination System

- 33-51-02A Exterior Emergency Illumination System
- 33-51-02B Exterior Emergency Illumination System

33-52-01 Cabin Standby Lighting System

- 33-52-01 Cabin Standby Lighting System

33-12-01 Master Warning Lights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
33-12-01	Master Warning Lights	Y	2	1	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-12-01	Affected Master Warning Light	____ Master Warning Light INOP

33-12-02 Master Caution Lights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
33-12-02	Master Caution Lights	Y	2	1 C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-12-02	Affected Master Caution Light	____ Master Caution Light INOP

33-13-01 Cockpit Flight Deck / Flight Compartment and Instrument Lighting Systems

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH		REPAIR CATEGORY	
FLIGHT CREW MAY PLACARD					REMARKS OR EXCEPTIONS	
ITEM NO.	NAME/DESCRIPTION					
33-13-01A	Cockpit Flight Deck / Flight Compartment and Instrument Lighting Systems	Y	-	-	C	Cockpit lights not controlled by the inner rheostats may be inoperative provided: 1. Remaining lighting system lights are sufficient to clearly illuminate all required instruments, controls, and other devices for which it is provided, 2. Overhead panel backlighting must be operative, 3. FCP backlighting must be operative, 4. Remaining lighting system lights are positioned so that direct rays are shielded from flight crewmembers eyes, and 5. Lighting configuration and intensity is acceptable to the flight crew. Note: Individual button/switch lights and/or annunciations/indications are excluded from this relief.
33-13-01B	Cockpit Flight Deck / Flight Compartment and Instrument Lighting Systems "DISP.APPR.REQ'D"	y	-	-	C	Cockpit lights controlled by the inner rheostats may be inoperative provided: 1. Aircraft is not operated at night, 2. Overhead panel backlighting must be operative, 3. FCP backlighting must be operative, 4. Remaining lighting system lights are sufficient to clearly illuminate all required instruments, controls, and other devices for which it is provided, 5. Remaining lighting system lights are positioned so that direct rays are shielded from flight crewmembers eyes, and 6. Lighting configuration and intensity is acceptable to the flight crew. Note: Individual button/switch lights and/or annunciations/indications are excluded from this relief.

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-13-01A	Associated Indicator, Panel or Lighting Switch	____ Cockpit / Instrument Lighting INOP
33-13-01B	Associated Indicator, Panel or Lighting Switch	____ Cockpit / Instrument Lighting INOP

33-20-01 Cabin/Handrail Illumination

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION					
33-20-01A	Cabin Interior Illumination	Y	-	-	C	May be inoperative provided: 1. No more than 10% of the cabin ceiling lights are inoperative, and 2. No two (2) adjacent lights may be inoperative. <i>Note: Sidewall lighting may be inoperative.</i>
33-20-01B	Handrail Lighting	Y	16	0	D	

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-20-01A	Flight Attendant Cabin Lighting Control Panel	___ Cabin Light(s) INOP
33-20-01B	Flight Attendant Cabin Lighting Control Panel	___ Handrail Lighting INOP

33-28-01 Passenger Notice System (No Smoking/ Fasten Seat Belt/ Return To Cabin)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
PASSENGER NOTICE SYSTEM (NO SMOKING/ FASTEN SEAT BELT/ RETURN TO CABIN)					
33-28-01A	Automatic System	Y	1	0	C (O) May be inoperative provided Manual Passenger Notice System is operative.
33-28-01B	Automatic System	Y	1	0	C (O) May be inoperative provided: - The Passenger Address System is operative, - Crew Call Chimes and Cabin Interphone Systems are operative, and - Procedures are used for alerting the Flight Attendants and notifying passengers by use of the Public Address System when seat belts should be fastened, and smoking is prohibited.
33-28-01C	Manual System	Y	1	0	C (O) May be inoperative provided: - The Passenger Address System is operative, - Crew Call Chimes and Cabin Interphone Systems are operative, and - Procedures are used for alerting the Flight Attendants and notifying passengers by use of the Public Address System when seat belts should be fastened, and smoking is prohibited.
33-28-01D	Lighted Signs "DISP.APPR.REQ'D"	Y	-	-	C (M) May be inoperative provided: - Associated passenger seat or lavatory is not occupied from which a passenger lighted information sign is not readily legible, and - Associated seat or lavatory is blocked and placarded, "DO NOT OCCUPY". Note: MCC must contact Flight Control for coordinating with ACS and ROC to remove seat from inventory until repaired. Note: These conditions are not intended to prohibit lavatory use or inspections by crewmembers.
33-28-01E	Lighted Signs	Y	-	-	C (O) May be inoperative and associated seat or lavatory may be occupied provided: - PA system operates normally, and - PA system is used to notify passengers and cabin crew when associated sign(s) are placed ON or OFF.

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-28-01A	Fasten Seat Belt Switch	Automatic Fasten Seat Belt / No Smoking Signs INOP
33-28-01B	Fasten Seat Belt Switch	Automatic Fasten Seat Belt / No Smoking Signs INOP
33-28-01C	Fasten Seat Belt Switch	Manual Fasten Seat Belt / No Smoking Signs INOP
33-28-01D	Fasten Seat Belt Switch	____ Fasten Seat Belt / No Smoking Sign(s) INOP
33-28-01E	Fasten Seat Belt Switch	____ Fasten Seat Belt / No Smoking Sign(s) INOP

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(O) PROCEDURES

33-28-01A

Operate the system manually.

33-28-01B, 33-28-01C, 33-28-01E

Pilots will alert the flight attendants by interphone and they will notify passengers by use of the Public Address System when seat belts should be fastened, and smoking is prohibited.

(M) PROCEDURES

33-28-01D

1. For passenger seats, placard the affected seat(s) inoperative per MEL 25-23-03A (M) PROCEDURE.
2. For lavatories, lock lavatory door closed and install an INOPERATIVE – DO NOT ENTER placard on the exterior of the associated lavatory door.

33-31-01 Cargo Compartment Lighting Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
33-31-01	Cargo Compartment Lighting Systems	Y	2	0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-31-01	Aircraft Logbook	____ Cargo Compartment Light Systems INOP

33-32-02 Maintenance Lights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
33-32-02	Maintenance Lights	Y	10	0 C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-32-02	Aircraft Logbook	____ Maintenance Light(s) INOP

MAINTENANCE NOTE:

This Item includes Wheel Well Lights, E&E Compartment Lights, Servicing Panel Lighting, and any other lighting systems not associated with flight operations.

33-41-01 Landing Lights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
	LANDING LIGHTS				
33-41-01A	Wing Lights "DISP.APPR.REQ'D"	Y	2	0	C (M) May be inoperative provided operations are not conducted from sunset to sunrise.
33-41-01B	Wing Lights	Y	2	0	C (M) May be inoperative provided both Nose Gear Lights are operative.
33-41-01C	Nose Gear Lights (Incandescent or LED) "DISP.APPR.REQ'D"	Y	2	0	C (M) May be inoperative provided operations are not conducted from sunset to sunrise. Note: For LED lights, refer to the AMM for operational requirements.
33-41-01D	Nose Gear Lights (Incandescent or LED)	Y	2	0	C (M) May be inoperative provided both Wing Lights are operative. Note: For LED lights, refer to the AMM for operational requirements.
33-41-01E	Wing Landing Light Extend/ Retract System	Y	2	0	C (M) May be inoperative provided: 1. Associated Landing Light is considered inoperative and deferred per MEL 33-41-01A or MEL 33-41-01B as appropriate, and 2. Associated Landing Light is secured RETRACTED.
33-41-01F	Wing Landing Light Extend/ Retract System "DISP.APPR.REQ'D"	Y	2	0	C (O) (M) May be inoperative provided: 1. Associated Landing Light is secured EXTENDED, and 2. Performance penalties are applied.
33-41-01G	FCC Wing Landing Light Auto Retract Function "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided 1. Wing Landing Lights remain RETRACTED, and 2. If airplane is to be operated at night, both Nose Gear Landing Lights are used and verified operative PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)
33-41-01H	FCC Wing Landing Light Auto Retract Function "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative and Wing Landing Lights EXTENDED for takeoff and landing provided performance penalties are applied.

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-41-01A	Associated Wing Light Switch(es)	___ Wing Landing Light(s) INOP
33-41-01B	Associated Wing Light Switch(es)	___ Wing Landing Light(s) INOP
33-41-01C	Nose Gear Light Switch	___ Nose Gear Landing Light(s) INOP
33-41-01D	Nose Gear Light Switch	___ Nose Gear Landing Light(s) INOP
33-41-01E	Associated Wing Light Switch(es)	___ Wing Landing Light(s) INOP Retracted
33-41-01F	Associated Wing Light Switch(es)	___ Wing Landing Light(s) INOP Extended
33-41-01G	Associated Wing Light Switch(es)	___ Wing Landing Light(s) INOP Retracted
33-41-01H	Associated Wing Light Switch(es)	___ Wing Landing Light(s) INOP Extended

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(O) PROCEDURES

33-41-01F

33-41-01F	WING LANDING LIGHT EXTEND/RETRACT SYSTEM
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	1. Reduce ENROUTE LIMIT WT by 2,420 lbs. per each light fixed extended. 2. Increase BLOCK FUEL by 1.5% per each light fixed extended.
NOTES	Make a notation on the Flight Plan when Enroute Climb is limiting.
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF 1. Reduce Structural Takeoff Weight Limit by 2,200 lbs. per each light fixed extended. 2. Reduce RATOW by 2,200 lbs. per each light fixed extended. 3. Reduce T/O CLIMB LIMIT by 2,200 lbs. per each light fixed extended. LANDING Reduce LANDING CLIMB LIMIT WT by 900 lbs per each light fixed extended. OTHER CORRECTIONS 1. Determine V-Speeds from ODM Takeoff Section charts for applicable Thrust / Flap setting and enter with the actual takeoff weight increased by the MEL takeoff weight penalty (2,200 lbs) for each light fixed extended. 2. Determine all other speeds at actual operating weight.

33-41-01G

If airplane is to be operated at night, assure both Nose Gear Landing Lights are used and verified operative prior to departure.

33-41-01H

33-41-01H	FCC WING LANDING LIGHT AUTO RETRACT FUNCTION
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	1. Reduce ENROUTE LIMIT WT by 2,420 lbs. per each light fixed extended. 2. Increase BLOCK FUEL by 1.5% per each light fixed extended.
NOTES	Make a notation on the Flight Plan when Enroute Climb is limiting.
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF 1. Reduce Structural Takeoff Weight Limit by 2,200 lbs. per each light fixed extended. 2. Reduce RATOW by 2,200 lbs. per each light fixed extended. 3. Reduce T/O CLIMB LIMIT by 2,200 lbs. per each light fixed extended. LANDING Reduce LANDING CLIMB LIMIT WT by 900 lbs per each light fixed extended. OTHER CORRECTIONS 1. Determine V-Speeds from ODM Takeoff Section charts for applicable Thrust / Flap setting and enter with the actual takeoff weight increased by the MEL takeoff weight penalty (2,200 lbs) for each light fixed extended. 2. Determine all other speeds at actual operating weight.

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(M) PROCEDURES

33-41-01A, 33-41-01B

If wing lights cracked or broken:

Note: *Wing lights that are inoperative due to cracked or broken glass: Maintenance is required and Flight Crew may not placard.*

1. Ensure the associated Wing Light is retracted.
2. Pull and secure the appropriate circuit breakers for the inoperative light:
 - A. For the left wing landing light:
 - LEFT WING LANDING LIGHT (B1-13) | Upper EPC | K-11
 - LEFT WING LANDING LIGHT CONT (B1-14) | Upper EPC | K-12
 - B. For the right wing landing light:
 - RIGHT WING LANDING LIGHT (B1-15) | Upper EPC | L-11
 - RIGHT WING LANDING LIGHT CONT (B1-16) | Upper EPC | L-12
3. Carefully remove or secure any loose pieces of broken glass.
4. Beginning on one side of the lamp, cover the cracked or broken lamp with strips of metalized 'speed' tape ensuring approximately 50% overlap between adjacent strips until the entire surface area of the lamp lens is covered. Apply one strip of tape, around the outside edge of the lamp and housing, ensuring there is approximately one inch of overlap to the lamp housing. It is acceptable to use preformed tape that provides the same coverage.

33-41-01C, 33-41-01D

ACCOMPLISH ONE OF THE FOLLOWING PROCEDURES:

PROCEDURE 1 Nose Gear Lights are NOT damaged

1. Pull and secure the appropriate circuit breaker(s) for the inoperative light(s)
 - A. For the Left NLG Light:
 - LEFT NOSE GEAR LANDING & TAXI LT (B1-19) | UPPER EPC | K-13
 - B. For the Right NLG Light:
 - RIGHT NOSE GEAR LANDING & TAXI LT (B1-20) | UPPER EPC | L-13

PROCEDURE 2 Nose Gear Light lenses ARE cracked or broken, or the retaining clamp is damaged or missing.

1. If the Nose Gear Lights are found with a cracked or broken lens or the retaining clamp is found damage or missing, ***maintenance is required, and flight crew may NOT placard.***
2. Pull and secure the appropriate circuit breaker(s) for the inoperative light(s)
 - A. For the Left NLG Light:
 - LEFT NOSE GEAR LANDING & TAXI LT (B1-19) | UPPER EPC | K-13
 - B. For the Right NLG Light:
 - RIGHT NOSE GEAR LANDING & TAXI LT (B1-20) | UPPER EPC | L-13
3. Carefully remove or secure any loose pieces of broken glass or retaining ring.
4. Beginning on one side of the lamp, apply subsequent layers of speed tape over the discrepant NLG lamp face to secure the lamp in place with the following provisions:
 - Ensure approximately 50% overlap between adjacent strips until the entire surface of the lamp lens is covered.
 - Extend and wrap the tape strips around the edges of the light housing to secure the lamp.
 - Apply one strip of tape, around the outside edge of the lamp and housing, with approximately 1 inch of overlap onto the lamp housing.
 - It is acceptable to use preformed tape that provides the same coverage.
 - Ensure the lamp is securely retained and is not loose inside the housing.
 - Ensure areas to be taped are clean and dry prior to speedtape application to ensure proper adhesion.

33-41-01E

1. Pull and secure the appropriate circuit breakers for the inoperative light:

- A. For the left wing landing light:

LEFT WING LANDING LIGHT (B1-13) | Upper EPC | K-11

LEFT WING LANDING LIGHT CONT (B1-14) | Upper EPC | K-12

- B. For the right wing landing light:

RIGHT WING LANDING LIGHT (B1-15) | Upper EPC | L-11

RIGHT WING LANDING LIGHT (B1-16) | Upper EPC | L-12

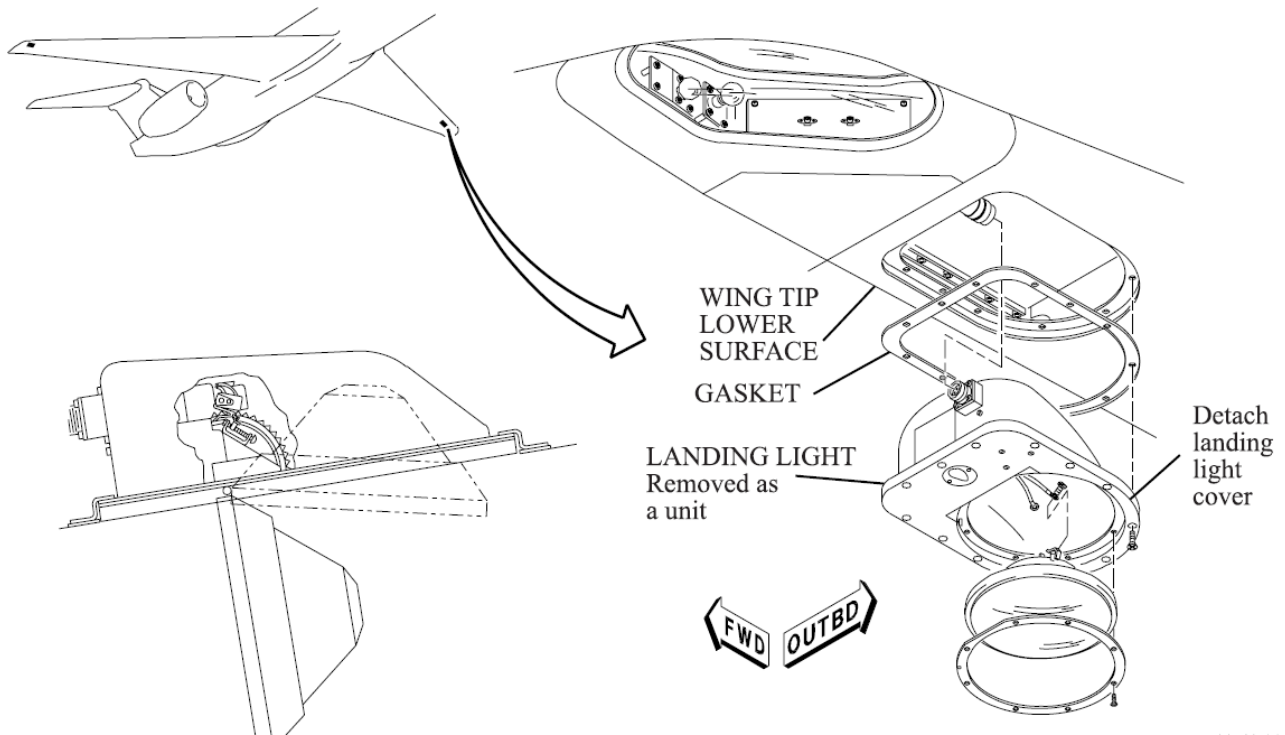
Note: *Lights that are failed in the extended position must be manually retracted. Maintenance is required and Flight Crew may not placard.*

2. For Light Assembly p/n 45-0057-1 or p/n 45-0057-3 failed in the extended position, secure the light assembly in the retracted position using the following procedure:

- A. Remove the 12 attaching screws and drop landing light assembly from wingtip. Disconnect electrical connector. Coil and stow aircraft wiring.
- B. Detach landing light cover from mounting plate assembly by removing appropriate screws and lockwashers.
- C. Loosen the six screws to the gear case assembly until the worm gear is free of the sector gear.
- D. Manually retract the landing light lamp and tighten the screws to the gear case assembly.
- E. Secure the landing light cover to the mounting plate assembly with screws and lockwashers.
- F. Re-install light assembly in wing tip with the 12 attaching screws.
- G. If lamp will not stay closed, install aluminum or stainless steel tab extending over the lamp retaining ring using the landing light assembly aft attaching screw.

3. For Lights Assembly p/n # 45-0057-9 or p/n 45-0057-11 failed in the extended position, secure the light assembly in the retracted position using the following procedure:

- A. Remove the 4 cap screws securing the pivot shafts to the housing.
- B. Move the canopy away from the housing sufficiently for the canopy drive gear to disengage (Do not allow the pivot shafts to fall out).
- C. Rotate the canopy to the retracted position and re-engage the canopy drive gear, making sure the drive gear is not jammed and that the pivot shafts lay flat in their pockets.
- D. Re-install the 4 cap screws.

Figure 33-1: Wing Lights


33-41-01F

1. With the wing landing light in the extended position, pull and secure the appropriate circuit breaker(s) for the inoperative light(s)
 - A. For the Left Wing Landing Light:
 LEFT WING LANDING LIGHT CONT (B1-14) | UPPER EPC | K-12
 - B. For the Right Wing Landing Light:
 RIGHT WING LANDING LIGHT CONT (B1-16) | UPPER EPC | L-12

33-42-01 Anti-Collision Beacon Lights (Fuselage-Red)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
33-42-01	Anti-Collision Beacon Lights (Fuselage-Red)	Y	2	0	C	May be inoperative provided wing tip Strobe / Anti-Collision Lights are operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-42-01	Anti-Collision Light Switch	____ Anti-Collision Light INOP

33-42-02 Fixed Forward Position Light Bulbs (Wing Tips)

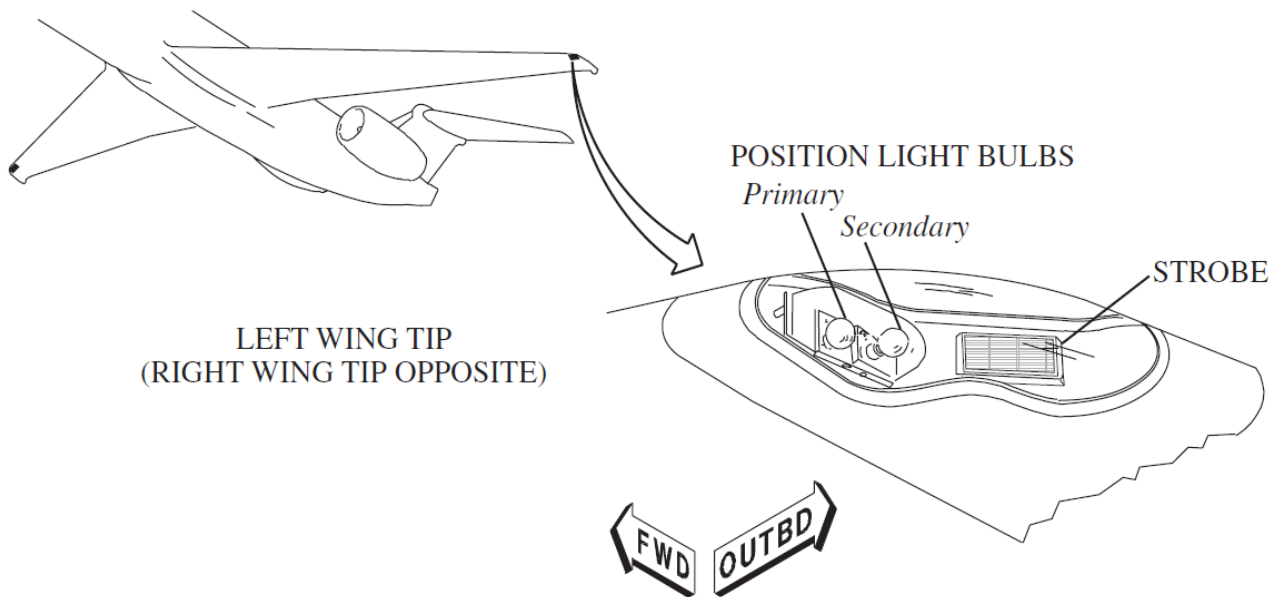
NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
33-42-02A	Fixed Forward Position Light Bulbs (Wing Tips)	Y	4	2	C One Bulb (red or green) must be operative on each wing tip for operations during the period from sunset to sunrise.
33-42-02B	Fixed Forward Position Light Bulbs (Wing Tips) "DISP.APPR.REQ'D"	Y	4	0	C May be inoperative provided airplane is not operated during the period from sunset to sunrise. Note: A Secondary Position Light Bulb that is not illuminated should be considered operative unless the associated Primary Light Bulb is also not illuminated.

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-42-02A	Position / Strobe Light Switch	____ Forward Position Light(s) INOP
33-42-02B	Position / Strobe Light Switch	____ Forward Position Light(s) INOP

MAINTENANCE NOTE:

The Forward Position Light assembly includes an automatic switching feature which normally allows only one Bulb in the assembly to illuminate. The Bulbs are designated as Primary and Secondary, as shown below. It should be noted that a Secondary Bulb that is not illuminated should be considered operative unless the associated Primary Bulb is also not illuminated.

Figure 33-2: Fixed Forward Position Light Bulbs (Wing Tips)



33-42-03 Fixed Aft Position Light Bulbs, (Wing Tips, White)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
33-42-03A	Fixed Aft Position Light Bulbs, Non LED , (Wing Tips, White) (Pre AA 3342-01019)	Y	4	2	C One Bulb must be operative on each wingtip for operations during the period from sunset to sunrise.
33-42-03B	Fixed Aft Position Light Bulbs, Non LED , (Wing Tips, White) (Pre AA 3342-01019) "DISP.APPR.REQ'D"	Y	4	0	C May be inoperative provided airplane is not operated during the period from sunset to sunrise. Note: A Secondary Position Light Bulb that is not illuminated should be considered operative unless the associated Primary Light Bulb is also not illuminated.
33-42-03C	Fixed Aft Position LED Lights, (Wing Tips, White) (Post AA 3342-01019) "DISP.APPR.REQ'D"	Y	2	0	C (M) Three or more LED segments may be inoperative per position light provided airplane is not operated during the period from sunset to sunrise. Note: With one or two LED segments inoperative, the position light is still considered operative. If three or more LED segments are inoperative, the position light is considered inoperative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-42-03A	Position / Strobe Light Switch	___ Aft Position Light(s) (Wing Tip) INOP
33-42-03B	Position / Strobe Light Switch	___ Aft Position Light(s) (Wing Tip) INOP
33-42-03C	Position / Strobe Light Switch	___ Aft Position Light(s) (Wing Tip) INOP

MAINTENANCE NOTE:

 [Link to Learning Video](#)

The Aft Position Light assembly includes an automatic switching feature which normally allows only one Bulb in the assembly to illuminate. The Bulbs are designated as Primary and Secondary, as shown below. It should be noted that a Secondary Bulb that is not illuminated should be considered operative unless the associated Primary Bulb is also not illuminated.

Figure 33-3: Fixed Aft Position Light Bulbs

Non LED Position Light

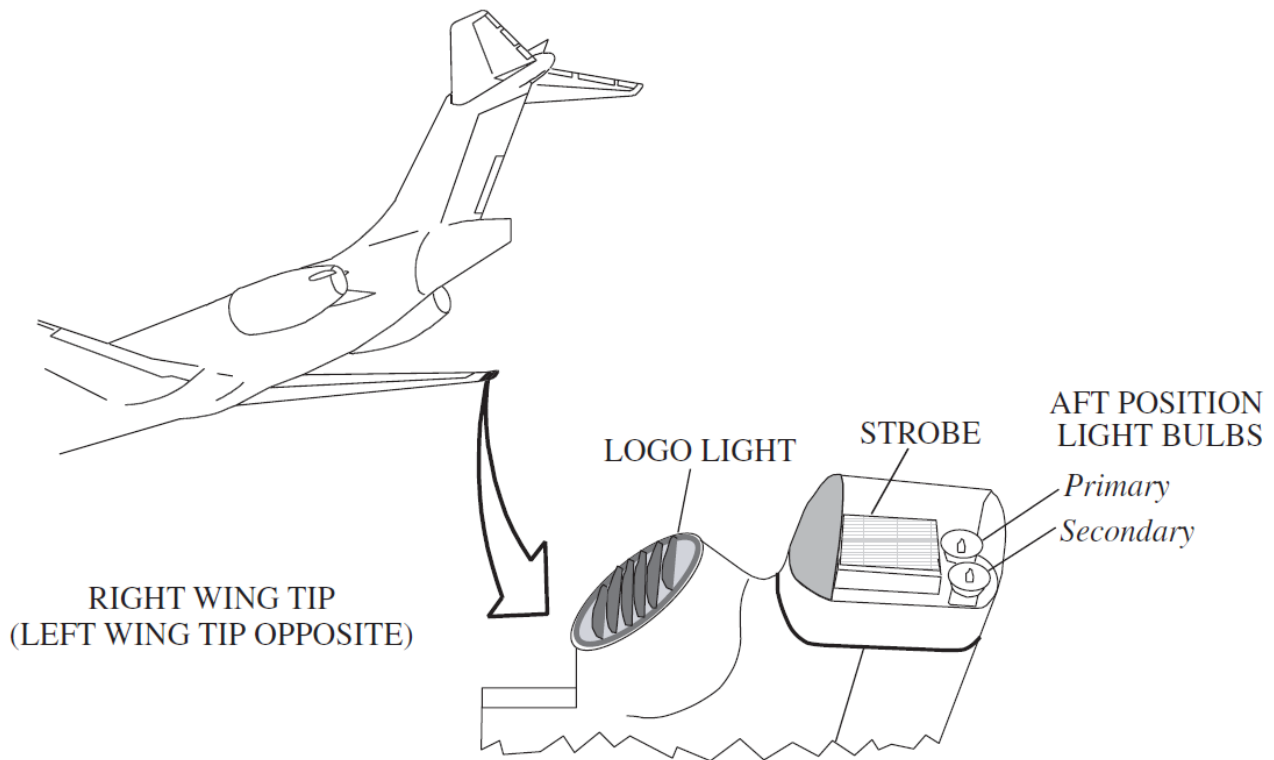
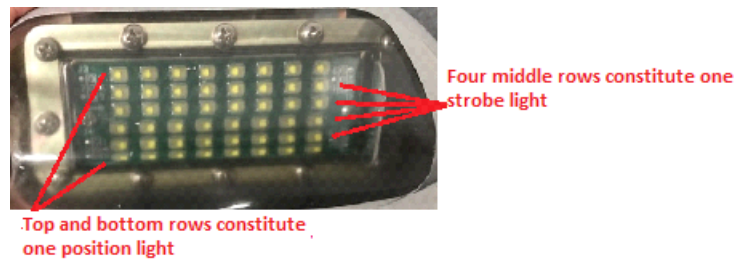


Figure 33-4: Fixed Aft Position Light Bulbs

LED Position light



(M) PROCEDURES

33-42-03C

Reference B-717 [AMM Task 33-42-06-710-801](#) for inspection procedure.

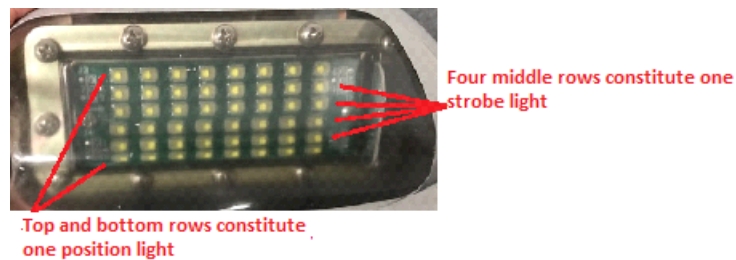
33-42-04 White Strobe / Anti-Collision Lights, (Forward and Aft Wing Tips)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
33-42-04A	White Strobe / Anti-Collision Lights, Non LED , (Forward and Aft Wing Tips) (Pre AA 3342-01019) "Contact MCC"	Y	4	0	C	May be inoperative provided all Fuselage Red Anti-Collision Beacon Lights are operative.
33-42-04B	White Strobe / Anti-Collision Lights, LED , (Forward and Aft Wing Tips) (Post AA 3342-01019)	Y	2	0	C	(M) Seven or more LED segments may be inoperative per Strobe Light provided all Fuselage Red Anti-Collision Beacon Lights are operative. Note: With six or less LED segments inoperative, the strobe light is still considered operative. If seven or more LED segments are inoperative, the strobe light is considered inoperative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-42-04A	Position / Strobe Light Switch	___ White Strobe / Anti-Collision Light(s) INOP
33-42-04B	Position / Strobe Light Switch	___ White Strobe / Anti-Collision Light(s) INOP

Figure 33-5: White Strobe / Anti-Collision Lights

LED Strobe light



(M) PROCEDURES

33-42-04B

Reference B-717 [AMM Task 33-42-06-710-801](#) for inspection procedure.

33-43-01 Wing and Nacelle Illumination Light Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		REPAIR CATEGORY	
FLIGHT CREW MAY PLACARD							
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS	
WING AND NACELLE ILLUMINATION LIGHT SYSTEMS							
33-43-01A	Wing Illumination Lights "DISP.APPR.REQ'D"	Y	2	0	C	(O) May be inoperative provided ground deicing procedures do not require their use.	
33-43-01B	Nacelle Illumination Lights	Y	2	0	C		

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-43-01A	Wing / Nacelle Light Switch	____ Wing Illumination Light(s) INOP
33-43-01B	Wing / Nacelle Light Switch	____ Nacelle Illumination Light(s) INOP

(O) PROCEDURES

33-43-01A

Utilize alternate lighting source for wing inspections as necessary.

33-43-02 Ground Floodlights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
33-43-02	Ground Floodlights	Y	2	0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-43-02	Associated Flood Light Switch(es)	____ Flood Light(s) INOP

33-44-01 Tail Illumination (Logo) Lights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY			
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS	
33-44-01	Tail Illumination (Logo) Lights	Y		2	0	D	

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-44-01	Associated (Logo) Light Switch(s)	____ Tail Illumination (Logo) Light(s) INOP

33-51-01 Emergency Escape Path Marking System Lights

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
EMERGENCY ESCAPE PATH MARKING SYSTEM LIGHTS					
33-51-01A	Photoluminescent Marking System and Lights	N	-	-	C
					Components may be inoperative as follows: - Up to 50% of the Cabin Ceiling Mounted Emergency Lights may be inoperative provided no two adjacent lights are inoperative, - Up to 2 of the 5 bulbs in each exit identifier may be inoperative, - Up to 6 of the 12 bulbs in each exit marker sign may be inoperative, - Up to 50% of the bulbs in each Exit Locator Sign may be inoperative, - Up to 5 inches of any 72 inch photoluminescent strip may be inoperative or missing, and - All photoluminescent strips in the zone from 4 feet forward of the centerline of the forward overwing exit to 4 feet aft of the centerline of the aft overwing exit must operate normally.
33-51-01B	Photoluminescent Marking System and Lights "Contact MCC" "DISP.APPR.REQ'D" "Revenue Passengers Not Authorized"	Y	-	0	C
					(O) May be missing or inoperative provided: - MCC must provide authorization prior to use, - No revenue passengers or cargo are carried, - A maximum of 19 persons are carried as authorized by 14 CFR for non-passenger carrying operations, and - Alternate procedures are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-51-01A	1. Flight Deck Emergency Light Switch 2. Fwd F/A Panel Emergency Light Switch	___ Emergency Escape Path Photoluminescent Strips or Lights INOP
33-51-01B	1. Flight Deck Emergency Light Switch 2. Fwd F/A Panel Emergency Light Switch	Emergency Escape Path Photoluminescent Strips or Lights INOP

(O) PROCEDURES

33-51-01B

- A maximum of 19 persons authorized by 14 CFR 121.583 may be carried. Persons being carried must be approved by the Flight Control Dispatcher.
- Captain will brief all persons authorized to be carried on safety/evacuation procedures.

33-51-02 Exterior Emergency Illumination System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
33-51-02A	Exterior Emergency Illumination System "DISP.APPR.REQ'D"	Y	1	0	B May be inoperative provided airplane is not operated at night.
33-51-02B	Exterior Emergency Illumination System "Contact MCC" "DISP.APPR.REQ'D" "Revenue Passengers Not Authorized"	Y	1	0	C (O) May be missing or inoperative provided: 1. MCC must provide authorization prior to use, 2. No revenue passengers or cargo are carried, 3. A maximum of 19 persons are carried as authorized by 14 CFR for non-passenger carrying operations, and 4. Alternate procedures are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-51-02A	Emergency Light Switch	Exterior Emergency Illumination System INOP
33-51-02B	Emergency Light Switch	Exterior Emergency Illumination System INOP

(O) PROCEDURES

33-51-02B

1. A maximum of 19 persons authorized by 14 CFR 121.583 may be carried. Persons being carried must be approved by the Flight Control Dispatcher.
2. Captain will brief all persons authorized to be carried on safety/evacuation procedures.

33-52-01 Cabin Standby Lighting System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
33-52-01	Cabin Standby Lighting System	Y	1	0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
33-52-01	Near Emergency Power Control Switch	Cabin Standby Lighting System INOP

34 Navigation

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- 34-51-01 VOR Navigation Systems

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- 34-52-01 Distance Measuring Equipment (DME) Systems

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VIA FLIGHT MANAGEMENT SYSTEMS

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34-00-00 LLM Category II/III Status (Lower Landing Minimums)

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION			REMARKS OR EXCEPTIONS
34-00-00L	LLM Category II/III Status (Lower Landing Minimums) "DISP.APPR.REQ'D" Administrative Control Item	Y	-	- (O) (M) Aircraft restricted to CAT I due to AMT is non-AFQ authorized. Note: This Item Number is only used for deferring the CAT II/III status of the aircraft and MCC authorization is required prior to its usage or removal. <u>CAT II/III</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-00-00L	Adjacent to each PFD	CAT II/III NOT AUTHORIZED

(O) PROCEDURES

CAT II - Not Authorized.

CAT III - Not Authorized.

(M) PROCEDURES

1. The AMT will be instructed by MCC to create a new log page in the aircraft logbook using MEL Item 34-00-00L.
2. The deferral code LLM must also be entered on the log page and the placard when deferring this item.
3. MCC authorization is required prior to its usage or removal.

34-24-01 Flight Director Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
34-24-01	Flight Director Systems "DISP.APPR.REQ'D"	Y	2	0	C May be inoperative provided: 1. Approach minimums do not require its use, and 2. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-24-01	Flight Director Command (FD CMD) Switch	____ Flight Director, Ch ____ INOP

34-25-01 Navigation

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
	NAVIGATION				
34-25-01A	Altitude Alerting System "DISP.APPR.REQ'D"	Y	1	0	A (O) May be inoperative provided: 1. Autopilot with altitude hold, and altitude capture operates normally, 2. Enroute operations, ie RVSM, do not require its use, 3. Airplane does not depart from a designated airport (ATL) where repair or replacement can be made, and 4. Repairs are made within 3 flight days. <u>RVSM</u> Not Authorized
34-25-01B	Aural Alert	Y	1	0	C May be inoperative provided: 1. Visual alert operates normally, and 2. Auto-pilot with altitude hold and altitude capture operates normally.
34-25-01C	Visual Alert	Y	1	0	C May be inoperative provided: 1. Aural alert operates normally, and 2. Auto-pilot with altitude hold and altitude capture operates normally.

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-25-01A	Below Captains N.D. Panel	Altitude Alerting System INOP
34-25-01B	Below Captains N.D. Panel	Aural Alert INOP
34-25-01C	Below Captains N.D. Panel	Visual Alert INOP

(O) PROCEDURES

34-25-01A

1. Enroute operations that require altitude alerting (such as RVSM) may not be conducted.
2. Use the autopilot and altitude hold for all operations for which it is appropriate.
3. Cross-check Captain's and First Officer's altitude displays upon reaching and departing assigned altitude. Periodically cross-check altitude indications when maintaining an assigned altitude.
4. Pilot monitoring should call out approaching and departing assigned altitudes.
5. Flight crew must be aware that the usual alerts for altitude deviations will not occur.

34-28-01 Integrated Standby Instrument System (ISIS)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
	INTEGRATED STANDBY INSTRUMENT SYSTEM (ISIS)				
34-28-01	Standby Attitude Indicator "DISP.APPR.REQ'D"	Y	1	0 B	May be inoperative provided: 1. Operations are conducted in VMC only, 2. Operations are not conducted at night, and 3. Operations are not conducted into known or forecast over-the-top conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-28-01	Standby Attitude Indicator	Standby Attitude Indicator INOP

34-29-01 Non-Stabilized Magnetic Compass (Standby)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
34-29-01	Non-Stabilized Magnetic Compass (Standby)	Y	1	0	C	May be inoperative provided two independent sources of heading information are operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-29-01	Aircraft Logbook	Standby Compass INOP

34-31-01 Marker Beacon System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
34-31-01	Marker Beacon System "DISP.APPR.REQ'D"	Y	1	0	C May be inoperative provided approach minimums do not require its use. Note: Approach minimums are not affected.

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-31-01	Aircraft Logbook	Marker Beacon INOP

34-32-01 Multi-Mode Receiver (MMR)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
	MULTI-MODE RECEIVER (MMR)				
34-32-01A	ILS Function "DISP.APPR.REQ'D"	Y	2	1	C May be inoperative provided approach minimums do not require its use, refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements . <u>CAT II/III ILS</u> Not Authorized <u>AUTOLAND</u> Not Authorized
34-32-01B	ILS Function "DISP.APPR.REQ'D"	Y	2	0	A (O) May be inoperative provided: 1. Approach minimums do not require its use, 2. Approval is obtained from arrival ATC facilities to conduct non-simultaneous parallel approach PRIOR TO DEPARTURE, 3. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements , and 4. Operation is limited to one flight to a Delta maintenance station with maintenance personnel on duty. <u>CAT I/II/III ILS, LOC, and LDA</u> Not Authorized <u>AUTOLAND</u> Not Authorized
34-32-01C	GPS Function "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided: 1. Alternate procedures are used, 2. If both GPS systems are inoperative, ADS-B Out Extended Squitter Transmissions are considered inoperative and deferred per item MEL 34-57-01B , and 3. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements .

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-32-01A	Left Of Captains P.F.D.	ILS Function INOP
34-32-01B	Left Of Captains P.F.D.	ILS Function INOP
34-32-01C	Left Of Captains P.F.D.	GPS Function INOP

CONTINUED ON NEXT PAGE

(O) PROCEDURES

34-32-01B

| PRIOR TO EACH DEPARTURE, Ensure the Flight Dispatch Release indicates that ATC approval has been obtained to conduct non simultaneous parallel approach. If ATC approval cannot be confirmed, contact Flight Control for confirmation and an amended Flight Dispatch Release if required.

34-32-01C

1. If GPS is installed as the primary source for long range navigation information, ensure route to be flown does not rely on long range navigation information unless an alternate source is available.
2. If both GPS systems are inoperative, extended Over Water (greater than 50 nm off shore) operations and GPS / RNAV approaches are not permitted.

34-41-01 Weather Radar Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
WEATHER RADAR SYSTEMS					
34-41-01A	Basic Weather Radar System "DISP.APPR.REQ'D"	Y	1	0	C No person may dispatch an aircraft in IFR or night VFR conditions when current weather reports indicate that thunderstorms, or other potentially hazardous weather conditions that can be detected with airborne weather radar, may be reasonably expected along the route to be flown.
34-41-01B	Predictive Windshear Mode	Y	1	0	B (O) May be inoperative provided alternate procedures are used. Note: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.
34-41-01C	Predictive Windshear Mode	Y	1	0	C (O) May be inoperative provided: 1. Alternate procedures are used, and 2. Windshear Warning and Guidance System (Reactive) operates normally.

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-41-01A	Below Captains N.D.	Weather Radar INOP
34-41-01B	Below Captains N.D.	Predictive Windshear INOP
34-41-01C	Below Captains N.D.	Predictive Windshear INOP

(O) PROCEDURES

34-41-01B, 34-41-01C

1. Refer to the [FCTM](#), [Maneuvers](#), [Windshear](#), for a review of windshear detection and escape/recovery techniques during takeoff/departure and approach/landing.
2. Flight crew must use Standard Callout procedures to assure flight crew awareness of airplane configuration, speed, altitude and flight path. Refer to [Operations Manual, Volume 1, Normal Procedures, Standard Callouts](#).
3. Assess and minimize the probability of encountering windshear during takeoff/departure and approach/landing.

34-42-01 Radio Altimeter Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
34-42-01	Radio Altimeter Systems "DISP.APPR.REQ'D"	Y	2	1	C (M) May be inoperative provided: - Approach minimums and/or operational procedures do not require its use, -903 FCC or subsequent installed, and - Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-42-01	Radio Altimeter Indicator(s)	____ Radio Altimeter R/T(s) INOP

(M) PROCEDURES

- Verify both FCC computer part numbers end in "-903" or higher number.

Procedure I

- To access FCC part number via the MCDU:

- Select the MENU key on the MCDU.
- Select CFDS>LRU Maintenance.
- Page over to FCC1(2) prompt.
- Select FCC1(2) Return to Service. The FCC part number will be displayed and the FCC will initiate a short return to service test.
- Press MENU on the MCDU after verifying part number.
- Repeat for the other FCC.

Note: After selecting MENU to exit the FCC, the FCC will reboot. The message FCC FAIL will be displayed until the FCC comes back online.

Procedure II

FCC part numbers may be viewed on the face of the FCC, located in the E&E compartment. Maintenance is Required and Flight Crew May **Not** Placard.

- Pull and secure the appropriate circuit breaker for the inoperative Radio Altimeter.

Row	Column	Number	Name
F	17	B1-1631	RADIO ALTIMETER-1
F	2	B1-1632	RADIO ALTIMETER-2

34-43-01 IRS NAV OFF Lights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
34-43-01	IRS NAV OFF Lights	Y	2	0	C	(O) May be inoperative provided alternate procedures are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-43-01	IRS NAV OFF Lights	IRS NAV OFF Light(s) INOP

(O) PROCEDURES

Verify all NAV displays are normal and no NAV fail indications are present.

34-45-01 Ground Proximity Warning System (GPWS)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
Ground Proximity Warning System (GPWS)					
34-45-01A	Ground Proximity Warning System "DISP.APPR.REQ'D"	Y	1	0	A (O) May be inoperative provided: 1. Alternate procedures are used, 2. Repairs are made within two flight days, and 3. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. Theater Restrictions May Apply
34-45-01B	Terrain Avoidance Modes (Modes 1 - 4) "DISP.APPR.REQ'D"	Y	4	0	A (O) May be inoperative provided: 1. Alternate procedures are used, and 2. Repairs are made within two flight days. Theater Restrictions May Apply
34-45-01C	Test Mode "Contact MCC"	Y	1	0	A (O) May be inoperative provided: 1. The GPWS is not used, and 2. Repairs are made within two flight days.
34-45-01D	Glideslope Deviation Mode Lights (Mode 5)	Y	2	1	C
34-45-01E	Glideslope Deviation Mode Lights (Mode 5)	Y	2	0	B
34-45-01F	Advisory Callouts (Mode 6)	Y	1	0	B (O) May be inoperative provided alternate procedures are used.
34-45-01G	Windshear Mode (Reactive)	Y	1	0	B (O) May be inoperative provided alternate procedures are used. Note: <i>Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.</i>
34-45-01H	Windshear Mode (Reactive)	Y	1	0	C (O) May be inoperative provided: 1. Alternate procedures are used, and 2. Windshear Detection and Avoidance System (Predictive) operates normally.
34-45-01I	Terrain Awareness and Warning System (TAWS) Forward Looking Terrain Avoidance (FLTA) and Premature Descent Alert (PDA) Functions "DISP.APPR.REQ'D"	Y	1	0	B (O) May be inoperative provided alternate procedures are used. Theater Restrictions May Apply
34-45-01J	Terrain Display Functions	Y	2	1	C
34-45-01K	Terrain Display Functions	Y	2	0	B

CONTINUED ON NEXT PAGE

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-45-01A	GPWS Test Switch	GPWS INOP
34-45-01B	GPWS Test Switch	GPWS Modes 1 thru 4 INOP
34-45-01C	GPWS Test Switch	GPWS Test Mode INOP
34-45-01D	GPWS Test Switch	___ GPWS Glideslope Deviation (Mode 5) INOP
34-45-01E	GPWS Test Switch	GPWS Glideslope Deviation (Mode 5) INOP
34-45-01F	GPWS Test Switch	GPWS Advisory Callouts (Mode 6) INOP
34-45-01G	Left Of Captains P.F.D	Windshear Mode (Reactive) INOP
34-45-01H	Left Of Captains P.F.D	Windshear Mode (Reactive) INOP
34-45-01I	Below Captains N.D.	Terrain Awareness and Warning System INOP
34-45-01J	Below Captains N.D.	Terrain Display System Functions INOP
34-45-01K	Below Captains N.D.	Terrain Display System Functions INOP

(O) PROCEDURES

34-45-01A, 34-45-01B, 34-45-01C, 34-45-01F, 34-45-01G, 34-45-01H, 34-45-01I

1. Review Standard Callout procedures. Refer to [Operations Manual, Volume 1, Normal Procedures, Introduction, Standard Callouts](#).
2. Preflight briefing to include review of terrain and/or obstructions in the proximity of the departure and arrival airports.
3. Pilot monitoring flight path during takeoff, approach and landing, will alert the pilot flying if any of the following conditions exist:
 - A. Excessive descent rate.
 - B. Excessive terrain closure rate.
 - C. Altitude loss after takeoff or go-around.
 - D. Unsafe terrain clearance while not in the landing mode.
 - E. Deviation from glideslope.

34-46-01 Traffic Alert / Collision Avoidance System (TCAS)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
TRAFFIC ALERT / COLLISION AVOIDANCE SYSTEM (TCAS)					
34-46-01A	TCAS System "DISP.APPR.REQ'D"	Y	1	0	A (M) May be inoperative provided: - Repairs are made within 3 Flight Cycles, - System is deactivated and secured, - Enroute or approach procedures do not require its use, and - TCAS must be operative on routes outbound from the Continental United States to destinations outside the Continental United State, except Canada. Theater Restrictions May Apply
34-46-01B	Combined TA and RA Dual Displays	Y	2	1	C (O) May be inoperative provided: - TA and RA visual display is operative on the flying pilot side, and - TA and RA audio function is operative on the flying pilot side.
34-46-01C	Resolution Advisory (RA) Display System(s)	Y	2	1	C (O) May be inoperative on non-flying pilot side.
34-46-01D	Resolution Advisory (RA) Display System(s) "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided: - Traffic Alert (TA) visual display and audio functions are operative, - TA only mode is selected by the crew, and - Enroute or approach procedures do not require its use.
34-46-01E	Traffic Alert Display System(s) "DISP.APPR.REQ'D"	Y	2	0	C (O) May be inoperative provided: - RA visual display and audio functions are operative, and - Enroute or approach procedures do not require its use.
34-46-01F	Audio Functions "DISP.APPR.REQ'D"	Y	1	0	B May be inoperative provided enroute or approach procedures do not require use of TCAS.

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-46-01A	TCAS/Transponder Control Panel	TCAS INOP
34-46-01B	Associated TA / VSI	____ TCAS Combined TA and RA Dual Display INOP
34-46-01C	Associated TA / VSI	____ TCAS RA Display System INOP
34-46-01D	Associated TA / VSI	____ TCAS RA Display System(s) INOP
34-46-01E	Associated TA / VSI	____ TCAS TA Display System(s) INOP
34-46-01F	TCAS/Transponder Control Panel	TCAS Audio INOP

CONTINUED ON NEXT PAGE

(O) PROCEDURES

34-46-01B

1. Verify TA and RA elements and audio functions are operative on flying pilot side.
2. Verify TA and RA displays are visible to the flying pilot.

34-46-01C

Corrective action based on Resolution Advisories should be implemented by pilot with the operative RA display.

34-46-01D

Operate with the Function Selector of the Mode S/TCAS Control Unit in the TA position.

34-46-01E

Verify all RA Display and audio functions are operative. (This is displayed on the Weather Radar Indicator).

(M) PROCEDURES

34-46-01A

1. Pull and secure TCAS COMPUTER circuit breaker (B1-1694: Upper EPC panel, location H-15).
 - A. On airplanes incorporating the -01 PSEU, pulling the TCAS COMPUTER circuit breaker will cause PSEU FAULT to appear. TCAS circuit breaker should be reset at least once each flight day and PSEU FAULT should then be verified to extinguish.
 - B. If PSEU FAULT alert does not extinguish, refer to [MEL Item 32-61-02](#) for guidance in dispositioning Proximity Sensing System faults
 - C. Upon completion of check, TCAS COMPUTER circuit breaker should again be pulled and secured.
2. Deselect TCAS on the EIS Control Panel.

34-47-01 Windshear Detection and Flight Guidance Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY			
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS	
WINDSHEAR DETECTION AND FLIGHT GUIDANCE SYSTEMS							
34-47-01A	Reactive "DISP.APPR.REQ'D"	Y	2	0	B	(O) May be inoperative provided alternate procedures are used. Note: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.	
34-47-01B	Reactive "DISP.APPR.REQ'D"	Y	2	0	C	(O) May be inoperative provided: 1. Alternate procedures are used, and 2. Windshear Detection and Avoidance System (Predictive) operates normally.	
34-47-01C	Predictive "DISP.APPR.REQ'D"	Y	-	0	B	(O) May be inoperative provided alternate procedures are used. Note: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.	
34-47-01D	Predictive "DISP.APPR.REQ'D"	Y	-	0	C	(O) May be inoperative provided: 1. Alternate procedures are used, and 2. Windshear Detection and Avoidance System (Reactive) operates normally.	

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-47-01A	Captain's Windshear Alert Light	Reactive Windshear INOP
34-47-01B	Captain's Windshear Alert Light	Reactive Windshear INOP
34-47-01C	Captain's Windshear Alert Light	Predictive Windshear INOP
34-47-01D	Captain's Windshear Alert Light	Predictive Windshear INOP

(O) PROCEDURES

34-47-01A, 34-47-01B, 34-47-01C, 34-47-01D

1. Refer to the [FCTM, Maneuvers, Windshear](#), for a review of windshear detection and escape/recovery techniques during takeoff/departure and approach/landing.
2. Flight crew must use Standard Callout procedures to assure flight crew awareness of airplane configuration, speed, altitude and flight path. Refer to [Operations Manual, Volume 1, Normal Procedures, Standard Callouts](#).
3. Assess and minimize the probability of encountering windshear during takeoff/departure and approach/landing.

34-51-01 VOR Navigation Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
34-51-01	VOR Navigation Systems	Y		2	1	C

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-51-01	VOR Navigation Systems	VOR Navigation System(s) INOP

34-52-01 Distance Measuring Equipment (DME) Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			1	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
34-52-01	Distance Measuring Equipment (DME) Systems "DISP.APPR.REQ'D"	Y	2	1	D May be inoperative provided: 1. Approach procedures do not require its use, and 2. RNAV operations do not require its use; refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements . Note: One or more of these systems may be required as noted on the approach plate.

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-52-01	DME Indicator	____ DME System(s) INOP

34-54-01 ATC Transponder and Automatic Altitude Reporting Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
34-54-01A	ATC Transponder and Automatic Altitude Reporting Systems "DISP.APPR.REQ'D"	Y	2	0	A May be inoperative provided: - Repairs are made within 3 Flight Cycles, - Enroute operations do not require its use, - Prior to flight, approval is obtained from ATC facilities having jurisdiction over the planned route of flight, and - ADS-B Out Extended Squitter Transmissions are considered inoperative and deferred per 34-57-01B - Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>RVSM</u> Not Authorized
34-54-01B	ATC Transponder and Automatic Altitude Reporting Systems "DISP.APPR.REQ'D"	Y	2	1	D May be inoperative provided ADS-B Out Extended Squitter Transmission is considered inoperative and deferred per 34-57-01A
34-54-01C	Elementary and Enhanced Downlink Aircraft Reportable Parameter not Required by 14 CFR	Y	-	0	A May be inoperative provided: - Enroute operations do not require its use, and - Repairs are made prior to completion of the next heavy maintenance check.

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-54-01A	Transponder Control Panel	___ ATC Transponder INOP
34-54-01B	Transponder Control Panel	___ ATC Transponder INOP
34-54-01C	Transponder Control Panel	___ ATC Transponder Downlink Parameters INOP

34-57-01 ADS-B Out Extended Squitter Transmissions

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
34-57-01A	ADS-B Out Extended Squitter Transmissions	Y	2	1	C (O) One may be inoperative. <i>Note:</i> Any ADS-B function that operates normally may be used.
34-57-01B	ADS-B Out Extended Squitter Transmissions "DISP.APPR.REQ'D"	Y	2	0	B (O) May be inoperative provided: 1. Alternate procedures are used, and 2. Authorization is obtained from ATC facilities having jurisdiction over planned route of flight. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements . <i>Note:</i> Any ADS-B function that operates normally may be used.

(O) PROCEDURES

34-57-01A

For one inoperative ADS-B Out, select transponder to operative side.

34-57-01B

Ensure the Flight Dispatch Release indicates that ATC approval has been obtained along the planned route.

If ATC approval cannot be confirmed, contact Flight Control for confirmation and an amended Flight Dispatch Release if required.

34-63-01 VIA Flight Management Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
VIA FLIGHT MANAGEMENT SYSTEMS					
34-63-01A	Computing Functions "DISP.APPR.REQ'D"	Y	2	1	C (O) May be inoperative provided: 1. The associated FMS is not required for operations being conducted, 2. ADS-B Out Extended Squitter Transmission is considered inoperative and deferred per MEL 34-57-01A, and 3. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements.
34-63-01B	Multi-Function Control Display Units (MCDU) "DISP.APPR.REQ'D"	Y	2	1	C (O) Individual portions (Line Select Keys, Remote Light Sensor, Switchlights, etc.) of one or both MCDUs may be inoperative provided: 1. At least one MCDU is capable of initializing IRU alignment, 2. Both MCDUs are capable of allowing manual tuning of Nav Radios, and 3. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. <u>CAT II/III</u> Not Authorized
34-63-01C	Navigation Database out of Currency "DISP.APPR.REQ'D" Administrative Control Item	Y	-	-	(O) (M) May be out of currency provided: 1. Current Aeronautical Charts or current Nav Database is used to verify Navigation Fixes PRIOR TO EACH DEPARTURE, 2. Pilot or Flight Control use charts or RVAL tool to verify status and suitability of Navigational Facilities used to define route of flight, 3. Approach Navigation Radios are manually tuned and identified, and 4. Dispatch will not file a flight plan requiring RNAV data that is identified as changed within the current database, and 5. Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements. Theater Restrictions May Apply Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
34-63-01A	On Associated MCDU	___ VIA FMS Computing Function INOP
34-63-01B	On Associated MCDU	___ MCDU Individual Component(s) INOP
34-63-01C	Adjacent Affected MCDU(S)	___ Navigation Database Out Of Currency

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(O) PROCEDURES

34-63-01A

An inoperative VIA will cause the associated primary navigation displays to be inoperative or unreliable.

For an inoperative VIA, the pilot flying should reference the operative side navigational system source.

■ If necessary select operable EIS with source select switch.

34-63-01B

Verify that Nav Radios can be tuned through both MCDUs.

34-63-01C

PRIOR TO EACH DEPARTURE:

1. Select the route to be flown from the (non-current) database.
2. Check the intended route against current Jeppesen pages for agreement of all navigation data, using applicable departure, enroute and arrival charts, or request Flight Control to validate the database using the RVAL tool.
3. If all information agrees, the FMS may be used normally on the route which was checked. Each new route must be checked in the same manner.
4. Any time the FMS data does not agree with current Jeppesen information, manual tuning and course selection must be accomplished.

(M) PROCEDURES

1. Verify database to be within twenty eight (28) days of the effective and active date range.
2. Enter a statement in the log book such as, "NAV Database out of currency."
3. MCC will CAR the log for update at next available RON.
4. MCC will close the applicable FMS Database Load forecast to prevent over-run.
5. MCC or Maintenance will create a MON log.

35 Oxygen

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35-11-01B	Oxygen Cylinder Discharge Indicator (Exterior Green Disc)
35-21-01	Passenger Oxygen System
	PASSENGER OXYGEN SYSTEM
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35-33-01	Protective Breathing Equipment (PBE) Seals

35-11-01 Crew Oxygen System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
CREW OXYGEN SYSTEM					
35-11-01A	Overhead Panel Oxygen Line Pressure Indicator	Y	1	0	C
(O) May be inoperative provided, PRIOR TO EACH DEPARTURE: 1. Direct reading gauge on the crew oxygen bottle verifies adequate pressure is above the minimum required for dispatch, and 2. Flight crew oxygen system is verified to operate normally. Repetitive Check Required (Yellow Placard)					
35-11-01B	Oxygen Cylinder Discharge Indicator (Exterior Green Disc)	Y	1	0	C
(O) May be damaged or missing.					

MEL ITEM	TAB LOCATION	PLACARD TEXT
35-11-01A	Oxygen Line Pressure Gauge	Oxygen Line Pressure Gauge INOP
35-11-01B	Aircraft Logbook	Green Oxygen Blow Out Disk Damaged or Missing

(O) PROCEDURES

35-11-01A

PRIOR TO EACH DEPARTURE:

1. Ensure crew oxygen cylinder contains sufficient oxygen supply for dispatch.
2. Verify that the direct reading gauge on the crew oxygen bottle indicates normal pressure.
3. Verify flight crew oxygen system operates normally.

35-11-01B

1. Ensure crew oxygen cylinder contains sufficient oxygen supply for dispatch.
2. Verify that the direct reading gauge on the crew oxygen bottle indicates normal pressure.

35-21-01 Passenger Oxygen System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
PASSENGER OXYGEN SYSTEM					
35-21-01A	Passenger Oxygen System "DISP.APPR.REQ'D"	Y	1	0	B (O) May be inoperative provided: 1. Flight is not conducted where the minimum enroute altitude is above 14,000 ft. MSL, 2. Both Air Conditioning Packs operate normally, 3. All other components of the pressurization system operate normally, 4. Airplane remains at or below FL 250, 5. Portable oxygen units are provided for 10% of the passengers, and 6. Passengers are appropriately briefed.
35-21-01B	Passenger Oxygen System "DISP.APPR.REQ'D"	Y	1	0	B (O) May be inoperative provided flight is conducted at or below 10,000 ft. MSL.
35-21-01C	Automatic Door Opening Function "DISP.APPR.REQ'D"	N	1	0	C (O) (M) May be inoperative provided: 1. All oxygen doors are CLOSED and LATCHED, 2. Manual Oxygen Door Deploy System is operative, and 3. The airplane remains at or below FL 300.
35-21-01D	Oxygen Compartment Doors "DISP.APPR.REQ'D"	N	46	-	B (O) (M) May be inoperative provided: 1. Associated oxygen door is UNLATCHED and masks secured, and 2. Associated seats are BLOCKED.
35-21-01E	Oxygen Compartment Doors "DISP.APPR.REQ'D"	N	46	-	B (O) (M) May be inoperative provided: 1. Associated oxygen door is UNLATCHED and masks secured, 2. Passengers occupying associated seats are briefed on access to oxygen masks, and 3. The airplane remains at or below FL 300.
35-21-01F	Oxygen Compartment Doors "DISP.APPR.REQ'D"	N	46	-	B (O) (M) May be inoperative provided: 1. Associated oxygen door is LATCHED, 2. Associated seats are BLOCKED, and 3. If two or more inoperative oxygen doors are adjacent (forward and aft, left and right), seat rows forward and aft of the inoperative oxygen doors are also BLOCKED.
35-21-01G	Lavatory Oxygen "DISP.APPR.REQ'D"	Y	3	0	C (M) May be inoperative provided: 1. Associated lavatory door is locked and closed and placarded "INOPERATIVE – DO NOT ENTER," and 2. Associated lavatory is not used for any purpose. Note: These provisos are not intended to prohibit lavatory inspections by crewmembers.

CONTINUED ON NEXT PAGE

MEL ITEM	TAB LOCATION	PLACARD TEXT
35-21-01A	Passenger Oxygen Mask Switch	Passenger Oxygen System INOP
35-21-01B	Passenger Oxygen Mask Switch	Passenger Oxygen System INOP
35-21-01C	Passenger Oxygen Mask Switch	Automatic Door Opening Function of Passenger Oxygen System INOP
35-21-01D	Passenger Oxygen Mask Switch	____ Oxygen Compartment Doors INOP
35-21-01E	Passenger Oxygen Mask Switch	____ Oxygen Compartment Doors INOP
35-21-01F	Passenger Oxygen Mask Switch	____ Oxygen Compartment Doors INOP
35-21-01G	Passenger Oxygen Mask Switch	____ Lavatory Oxygen System INOP

(O) PROCEDURES

35-21-01A

1. Verify minimum enroute altitude is not greater than 14,000 feet MSL.
2. Limit flight altitude to a maximum of FL 250.
3. Modify passenger briefing to accommodate this operation.

35-21-01B

Limit flight altitude to a maximum of 10,000 ft. MSL.

35-21-01C

Limit flight altitude to a maximum of FL 300.

35-21-01D, 35-21-01F

No procedure required if affected seats are blocked.

35-21-01E

If affected seats are occupied, limit flight altitude to a maximum of FL 300, and brief passengers on access to masks.

(M) PROCEDURES

35-21-01C

Auto opening feature inoperative – all doors open with cabin altitude less than 14,000 feet MSL:

1. With aircraft unpowered, disconnect wire from the Mask Eject Timer Relay R2-315.
2. Close doors and reopen using OXY MASK EJECT switch located on the forward side of the cockpit galley bulkhead.
3. Block seats for any doors which did not open.
4. Close remaining doors.

Auto opening feature inoperative - doors will not open automatically:

1. Open doors using OXY MASK EJECT Switch.
2. Block seats for any doors which did not open.
3. Close remaining doors.

35-21-01D, 35-21-01E

Individual oxygen compartment door(s) will not remain latched:

1. Block off affected seat(s) if the aircraft is to be flown above FL 300.
2. If affected seat(s) are to be occupied, secure door(s) in the closed position in a manner which will allow quick access by passengers in the event oxygen is required.

35-21-01F

If individual oxygen compartment door(s) will not unlatch, block off affected seats, as required.

35-21-01G

Close and lock the lavatory door and placard: INOPERATIVE – DO NOT ENTER.

35-22-01 No Mask Annunciation System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
35-22-01	NO MASK Annunciation System "DISP.APPR.REQ'D"	Y	1	0	C	May be inoperative provided flight is not conducted above 14,000 feet MSL.

MEL ITEM	TAB LOCATION	PLACARD TEXT
35-22-01	Cabin Oxygen On Light	CABIN OXYGEN ON Light INOP

35-31-01 Portable Oxygen Dispensing Units (Bottle and Mask)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	Y		D	REMARKS OR EXCEPTIONS
35-31-01	Portable Oxygen Dispensing Units (Bottle and Mask) "Contact MCC"	Y	4	3	(O) (M) May be unserviceable or missing provided: 1. Required distribution of serviceable bottles is maintained throughout aircraft, 2. Bottles not properly serviced are replaced, serviced, or removed at the next available maintenance facility, and 3. Maintenance Control Center is notified regarding the unserviceable bottle and location of the removed mask assembly.

MEL ITEM	TAB LOCATION	PLACARD TEXT
35-31-01	Aircraft Logbook	____ Portable Oxygen Dispensing Unit(s) INOP

(O) PROCEDURES

Ensure cabin crew (Flight Leader) is notified of the unserviceable bottle and its location.

(M) PROCEDURES

1. Ensure serviceable bottles are evenly distributed throughout the cabin.
2. Remove mask/hose assembly from the unserviceable unit. Secure the mask in a location where it can be retrieved at the next available maintenance facility.
3. Disconnect mask assembly from oxygen cylinder outlet by rotating hose fitting $\frac{1}{2}$ turn and pulling out.
4. Place a piece of masking tape over the pressure gage of oxygen bottle and label INOP. Position the unserviceable cylinder in holder and fasten quick release clamp.
5. Using tape or other suitable means, cover affected oxygen bottle location placarding.

Note: For locations with two oxygen bottles installed, do **not** cover placard if remaining bottle is operative.

35-33-01 Protective Breathing Equipment (PBE) Seals (Smoke Hood Red Rope Seals)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
35-33-01	Protective Breathing Equipment (PBE) Seals (Smoke Hood Red Rope Seals)	Y	-	0	C (M) Seals may be broken or missing.

MEL ITEM	TAB LOCATION	PLACARD TEXT
35-33-01	Aircraft Logbook	____ PBE Smoke Hood Seal(s) Broken / Missing

(M) PROCEDURES

ONE SEAL BROKEN OR MISSING:

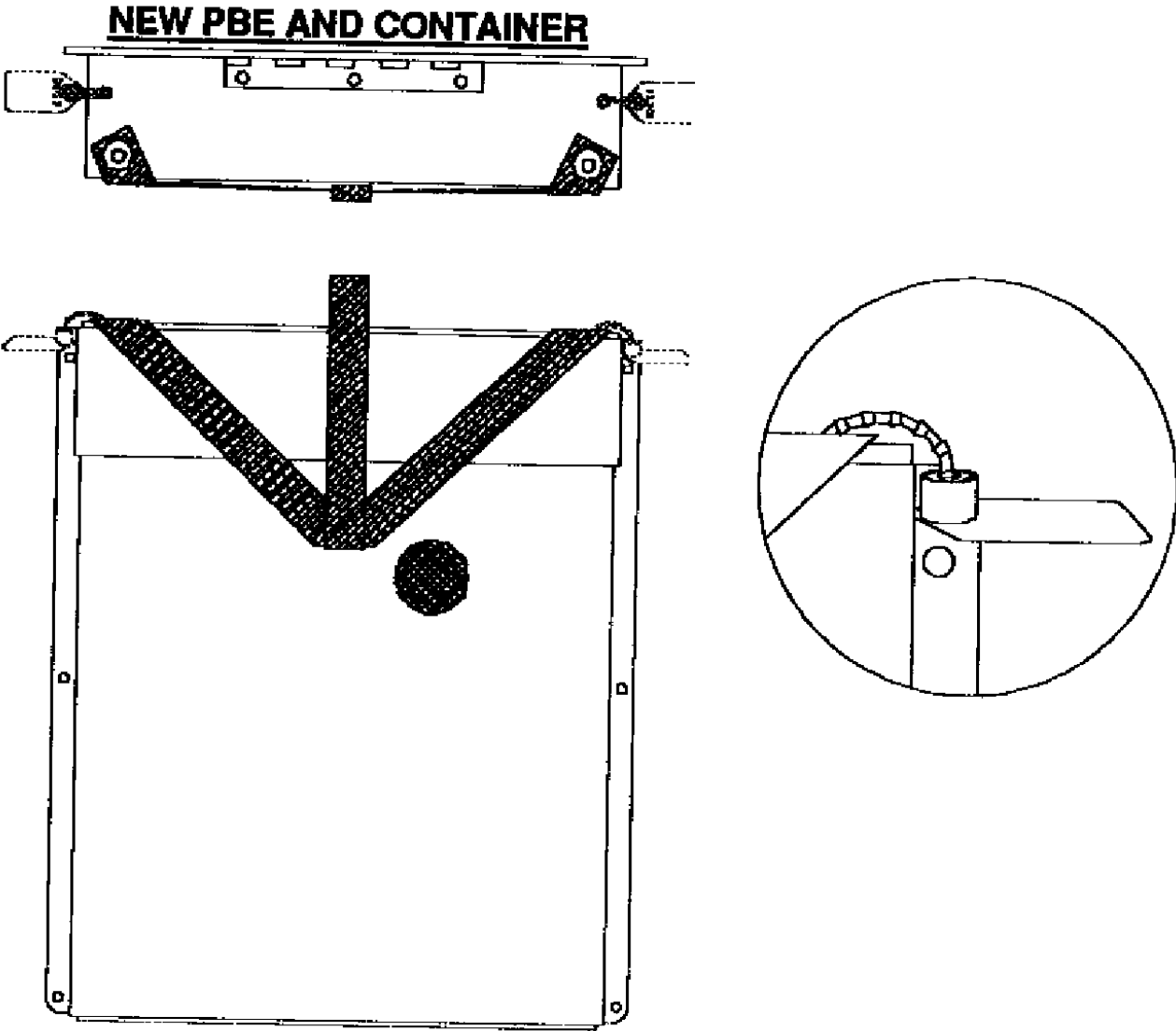
With one seal missing, the PBE is airworthy.

Note: Install the 2nd seal prior to flight or at the next available Delta maintenance station. (Spare seals are located in the Aircraft Log Book.)

BOTH SEALS BROKEN OR MISSING: Maintenance is required and Flight Crew may **not** placard.

1. With both seals missing, remove the PBE and inspect for obvious damage (punctures, rips in pouch etc.).
2. If no obvious damage is found, reseal PBE in container.
3. If PBE is damaged, replace the PBE.

Figure 35-1: Protective Breathing Equipment (PBE) Seals



36 Pneumatic

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36-11-01 Fan Air Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
36-11-01	Fan Air Valve "DISP.APPR.REQ'D"	N	2	0	C
					(O) (M) May be inoperative with the Valve in the OPEN position provided: - Flight is not made in known or forecast icing conditions, and Performance penalties are applied.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-11-01	Below Synoptic Display	____ Fan Air Valve(s) INOP OPEN

(O) PROCEDURES

For dispatch with one or both Valves inoperative refer to ODM - Abnormal Section.

36-11-01	FAN AIR VALVE
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	- Reduce ENROUTE LIMIT WT by 3,400 lbs. per affected valve. - Increase BLOCK FUEL by 1.1% per affected valve.
NOTES	Make a notation on the Flight Plan when Enroute Climb is limiting.
AWABS	- Performance weight corrections have been applied. - Refer to ODM – Abnormal Section to obtain an equivalent weight for adjusted V-Speeds. Use this weight to reference the ODM Takeoff Section to obtain V-Speeds for the applicable Thrust/Flap setting. - Determine all other speeds at actual operating weight.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF - Reduce RATOW by 4,250 lbs. per affected valve. - Reduce T/O CLIMB LIMIT by 4,250 lbs. per affected valve. LANDING - Reduce RWY ALLOWABLE LANDING WT by 2,860 lbs. per affected valve. - Reduce LANDING CLIMB LIMIT WT by 2,860 lbs per affected valve. OTHER CORRECTIONS - Refer to ODM – Abnormal Section to obtain an equivalent weight for adjusted V-Speeds. - Use this weight to reference the ODM Takeoff Section to obtain V-Speeds for the applicable Thrust/Flap setting. - Determine all other speeds at actual operating weight.

(M) PROCEDURES

Disconnect, cap and stow electrical connector from the affected Fan Air Pilot Valve Pilot Pressure Regulator.

36-11-03 High Stage Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
36-11-03	High Stage Valve "DISP.APPR.REQ'D"	N	2	1	C	(O) (M) May be inoperative CLOSED provided flight is not made in known or forecast icing conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-11-03	Below Captains N.D. Panel	____ High Stage Valve INOP CLOSED

(O) PROCEDURES

1. Operate Air Conditioning System associated with the inoperative High Stage Valve as normal.
2. If AIR SYS L/R PRES LO alert for the affected system appears, attempt to extinguish alert by advancing the throttle on the affected engine, conditions permitting.

(M) PROCEDURES

 [Link to Learning Video](#)

1. Gain access to the associated High Stage Pilot Valve located in the nacelle near the Precooler.
2. Disconnect the electrical connector from the High Stage Pilot Valve.
3. Secure the affected High Stage Valve inoperative in the CLOSED position as follows:
 - A. Loosen the Position Indicator screw.
 - B. Slide the Lock into the cover plate recess.
 - C. Tighten the Position Indicator screw.
4. Verify position indicator on the High Stage Valve indicates the Valve is CLOSED.

36-11-04 High Stage Pilot Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD			N	2	1	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION						REMARKS OR EXCEPTIONS
36-11-04	High Stage Pilot Valve “DISP.APPR.REQ’D”				C	(M)	May be inoperative provided: 1. The associated High Stage Valve is secured in the CLOSED position, and 2. Flight is not made in known or forecast icing conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-11-04	Below Captains N.D. Panel	____ High Stage Pilot Valve INOP

OPERATIONS NOTE:

- Operate Air Conditioning System associated with the inoperative High Stage Valve as normal.
- If AIR SYS L/R PRES LO alert for the affected system appears, attempt to extinguish alert by advancing the throttle on the affected engine, conditions permitting.

(M) PROCEDURES

- Gain access to the associated High Stage Pilot Valve located in the nacelle near the Precooler.
- Disconnect the electrical connector from the High Stage Pilot Valve.
- Secure the affected High Stage Valve inoperative in the CLOSED position as follows:
 - Loosen the Position Indicator screw.
 - Slide the Lock into the cover plate recess.
 - Tighten the Position Indicator screw.
- Verify position indicator on the High Stage Valve indicates the Valve is CLOSED.

36-13-01 Pressure Regulator and Shutoff Valve (PRSOV)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
36-13-01	Pressure Regulator and Shutoff Valve (PRSOV) "DISP.APPR.REQ'D"	N	2	1	C	(O) (M) May be inoperative provided: 1. Associated Valve is secured CLOSED, 2. Associated Air Conditioning System is not used, 3. Flight is not made in known or forecast icing conditions, and 4. Aircraft remains at or below FL 250

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-13-01	Associated Bleed Air Switch	____ Pressure Regulator and Shutoff Valve (PRSOV) INOP

(O) PROCEDURES

1. Verify that the isolation valve is selected to AUTO position.
2. Verify the L or R BLEED Switch is in the OFF position.

Note: Higher engine idle speeds will be noted with one pack switch in the off position (Pack Inoperative), therefore careful planning must be done during the descent phase in order to meet altitude crossing restrictions.

3. For takeoff with operative pack ON:
 - A. Verify the inoperative Pack is producing no flow.
 - i. Select the Gasper Fan Switch to the OFF position.
 - ii. Select the inoperative Pack Switch to AUTO.
 - iii. Select the operative Pack Switch to OFF.
 - iv. Verify by feel at the open Air Conditioning outlets that there is no air flow.
 - B. Prior to takeoff, select both Pack Switches to AUTO.

Note: The PACK L/R FLOW DISAG alert associated with the inoperative Pack will be displayed on the EAD Air Page.

- C. After first power reduction following takeoff, select Pack Switch for the inoperative Pack to OFF. Verify associated PACK L/R FLOW DISAG Alert extinguishes.
 - D. Limit flight altitude to a maximum of FL250.
4. For takeoff with both Packs OFF:

Note: The following procedures will automatically shut down the operative Pack upon throttle advancement for takeoff, and restore its operation at 3,000 feet AGL.

Note: Performance data for either Packs OFF or Packs ON may be used. The Packs ON data will be conservative.

- A. Prior to takeoff, verify the Pack Switch for the operative Air Conditioning System is selected to AUTO, and the Pack Switch for the inoperative Air Conditioning System is selected to OFF.
 - B. On the FMS Takeoff Data Entry Page, select/toggle the PACKS option (Line Select Key 2L) to OFF.
 - C. After takeoff, verify operative Air Conditioning System restores normal operation at 3000 feet AGL.
 - D. Limit flight altitude to a maximum of FL250.

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(M) PROCEDURES

 [Link to Learning Video](#)

1. Select the affected L or R BLEED Switch on the Cockpit Overhead Panel to OFF.
2. Gain access to the associated PRSOV located in the nacelle near the Precooler.
3. Secure the affected PRSOV inoperative in the CLOSED position as follows:
 - A. Loosen the Position Indicator screw.
 - B. Slide the Lock into the cover plate recess.
 - C. Tighten the Position Indicator screw.
4. Select the affected L or R Pack Switch on the Cockpit Overhead Panel to OFF.
5. Secure the Flow Control Valve for the affected Air Conditioning System in the CLOSED position as follows:
 - A. Gain access to the valve, located in the aft accessory compartment upstream of the Primary Heat Exchanger.
 - B. Remove the lanyard cap from the Flow Control Valve Actuator tee fitting.
 - C. Verify position indicator on the Valve body indicates the Valve is CLOSED.

36-13-02 Pressure Regulator and Shutoff Pilot Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
36-13-02	Pressure Regulator and Shutoff Pilot Valve "DISP.APPR.REQ'D"	N	2	1	C (O) (M) May be inoperative provided: 1. Associated PRSOV valve is secured CLOSED, 2. Associated Air Conditioning System is not used, 3. Flight is not made in known or forecast icing conditions, and 4. Aircraft remains at or below FL 250.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-13-02	Associated Bleed Air Switch	____ Pressure Regulator and Shutoff Pilot Valve INOP

(O) PROCEDURES

1. Verify that the isolation valve is selected to AUTO position.
2. Verify the L or R BLEED Switch is in the OFF position.

Note: Higher engine idle speeds will be noted with one pack switch in the off position (Pack Inoperative), therefore careful planning must be done during the descent phase in order to meet altitude crossing restrictions.

3. For takeoff with operative pack ON:

- A. Verify the inoperative Pack is producing no flow.
 - i. Select the Gasper Fan Switch to the OFF position.
 - ii. Select the inoperative Pack Switch to AUTO.
 - iii. Select the operative Pack Switch to OFF.
 - iv. Verify by feel at the open Air Conditioning outlets that there is no air flow.
- B. Prior to takeoff, select both Pack Switches to AUTO.

Note: The PACK L/R FLOW DISAG alert associated with the inoperative Pack will be displayed on the EAD Air Page.

- C. After first power reduction following takeoff, select Pack Switch for the inoperative Pack to OFF. Verify associated PACK L/R FLOW DISAG Alert extinguishes.
- D. Limit flight altitude to a maximum of FL250.

4. For takeoff with both Packs OFF:

Note: The following procedures will automatically shut down the operative Pack upon throttle advancement for takeoff, and restore its operation at 3,000 feet AGL.

Note: Performance data for either Packs OFF or Packs ON may be used. The Packs ON data will be conservative.

- A. Prior to takeoff, verify the Pack Switch for the operative Air Conditioning System is selected to AUTO, and the Pack Switch for the inoperative Air Conditioning System is selected to OFF.
- B. On the FMS Takeoff Data Entry Page, select/toggle the PACKS option (Line Select Key 2L) to OFF.
- C. After takeoff, verify operative Air Conditioning System restores normal operation at 3000 feet AGL.
- D. Limit flight altitude to a maximum of FL250.

(M) PROCEDURES

1. Select the affected L or R BLEED Switch on the Cockpit Overhead Panel to OFF.
2. Gain access to the associated PRSOV located in the nacelle near the Precooler.
3. Secure the affected PRSOV inoperative in the CLOSED position as follows:
 - A. Loosen the Position Indicator screw.
 - B. Slide the Lock into the cover plate recess.
 - C. Tighten the Position Indicator screw.
4. Select the affected L or R Pack Switch on the Cockpit Overhead Panel to OFF.

36-13-03 Pneumatic System Controller (PSC) Channels

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION			REMARKS OR EXCEPTIONS
36-13-03	Pneumatic System Controller (PSC) Channels "DISP.APPR.REQ'D"	N 2	1 C	(O) (M) One channel of the PSC may be inoperative provided: 1. Power is removed from the affected PSC Channel, 2. Performance penalties for the associated Fan Air Valve inoperative in the OPEN position are applied, 3. Flight is not made in known or forecast icing conditions, and 4. Associated Air Conditioning System is considered inoperative and deferred per MEL 21-51-01.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-13-03	Below Synoptic Display	____ PSC Controller INOP

(O) PROCEDURES

Refer to (O) PROCEDURES for one Air Conditioning System inoperative given in **MEL 21-51-01**.

36-13-03	PNEUMATIC SYSTEM CONTROLLER (PSC) CHANNELS
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	1. Reduce ENROUTE LIMIT WT by 3,400 lbs. 2. Increase BLOCK FUEL by 1.1%.
NOTES	Make a notation on the Flight Plan when Enroute Climb is limiting.
AWABS	1. Performance weight corrections have been applied. 2. Refer to ODM – Abnormal Section to obtain an equivalent weight for adjusted V-Speeds. Use this weight to reference the ODM Takeoff Section to obtain V-Speeds for the applicable Thrust/Flap setting. 3. Determine all other speeds at actual operating weight.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF 1. Reduce RATOW by 4,250 lbs. 2. Reduce T/O CLIMB LIMIT by 4,250 lbs. LANDING 1. Reduce RWY ALLOWABLE LANDING WT by 2,860 lbs. 2. Reduce LANDING CLIMB LIMIT WT by 2,860 lbs. OTHER CORRECTIONS 1. Refer to ODM – Abnormal Section to obtain an equivalent weight for adjusted V-Speeds. Use this weight to reference the ODM Takeoff Section to obtain V-Speeds for the applicable Thrust/Flap setting. 2. Determine all other speeds at actual operating weight.

(M) PROCEDURES

1. Pull and secure the appropriate circuit breaker for the affected PSC Channel:

Circuit Breaker (CB#)	Panel	Location
PNEUMATIC SYSTEM CONTROL RIGHT (B1-1032)	Lower EPC	T-25
PNEUMATIC SYSTEM CONTROL LEFT (B1-1034)	Lower EPC	S-25

36-13-04 Intermediate Bleed Pressure Sensor (PI)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
36-13-04	Intermediate Bleed Pressure Sensor (Pi) "DISP.APPR.REQ'D"	N	2	1	C (M) May be inoperative provided: 1. The associated High Stage Valve is considered inoperative, deactivated in the CLOSED position and deferred per MEL 36-11-03, and 2. Flight is not made in known or forecast icing conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-13-04	Below Synoptic Display	Intermediate Bleed Pressure Sensor INOP

OPERATIONS NOTE:

- Operate Air Conditioning System associated with the inoperative High Stage Valve as normal.
- If AIR SYS L/R PRES LO alert for the affected system appears inflight, attempt to extinguish alert by advancing the throttle on the affected engine, conditions permitting.

(M) PROCEDURES

- Gain access to the associated High Stage Pilot Valve located in the nacelle near the Precooler.
- Disconnect, cap and stow the electrical connector from the High Stage Pilot Valve.
- Verify position indicator on the High Stage Valve indicates the Valve is CLOSED.

36-13-05 Manifold Pressure Sensor (PM)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
36-13-05	Manifold Pressure Sensor (Pm) "DISP.APPR.REQ'D"	N	2	1	C	(O) May be inoperative provided: 1. Flight is not made in known or forecast icing conditions, and 2. Associated Air Conditioning System is considered inoperative and deferred per MEL 21-51-01.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-13-05	Below Synoptic Display	Manifold Pressure Sensor INOP

(O) PROCEDURES

Refer to (O) PROCEDURES for one Air Conditioning System inoperative given in [MEL 21-51-01](#)

36-13-06 Intermediate Bleed Temperature Sensor (TI)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
36-13-06	Intermediate Bleed Temperature Sensor (Ti) "DISP.APPR.REQ'D"	N	2	0	C May be inoperative provided flight is not made in known or forecast icing conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-13-06	Below Synoptic Display	Intermediate Bleed Temperature Sensor INOP

36-13-07 Manifold Temperature Sensor (TM)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD			N	2	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION						REMARKS OR EXCEPTIONS
36-13-07	Manifold Temperature Sensor (Tm) "DISP.APPR.REQ'D"				C	(O) (M)	May be inoperative provided: 1. Associated Sensor is deactivated, 2. Performance penalties for the associated Fan Air Valve inoperative in the OPEN position are applied, and 3. Flight is not made in known or forecast icing conditions.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-13-07	Below Synoptic Display	Manifold Temperature Sensor INOP

(O) PROCEDURES

For dispatch with one or both Manifold Temperature Sensors inoperative refer to [ODM](#) - Abnormal Section.

36-13-07	MANIFOLD TEMPERATURE SENSOR (TM)
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	1. Reduce ENROUTE LIMIT WT by 3,400 lbs. per affected valve. 2. Increase BLOCK FUEL by 1.1% per affected valve.
NOTES	Make a notation on the Flight Plan when Enroute Climb is limiting.
AWABS	1. Performance weight corrections have been applied. 2. Refer to ODM – Abnormal Section to obtain an equivalent weight for adjusted V-Speeds. Use this weight to reference the ODM Takeoff Section to obtain V-Speeds for the applicable Thrust/Flap setting. 3. Determine all other speeds at actual operating weight.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF 1. Reduce RATOW by 4,250 lbs. per affected valve. 2. Reduce T/O CLIMB LIMIT by 4,250 lbs. per affected valve. LANDING 1. Reduce RWY ALLOWABLE LANDING WT by 2,860 lbs. per affected valve. 2. Reduce LANDING CLIMB LIMIT WT by 2,860 lbs per affected valve. OTHER CORRECTIONS 1. Refer to ODM – Abnormal Section to obtain an equivalent weight for adjusted V-Speeds. Use this weight to reference the ODM Takeoff Section to obtain V-Speeds for the applicable Thrust/Flap setting. 2. Determine all other speeds at actual operating weight.

(M) PROCEDURES

1. Gain access to the Tm sensor in the cowl located in the Pneumatic Supply Duct downstream from the Precooler.
2. Disconnect, cap and stow the electrical connector from the Tm Sensor.

36-13-08 Supply Temperature Sensor (TS)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
36-13-08	Supply Temperature Sensor (Ts)	N	2	0	C	(M) May be inoperative provided associated Sensor is deactivated.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-13-08	Below Synoptic Display	Supply Temperature Sensor INOP

(M) PROCEDURES

1. Gain access to the affected Ts sensor in the aft accessory compartment located in the Pneumatic Supply Duct at the pylon interface.
2. Disconnect, cap and stow the electrical connector from the Ts Sensor.

36-14-01 Ground Connection and Pneumatic Check Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
36-14-01A	Ground Connection and Pneumatic Check Valve "DISP.APPR.REQ'D"	N	1	0	C (O) May be inoperative OPEN provided: 1. Flight is not made in known or forecast icing conditions, 2. Right Air Conditioning System is considered inoperative and deferred per MEL 21-51-01, 3. APU Air is not used, 4. Engine Start on both engines is accomplished using ground Start Air, and 5. Pneumatic Isolation Valve is verified CLOSED after engine start.
36-14-01B	Ground Connection and Pneumatic Check Valve	Y	1	0	C May be inoperative CLOSED.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-14-01A	1. Left of APU AIR Switch 2. Above Isolation Valve Switch 3. Left of Air Data Heat Switch	1. Air/Ground Connection INOP 2. Do Not Use APU Air 3. Do Not Use Airfoil Ice
36-14-01B	Below Captains N.D. Panel	Ground Connection / Pneumatic Check Valve INOP

(O) PROCEDURES

36-14-01A

1. Verify R BLEED Switch is in OFF position.
2. Refer to (O) PROCEDURES for one Air Conditioning System inoperative given in [MEL 21-51-01](#).

MAINTENANCE NOTE:

Utilize ground cart or other pneumatic source for engine start on both engines.

36-15-01 Manifold Isolation Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD			N	1	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION						
36-15-01	Manifold Isolation Valve “DISP.APPR.REQ’D”					C	(O) (M) May be inoperative provided: 1. Flight is not made into known or forecast icing conditions, and 2. Isolation Valve is CLOSED after engine start PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-15-01	Manifold Isolation Valve Switch	____ Manifold Isolation Valve INOP

(O) PROCEDURES

PRIOR TO EACH DEPARTURE:

Coordinate left engine start with maintenance personnel, as follows:

1. With pneumatic supply available, instruct maintenance personnel to open the Manifold Isolation Valve.
2. Start left engine as normal.
3. After engine start is complete, instruct maintenance personnel to close the Manifold Isolation Valve.

Note: The AIR ISOL DISAG alert may be displayed in this dispatch condition. This is normal.

Note: With the APU Air Switch on, the left bleed air will not be available until the APU Air Switch is selected OFF.

(M) PROCEDURES

 [Link to Learning Video](#)

1. Pull and secure the ISOLATION VALVE and ISOLATION VALVE CONTROL circuit breakers (B1-1639 and B1- 1640, located at C-14 and C-15 of the Upper EPC Panel).
2. Gain access to the Manifold Isolation Valve, located in the aft accessory compartment, through aft compartment access panel or through passenger cabin aft bulkhead door.
3. Disconnect, cap and stow the electrical connector from the Manifold Isolation Valve.
4. Coordinate left engine start with flight crew. Upon receiving signal from flight crew, open the Manifold Isolation valve using the manual control knob on the side of the valve.
5. After left engine start is complete, close the valve again.

36-15-02 Manifold Check Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			2	REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION	N				REMARKS OR EXCEPTIONS
36-15-02	Manifold Check Valve	N		0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-15-02	Logbook	Manifold Check Valve INOP

36-23-01 Pneumatic Overheat Detection System (PODS)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
36-23-01A	Pneumatic Overheat Detection System (PODS) "Contact MCC"	Y	1	- C	May be dispatched with PODS FAULT displayed on the STATUS Page provided PODS FAIL alert is not present on the EAD. Note: The PODS will continue to operate normally in this condition.
36-23-01B	Pneumatic Overheat Detection System (PODS) "DISP.APPR.REQ'D"	Y	1	- C	May be dispatched with PODS A-ICE FAULT displayed on the STATUS Page provided: 1. Flight is not made into known or forecast icing conditions, and 2. PODS FAIL alert is not present on the EAD. Note: Tail Compartment overheat detection is not affected in this dispatch condition.

MEL ITEM	TAB LOCATION	PLACARD TEXT
36-23-01A	Below Synoptic Display	Verify No PODS Fail
36-23-01B	Below Synoptic Display	Verify No PODS Fail

38 Water / Waste

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38-30-01	Potable Water Systems
38-30-01A	Potable Water Systems
38-30-01B	Potable Water Systems
38-30-02	Lavatory Systems
38-30-02A	Lavatory Systems
38-30-02B	Lavatory Systems
38-30-02C	Lavatory Systems- Vacuum Blower
38-33-01	Vacuum Waste Logic Control Module
38-33-01	Vacuum Waste Logic Control Module

38-30-01 Potable Water Systems

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
38-30-01A	Potable Water Systems	N	-	-	C
					(O) (M) Individual components may be inoperative provided: - Associated components are deactivated or isolated, and - Associated system components are verified not to have leaks. Note: Any portion of the system which operates normally may be used..
38-30-01B	Potable Water Systems "DISP.APPR.REQ'D"	N	-	0	C
					(O) (M) May be inoperative provided: - System is drained, - Procedures are used to ensure that system is not serviced, and - Performance corrections are applied.

MEL ITEM	TAB LOCATION	PLACARD TEXT
38-30-01A	Aircraft Logbook	Potable Water System ____ INOP
38-30-01B	Aircraft Logbook	Potable Water System INOP

(O) PROCEDURES

38-30-01A

If lavatory water supply has been secured, Captain to brief Lead Flight Attendant/Purser to pour a small bottle/cup of water down each affected lavatory toilet hourly to prevent clogging.

38-30-01B

38-30-01B	POTABLE WATER SYSTEMS
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	None
NOTES	None
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	TAKEOFF None LANDING None OTHER CORRECTIONS (Manual Weight & Balance Worksheet only) Use OEW correction for "Drain Potable Water System" in ALM – Weight and Balance Section .

Captain to brief Lead Flight Attendant/Purser:

- To pour a small bottle/cup of water down each lavatory toilet hourly to prevent clogging.
- Galley water and coffee makers cannot be used.

(M) PROCEDURES

Note: *The Potable Water System may still be used with failed components provided affected components are isolated from the rest of the system.*

Note: *Any system leakage must be cleaned up prior to subsequent dispatch.*

38-30-01A

1. Deactivate inoperative system component(s) and isolate as necessary.
2. Ensure system is properly serviced.
3. Check deactivated component(s) for leakage. No leakage is allowed.
4. If lavatory water heater is too hot, pull and secure the associated lavatory water heater circuit breaker.

Circuit Breaker (CB#)	Panel	Location
LEFT LAVATORY WATER HEATER - FWD (B1-527)	EPC CBP	X-30
LEFT LAVATORY WATER HEATER - AFT (B1-369)	EPC CBP	X-29
RIGHT AFT LAVATORY WATER HEATER (B1-390)	EPC CBP	Z-28

38-30-01B

1. Drain potable water system.
2. Install INOPERATIVE - DO NOT SERVICE placard on the inside of exterior service panel to ensure system is not serviced.

38-30-02 Lavatory Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
38-30-02A	Lavatory Systems	N	3	0	C (M) Individual components may be inoperative provided: 1. Associated components are deactivated or isolated, and 2. Associated system components are verified not to have leaks. <i>Note: Any portion of the system which operates normally may be used.</i>
38-30-02B	Lavatory Systems “DISP.APPR.REQ’D”	N	3	0	C (M) Associated lavatory system (s) may be inoperative provided: 1. Associated components are deactivated or isolated to prevent leaks, 2. The Pilot-in-Command will determine if flight duration is acceptable with a FWD lavatory unusable, and 3. Associated lavatory door(s) is secured closed and placarded “INOPERATIVE – DO NOT ENTER”. <i>Note: These provisions are not intended to prohibit inspections by crewmembers.</i>
38-30-02C	Lavatory Systems- Vacuum Blower “DISP.APPR.REQ’D”	N	1	0	C (O) (M) May be inoperative provided: 1. Vacuum Blower is deactivated, 2. Associated system components are verified not to have leaks, and 3. Lavatories are not used below 18,000 feet.

MEL ITEM	TAB LOCATION	PLACARD TEXT
38-30-02A	Aircraft Logbook	Lavatory System ____ INOP
38-30-02B	Aircraft Logbook	____ Lavatory INOP
38-30-02C	Aircraft Logbook	Lavatory Vacuum Blower INOP Lavatories INOP Below 18,000 Feet

(O) PROCEDURES

38-30-02C

Captain to brief Flight Attendants to lock affected lavatories below 18,000 ft and unlock affected lavatories above 18,000 ft.

(M) PROCEDURES

38-30-02A

Inoperative individual lavatory waste system components:

1. Deactivate inoperative system component(s) and isolate as necessary.
2. Ensure system is properly serviced.
3. Check deactivated components(s) for leakage. No leakage is allowed.
4. If lavatory servicing panel leak is known, suspected, or reported, use MEL item 38-00-02B.

38-30-02B

Inoperative lavatory waste system:

1. Deactivate inoperative system components(s) as necessary. No leakage is allowed.
2. For known, suspected, or reported leaks at a lavatory servicing panel, associated waste tank(s) must be drained per AMM.
3. Lock lavatory door closed and install an INOPERATIVE – DO NOT ENTER placard on the exterior of the associated lavatory door.
4. Install additional INOPERATIVE – DO NOT SERVICE placard on the interior of exterior service panel, as applicable.

38-30-02C

Deactivate the Vacuum Waste Blower by opening and securing Vacuum Blower circuit breaker on the 115VAC Ground Service Bus.

38-33-01 Vacuum Waste Logic Control Module

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
38-33-01	Vacuum Waste Logic Control Module "DISP.APPR.REQ'D"	Y	1	0	C (O) (M) May be inoperative provided: 1. Procedures are used to override the Vacuum Waste LCM, if necessary and 2. Alternate waste tank servicing procedures are used to preclude overflow PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
38-33-01	Aircraft Logbook	Vacuum Waste Logic Control Module INOP

(O) PROCEDURES

1. PRIOR TO EACH DEPARTURE:

- A. Verify with dispatch the availability of lavatory servicing in destination station.

(M) PROCEDURES

 [Link to Learning Video](#)

1. Pull and secure Logic Control Module circuit breaker (located on the AC GROUND SERVICE BUS B1-1149 CBP, mounted in the left console).
2. **PRIOR TO EACH DEPARTURE:**
 - A. Lavatory waste tank must be empty for departure. Verify with ACS that the waste tank has been emptied.

45 Onboard Maintenance Systems (OMS)

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45-45-01	Central Fault Display Interface Unit (CFDIU)
45-45-01	Central Fault Display Interface Unit

45-45-01 Central Fault Display Interface Unit (CFDIU)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
45-45-01	Central Fault Display Interface Unit (CFDIU)	N	1	0	C	(M) May be inoperative provided alternate maintenance procedures are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
45-45-01	Aircraft Logbook	Central Fault Display Interface Unit INOP

(M) PROCEDURES

If a Return-to-Service test is required for any LRU, refer to alternate RTS instructions in the Aircraft Maintenance Manual and/or Component Maintenance Manual, as required.

46 Information Systems

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46-11-01	Electronic Flight Bag (EFB) System
	ELECTRONIC FLIGHT BAG (EFB) SYSTEM
46-11-01A	Power Connection
46-11-01B	Power Connection
46-11-01C	Mounting Device (Permanent)

46-11-01 Electronic Flight Bag (EFB) System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
ELECTRONIC FLIGHT BAG (EFB) SYSTEM					
46-11-01A	Power Connection	Y	-	0	C (O) May be inoperative provided alternate procedures are used.
46-11-01B	Power Connection	Y	-	0	D (O) May be inoperative provided procedures do not require its use.
46-11-01C	Mounting Device (Permanent)	N	-	0	C (O) (M) May be inoperative or missing provided: - Associated EFB and hardware is secured by an alternate means or removed from aircraft, and - Alternate procedures are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
46-11-01A	Logbook	EFB Power Connection INOP
46-11-01B	Logbook	EFB Power Connection INOP
46-11-01C	Logbook	EFB Mount INOP

(O) PROCEDURES

46-11-01A

Refer to [FOM, Chapter 16, Section 1](#) for EFB battery and backup battery requirements.

46-11-01B

Monitor EFB battery power to assure functionality.

46-11-01C

Utilize spare PED Mount.

(M) PROCEDURES

46-11-01C

See [TOPP 40-40-05](#) for reconfiguration instructions using spare PED Mount.

49 Airborne Auxiliary Power (APU)

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49-11-01	APU Enclosure and Access Door Acoustic Blankets
49-17-01	APU Air Inlet Door Actuator System
49-17-01A	APU Air Inlet Door Actuator System
49-17-01B	APU Air Inlet Door Actuator
49-44-01	APU Starter Relay
49-44-01	APU Starter Relay
49-51-01	APU Pneumatic Check Valve
49-51-01A	APU Pneumatic Check Valve
49-51-01B	APU Pneumatic Check Valve
49-51-02	APU Bleed Air Modulating Valve
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49-51-02B	APU Bleed Air Modulating Valve Position Synoptic Indication "APU VALVE DISAG"
49-72-01	APU EGT Probes
49-72-01A	APU EGT Probes
49-72-01B	APU EGT Probes
49-75-01	APU Data Memory Module (DMM)
49-75-01A	APU Data Memory Module (DMM)
49-75-01B	APU Data Memory Module (DMM)

49-10-01 Auxiliary Power Unit (APU)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
49-10-01	Auxiliary Power Unit (APU) "DISP.APPR.REQ'D"	Y	1	0	C (M) May be inoperative provided: 1. APU Control Circuit Breaker is pulled and secured, and 2. APU is not required for Electrical Power.

MEL ITEM	TAB LOCATION	PLACARD TEXT
49-10-01	APU Start Switch	APU INOP

OPERATIONS NOTE:

1. Coordinate with ground crew in advance to ensure ground electrical power from an external source is available, if necessary.
2. Utilize alternate source of pneumatic supply for engine starting (e.g., ground cart, pressurized air bottle, etc.).
3. If electrical power from an external source is not available for initial engine start, proceed as follows:
 - A. Refer to [Operations Manual, Volume 1, Supplementary Procedures, Engines / APU, "Battery Start"](#).
 - B. Note that Autostart feature is not available for a Battery Start, and "Manual Engine Start" Procedure should be used. Also, the EEC will not abort the start for TGT or start timer exceedences, and the flight crew must observe all starting limits and be prepared to abort the start attempt if necessary.

(M) PROCEDURES

1. Visually verify APU Inlet Door is closed. APU Inlet Door may be manually repositioned if it is not fully closed as per the (M) PROCEDURES given in MEL [49-17-01A](#). *If door repositioning is required, Flight Crew may **NOT** placard. Maintenance is required.*
2. Pull and secure APU CONTROL circuit breaker (B1-291: Lower EPC CBP, location Z-39). The "ENG" and "APU FAULT" alert will be displayed on the EAD.

49-11-01 APU Enclosure and Access Door Acoustic Blankets

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
49-11-01	APU Enclosure and Access Door Acoustic Blankets	N		5	0	D

MEL ITEM	TAB LOCATION	PLACARD TEXT
49-11-01	Aircraft Log Book	____APU Enclosure and Access Door Acoustic Blankets damaged or missing.

(M) PROCEDURES

1. Gain access to APU enclosure through fuselage doors underneath APU support structure.
2. Inspect Acoustic Blankets and verify loose or damaged blankets do not present a FOD hazard to the APU. If necessary, remove torn or damaged blankets.

49-17-01 APU Air Inlet Door Actuator System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
49-17-01A	APU Air Inlet Door Actuator System	N	1	0	C (O) (M) May be inoperative and the APU used provided APU Air Inlet Door is secured OPEN or removed. Note: Recommended maximum altitude for starting APU with Inlet Door secured open is 20,000 feet and 225 KIAS or slower.
49-17-01B	APU Air Inlet Door Actuator	N	1	0	C (M) May be inoperative provided: 1. APU Air Inlet door is secured CLOSED, 2. APU Control Circuit Breaker is pulled and secured, and 3. APU is considered inoperative and deferred per MEL 49-10-01.

MEL ITEM	TAB LOCATION	PLACARD TEXT
49-17-01A	APU Start Switch	APU Air Inlet Door Actuator INOP
49-17-01B	APU Start Switch	1. APU Air Inlet Door Actuator INOP 2. Do Not Use APU

(O) PROCEDURES

49-17-01A

1. Verify APU Air Inlet Door is secured OPEN prior to starting APU.
2. If APU is to be started inflight, observe recommended ceiling of 20,000 feet and 225 knots for accomplishing start.

(M) PROCEDURES

 [Link to Learning Video](#)

49-17-01A

1. If APU is not to be used, pull and secure APU CONTROL circuit breaker (B1-291; Lower EPC CBP, location Z-39). The "APU FAULT" alert will be displayed on the EAD.
2. APU Air Inlet Door may be repositioned OPEN as follows:
 - A. Pull and secure APU DOOR CONTROL circuit breaker (B1-290; Lower EPC CBP, location U-39).
 - B. Gain access to the APU Air Inlet Door Actuator located on the upper air inlet duct assembly in the aft accessory compartment.
 - C. Disconnect, cap and stow the electrical connector from the Actuator.
 - D. Remove the Actuator from the APU Air Inlet Door Shaft by removing the four screws holding the Actuator to the APU Air Inlet Duct, and sliding the Actuator off the Door Shaft. After removal of the Actuator, reinstall the bushing and adapter plate assembly, using the original screws and additional washers to ensure the screws do not interfere with the Air Inlet Door. Stow the Actuator in the MCO pouch of fwd cargo.
 - E. A lockout pin is mounted on the top of the APU Inlet Air Duct that is to be used to secure the Air Inlet Door in the open position. Remove the pin from its stowage bracket, invert and reinstall with long side down into door assembly locking door into open position.
 - F. Select the "Door Pinned Open" status through the CFDS:
 - i. From the APU MAINTENANCE PAGE, select the line select key adjacent to More setting the second APU MAINTENANCE page.
 - ii. Select the line select key adjacent to APU DOOR STATUS OPTIONS to show the APU DOOR MENU page.
 - iii. Select DOOR PINNED OPEN.
3. If the APU Air Inlet Door is removed per CDL 49-17-01, perform the following:

CONTINUED ON NEXT PAGE

- A. Pull and secure APU DOOR CONTROL circuit breaker (B1-290: Lower EPC CBP, location U-39).
- B. Gain access to the APU Air Inlet Door Actuator located on the upper air inlet duct assembly in the aft accessory compartment.
- C. Disconnect, cap and stow the electrical connector from the Actuator.
- D. Remove the Actuator from the APU Inlet Door Shaft by removing the four screws holding the Actuator to the APU Air Inlet Duct, and sliding the Actuator off the Door Shaft. Stow the Actuator and bushing.
- E. Install a BACP20B45 plug (or equivalent) into the 0.625 in. diameter hole in the 1911700-2 threaded adapter bearing plug, upper face of the box.
- F. Install a BACP20B49 plug (or equivalent) into the 0.875 in. diameter hole in the 1911700-3 adapter plate for the Actuator side.
- G. Secure the plugs using a bead of PR1422-B1/2 or PR2001-B1/2 sealant or equivalent per AMS-S- 8802, Class B, over and around the plug periphery and metal fingers of the button plugs. The sealant should completely encapsulate the metal fingers to prevent the possibility of FOD.
- H. Reinstall the adapter bearing plug and adapter plate using original screws and additional washers to ensure the screw shanks do not bottom out on the adapter plate nutplate threads.
- I. Select "Door Pinned Open" status through the CFDS:
 - i. From the APU MAINTENANCE PAGE, select the line select key adjacent to More setting the second APU MAINTENANCE page.
 - ii. Select the line select key adjacent to APU DOOR STATUS OPTIONS to show the APU DOOR MENU page.
 - iii. Select DOOR PINNED OPEN.

Figure 49-1: APU Air Inlet Actuator



Figure 49-2: Lockout pin



49-17-01B

APU Air Inlet Door may be repositioned CLOSED as follows:

1. Pull and secure APU DOOR CONTROL circuit breaker (B1-290; Lower EPC CBP, location U-39).
2. Gain access to the APU Air Inlet Door Actuator located on the upper air inlet duct assembly in the aft accessory compartment.
3. Disconnect, cap and stow the electrical connector from the Actuator.
4. Remove the Actuator from the APU Air Inlet Door Shaft by removing the four screws holding the Actuator to the APU Air Inlet Duct, and sliding the Actuator off the Door Shaft. After removal of the Actuator, reinstall the bushing and adapter plate, using the original screws and additional washers to ensure the screws do not interfere with the Air Inlet Door. Stow the Actuator.
5. A lockout pin is mounted on the APU Inlet Air Duct that is to be used to secure the Air Inlet Door in the closed position. Remove the pin from its stowage bracket, reposition the Air Inlet Door to the closed position, and insert the lockout pin to secure the Air Inlet Door in position.

49-44-01 APU Starter Relay

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
49-44-01	APU Starter Relay	N	1	0	C (M) May be inoperative provided APU is considered inoperative and deferred per MEL 49-10-01.

MEL ITEM	TAB LOCATION	PLACARD TEXT
49-44-01	APU MASTER Switch	APU STARTER RELAY INOP

(M) PROCEDURES

1. Pull and secure APU STARTER circuit breaker in the E & E Compartment (B1-1586) location station 120 left hand side.
2. Pull and secure APU CONTROL circuit breaker (B1-291: Lower EPC CBP, location Z-39).

49-51-01 APU Pneumatic Check Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	Y	1	0	C
49-51-01A	APU Pneumatic Check Valve	Y	1	0	C
(M) May be inoperative OPEN provided it is verified that the APU Bleed Air Valve is functioning normally if the APU Air is to be used.					
49-51-01B	APU Pneumatic Check Valve "DISP.APPR.REQ'D"	Y	1	0	C
(O) May be inoperative CLOSED provided APU Pneumatic Distribution System is considered inoperative and is not used.					

MEL ITEM	TAB LOCATION	PLACARD TEXT
49-51-01A	APU AIR Switch	APU Pneumatic Check Valve INOP OPEN
49-51-01B	APU AIR Switch	1. APU Pneumatic Check Valve INOP CLOSED 2. Do Not Use APU Air

(O) PROCEDURES

49-51-01B

If Pneumatic Check Valve is failed CLOSED:

1. Utilize alternate source of pneumatic supply for engine starting (e.g., ground cart, pressurized air bottle, etc.).
 - A. Refer to Vol 1 5.7.9, Starting with APU Air INOP (External AC Electrical Power Available).

(M) PROCEDURES

49-51-01A

If Pneumatic Check Valve is failed OPEN, verify APU Bleed Air Modulating Valve is functioning normally as follows:

1. Start APU, and select APU AIR Switch to ON.
2. On the AIR Synoptic page, observe APU Bleed Air Valve to be in the OPEN position, and verify corresponding pressure in right pneumatic manifold.
3. Select APU AIR Switch to OFF, and verify APU Bleed Air Valve indicates CLOSED.

49-51-02 APU Bleed Air Modulating Valve

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD					REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
49-51-02A	APU Bleed Air Modulating Valve "DISP.APPR.REQ'D"	N	1	0	C	(M) May be inoperative CLOSED provided: 1. APU Air Control Switch remains OFF, and 2. APU Pneumatic Distribution System is placarded inoperative and is not used.
49-51-02B	APU Bleed Air Modulating Valve Position Synoptic Indication "APU VALVE DISAG"	Y	1	0	C	(O) May be illuminated provided Bleed Air Modulating position is verified CLOSED PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
49-51-02A	APU AIR Switch	1. APU Air On/Off Valve INOP CLOSED 2. Do Not Use APU Air
49-51-02B	APU AIR Switch	APU Bleed Air Valve Synoptic Indication Inop

(O) PROCEDURES

49-51-02B

PRIOR TO EACH DEPARTURE:

After use of APU bleed air source, verify that the APU Bleed Air Modulating Valve is closed as follows:

1. Ensure that the APU has been running at governed speed for at least one minute.
2. Place L BLEED and R BLEED engine bleed air supply switches to OFF.
3. Place APU AIR switch to OFF.
4. On EID AIR page display, ensure that system bleed air pressure reads zero.
5. Return L BLEED and R BLEED engine bleed air supply switches to AUTO.

(M) PROCEDURES

49-51-02A

1. Ensure in advance an alternate source of pneumatic supply for engine starting (e.g., ground cart, pressurized air bottle, etc.) is available.
2. Gain access to the APU Bleed Air Modulating Valve through the Right APU Access Door.
3. Disconnect, cap and stow the electrical connector from the valve.

OR

4. Pin the Bleed Modulating Valve in the closed position using the attached ball lock pin.
5. Verify valve position indicator on the valve housing indicates the valve is closed.

Note: Valve is normally spring-loaded closed when power is removed.

49-72-01 APU EGT Probes

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
49-72-01A	APU EGT Probes	Y	2	0	C	(M) May be inoperative provided Auxiliary Power Unit (APU) is considered inoperative and deferred per MEL 49-10-01.
49-72-01B	APU EGT Probes	Y	2	1	C	(M) May be inoperative provided inoperative EGT Probe does not auto shutdown APU.

MEL ITEM	TAB LOCATION	PLACARD TEXT
49-72-01A	Below Synoptic Display	APU EGT Probes INOP Do Not Use APU
49-72-01B	Below Synoptic Display	One APU EGT Probe INOP

(M) PROCEDURES

49-72-01A

Refer to (M) PROCEDURES for Auxiliary Power Unit (APU) inoperative given in MEL 49-10-01.

49-72-01B

 [Link to Learning Video](#)

Note: Auto shutdown of the APU on the ground may result from various failures or exceedances of the APU. A failed or intermittent EGT Probe may result in an auto shutdown of the APU while on the ground.

1. No maintenance action required if inoperative APU EGT Probe does not auto shutdown APU.
2. If auto shutdown of the APU is experienced proceed as follows:
 - A. Gain access to APU EGT Probe through the Right APU Access Door.
 - B. Disconnect, cap and stow the affected EGT Probe Connector on the APU wire harness.
 - C. Confirm APU no longer exhibits auto shutdown.

49-75-01 APU Data Memory Module (DMM)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
49-75-01A	APU Data Memory Module (DMM)	N	1	0	D	May be inoperative provided alternate procedures are used.
49-75-01B	APU Data Memory Module (DMM)	N	1	0	D	May be inoperative or missing provided maintenance programs do not require its use.

MEL ITEM	TAB LOCATION	PLACARD TEXT
49-75-01A	Aircraft Logbook	APU Data Memory Module INOP
49-75-01B	Aircraft Logbook	APU Data Memory Module INOP / Missing

MAINTENANCE NOTE:

49-75-01A

If the APU Data Memory Module (DMM) is inoperative, any information stored by the DMM can be retrieved by performing a fault review of the APU on the CFDS.

52 Doors

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52-11-01	Door Stop Screws
52-11-01	Door Stop Screws
52-54-04	Enhanced Security Flight Deck Door Automatic Locking System
	ENHANCED SECURITY FLIGHT DECK DOOR AUTOMATIC LOCKING SYSTEM
52-54-04A	Boeing / TIMCO Enhanced Security Flight Deck Door Automatic Locking / Access / Control System(s)
52-54-04B	Flight Deck Access Panel System (Keypad)
52-54-04C	Flight Deck Access Panel (Keypad) LEDs
52-54-04D	Flight Deck Access Panel (Keypad) Aural Tone (Beep)
52-54-04E	Flight Deck Door Control Panel Door LOCK FAIL Light
52-54-04F	Flight Deck Door Control Panel Door AUTO UNLK Light
52-54-04G	Flight Deck Door Lock Control Switch
52-54-04H	Flight Deck Door Control Sonalert
52-54-04I	Flight Deck Door Strike Assembly
52-54-05	Boeing / TIMCO Enhanced Security Flight Deck Door Dead Bolt
52-54-05	Boeing / TIMCO Enhanced Security Flight Deck Door Dead Bolt
52-70-01	Door Warning Annunciation Systems
52-70-01	Door Warning Annunciation Systems

52-11-01 Door Stop Screws

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
52-11-01	Door Stop Screws "DISP.APPR.REQ'D"	N	26	25	C (O) One door stop screw may be inoperative or missing on any passenger, service, or cargo door provided flight is conducted in an unpressurized configuration.

MEL ITEM	TAB LOCATION	PLACARD TEXT
52-11-01	Below Captains N.D. Panel	____ Door Stop Screw INOP/Missing

(O) PROCEDURES

1. Refer to [QRH](#), Non-Normal Checklist, Section 2, Air Systems, Unpressurized Flight.
2. Limit flight altitude to 10,000 feet MSL or below.

52-54-04 Enhanced Security Flight Deck Door Automatic Locking System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
ENHANCED SECURITY FLIGHT DECK DOOR AUTOMATIC LOCKING SYSTEM					
52-54-04A	Boeing / TIMCO Enhanced Security Flight Deck Door Automatic Locking / Access / Control System(s)	Y	1	0	C (O) (M) May be inoperative provided: - Automatic locking system is deactivated, - Door dead bolt operates normally and is used to lock the door, and - Alternate procedures are used for locking and unlocking the door using the dead bolt.
52-54-04B	Flight Deck Access Panel System (Keypad)	Y	1	0	C (O) (M) May be inoperative provided: - Automatic locking system is deactivated, - Door dead bolt operates normally and is used to lock the door, and - Alternate procedures are used for locking and unlocking the door using the dead bolt.
52-54-04C	Flight Deck Access Panel (Keypad) LEDs	Y	3	0	C (O) May be inoperative provided alternate procedures are used.
52-54-04D	Flight Deck Access Panel (Keypad) Aural Tone (Beep)	Y	1	0	C (O) May be inoperative provided alternate procedures are used.
52-54-04E	Flight Deck Door Control Panel Door LOCK FAIL Light	Y	1	0	C (M) May be inoperative provided: - Automatic lock controls are verified to operate normally, and - Door Sonalert operates normally.
52-54-04F	Flight Deck Door Control Panel Door AUTO UNLK Light	Y	1	0	C (M) May be inoperative provided: - Automatic lock controls are verified to operate normally, and - Door Sonalert operates normally.
52-54-04G	Flight Deck Door Lock Control Switch	Y	1	0	C (O) (M) May be inoperative provided: - Automatic locking system is deactivated, - Door dead bolt operates normally and is used to lock the door, and - Alternate procedures are used for locking and unlocking the door using the dead bolt.
52-54-04H	Flight Deck Door Control Sonalert	Y	1	0	C (O) (M) May be inoperative provided: - Automatic locking system is deactivated, - Door dead bolt operates normally and is used to lock the door, and - Alternate procedures are used for locking and unlocking the door using the dead bolt.
52-54-04I	Flight Deck Door Strike Assembly	Y	1	0	C (O) (M) May be inoperative provided: - Automatic locking system is deactivated, - Door dead bolt operates normally and is used to lock the door, and - Alternate procedures are used for locking and unlocking the door using the dead bolt.

CONTINUED ON NEXT PAGE

MEL ITEM	TAB LOCATION	PLACARD TEXT
52-54-04A	Flight Deck Door Control Panel	Flight Deck Door Automatic Locking / Access / Control Systems INOP
52-54-04B	Flight Deck Door Control Panel	Flight Deck Access Panel System (Keypad) INOP
52-54-04C	Flight Deck Door Control Panel	Flight Deck Access Panel (Keypad) ____ LED(s) INOP
52-54-04D	Flight Deck Door Control Panel	Flight Deck Access Panel (Keypad) Aural Tone (Beep) INOP
52-54-04E	Flight Deck Door Control Panel	Flight Deck Door Control Panel Door LOCK FAIL Light INOP
52-54-04F	Flight Deck Door Control Panel	Flight Deck Door Control Panel AUTO UNLK Light INOP
52-54-04G	Flight Deck Door Control Panel	Flight Deck Door Lock Control Selector Switch INOP
52-54-04H	Flight Deck Door Control Panel	Flight Deck Door Control Sonalert INOP
52-54-04I	Flight Deck Door Control Panel	Flight Deck Door Strike Assembly INOP

(O) PROCEDURES

52-54-04A, 52-54-04B, 52-54-04G, 52-54-04H, 52-54-04I

The red LOCK FAIL light remains illuminated when the door locking solenoid is retracted either because of a solenoid or keypad failure.

1. To silence the aural alert, remove power from the system by pulling the following circuit breakers:

Row	Column	Number	Name
R	23	B1-998	RAS CONTROL
P	23	B1-999	RAS KEYPAD

2. Position the deadbolt to the Locked, Key Inoperable position.

Note: The deadbolt position allowing access with a key is to be used on the ground only. A second crew member is required to secure the flight deck door deadbolt when a pilot leaves the flight deck. Reference [FOM Chapter 31, Security for flight deck entry/exit procedures](#).

Note: The FAA has authorized the use of the deadbolt for MEL dispatch. Disregard "For Ground Use Only" placard when dispatched under this MEL and procedures require use of the deadbolt.

52-54-04C

During preflight briefing, advise the Purser / FL of the following procedures for inoperative keypad LEDs:

1. Red LED Inoperative - Cabin crew to communicate with flight crew via interphone to confirm cockpit door lock status as necessary.
2. Green LED Inoperative - Sixty seconds after the correct access code is entered on the keypad, the yellow LED will extinguish and the door will unlock for 5 seconds.
3. Yellow LED Inoperative - Sixty seconds after correct access code is entered on the keypad, the green LED will illuminate and the door will unlock for 5 seconds.

Note: Door is set to open 60 seconds after correct access code is entered on keypad. Depending on keypad LED failure, a watch may be used to time the unlock sequence.

Note: Audible cues, such as cockpit door unlock tone and mechanical sounds of the door lock solenoid releasing, may aid in determining when cockpit door is unlocked.

52-54-04D

Ensure Purser / FL is notified of the inoperative function of the keypad. Aural feedback (beep) will not be provided when making keypad entries.

(M) PROCEDURES

52-54-04A, 52-54-04B, 52-54-04G, 52-54-04H, 52-54-04I

Automatic Locking System Deactivation:

Pull and secure the following circuit breakers located on the upper half of the lower EPC:

Row	Column	Number	Name
R	23	B1-998	RAS CONTROL
P	23	B1-999	RAS KEYPAD

52-54-04E, 52-54-04F

Automatic Lock controls and Sonalert operations check:

1. With the flight deck door open supply electrical power to the aircraft.
2. Ensure RAS KEYPAD and RAS CONTROL circuit breakers are closed (ref. step1 table).
3. Verify the door solenoid, located in the ceiling header is in the locked position, pin down.
4. Verify no keypad LED's are illuminated.
5. Enter 4-digit keypad access code:
 - A. An audible "beep" is heard at each keystroke and the amber LED will blink.
 - B. The amber LED illuminates after the correct access code is entered.
 - C. The flight deck Sonalert emits a tone for 2 seconds.
6. At approximately 30 seconds the flight deck Sonalert emits a tone for 2 seconds.
7. At approximately 40 seconds the flight deck Sonalert emit a constant pulsed tone.
8. At approximately 60 seconds:
 - A. The keypad amber LED will extinguish and the green LED will illuminate.
 - B. The flight deck sonalert will emit a constant tone.
 - C. The door solenoid moves to the unlocked position, pin up.
9. At approximately 65 seconds:
 - A. No keypad LED's are illuminated.
 - B. The flight deck Sonalert tone is silenced.
 - C. The door solenoid moves to the locked position, pin down.

52-54-05 Boeing / TIMCO Enhanced Security Flight Deck Door Dead Bolt

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
52-54-05	Boeing / TIMCO Enhanced Security Flight Deck Door Dead Bolt	Y	1	0	C	May be inoperative provided automatic lock controls operate normally.

MEL ITEM	TAB LOCATION	PLACARD TEXT
52-54-05	Flight Deck Door Control Panel	Flight Deck Door Dead Bolt INOP

52-70-01 Door Warning Annunciation Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
52-70-01	Door Warning Annunciation Systems (except TAILCONE UNSAFE)	Y	1	0	C (O) May be inoperative for all associated doors except the forward stairwell door provided it is verified by visual inspection that the associated door is CLOSED and LOCKED PRIOR TO EACH DEPARTURE. Note: The actual cockpit Warning Annunciation may show "UNLOCKED". Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
52-70-01	Below Captains N.D. Panel	____ Door Warning Annunciation INOP

(O) PROCEDURES

PRIOR TO EACH DEPARTURE:

Visually inspect affected door(s) prior to flight to ensure door is properly closed and latched.

53 Fuselage

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53-00-00	RVSM Status
53-00-00V	RVSM Status
53-52-01	Tailcone Unsafe Annunciation System
53-52-01A	TAILCONE UNSAFE Annunciation System
53-52-01B	TAILCONE UNSAFE Annunciation System

53-00-00 RVSM Status

NUMBER INSTALLED			NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION			REMARKS OR EXCEPTIONS
53-00-00V	RVSM Status "DISP.APPR.REQ'D" Administrative Control Item	N	-	- (O) (M) Authorized by MCC when: 1. RVSM operations cannot be used, 2. No specific MEL item applies, and 3. The aircraft must remain at or below FL280. Note: This Item Number is only used for deferring the RVSM status of the aircraft and MCC authorization is required prior to its usage or removal. <u>RVSM Operations</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
53-00-00V	Adjacent to each PFD.	RVSM OPERATIONS NOT AUTHORIZED

(O) PROCEDURES

1. RVSM operations - Not Authorized.
2. Limit altitude to FL280 or below.

(M) PROCEDURES

1. The AMT will be instructed by MCC to create a new log page in the aircraft logbook using the appropriate MEL Item Number.
2. The deferral code VSM must also be entered on the log page and the placard when deferring this item.
3. MCC authorization is required prior to its usage or removal.

53-52-01 Tailcone Unsafe Annunciation System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY			
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS	
53-52-01A	TAILCONE UNSAFE Annunciation System (Tailcone Unlock message) "DISP.APPR.REQ'D"	Y	1	0	C	May be inoperative provided: 1. Airplane is operated in VMC only, and 2. Airplane is not operated at night. Note: System includes Tailcone Release Mechanism Decal located over slot in tail exit door frame.	
53-52-01B	TAILCONE UNSAFE Annunciation System (Tailcone Unlock message) "Contact MCC"	N	1	0	A	(M) May be inoperative provided: 1. Tailcone lockpins are verified to be properly seated and rotated within the lock, 2. Tailcone locking cable is properly secured, 3. No evidence of slack exists on cable leads, and 4. System is repaired within 2 flight days. Note: System includes Tailcone Release Mechanism Decal located over slot in tail exit door frame.	

MEL ITEM	TAB LOCATION	PLACARD TEXT
53-52-01A	Below Captains N.D. Panel	TAILCONE UNSAFE Annunciator INOP
53-52-01B	Below Captains N.D. Panel	TAILCONE UNSAFE Annunciator INOP

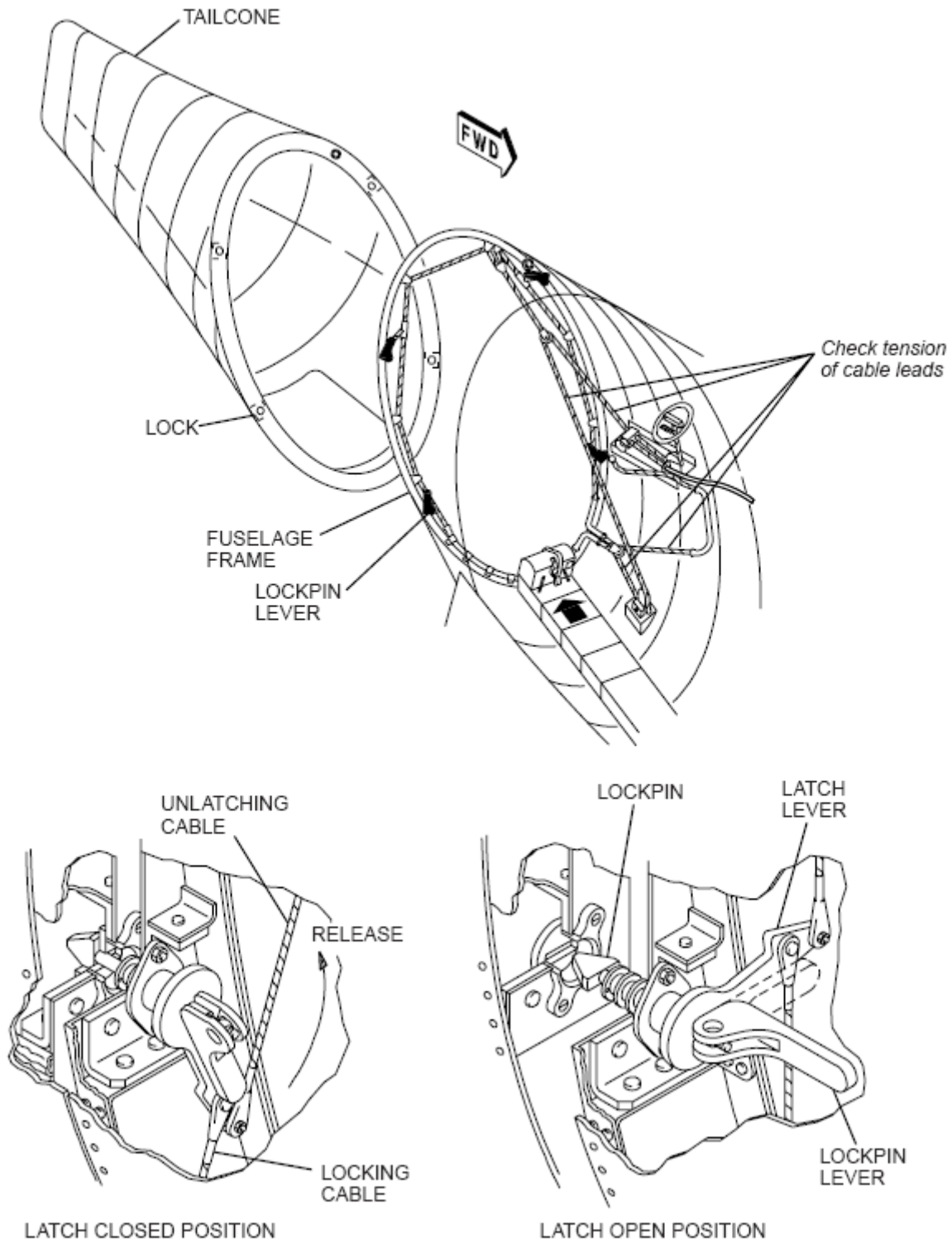
(M) PROCEDURES

 [Link to Learning Video](#)

53-52-01B

If airplane is to dispatch with TAILCONE UNSAFE Annunciation System inoperative in other than Day VMC conditions, verify tailcone is properly secured as follows:

1. Gain access to tailcone compartment.
2. Visually verify lockpins are seated and rotated within the lock.
3. Verify that locking cable is properly secured by pulling sharply on all four leads of the cable at the clamp located just to the left of centerline and forward of the tailcone mating flange. All four leads must be taut with no evidence of slack or tendency to uncouple.

Figure 53-1: TAILCONE UNSAFE Annunciation System


73 Engine Fuel and Control

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73-21-01	Electronic Engine Control Systems (EEC) Short Term Dispatch (STD) Fault
73-21-01A	Electronic Engine Control Systems (EEC) Short Term Dispatch (STD) Fault
73-21-01B	Electronic Engine Control Systems (EEC) Long Term Dispatch (LTD) Fault
73-21-02	EEC Normal (EPR) Mode
73-21-02	EEC Normal (EPR) Mode
73-21-03	FADEC MODE Switchlights
	FADEC MODE SWITCHLIGHTS
73-21-03A	SELECT Lights
73-21-03B	ALTN Lights
73-21-16	Electronic Engine Control (EEC) Cooling Fan
73-21-16	Electronic Engine Control (EEC) Cooling Fan
73-31-01	Fuel Flow / Fuel Used Indication Systems
73-31-01	Fuel Flow / Fuel Used Indication Systems
73-33-01	ENG L/R FUEL PRES Annunciation System
73-33-01	ENG L/R FUEL PRES Annunciation System
73-34-01	L/R FUEL FIL PRES DROP Annunciation Systems
73-34-01	L/R FUEL FIL PRES DROP Annunciation Systems

73-21-01 Electronic Engine Control Systems (EEC) Short Term Dispatch (STD) Fault

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
73-21-01A	Electronic Engine Control Systems (EEC) Short Term Dispatch (STD) Fault	Y	2	- A	(M) May be dispatched with EEC STD faults provided repairs are made within a maximum of 150 elapsed flight hours.
73-21-01B	Electronic Engine Control Systems (EEC) Long Term Dispatch (LTD) Fault	Y	2	- A	(M) May be dispatched with EEC LTD faults provided repairs are made within a maximum of 500 elapsed flight hours.

MEL ITEM	TAB LOCATION	PLACARD TEXT
73-21-01A	Below Captains N.D. Panel	STD EEC Fault
73-21-01B	Below Captains N.D. Panel	LTD EEC Fault

(M) PROCEDURES

73-21-01A, 73-21-01B

Note: If STD or LTD fault causes EEC to operate in ALTN (N1) mode, [MEL 73-21-02](#) must also be applied.

1. Appearance of ENG L SYS FAIL or ENG R SYS FAIL on the EAD indicates presence of a nondispatchable fault.
2. Short-term dispatchable EEC Faults are indicated by ENG L SYS FAULT or ENG R SYS FAULT on the EAD. Short-term dispatch (STD) faults are to be logged and repaired within a maximum of 150 elapsed flight hours.
3. There are no immediate maintenance actions required for dispatch with Long Term Dispatch (LTD) EEC faults. LTD faults are managed as part of the operator's scheduled maintenance program. All LTD faults must be repaired within a maximum periodic inspection/repair interval not to exceed 500 flight hours. If LTD faults are found during Non-EEC scheduled inspection, the LTD faults may be deferred and treated as if it was found at the next periodic inspection for LTD faults.

73-21-02 EEC Normal (EPR) Mode

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
73-21-02	EEC Normal (EPR) Mode "DISP.APPR.REQ'D"	Y	2	0	C
					(O) EEC Normal Mode may be inoperative provided: - Both EEC's are selected to the ALTN mode, - Autothrottles are deferred per MEL 22-30-01, Performance corrections are applied, - Refer to Navigation Equipment Tab (this manual) for RNAV/LLM Minimum Requirements, and - The airplane remains at or below FL 250. <u>CAT I/II/III</u> Not Authorized <u>AUTOLAND</u> Not Authorized

MEL ITEM	TAB LOCATION	PLACARD TEXT
73-21-02	Below Captains N.D. Panel	EEC Normal (EPR) Mode INOP

(O) PROCEDURES

- AWABS takeoff data is provided for DRY or SLP-WET runway conditions only. Takeoffs using ICY, QCTR, or HTRC data are not authorized.
- Use of Reduced Takeoff Thrust (TO FLX) or Derated Takeoff Thrust is prohibited.
- Select both FADEC MODE Switches (Overhead Panel) to ALTN. Observe ENG L FADEC ALTN and ENG R FADEC ALTN Alerts displayed on SED.
- Autothrottles will be inoperative.
- Takeoff procedures remain unchanged, except that thrust is to be set manually using N1 power settings in lieu of EPR. Refer to ODM Abnormal Section.
- Takeoff thrust setting technique is as follows:
 - Prior to brake release, advance throttles to obtain approximately 75% N1.
 - Release brakes, and advance throttles to reach target N1 prior to 60 knots.
- At least one MCDU should be selected to the PROGRESS page for the duration of the flight. Access this page by pressing the PROG key on the MCDU. Note that the N1 thrust setting tables of ODM Abnormal Section are a function of Ambient Temperature (SAT). Refer to TEMP display on the PROGRESS Page to determine target N1 setting for Max Continuous Thrust and for Go-Around Thrust.
- For Climb and Cruise Thrust, the EPR Limit values may be set on the engine with the valid EPR display, and the N1 values then matched between the two engines. Flight crew must ensure that N1 does not exceed the appropriate N1 limits for Max Continuous Thrust. Refer to ODM, Abnormal Section for information.

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73-21-02	EEC NORMAL (EPR) MODE
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	Reduce ENROUTE LIMIT WT by 13,450 lbs.
NOTES	Make a notation on the Flight Plan when Enroute Climb is limiting. 1. AWABS Takeoff data is provided for DRY or SLP-WET runway conditions only. Takeoffs using ICY, QCTR or HCTR data are not authorized. This item uses the EEC TPI which can be very restrictive. Be aware that any flight using this relief is likely to be weight limited. There are also RALW and LCLW penalties of 10,560 lbs. each associated with this item.
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	Manual WDR not permitted with MEL 73-21-02.

73-21-03 FADEC MODE Switchlights

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
	FADEC MODE SWITCHLIGHTS					
73-21-03A	SELECT Lights	Y	2	0	C	May be inoperative provided associated ALTN Light is operative.
73-21-03B	ALTN Lights	Y	2	0	C	May be inoperative provided associated SELECT Light is operative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
73-21-03A	Near Affected FADEC MODE Push Button Switch	____ FADEC SELECT Light(s) INOP
73-21-03B	Near Affected FADEC MODE Push Button Switch	____ FADEC ALTN Light(s) INOP

73-21-16 Electronic Engine Control (EEC) Cooling Fan

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
73-21-16	Electronic Engine Control (EEC) Cooling Fan	Y	2	0	A May be inoperative provided repairs are made prior to the completion of the next heavy maintenance visit.

MEL ITEM	TAB LOCATION	PLACARD TEXT
73-21-16	Below Captains N.D. Panel	Electronic Engine Control (EEC) Cooling Fan INOP

73-31-01 Fuel Flow / Fuel Used Indication Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
73-31-01	Fuel Flow / Fuel Used Indication Systems	Y	2	1	B	(O) May be inoperative provided N1, EPR, Engine Vibration, and Main Tank Fuel Quantity Indicating Systems are operative for associated engine.

MEL ITEM	TAB LOCATION	PLACARD TEXT
73-31-01	Below Captains N.D. Panel	____ Fuel Flow Rate / Fuel Used INOP

(O) PROCEDURES

Closely monitor EGT during engine start.

73-33-01 ENG L/R FUEL PRES Annunciation System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
73-33-01	ENG L/R FUEL PRES Annunciation System	Y	2	1	C (O) May be inoperative provided adequate pump pressure for all operative fuel pumps is verified PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
73-33-01	Near Fuel Boost Pump Switches	___ INLET FUEL PRES LOW Annunciation INOP

(O) PROCEDURES

PRIOR TO EACH DEPARTURE:

The following procedure should be performed to verify proper operation of the affected main tank boost pumps:

1. Verify all Boost Pumps are OFF.
2. Select FUEL XFEED Lever to ON.
3. For an inoperative ENG L FUEL PRES Annunciation System, proceed as follows:
 - A. Select the Left Fwd Boost Pump ON.
 - B. Select the Right Fuel Switch to ON.
 - C. Verify ENG R FUEL PRES alert is not present on EAD.
 - D. Repeat these steps using the Left Aft Boost Pump to verify its operation.
 - E. Select the Right Fuel Switch to OFF.
4. For an inoperative ENG R FUEL PRES Annunciation System, proceed as follows:
 - A. Select the Right Fwd Boost Pump ON.
 - B. Select the Left Fuel Switch to ON.
 - C. Verify ENG L FUEL PRES alert is not present on EAD.
 - D. Repeat these steps using the Right Aft Boost Pump to verify its operation.
 - E. Select the Left Fuel Switch to OFF.
5. Select the FUEL XFEED Lever to OFF.

73-34-01 L/R FUEL FIL PRES DROP Annunciation Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
73-34-01	L/R FUEL FIL PRES DROP Annunciation Systems	N	2	1	B	(M) May be inoperative provided it is verified that the malfunction is in the annunciation system.

MEL ITEM	TAB LOCATION	PLACARD TEXT
73-34-01	Below Captains N.D. Panel	____ L/R FUEL FIL PRES DROP Annunciation INOP

(M) PROCEDURES

Inspect Fuel Filter element. Verify filter element is clean per AMM.

74 Engine Ignition

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74-20-01	Ignition Systems
74-20-01	Ignition Systems

74-20-01 Ignition Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
74-20-01	Ignition Systems	Y	4	2 C	(O) Two may be inoperative provided both inoperative systems are not on the same engine.

MEL ITEM	TAB LOCATION	PLACARD TEXT
74-20-01	Next To Ignition Switch	____ Ignition System(s) INOP

(O) PROCEDURES

If circuit breaker for affected Ignitor is pulled, Auto Engine Start will be available. If circuit breaker for affected Ignitor is not pulled, Auto Engine Start will not be available; start engine per manual. Engine start procedure in [Vol I, Supplementary Procedures, Chapter 5, Engines/APU](#).

MAINTENANCE NOTE:

If Auto Engine Start is to be used, pull and secure appropriate circuit breaker for the affected Ignitor:

Circuit Breaker (CB#)	Panel	Location
LEFT ENGINE IGNITOR-1 (B1-1500)	EPC CBP	S-39
LEFT ENGINE IGNITOR-2 (B1-1501)	Emergency CBP (Overhead)	B-8
RIGHT ENGINE IGNITOR-1 (B1-1502)	EPC CBP	X-39
RIGHT ENGINE IGNITOR-2 (B1-1503)	Emergency CBP (Overhead)	B-9

77 Engine Indicating

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77-32-01	Engine Vibration Monitoring Systems
77-32-01	Engine Vibration Monitoring Systems

77-32-01 Engine Vibration Monitoring Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
77-32-01	Engine Vibration Monitoring Systems	Y	2	0	D	May be inoperative provided N1, EPR and Fuel Flow Indicating Systems are operative for associated engine.

MEL ITEM	TAB LOCATION	PLACARD TEXT
77-32-01	Below Synoptic Display	Engine Vibration Monitoring System(s) INOP

78 Engine Exhaust

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78-33-01	Thrust Reverser System Accumulators
78-33-02	Thrust Reverser Linear Variable Transformers (LVT)
78-33-02	Thrust Reverser Linear Variable Transformers (LVT)
78-33-03	Thrust Reverser Proximity Sensors
78-33-03	Thrust Reverser Proximity Sensors
78-33-04	Maintenance Test Enable Switch (MTES)
78-33-04	Maintenance Test Enable Switch (MTES)

78-32-01 Thrust Reverser Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
78-32-01	Thrust Reverser Systems "DISP.APPR.REQ'D"	N	2	1	C (O) (M) May be inoperative provided: 1. Inoperative reverser is STOWED and DEACTIVATED, 2. Antiskid System is operative, and 3. Performance corrections are applied.

MEL ITEM	TAB LOCATION	PLACARD TEXT
78-32-01	Below Engine Display	____ Thrust Reverser INOP

(O) PROCEDURES

- Verify amber U/L indication is not present on EAD.
- During walk around, verify red lockout pins are visibly protruding from the aft lockout covers (outboard and inboard) of the inoperative reverser.
- Crew should be advised that movement of the inoperative reverser handle will cause associated ENG L/ R SYS FAULT alert to appear.
- Unaffected reverser may be used within limits of directional controllability.
- Tailwind takeoffs are only permitted from DRY runways.

78-32-01	THRUST REVERSER SYSTEMS
EFFECTIVITY	All
FLIGHT PLANNING REQUIREMENTS	None
NOTES	1. AWABS Takeoff data is provided for DRY or SLP-WET runway conditions only. Takeoffs using ICY, QCTR or HCTR data are not authorized. AWABS Takeoff data is provided for SLP-WET runways with a slope greater than -1.0%. Tailwind takeoffs are not allowed from SLP-WET runways. There are no such restrictions from DRY runways.
AWABS	Performance corrections have been applied.
TAKEOFF PERFORMANCE WORKSHEET	Manual WDR not permitted with MEL 78-32-01.

(M) PROCEDURES

Deactivate affected Thrust Reverser (with engine shut down) as follows:

1. Visually verify Thrust Reverser Doors are fully closed. When properly closed, upper and lower doors should be faired at the leading edge, and amber U/L indication will be extinguished.
2. If the thrust reverser is not fully stowed or if the amber U/L indication is present, attempt to re-stow the affected Thrust Reverser by cycling it as follows:
 - A. Turn on the Aux Hydraulic Pump and, if necessary, Hydraulic Power Transfer Unit (PTU).
 - B. Open the Thrust Reverser Hydraulic Control Unit (HCU) access panel and locate the red-guarded Maintenance Test Enable Switch (MTES).
 - C. Establish communication between a person at the MTES and the flight deck. From the flight deck, coordinate with person at MTES to push and hold Switch.
 - D. Cycle the affected reverser handle from forward idle to reverse thrust position.
 - E. Return reverser handle to forward idle, and allow MTES to be released.

Note: *If Thrust Reverser Doors cannot be closed using the above steps, or if amber U/L indication is still present, refer to FIM for further troubleshooting.*

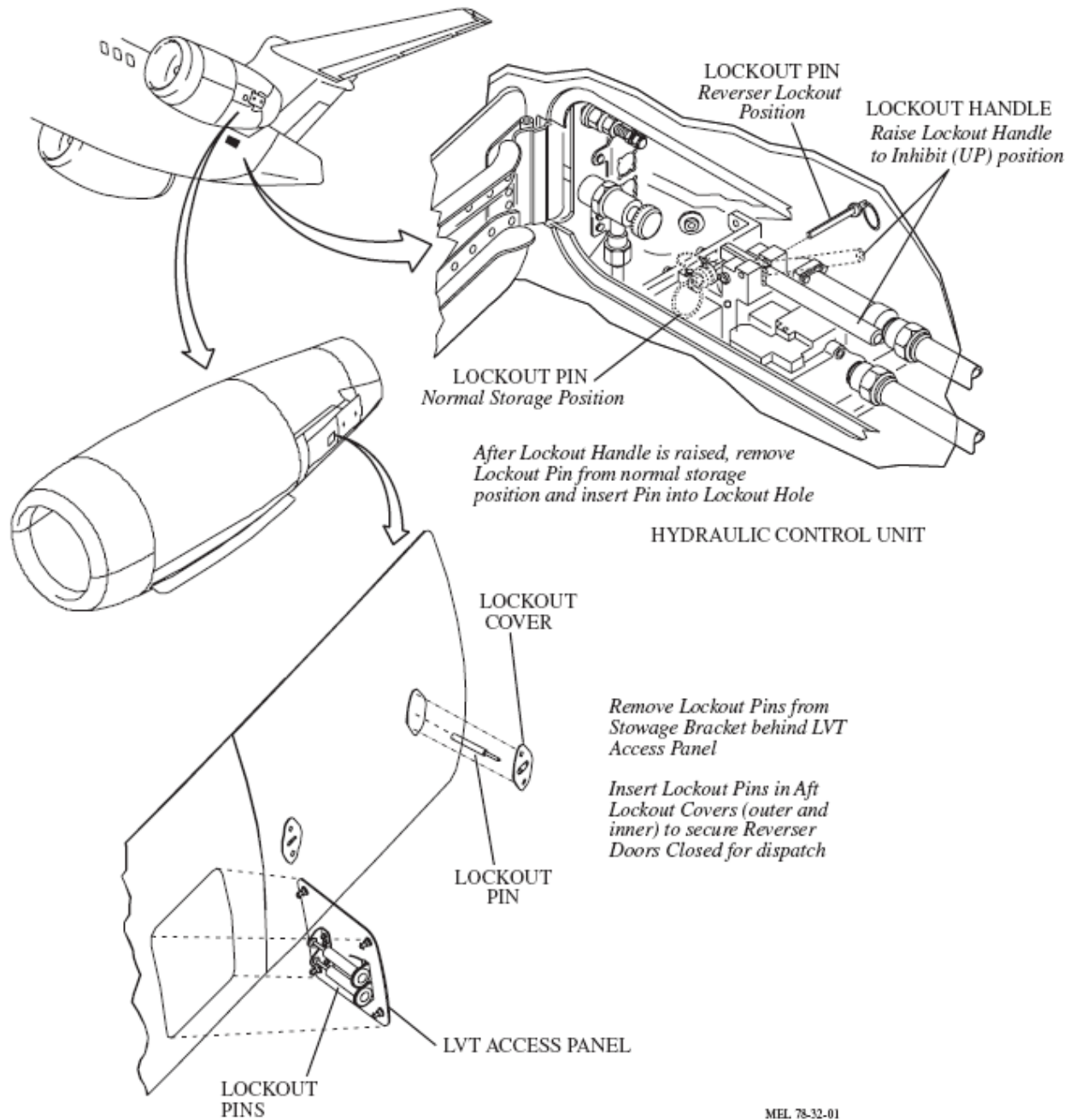
3. After Thrust Reverser Doors are fully closed:

- A. Pull and secure the appropriate circuit breaker for the affected reverser, as follows:

Circuit Breaker (CB#)	Panel	Location
LEFT ENGINE THRUST REVERSER (B1-1508)	EPC CBP	S-38
RIGHT ENGINE THRUST REVERSER (B1-1511)	EPC CBP (Overhead)	B-15

- B. Open the Thrust Reverser Hydraulic Control Unit (HCU) access panel.
 - C. Move the lockout lever on the HCU to the locked position.
 - D. Remove the HCU Lockout Pin from the stowage clip and install it in the lockout hole of the HCU.
 - E. Lock the upper and lower reverser doors closed as follows (refer to [Figure 78-1](#) on next page):
 - i. Remove the outboard LVT Access Panel on the associated engine.
 - ii. Remove the two Reverser Lockout Pins from their stowage bracket mounted behind the LVT Access Panel.
 - iii. Remove the Aft Lockout Covers at the outboard and inboard locations.
 - iv. Install the larger diameter end of the lockout pin into the round hole of the actuator carriage (outboard and inboard) behind each Aft Lockout Cover. Note that the round lockout hole should align properly with the center of the Lockout Cover when the reverser is properly stowed, and the lockout pin should fit securely into the hole.
 - v. Attach and secure the Aft Lockout Covers (outboard and inboard).
 - vi. Close and secure the LVT Access Panel.

Figure 78-1: Thrust Reverser Systems



78-33-01 Thrust Reverser System Accumulators

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
78-33-01	Thrust Reverser System Accumulators	N	2	0	C (M) May be inoperative provided it is verified that no external leakage exists.

MEL ITEM	TAB LOCATION	PLACARD TEXT
78-33-01	Below Engine Display	____ T/R Accumulator(s) INOP

(M) PROCEDURES

1. Open the Thrust Reverser Hydraulic Control Unit (HCU) access door.
2. Inspect the HCU area for hydraulic fluid leakage.

78-33-02 Thrust Reverser Linear Variable Transformers (LVT)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
78-33-02	Thrust Reverser Linear Variable Transformers (LVT)	N	4	2	C	One LVT may be inoperative on each reverser.

MEL ITEM	TAB LOCATION	PLACARD TEXT
78-33-02	Below Engine Display	____ LVDT(s) INOP

78-33-03 Thrust Reverser Proximity Sensors

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
78-33-03	Thrust Reverser Proximity Sensors	Y	8	4	C	One per door may be inoperative.

MEL ITEM	TAB LOCATION	PLACARD TEXT
78-33-03	Below Engine Display	____ Thrust Reverser Proximity Sensor(s) INOP

OPERATIONS NOTE:

1. Do not dispatch if amber U/L indication is present on EAD.
2. If ENG SYS L/R FAULT alert appears, contact MCC for guidance.

MAINTENANCE NOTE:

With certain combinations of more than one Proximity Sensor inoperative on an engine, the associated ENG L/R SYS FAULT alert will be present and must be deferred per [MEL 73-21-01](#).

78-33-04 Maintenance Test Enable Switch (MTES)

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY			
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
78-33-04	Maintenance Test Enable Switch (MTES)	N	2	0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
78-33-04	Maintenance Test Enable Switch	Maintenance Test Enable Switch INOP

79 Engine Oil

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79-36-01	ENG L/R OIL FILTER Annunciation Systems

79-31-01 Engine Oil Quantity Indicating Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH			
FLIGHT CREW MAY PLACARD			N	2	0	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION						REMARKS OR EXCEPTIONS
79-31-01	Engine Oil Quantity Indicating Systems "Contact MCC"					B	(O) (M) May be inoperative provided: 1. Associated engine oil quantity is verified (visual or dip stick check) filled to recommended capacity at each refueling, PRIOR TO EACH DEPARTURE (Log Book Entry Required), and 2. Associated Engine Oil Temperature and Oil Pressure Indicating Systems are operative. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
79-31-01	Below Captains N.D. Panel	____ Engine Oil Quantity Indicator(s) INOP

Note: The AMM specifies time limits for oil servicing after engine shutdown. If it is necessary to run or dry motor the engine prior to servicing, the flight crew may accomplish this task if qualified maintenance personnel are not available.

(O) PROCEDURES

1. Verify that engine oil quantity has been checked.
2. Periodically monitor engine oil pressure and temperature indications on Secondary Engine Display.

(M) PROCEDURES

PRIOR TO EACH DEPARTURE, (Log Book Entry Required):

At each refueling, visually verify oil quantity within 30 minutes of engine shutdown using the sight glass on the engine oil tank.

79-36-01 ENG L/R OIL FILTER Annunciation Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD			1	REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION	N		REMARKS OR EXCEPTIONS	
79-36-01	ENG L/R OIL FILTER Annunciation Systems	N	2	1	B (M) May be inoperative provided: 1. It is verified that the Oil Pressure Filter is not bypassed, and 2. Oil Pressure Filter (Main Filter) is inspected ONCE EACH FLIGHT DAY (Log Book Entry Required), and verified to be CLEAN. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
79-36-01	Below Captains N.D. Panel	____ Engine OIL STRAINER CLOG Light INOP

(M) PROCEDURES

1. Verify Oil Filter Poppet is not extended ([AMM 79-34-02-200-801](#)).
2. At initial MEL application and thereafter ONCE EACH FLIGHT DAY:
 - A. Remove the Oil Pressure Filter (Main Oil Filter) ([AMM 79-21-02-000-801](#)).
 - B. Inspect the Oil Pressure Filter and verify it is clean ([AMM 79-00-00-200-802](#)).
 - C. Install Oil Pressure Filter ([AMM 79-21-02-400-801](#)).
 - D. Accomplish required logbook entry.

80 Engine Starting

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80-11-01	START VALVE L/R OPEN Annunciation Systems
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	STARTER SHUTOFF SYSTEM
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80-11-02B	Solenoid Coils
80-11-03	Auto Starter Control System
80-11-03	Auto Starter Control System
80-11-04	L/R Engine Start Switch Internal Switchlight
80-11-04	L/R Engine Start Switch Internal Switchlight

80-11-01 START VALVE L/R OPEN Annunciation Systems

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS	
80-11-01	START VALVE L/R OPEN Annunciation Systems	N	2	0	C	(M) May be inoperative provided it is verified that the associated Starter Shutoff Valve is CLOSED after starting and PRIOR TO EACH DEPARTURE. Repetitive Check Required (Yellow Placard)

MEL ITEM	TAB LOCATION	PLACARD TEXT
80-11-01	Below Captains N.D. Panel	___ START VALVE OPEN Light(s) INOP

(M) PROCEDURES

PRIOR TO EACH DEPARTURE:

For engine start, coordinate with flight crew as follows:

1. Gain access to the affected Start Valve prior to engine start.
2. Advise flight crew to start engine as normal.
3. When engine has reached stabilized idle, visually verify Start Valve position indicator on the valve housing shows the valve is CLOSED.

80-11-02 Starter Shutoff System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH	
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY	
ITEM NO.	NAME/DESCRIPTION				REMARKS OR EXCEPTIONS
STARTER SHUTOFF SYSTEM					
80-11-02A	Valves "Contact MCC"	N	2	0	C
(M) Valve may be inoperative CLOSED provided, PRIOR TO EACH DEPARTURE: 1. Alternate starting procedures are used, and 2. Associated Igniter System operates normally. Repetitive Check Required (Yellow Placard)					
80-11-02B	Solenoid Coils	N	4	2	C
Note: If two Solenoid Coils are inoperative on the same Valve, the Valve will be considered inoperative.					

MEL ITEM	TAB LOCATION	PLACARD TEXT
80-11-02A	Associated Engine Start Switch	____ Starter Valve(s) INOP
80-11-02B	Associated Engine Start Switch	____ Starter Valve Solenoid Coil(s) INOP

(M) PROCEDURES

80-11-02A

PRIOR TO EACH DEPARTURE:

For engine start, coordinate with flight crew as follows:

1. Gain access to the affected Start Valve prior to engine start.
2. Advise flight crew to command (pull) ENG START Switch to ON.
3. Manually open Start Valve by rotating the manual drive fitting on the valve body.
4. Advise flight crew to instruct when engine has reached maximum motoring speed (approximately 20% N2).
5. Advise flight crew to select FUEL Switch to ON.
6. Upon instruction from the flight crew, close the Start Valve using the manual drive fitting.

80-11-03 Auto Starter Control System

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD				REPAIR CATEGORY		
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
80-11-03	Auto Starter Control System	Y	2	0	C	(O) Automatic Start System may be inoperative provided manual start procedures are used.

MEL ITEM	TAB LOCATION	PLACARD TEXT
80-11-03	Ignition Panel	____ Auto Starter Control System INOP

(O) PROCEDURES

For engine start, utilize manual start procedure from [Vol I, Supplementary Procedures, Engines/APU](#).

80-11-04 L/R Engine Start Switch Internal Switchlight

NUMBER INSTALLED				NUMBER REQUIRED FOR DISPATCH		
FLIGHT CREW MAY PLACARD			REPAIR CATEGORY			
ITEM NO.	NAME/DESCRIPTION					REMARKS OR EXCEPTIONS
80-11-04	L/R Engine Start Switch Internal Switchlight	Y	2	0	C	

MEL ITEM	TAB LOCATION	PLACARD TEXT
80-11-04	Associated Engine Start Switch	____ ENG MANUAL START Light(s) INOP